Electromagnetic Torque in Tokamaks with Toroidal Asymmetries

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For my grandfather George Fredrick Logan Sr.

Abstract

Toroidal rotation and rotation shear strongly influences stability and confinement in tokamaks. Breaking of the toroidal symmetry by fields orders of magnitude smaller than the axisymmetric field can, however, produce electromagnetic torques that significantly affect the plasma rotation, stability and confinement. These electromagnetic torques are the study of this thesis.

There are two typical types of electromagnetic torques in tokamaks: 1) "resonant torques" for which a plasma current defined by a single toroidal and single poloidal harmonic interact with external currents and 2) "nonresonant torques" for which the global plasma response to nonaxisymmetric fields is phase shifted by kinetic effects that drive the rotation towards a neoclassical offset. This work describes the diagnostics and analysis necessary to evaluate the torque by measuring the rate of momentum transfer per unit area in the vacuum region between the plasma and external currents using localized magnetic sensors to measure the Maxwell stress. These measurements provide model independent quantification of both the resonant and nonresonant electromagnetic torques, enabling direct verification of theoretical models.

Measured values of the nonresonant torque are shown to agree well with the perturbed equilibrium nonambipolar transport (PENT) code calculation of torque from cross field transport in nonaxisymmetric equilibria. A combined neoclassical toroidal viscosity (NTV) theory, valid across a wide range of kinetic regimes, is fully implemented for the first time in general aspect ratio and shaped plasmas. The code captures pitch angle resonances, reproducing previously inaccessible collisionality limits in the model. The complete treatment of the model enables benchmarking to the hybrid kinetic MHD stability codes MARS-K and MISK, confirming the energy-torque equivalency principle in perturbed equilibria. Experimental validations of PENT results confirm the torque applied by nonaxisymmetric coils is often proportional to the energy put into the dominant ideal MHD kink mode. This reduces the control of nonresonant torque to a single mode model, enabling efficient feed forward optimization of applied fields. Initial results including the anisotropic kinetic pressure tensor directly in the plasma eigenmode calculations are presented here, and may eventually provide accurate metrics for multimodal coupling similar to the established single mode metrics.

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Chapter 1

Introduction

A tokamak is a toroidal magnetic confinement fusion device, in which magnetic fields are used to confine a quasi-neutral plasma. The field lines in a tokamak plasma lie on surfaces of constant flux, being fully comprised of a toroidal and poloidal component parallel to the surface. The toroidal component of this field is produced primarily by electromagnets, large coils encompassing the plasma poloidally. Typical values of the toroidal field in a tokamak plasma are on the order of 1-10 Tesla. The poloidal component is produced primarily by toroidal currents within the plasma itself. These currents are typically driven by more external coils, using the plasma as the secondary in a transformer. The amount of plasma current is restricted by stability considerations, and the poloidal field produced is typically an order of magnitude below the toroidal field. Vertical elongation of tokamak plasmas has advantages for stability, and additional poloidal field coils can be used to shape the plasma. The term tokamak has thus come to refer to a wide variety of devices, ranging from large aspect ratio, limited, circular cross section configurations to highly elongated, diverted, and nearly spherical devices [1].

Tokamaks, and tokamak plasmas, are nominally symmetric in the direction along the toroidal axis. In practice the discrete toroidal field coils, small distortions and tilts in the toroidal and vertical field coils, and unplanned field sources such as power lead busworks create unavoidable nonaxisymmetric fields on the order of 10^{-4} times the axisymmetric toroidal field component. These nonaxisymmetries can be amplified by the plasma. Ideal kink modes (KMs), resistive wall modes (RWMs) and neoclassical tearing modes (NTMs) are all common in modern tokamaks and can create relatively large nonaxisymmetric fields ($\sim 10^{-3}$ times the axisymmetric field) in the plasma.

The stability of tokamak plasmas has been found to be sensitive to these small, but unavoidable, nonaxisymmetries due to their large influence on the toroidal plasma rotation. Rotation plays a central role in the shielding of external error fields [2], stability of resistive wall modes [3, 4, 5], stability of neoclassical tearing modes [6], and the formation of transport barriers [7]. Rotation can be intrinsic [8], or it can be induced by external sources of torque such as tangential neutral beam injection (NBI). Toroidal symmetry breaking creates Maxwell stresses in the plasma that do not integrate to zero, forming a new electromagnetic channel of momentum exchange between the plasma and external world.

In order to better control tokamak plasmas, it is thus of great importance to understand these torques. This dissertation presents modeling and measurement of various electromagnetic torques and the corresponding effects on tokamak rotation.

1.1 Electromagnetic Torque

In the transport timescale equation of motion for a rotating toroidal plasma perturbed from equilibrium by nonaxisymmetric fields, the change in the plasma angular momentum density L_φ with time is the sum of the second order toroidal torques on each flux surface [9],

$$\begin{aligned} \frac{\partial L_\varphi}{\partial t} &= \langle \mathbf{e}_\varphi \cdot (\delta \mathbf{J} \times \delta \mathbf{B}) \rangle - \langle \mathbf{e}_\varphi \cdot \nabla \cdot \mathbf{\Pi} \rangle \\ & -\frac{1}{V'} \frac{\partial}{\partial \psi} \left(V' \Pi_{i\psi\varphi} \right) + \left\langle \mathbf{e}_\varphi \cdot \sum_s \mathbf{S}_s \right\rangle . \end{aligned} \tag{1.1}$$

The left hand side of this equation represents the plasma inertia, while the terms on the right hand side represent flux surface averages of toroidal torques. These torques represent, from left to right, (1) electromagnetic torque from resonant field perturbations on low order rational surfaces in the plasma; (2) nonambipolar induced classical, neoclassical and paleoclassical viscosity torque; (3) micro-turbulence stresses; and (4) external momentum input such as neutral beam injection. Here $\mathbf{J}$ is the current density, $\mathbf{B}$ is the magnetic field, $\mathbf{\Pi}$ is the pressure tensor, V is the volume, $\mathbf{S}$ are generic momentum input sources, δ indicates perturbed quantities, prime indicates a derivative with respect to ψ and magnetic coordinates $(\psi, \vartheta, \varphi)$ are used. The first two terms arise only in the presence of nonaxisymmetric magnetic fields, and are sensitive to perturbations on the order of 10^{-4} times the axisymmetric equilibrium fields. The perturbations may be generated internally by a plasma instability, or externally by nonaxisymmetric coils and irregularities in the nominally axisymmetric magnet coils. Note that the second term is the equivalent of electromagnetic torque from cross field currents associated with nonambipolar transport [9]. Depending on the details of plasma equilibrium, the electromagnetic torque can slow, maintain, or increase the plasma rotation [10, 11].

A fundamental understanding of electromagnetic torques is essential for the most commonly held vision of magnetic confinement fusion reactors: large, ITER-like, tokamaks. In these large fusion grade devices, the relative angular momentum input from neutral beam injection will be small, while nonaxisymmetric field perturbations on the order of 10^{-4} times the equilibrium field will be unavoidable. It has been predicted that toroidal symmetry breaking may generate nonresonant torque on the order of or larger than the neutral beam injection torque [12], while the scaling for resonant error field penetration thresholds remain dangerously close to the expected $\delta B/B \sim 10^{-4}$ level [13, 14]. The electromagnetic torques, the first two terms in Eq. (1.1), thus have the potential to dominate the momentum balance in fusion devices.

The torque produced by nonaxisymmetric fields is often split into two categories: resonant and nonresonant torques. The first, is associated with fields with the same pitch as the helical fields laying on a flux surface inside the plasma. These resonant fields cause reconnection of field lines across flux surfaces near the resonance, opening magnetic islands that enable fast radial transport of particles and energy, and degrade the confinement properties of the tokamak [15, 2]. These macroscopic islands exchange momentum with the intrinsic error field [16], as well as eddy currents that flow in the surrounding structures in response to mode [17, 18, 11, 19]. The electromagnetic torque on island structures has attracted much attention due to its propensity to bifurcate [20], often causing a sudden halt of the plasma rotation, further growth of the island structure, and a total loss of confinement known as a disruption [21]. The second, "nonresonant", torque is associated with small nonaxisymmetric displacements of the plasma flux surfaces that do not cause reconnection or have a direct macroscopic effect on the equilibrium energy confinement properties of the plasma. These nonaxisymmetries give rise to an anisotropic pressure tensor, nonambipolar transport, and a second order electromagnetic torque however, which can significantly effect the plasma stability and confinement through its rotation. This torque is nonlinear with respect to the nonaxisymmetric fields, and has been modeled under various assumptions using semi-analytic models [22, 23, 24, 25] as well as δf Monte Carlo codes [26, 27]. It recently received much attention for its ability to drive rotation [10, 12, 28], as well as its importance in error field correction [29]. Both resonant and nonresonant torques have ultimately been observed to have significant effects on the stability and confinement, and are thus included in this dissertation.

The integral electromagnetic torque can be obtained directly using magnetic sensors to measure the Maxwell stress tensor on a surface surrounding the plasma. This technique allows comparison between measured torques and models of the resonant and nonresonant electromagnetic torques. The plasma angular momentum is also readily observable quantity in tokamaks, and specific torques are often inferred from the simplified momentum balance that occurs when only one or two terms dominate the right hand side of Eq. (1.1). The addition of torque inferred from the angular momentum measurements, with vetted models for the non-electromagnetic sources when non-negligible, provides a three way comparison between data and models. This dissertation compares new measurement techniques and new torque models, with applications optimizing control of electromagnetic torques and tokamak stability. Emphasis is placed on the nonresonant torques, for which a new model in perturbed equilibria has been developed and used to predict the coupling between applied nonaxisymmetric fields and the resultant toroidal torque.

1.2 Perturbed Equilibria

Tokamak plasmas naturally evolve to the state that minimizes the potential energy of the system subject to certain constraints. The ideal magnetohydrodynamics (MHD) equation of motion relates the fluid density (ρ) and velocity ($\mathbf{v}$), to the electromagnetic and pressure forces in the plasma,

$$\rho\left[\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v}\cdot\nabla\mathbf{v})\right] = \mathbf{J}\times\mathbf{B} - \nabla\cdot\mathbf{\Pi}. \tag{1.2}$$

The ideal MHD plasma evolves according to this, the continuity equation, energy equation, and Maxwell's equations using a generalized Ohm's law [30]. In a stable and axisymmetric plasma equilibrium, the first term on the left hand side of the momentum equation is identically zero and the second is typically negligible for thermal particles in large tokamaks (the ratio to the electromagnetic force is approximately ordered as the ion skin depth to the system gradient scale length). If the plasma distribution function is near Maxwellian in velocity space, the pressure force is well approximated by the gradient of an isotropic scalar pressure, ∇P. These approximations, together with the approximations such as charge neutrality and negligible displacement current inherent in the original ideal MHD equations, result in currents and fields that lie on simply nested flux surfaces of constant pressure ($\mathbf{J}\cdot\nabla P, \mathbf{B}\cdot\nabla P = 0$) in an axisymmetric equilibrium. The 1D force balance, $(\mathbf{J}\times\mathbf{B})_\psi = \partial P/\partial\psi$, gives the Grad-Shafranov equation [31, 32] used extensively to reconstruct two dimensional equilibria [33, 34, 35]. Note that a one dimensional force balance can also be obtained keeping the ψ components of the inertial term in Eq. 1.2, including the Coriolis and centrifugal forces if the plasma rotation is high enough. Although methods exist [36, 37, 38, 39, 40, 41, 42], the full three dimensional force balance ($\mathbf{f} = \mathbf{J} \times \mathbf{B}$ for general force $\mathbf{f}$) is a much more difficult to computational challenge.

The full three dimensional force balance, $\mathbf{f} = \mathbf{J} \times \mathbf{B}$, is much more difficult to solve.

Nonaxisymmetric fields are small enough in stable tokamak equilibria that the force balance is strongly ordered, and the nonaxisymmetric equilibrium can be determined using a perturbative solution [43, 44, 45]. Neglecting the inertial terms, the Grad-Shafranov equation solves the zeroth order axisymmetric equilibrium described by nested flux surfaces and flux function pressure and current profiles. Using these equilibrium quantities, a first order force balance is then solved,

$$\nabla \cdot \mathbf{\Pi} = \delta \mathbf{J} \times \mathbf{B} + \mathbf{J} \times \delta \mathbf{B}. \tag{1.3}$$

Again, the assumption of a Maxwellian distribution function reduces the pressure tensor to a scalar pressure and reduces the vector force balance to the balance of a single normal component. This new force balance, in the ideal limit, is equivalent to a generalized Newcomb's equation [46].

1.2.1 The Direct Criterion of Newcomb

The Direct Criterion of Newcomb (DCON) code determines the stability of nonaxisymmetric equilibria using ideal MHD and a generalization of the method developed by Newcomb for cylindrical plasmas [47, 46]. The method utilizes adaptive numerical integration of M coupled second order ordinary differential equations, minimizing the potential energy of small perturbations about the axisymmetric equilibrium subject to the ideal MHD constraints. Here M is the number of coupled poloidal harmonics used to represent the displacement in Fourier space of a single toroidal harmonic n. This matrix Euler-Lagrange equation provides

M neighboring equilibria, characterized by the M eigenmode displacements that minimize the perturbed energy δW.

Coupling to a vacuum region outside the plasma includes free boundary modes and has the ability to include the stabilizing effects of a conducting surface surrounding the plasma. The DCON code does not, however, couple to externally applied nonaxisymmetric fields. In order to find stable nonaxisymmetric equilibria produced by the presence of nonaxisymmetric fields in a tokamak, the ideal perturbed force balance must be solved using the known external fields.

1.2.2 Ideal Perturbed Equilibrium Code

The Ideal Perturbed Equilibrium Code (IPEC) solves the linearized ideal force balance when stable axisymmetric equilibria are perturbed by external nonaxisymmetric fields [44, 45]. Each of the M eigenmodes calculated by DCON corresponds to a perturbed equilibrium if an external magnetic field produces the required force to support its distinct displacement of the plasma surface. Using the M neighboring equilibria calculated by DCON, IPEC constructs the plasma inductance and surface inductance linear matrix operators relating the external magnetic perturbation on a control surface to the total and external fluxes respectively. The plasma permeability, a third linear operator constructed from the two inductance matrices, relates the total and external flux. If the external field is then specified on this control surface, the total flux can then be determined and expanded in the DCON eigenmode basis. The perturbed equilibrium is then completely defined by the expansion of the corresponding combination of DCON eigenfunction displacements.

IPEC has been used extensively as a predictive and analysis tool in multiple machines [48, 49, 50, 51, 52, 53, 29], as well as for sensitivity projections for ITER [13, 54]. It has been widely used for the ability to predict plasma amplification and shielding of applied fields [50, 49], and to predict optimal correction of intrinsic or proxy error fields [14, 55, 13, 29].

It has also been used extensively as the input for kinetic calculations of the torque due to neoclassical toroidal viscosity (NTV), and has provided the equilibria used in the Perturbed Equilibrium Nonambipolar Transport code [23, 51, 28, 53, 29, 56, 57].

1.2.3 Perturbed Equilibrium Nonambipolar Transport Code

The Perturbed Equilibrium Nonambipolar Transport (PENT) code calculates the cross field transport induced by nonaxisymmetric field perturbations and the resultant toroidal torque [56, 57, 25]. The torque can be derived from the perturbed anisotropic pressure tensor, and is referred to as torque due to Neoclassic Toroidal Viscosity (NTV). Physically, the torque is a result of electromagnetic forces that arise from the cross field current produced by nonambipolar transport.

The PENT code, like IPEC and DCON, uses the lowest order axisymmetric equilibrium pressure, safety factor, and other flux functions. It assumes a strongly ordered force balance, and uses the perturbed equilibrium displacements to calculate an additional second order force ($\sim \delta B^2$) that is not self consistently accounted for. That is, it does not re-solve the perturbed equilibrium force balance equation. In terms of the perturbed energy, it is applicable only in the limit in which the ratio of kinetic perturbed energy to the MHD perturbed energy is small $|\delta W_k/\delta W| \ll 1$. Beyond this limit the currents associated with the force are no longer negligible in the first order perturbed equilibrium. Within this limit, the PENT code is a useful model of the toroidal torque in a stable perturbed equilibrium such as calculated by DCON and IPEC. These codes are now packaged together, and have been validated in a wide variety of experimental and theoretical applications.

The PENT code also produces semi-analytic coefficient matrices for the kinetic modification of the ideal MHD Euler-Lagrange equation used in DCON. These coefficients depend only on the axisymmetric equilibrium properties, and can be self consistently incorporated into the DCON adaptive solver in order to calculate plasma eigenfunctions with kinetic effects self consistently included in the perturbations. This lends itself to a new generalized perturbed equilibrium solver, using the external field coupling techniques developed in IPEC to calculate self consistent perturbed equilibrium including the anisotropic kinetic pressure.

1.3 Overview of Dissertation

Chapter 1 motivates the need to understand electromagnetic torques in tokamaks and briefly outlines the fundamental tools used to study them in this dissertation. These tools include both experimental measurement techniques and theoretical models developed for this purpose.

Chapter 2 describes the magnetic diagnostics and data analysis techniques used in the measurement of electromagnetic torques in the DIII-D tokamak. Section 2.1 presents the Maxwell stress tensor technique for measuring electromagnetic torque in detail, outlining the basic requirements for magnetic sensors to make the measurement. Common attributes of the plasma response eigenstructures expected in DIII-D add additional constraints. The structures predicted by IPEC and the ability of the diagnostics to accurately reconstruct them are described in Section 2.2. This is followed by a brief overview of the design, fabrication, and distribution of the working DIII-D magnetic diagnostic suite in Section 2.3 before moving directly to data analysis techniques. Many of the analysis techniques, presented in Section 2.4, were applied to the DIII-D magnetic data for the first time to enable isolation of the coherent plasma response structures, extrapolation of the structures across the measurement surface, and ultimately the calculation of the Maxwell stress between the plasma and external field sources.

Chapter 3 presents electromagnetic (EM) torque measurements in DIII-D. Two distinct measurement techniques are defined and their uses presented in a number of practical examples. Section 3.1 reviews the general Maxwell stress tensor measurement technique, applicable for long wavelength magnetic fields structures at the vessel wall such as low mode number resonant fields and their associated eddy currents. Section 3.2, in contrast, shows torque between the poorly constrained applied field from in-vessel coils and the resultant plasma response can be recovered through a unique use of the vessel and coil geometries. Finally, Section 3.3 demonstrates the direct measurement of the EM torque in nonaxisymmetric perturbed equilibria. This observation provides motivation for the development of an accurate model for the measured torque and its dependencies on the applied fields.

Chapter 4 presents a generalized semi-analytic model describing NTV torque in perturbed tokamak equilibria. The established derivation from a perturbed anisotropic pressure tensor is presented in detail in Section 4.1, followed by a discussion of the root nonambipolar transport and a novel derivation of the NTV as electromagnetic torque in Section 4.2. Section 4.3 then outlines the importance of maintaining general geometry in the semi-analytic theory. The Perturbed Equilibrium Nonambipolar Transport code (PENT) is introduced, providing new computational modeling for theory. The chapter finishes with a detailed discussion of new physics captured by PENT in Section 4.4, properly treating the pitch angle dependencies in general geometry.

Chapter 5 gives some context for the PENT code, comparing and contrasting it with other computational models that include kinetic modifications of MHD equilibria. This chapter contains extensive benchmarking of PENT with two other codes, MISK[58] and MARS-K[59], using the equivalency principle [60] to relate the stability codes' kinetic energy and PENT's torque calculations. The application to resistive wall mode (RWM) stability used for the benchmarks serves to emphasize the fundamental differences between PENT and other computational tools as well as demonstrating the flexibility of the PENT code. After the equilibria and profiles used throughout the chapter are established in Section 5.1, the fundamental kinetic frequencies, energy integrated kinetic resonance operator, and perturbed action calculated by the various codes are compared in Section 5.2-Section 5.4. Finally, Section 5.5 compares the computed perturbed energies and NTV torques across kinetic regimes using extensive parametric scans.

Chapter 6 returns to experimental measurements of electromagnetic torque from DIII-D with comparisons to PENT torque predictions. It is shown that the model predicts NTV torque dominated by the coupling of external field perturbations to a dominant plasma response spectrum. The details and experimental validation of this single mode model are presented in Section 6.1. The reduction of the nonlinear NTV physics to a linear single mode model allows feed forward correction of unwanted nonaxisymmetric "error field" perturbations. Section 6.2 shows that the model accurately predicts the optimal error field correction coil currents across a range of scenarios. The control of the torque profile, however, requires multimodal effects. Explicit measurements of multimodal effects are then presented in Section 6.3, motivating the continued effort to more accurately model and predict the detailed coupling between applied fields and the NTV torque.

Chapter 7 presents initial results in the next generation of predictive nonaxisymmetric equilibrium and NTV modeling that will eventually improve the NTV coupling metrics as well as extend them to new regimes of applicability. The kinetic effects calculated by PENT are included, for the first time, directly in the perturbed equilibrium eigenfunction calculations. The new calculation enables the study of drift-kinetic MHD perturbed equilibria, self-consistently including the anisotropic kinetic pressure tensor in the perturbed force balance.

Finally, Chapter 8 provides a brief summary of the dissertations major results and discussion of important future applications.

Chapter 2

Nonaxisymmetric Magnetic Diagnostics in the DIII-D Tokamak

The DIII-D nonaxisymmetric magnetic diagnostic set was upgraded in 2013, greatly enhancing the capability to measure electromagnetic torques on both rotating and stationary plasma modes [61]. The previously existing magnetic diagnostics included roughly 125 sensors used to inform axisymmetric equilibrium reconstruction and plasma control as well as over 150 sensors for the detection of rotating and nonrotating nonaxisymmetric plasma modes [62]. The upgrade added 101 local field sensors for the detection and classification of nonaxisymmetric modes. Theses additional sensors extended the toroidal resolution of the existing nonaxisymmetric sensor set as well as the poloidal resolution and extent, allowing new physics to be addressed. The design and placement of these sensors, the unprecedented arrays on the High Field Side (HFS) in particular, relied extensively on predictive modeling and synthetic diagnostic optimization using the IPEC modeled plasma response in order to meet the desired physics goals. Once operational, improved data analysis techniques were needed in order to accurately isolate the physics of interest. This modeling and analysis work, as well as an overview of the hardware installed by DIII-D, is presented here.

The top physics objectives of the upgrade to the nonaxisymmetric magnetic sensor set in DIII-D included [61],

- Measurement of the plasma response to external fields:
  - Measurement of the response to applied Resonant Magnetic Perturbations (RMP) from external coils.
  - Measurement of the effects of intrinsic error fields.
- Direct measurement of Maxwell stress:
  - Measurement of resonant and nonresonant electromagnetic torque.
  - Measurement of the poloidal structure, including direct measurement on the High Field Side (HFS).
- Measurement of low frequency, long wavelength unstable plasma modes:
  - Measurement of the structure of rotating and stationary modes.
  - Measurement of the mode locking dynamics.
- Improved axisymmetric (averaged) and nonaxisymmetric equilibrium construction.
- Detection of high frequency, short wavelength instabilities.

The first three priorities are inseparably linked, with the second being the primary focus of this chapter. Design points devoted to the final two are reviewed briefly in [61], and are beyond the scope of this dissertation.

This chapter is organized as follows: Section 2.1 gives the fundamentals of the Maxwell stress tensor measurement technique, including a description of the limits and requirements of the measurement in DIII-D. The plasma response to applied nonaxisymmetric fields, of primary importance to the upgrade objectives and Maxwell stress requirements, is presented in Section 2.2. The design of the sensors is motivated by model projections of the plasma modes, addressing concerns such as the signal to noise ratio, spacing required to capture the poloidal structure, and sensitivity to sensor misalignment. The final sensor characteristics, fabrication and distribution are then detailed in Section 2.3. Section 2.4 describes the data analysis techniques developed to meet the requirements of the magnetics upgrade objectives. Vacuum compensation and mode identification techniques are presented in detail, concluding in an overview of their application to Maxwell stress measurements.

2.1 Electromagnetic Torque Measurements

Magnetic diagnostics can be used to directly measure the momentum transfer between the plasma and external nonaxisymmetric field sources. The requirements for this measurement combine the difficulty of nonaxisymmetric plasma mode measurements [63] with multi-component requirements of equilibrium reconstruction [33]. Although the method has been known for some time [18], it has only recently been developed in DIII-D [11] due to the difficult nature of the measurement. Enhancing the ability to measure the Maxwell stress was a major physics objective of the upgrade, and this section details the basic requirements of the measurement that influenced the design and distribution of the upgrade sensors.

At the foundation of the electromagnetic torque measurements is the Maxwell stress tensor. The Maxwell stress tensor $\boldsymbol{\sigma}$ provides a general method of calculating the angular momentum transfer between the plasma and surrounding electromagnetic sources such as external coils [18],

$$\frac{\partial L_\phi}{\partial t} = \int d\mathbf{s} \cdot \boldsymbol{\sigma} \cdot \mathbf{e}_\phi R. \tag{2.1}$$

Here L_ϕ is the toroidal angular momentum of the plasma, $\int d\mathbf{s}$ is a surface integral over any surface surrounding the plasma and excluding the external electromagnetic sources of interest, R is the major radius of each surface area element, $\mathbf{e}_\phi$ is the toroidal unit vector and $\boldsymbol{\sigma}$ is the maxwell stress tensor.

Evaluated across a surface such as the inner side of a tokamak vessel wall, Eq. (2.1) gives the volume integrated value of all electromagnetic torques inside the plasma. At any given point, the force in direction $\mathbf{e}_j$ per unit area across an infinitesimal area element with normal vector $\mathbf{e}_i$ is given by $\sigma_{ij} \equiv \mathbf{e}_i \cdot \boldsymbol{\sigma} \cdot \mathbf{e}_j$, where,

$$\sigma_{ij} = \epsilon_0 \left(E_i E_j - \frac{1}{2}\delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left(B_i B_j - \frac{1}{2}\delta_{ij} B^2 \right). \tag{2.2}$$

Here E is the electric field, B is the magnetic field, and δ_{ij} is the Kronecker delta.

2.1.1 Frequency Requirements

Electromagnetic torque measurements are naturally low frequency measurements. At high frequency the vessel wall acts as a near perfect conductor, shielding rotating nonaxisymmetric plasma perturbations. Long wavelength fast plasma instabilities are largely suppressed, leaving instabilities such as Resistive Wall Modes (RWMs) that develop on the vessel current resistive decay time to grow and dominate the global torque balance [64, 65, 66, 67]. Similarly, high frequency external fields are filtered by the vessel before they can act on the plasma. The systems of interest, in which momentum is exchanged between the plasma and external field sources through the electromagnetic fields, are thus constricted by this resistive time scale. The electric components of the Maxwell stress tensor are negligible for these systems,

$$\frac{\sigma^E}{\sigma^B} \sim \epsilon_0 \mu_0 \left(\frac{\omega}{k} \right)^2 \sim \frac{1}{c^2} \left(3\text{km/s} \right)^2 \ll 1, \tag{2.3}$$

reducing the stress measurement to a measurement of the magnetic field only. Here an approximate DIII-D wall time $\omega \sim 2\pi/\tau_w \sim 2 \times 10^3$rad/s and tearing mode wavenumber $k \sim 1/R_0 \sim 0.6\text{m}^{-1}$ were used as representative estimates.

In addition to the above considerations, vacuum calibration measurements have shown that the graphite tiles between the plasma and the sensors in DIII-D have an effective time constant of 0.14ms [11]. In the absence of external error fields, the eddy currents in the graphite tiles dominate the torque transfer above $\sim$ 1kHz and prevent the direct measurement of the Maxwell stress. As a result, the magnetic diagnostics and analysis techniques developed for the measurement of electromagnetic torque have been designed for low frequency ($<$ 1kHz) and stationary mode detection.

2.1.2 Field Component Requirements

In the low frequency limit the toroidal torque on the left side of Eq. (2.1) becomes,

$$\begin{aligned} T_{EM} &= \int d\mathbf{s}\cdot\boldsymbol{\sigma}\cdot\mathbf{e}_\phi R = \int dAR\frac{1}{\mu_0}B_\phi B_\perp \\ &= \sum_n \int d\theta\pi r R^2\frac{1}{\mu_0}\left[\Re\left(B_{n,\phi}\right)\Re\left(B_{n,\perp}\right)+\Im\left(B_{n,\phi}\right)\Im\left(B_{n,\perp}\right)\right], \end{aligned} \tag{2.4}$$

where B_ϕ and $B_\perp$are the field components in the toroidal and surface-normal directions respectively, r is the minor radius measured from the plasma center, and $\theta = \tan^{-1}(z/r)$ is the poloidal angle of a point at height z. The field components have been decomposed in toroidal Fourier harmonics,

$$B_i = \sum_n B_{n,i}\left(r,\theta\right)e^{in\phi}. \tag{2.5}$$

Measurement of the electromagnetic torque from the Maxwell stress requires simultaneous measurement of two field components across the surface $\mathbf{s}$. In the DIII-D tokamak this surface is the vessel's inner wall, on which the local magnetic sensors are located (see Section 2.3 for detailed description). The normal component is measured by inductive loop sensors. The component parallel to the surface perpendicular to the toroidal direction, rather than the toroidal component itself, is measured due to the relative sensitivity of the toroidal measurement to axisymmetric pickup. The measurement of these two components $(B_\perp, B_\parallel)$ is sufficient to completely describe the field in the vacuum region at the vessel wall.

Ampere's law relates the toroidal component of the vacuum magnetic field to the components typically measured at the tokamak vessel wall. On the infinitesimal surface element $d\mathbf{s}$, the magnetic field can be described by its components in orthogonal coordinates $\left(x_{\perp}, x_{\parallel}, x_{\phi}\right)$ in the generic form $B_i = \tilde{B}_i e^{i(k_\phi x_\phi + k_\parallel x_\parallel + k_\perp x_\perp)}$. Ampere's law, $\nabla \times \mathbf{B} = 0$, gives

$$B_\phi = \frac{k_\phi}{k_\parallel} B_\parallel. \tag{2.6}$$

If the field across the total surface is further decomposed into Fourier harmonics (m) of the poloidal angle,

$$B_i = \sum_{n,m} B_{n,m,i} e^{i(n\phi - m\theta)}, \tag{2.7}$$

Eq. (2.6) maps the measured parallel field to the toroidal component,

$$B_{n,m,\phi} = \frac{nr\left(\theta\right)}{mR\left(\theta\right)} B_{n,m,\parallel}. \tag{2.8}$$

The toroidal electromagnetic torque is then given directly from the measured Fourier harmonics $B_{n,m,\perp}$ and $B_{n,m,\parallel}$,

$$T_{EM} = \frac{1}{\mu_0} \sum_{n,m,m'} \frac{n}{m} B_{n,m,\parallel} B^*_{n,m',\perp} \int d\theta \pi R r^2 e^{i(m'-m)\theta}. \tag{2.9}$$

Note that the integral in poloidal angle can be completed analytically for a cylindrical vessel surface [68], but is left in this form to retain geometric effects of the vessel shaping (see Fig. 2.18). It is also possible to obtain an expression for the electromagnetic torque dependent on only a single measured field component by analytically relating components using a model of the external field source such as a thin wall model for the eddy currents induced by a rotating plasma mode [69, 17, 18, 68, 70]. Two components of the magnetic field are required calculate the torque independent of any plasma or external field model that may not capture all the important physics.

2.1.3 Mode Resolution Requirements

The spatial resolution, or equivalently the mode spectrum resolution, requirements of Maxwell stress measurements are determined by the physics of interest. There is no toroidal torque associated with the axisymmetric magnetic field, making the Maxwell stress tensor inherently a nonaxisymmetric measurement. This, combined with the frequency limit discussed above, means that the fields of interest are nonaxisymmetric fields from low frequency and stationary modes. These are the driven and unstable plasma modes and their associated external fields described in the top priorities of the magnetics upgrade.

The large instabilities in DIII-D responsible for dramatic changes in the angular momentum are long wavelength MHD modes such as neoclassical tearing modes [69, 17, 18, 71] and resistive wall modes [64, 65, 66, 67]. These have low toroidal mode numbers ($n \lesssim 3$) and poloidal spectra dominated by low mode numbers ($|m| \lesssim 12$) determined by either a single resonant surface pitch ($q = m/n$) or a single kink mode spectrum [72, 73, 74, 13, 75, 29].

The dominant mode spectra Fourier decomposed in straight field line coordinates using the geometric toroidal angle (Princeton Equilibrium, Stability, and Transport PEST coordinates [76]) at the plasma surface for a number of widely varied DIII-D scenarios are shown in Fig. 2.1. These are the external spectra that IPEC predicts have the strongest resonant plasma response [52]. Some of them do have significant components at larger m. A combination of modified Bessel function $\sim K_m(nr/R_0)$ decay [18] between the plasma and measurement surface and plasma shaping that tends to expand the magnetic coordinates near the midplane results in measurements dominated by lower cylindrical poloidal mode numbers. The detailed geometric coupling of the measurement surface to common plasma modes is presented in the following section.

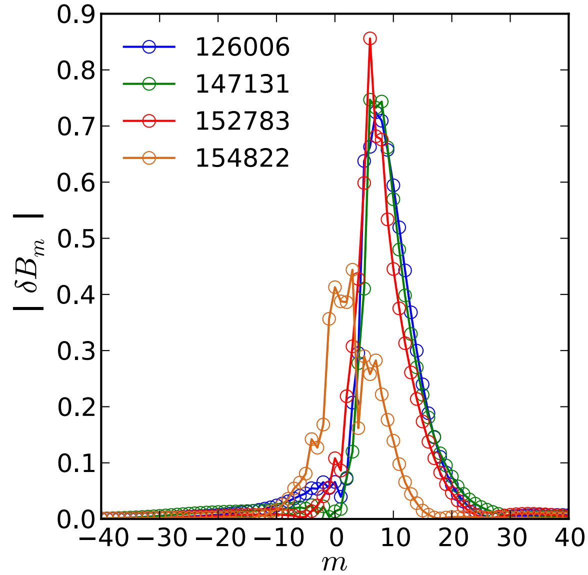

Figure 2.1: The dominant mode external field spectra as calculated by IPEC for a a variety of DIII-D applications including ELM suppression (126006), H-mode EFC (147131), Ohmic EFC (152783), and low-q stability control (154822). The nonaxisymmetric perturbations are on the plasma surface, have mode number $n = 1$, and have been decomposed in poloidal modes m using PEST coordinates.

2.2 Plasma Response Measurements

Each objective of the magnetic field diagnostics upgrade is dependent on the same fundamental challenges associated with measuring a specific signal amplitude and spacial structure in the presence of background and noise. Nonaxisymmetric plasma equilibrium modeling was used extensively throughout the design process to quantify the measurement requirements. This section presents the Ideal Perturbed Equilibrium Code (IPEC) modeling of the plasma response to applied nonaxisymmetric perturbations used in the design of the magnetic diagnostics. Although no net electromagnetic torque exists in ideal MHD, the predicted perturbations from IPEC have been shown to agree with previously available measurements for plasmas below the no-wall stability limit [48, 77, 63]. IPEC solves the linearized and first order equilibrium force balance, while the toroidal torque balance from Eq. 1.1 is second order in the small perturbation expansion. Thus, the results in this section can be taken as a guide for the field measurement requirements necessary to infer electromagnetic torque from applied nonaxisymmetric fields in DIII-D but not a self consistent prediction of the torque itself.

2.2.1 Applied Nonaxisymmetric Spectrum

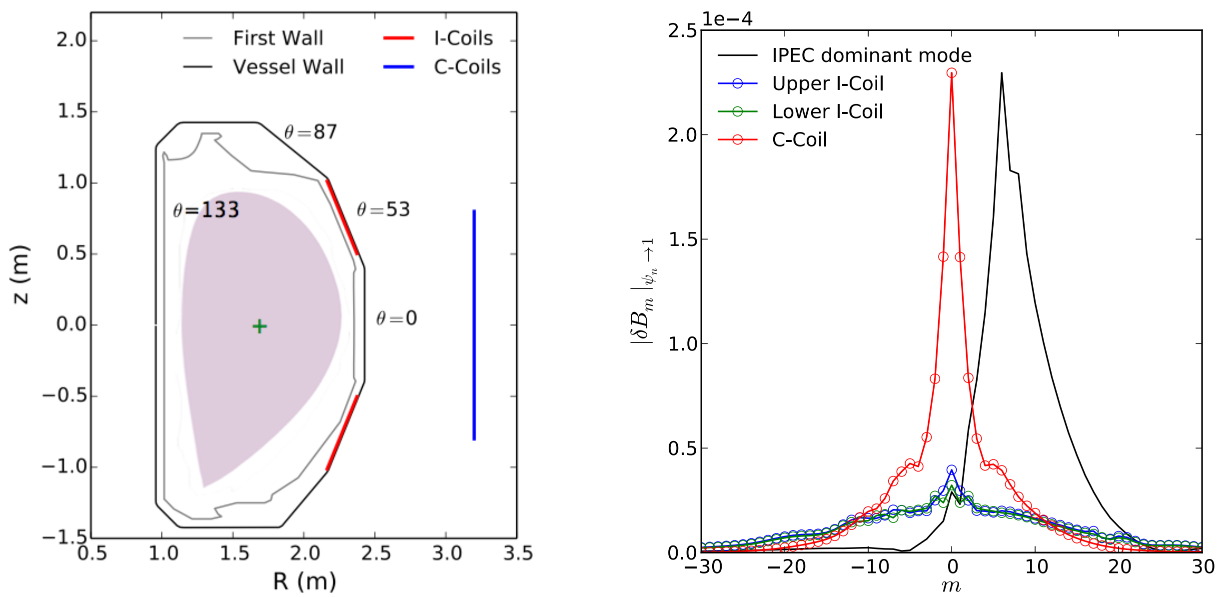

Figure 2.2: Poloidal cross-section of the DIII-D vessel (left), including the positions of the in-vessel (I) and external correction (C) coil arrays. The nominal major radius ($R_0 = 1.69$m) is shown by a green "+", a typical low single null plasma shape is shown in purple and select poloidal angles are given for reference. Vacuum poloidal spectra (right) at the plasma surface of a lower single null shot 152783 for 1kA $n = 1$ waveforms in each of the coil sets show the applied spectra are dominantly low m. Also shown is the IPEC dominant mode spectrum for this equilibrium (black).

Toroidal arrays of error field correction (EFC) coils are regularly used for a wide variety of applications including RWM stability control [78, 66, 79, 80], error field correction [81, 66, 50, 14, 29], entrainment of magnetic islands [82, 83, 84] and application of nonresonant magnetic torque [10, 12, 51]. It is the goal of the magnetics upgrade to study the physics of each of these applications, and the design of the diagnostic set is optimized to measure the plasma response to the vacuum fields applied by the DIII-D coil sets.

The DIII-D tokamak has three toroidal arrays of nonaxisymmetric field coils shown in Fig. 2.2: two in-vessel arrays mounted to the vessel wall above and below the midplane and a single external array centered on the midplane. Each consists of six large picture-frame coils extending approximately 60° in the toroidal direction. This enables clean toroidal spectra for mode numbers $n \leq 3$, although some pollution of the sidebands has been observed [85].

The poloidal spectra from the individual nonaxisymmetric field coil sets are shown on the surface of a representative plasma in Fig. 2.2, and stay relatively consistent despite dramatic variations in the equilibrium properties and shaping [29]. These spectra at the plasma surface are calculated using Biot-Savart law calculations for the vacuum field from line currents following the geometric path of the coils. The three arrays can be linearly combined to create a wide variety of spectra, although the physically relevant modes are the low m spectrum that interacts most strongly with the plasma. Thus, the measurement of plasma response to applied nonaxisymmetric fields in DIII-D has similar resolution requirements ($n \leq 3, |m| \leq 12$) as the measurement of low frequency unstable MHD.

2.2.2 Predicted Structure

In designing the diagnostics required to measure new elements of the plasma response to nonaxisymmetric fields, extensive use was made of IPEC to predict the sensor requirements from the modeled field structures of interest. The existing vessel geometry formed the obvious basis on which any sensor set would be built, and thus a mapping of the plasma fields to the vessel wall provides a high resolution synthetic diagnostic. On this surface, the field is decomposed into the two measured components: one normal to the wall ($b_\perp$) and a second in the poloidal plane parallel to the vessel cross-section ($b_\parallel$). The amplitude and structure of these components on the vessel wall determine the required size and spacing of the diagnostic set needed to reconstruct the plasma response.

A typical plasma response to identical 1kA $n = 1$ toroidally sinusoidal waves in the upper

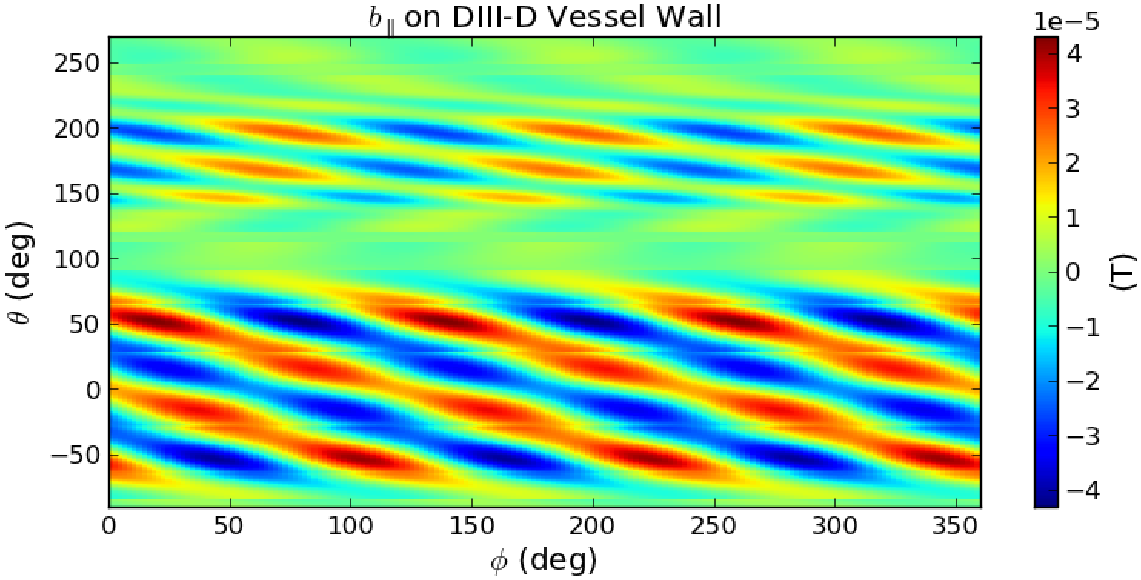

Figure 2.3: Contour plot of the plasma response poloidal field parallel to the vessel wall as calculated by IPEC for a representative RMP equilibrium, shot 126006, with 1kA even parity $n = 1$ I-coil currents. The wall is characterized by the toroidal angle (ϕ) and poloidal angle (θ) defined from machine major radius $R_0 = 1.69$m. The poloidal angle is 0 at the LFS midplane $z = 0$m. The model shows large 60° sections on the top and bottom of the vessel are dead-zones for plasma response measurements.

and lower I-coil arrays is shown in Fig. 2.3. The current distributions are identical in the sense that they have the same amplitude and toroidal phase (0° phasing or "even parity"). The modeled plasma response can be immediately divided into two poloidal regions of coherent structure: the large LFS region extending $\sim 180°$ from the bottom to the top of the vessel, and the $\sim 60°$ HFS region centered around the mid-plane. Note that the poloidal angle in Fig. 2.3 has been rotated by $-90°$ from the standard reference angle in order to smoothly connect these regions. Between the LFS and HFS regions lie dead zones corresponding to the top and bottom of the vacuum vessel. Here, the falloff over increased distances between the plasma and the measurement combines with phase mixing from short cylindrical wavelengths corresponding to compressed magnetic coordinates near the elongated ends of the plasma to effectively eliminate a measurable magnetic structure at the vessel wall.

The consistent prediction of de-correlated, small amplitude signals in the top and bottom of the vessel combined with the engineering constraints of limited access due to limiter structures resulted in a design concentrated on the two distinct mid-plane centered LFS and HFS regions. In each of these, synthetic diagnostic sets of local sensors were constructed and evaluated using the IPEC model's plasma response predictions. The remainder of this section will review the validation of these synthetic diagnostics necessary to accurately measure such plasma response spectra.

Examples in this section will exclusively show the plasma response to even parity 1kA sinusoidal currents in the I-coil arrays. Although each single array and many combinations of coil phasings were considered in the physics validation review of the magnetics upgrade, this scenario represents one of the most challenging studied and exemplifies the main challenges addressed in the design of the new diagnostic set.

2.2.2.1 Low Field Side Plasma Response

The pre-upgrade nonaxisymmetric magnetic diagnostics on the DIII-D LFS consisted of primarily three toroidal arrays on the inner vessel wall at $\theta \approx -53°, 0°, 53°$ for both components $(b_\perp, b_\|)$ as well as poloidal arrays for the classification of rotating instabilities [62]. The toroidal arrays were optimized for $n = 1$, and hardware differencing of 180° separated pairs eliminated coupling to $n = 0$ as well as all even toroidal mode numbers. Only the midplane array contained 4 pairs, enough to simultaneously fit the $n = 1$ and $n = 3$ modes. A primary mission of the upgrade was to fill in these arrays, adding toroidal locations and asymmetric differencing schemes to optimize toroidal mode fitting for simultaneous detection of multiple (quasi-)stationary modes $n = 1 - 3$.

The three poloidal locations of the original sensor arrays correspond to symmetry points for the I-coil vacuum fields, optimizing the signal-to-noise for the parallel field magnetic probe arrays. This can be seen by the ratio of the vacuum to plasma response field at these points in Fig. 2.4. Locations between the pre-existing arrays are characterized by steep gradients

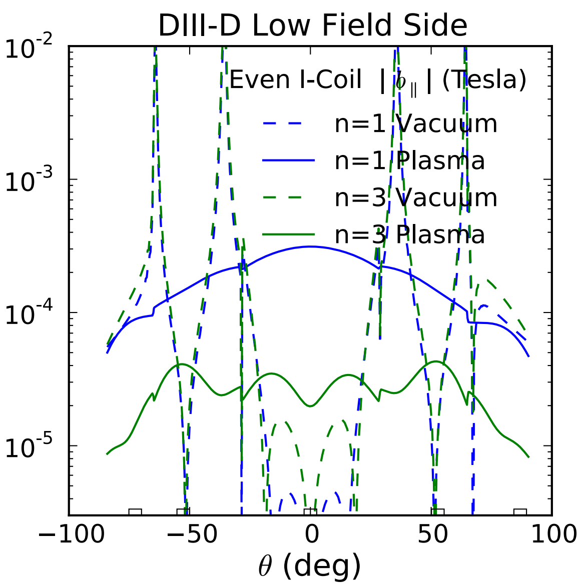

Figure 2.4: The LFS vacuum field and plasma response field parallel to the DIII-D vacuum vessel wall as modeled by IPEC for shot 126006 with 1kA $n = 1$ and $n = 3$ even parity I-coil currents. The field amplitudes in Tesla are shown as a function of poloidal angle (left) and approximate sensor array locations are indicated (black squares). The plasma response field poloidal spectrum in wavenumber (right) is calculated using an FFT of the parallel field in this region of the vessel wall.

in the applied vacuum fields, and thus heightened sensitivity to misalignment errors as well as decreased signal-to-noise. Relatively low gradients, and high signal-to-noise is recovered near $\theta = \pm 87°$, where new arrays were added in the upgrade to extend the poloidal range of measurements.

Fig. 2.4 also shows that the poloidal spacing of roughly 0.75m between the original arrays is comparable to the expected poloidal wavelength of the $n = 3$ plasma response. Here, a Fast Fourier Transform (FFT) has been performed on the parallel field projection across the length of LFS poloidal cross-section. The vast majority of the mode structures are contained in $k_\parallel \lesssim 12$, which corresponds to a wavelength of $\sim$ 1m. The main wavefront, however, is characterized by a dominant wavelength $\lambda_{dom} \lesssim$ 2m. Spatial aliasing is expected when the measurement spacing is greater than the Nyquist criterion ($\lambda_{sample} \geq \lambda/2 \sim$ 1m). This allows confident measurement of the dominant wavefront, although the shorter wavelength behavior associated with sharp kinks in the vessel surface are sacrificed in reconstruction (see Sec. 2.2.3). Note that any remaining spacial aliasing of short wavelength contributions is further eliminated in experiments using SVD analysis that identifies coherent mode structures in space and time (see Sec. 2.4.2).

2.2.2.2 High Field Side Plasma Response

The pre-existing magnetic diagnostics on DIII-D did not have the capability to measure low frequency or stationary plasma modes on the HFS, leaving the design of the magnetics upgrade diagnostics in this region heavily dependent on predictive modeling capabilities. IPEC predicts the highest plasma response signal is concentrated in a $\sim$ 1m band centered on the mid-plane. For the plasma response to $n = 3$ even parity I-coil fields, the signal beyond this limited band often became unacceptably low (below $\sim 3 \times 10^{-5}$T/kA). Fig. 2.5 shows the predicted plasma response at the LFS wall for examples applying $n = 1$ and $n = 3$ even parity I-coil fields. The corresponding Fourier spectrum is also shown. Typical poloidal wavelengths in the measurable band are in the range of $0.2 - 2.0$m, with a corresponding Nyquist condition of approximately 0.1m vertical spacing between measurements.

The compact structure of the plasma response on the HFS motivated the design of new compact saddle loop coils that could be mounted together with $b_{\parallel}$ sensors in dense arrays. Although effective area requirements and limited space between the legs of protective graphite tiles mandated that these have a vertical length of $\sim$ 0.15m, the solution of Laplace's equation in the vacuum region can be used to combine the $\parallel$ and $\perp$ component measurements to reduce the effective measurement spacing to $\sim$ 0.075m [61]. The corresponding wavenumber limit $k_{\parallel} \lesssim 42$ safely encompasses the predicted spectra. This technique is dependent on the source being completely external to the measurement surface (the vertical column of the center stack) however, and cannot be applied for Maxwell stress measurements that require deviations from the 90° phase difference otherwise assumed between $b_{\perp}$ and $b_{\parallel}$.

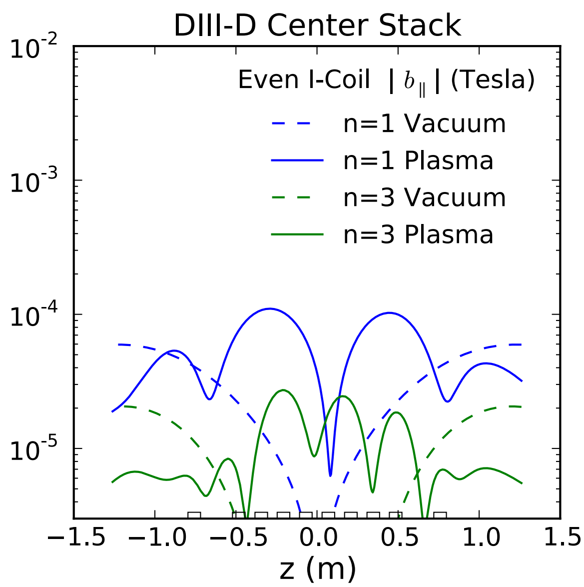

Figure 2.5: The HFS vacuum field and plasma response field parallel to the DIII-D vacuum vessel wall as modeled by IPEC for shot 126006 with 1kA $n = 1$ and $n = 3$ even parity I-coil currents. The field amplitudes in Tesla are shown as a function of poloidal angle (left) and approximate sensor array locations are indicated (black squares). The plasma response field poloidal spectrum in wavenumber (right) is calculated using an FFT of the parallel field in this region of the vessel wall.

2.2.3 Synthetic Diagnostic Reconstruction

The power of synthetic diagnostics is that they can be manipulated to test the ability of a discrete set of local magnetics sensors to reconstruct the known surface distribution. Local synthetic diagnostics were developed in IPEC for the DIII-D upgrade effort, consisting of two dimensional surfaces defined by their center point (r, z, ϕ), toroidal extent, cross-sectional length, and cross-sectional tilt. The full three dimensional field is then mapped to and integrated across these surfaces in order to get the appropriate signal value $\bar{b}_\parallel$ or $\bar{b}_\perp$. From a set of P such discrete signal values, the model field can then be reconstructed using the analysis techniques used in the experiment (see Sec. 2.4.2). Comparison between the model and reconstructed field then gives a direct test of the diagnostic set's ability to accurately capture the plasma response.

The magnetics upgrade sensor sets were extensively tested using synthetic diagnostics

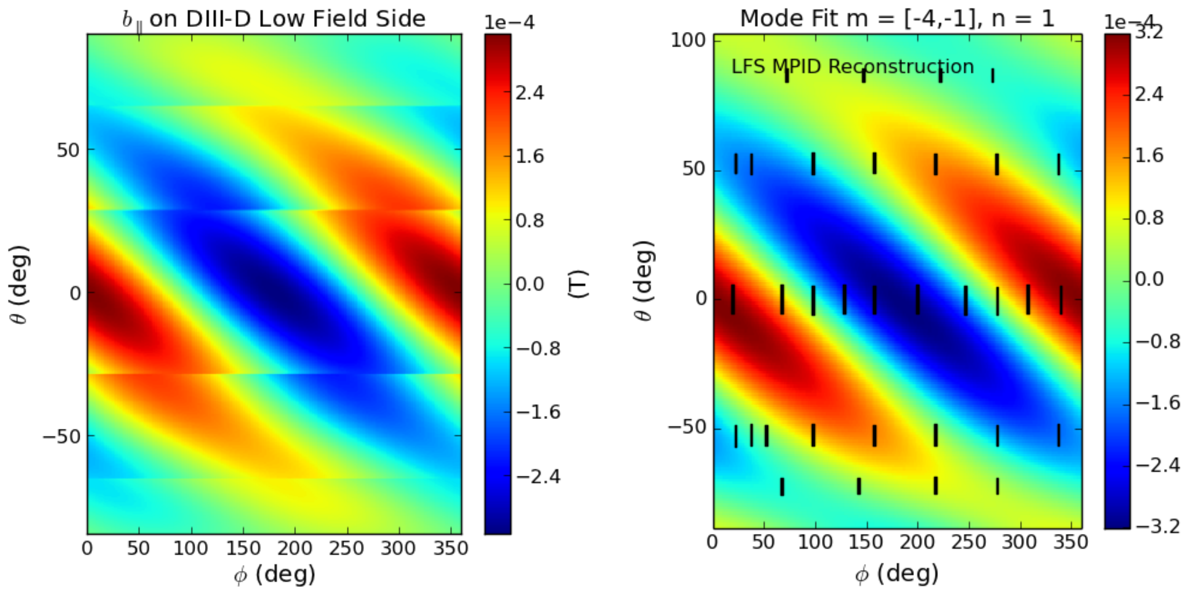

Figure 2.6: The plasma response to 1kA $n = 1$ even parity I-coil currents calculated by IPEC at the DIII-D LFS vessel wall (left) and the field reconstructed from a discrete set of synthetic diagnostics (right) shown in black. The reconstruction assumes a single $n = 1$ toroidal harmonic and includes low, single helicity poloidal mode numbers $-4 \leq m \leq -1$.

prior to their fabrication and installation in DIII-D. Fig. 2.6 shows a synthetic reconstruction of the plasma response field using the toroidal arrays of magnetic sensors now installed on the LFS vessel wall. Only the plasma response to 1kA $n = 1$ even parity I-coil currents is considered, corresponding to perfect vacuum compensation. The synthetic diagnostic set, including measured mis-alignments in the sensor locations and tilts, is able to accurately reconstruct the major wavefronts of the field on the vessel surface. The sharp transitions at the corners of the vessel are not expected to ever be accurately captured. The longer wavelength characteristics are expected to contain the important physics, including the majority of the electromagnetic torque. The additional arrays furthest from the mid-plane do well to accurately constrain the amplitude falloff near the top and bottom of the vessel and are crucial for the accurate decomposition using multiple poloidal modes.

Fig. 2.7 shows a more challenging reconstruction, requiring many more poloidal modes to capture the main characteristics of the HFS field structure. Here, the plasma response

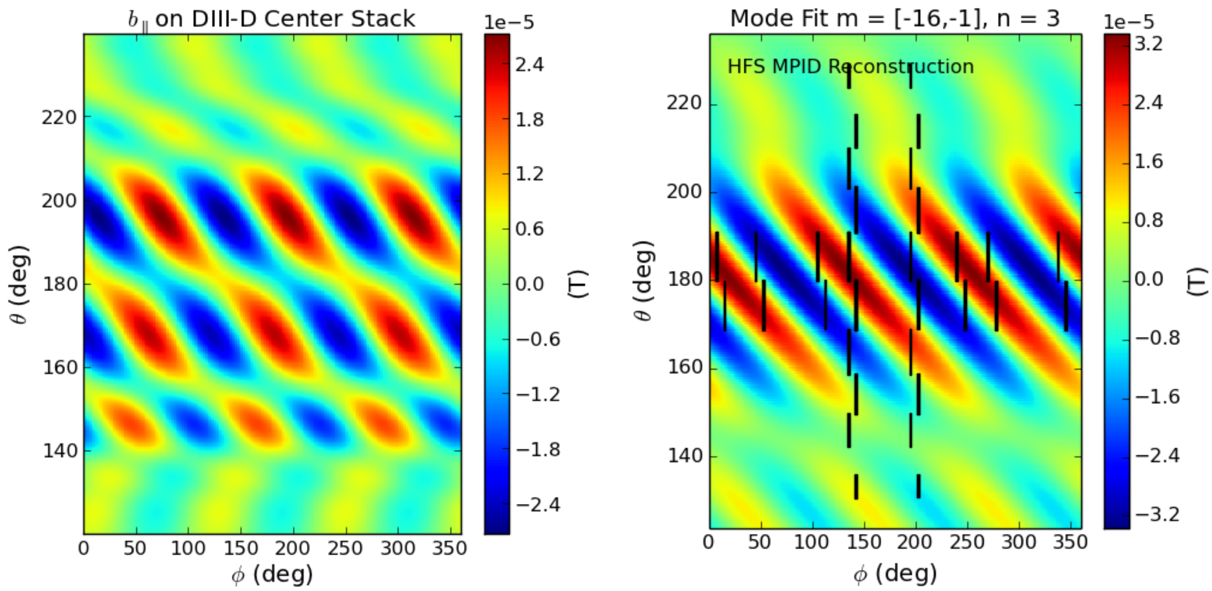

Figure 2.7: Plasma response to 1kA $n = 3$ even parity I-coil currents calculated by IPEC at the DIII-D HFS vessel wall (left) and the field reconstructed from a discrete set of synthetic diagnostics (right) shown in black. The reconstruction assumes a single $n = 3$ toroidal harmonic and includes the single helicity poloidal mode numbers $-16 \leq m \leq -1$. Note that the reconstruction captures the main wavefronts of the model, but fails to capture the short wavelength structure.

to 1kA $n = 3$ even parity I-coil currents is reconstructed using the final design of the new HFS parallel field sensors. Although there are no sharp corners in the center stack wall, some fine structure is ignored when decomposing in only the modes with helicity matching that of the plasma. This case was the most extreme example of this observed in the modeled predictions, and the reconstruction accurately describes the pitch of the primary wavefront near the mid-plane where the amplitude is strongest.

The Maxwell stress can be immediately calculated from these reconstructions using the corresponding Fourier amplitudes in Eq. (2.9). There is no electromagnetic torque, however, on a flux surface in Ideal MHD equilibrium and the net torque from the IPEC fields will always integrate to zero. The ability of reconstruction using Fourier decomposition and the discrete signal set to accurately describe the measured field across the vessel surface is presented here as a validation of the diagnostic set's design capability as well as a motivation for the approach taken a-priori for the experimental calculation of Maxwell torque in Section 2.1.

2.2.4 Error Analysis

Analysis of synthetic diagnostics in IPEC allows for fast manipulation of many probe layouts, including ones with applied errors in the position of individual diagnostics. Monte Carlo analysis applied to the center position and cross sectional angle of probes shows that plasma response mode identification is not very sensitive to probe position. For identification of the $n = 1$ plasma response amplitude and phase at the LFS mid-plane, Fig. 2.8 shows that 1% error in the mode amplitude or 1° error in the phase requires positional accuracy of ~5mm and tilt accuracy of $\sim 4°$. The use of a coordinate measurement machine enabled $\sim$ 0.5mm precision in aligning the sensors [61], more than meeting these requirements.

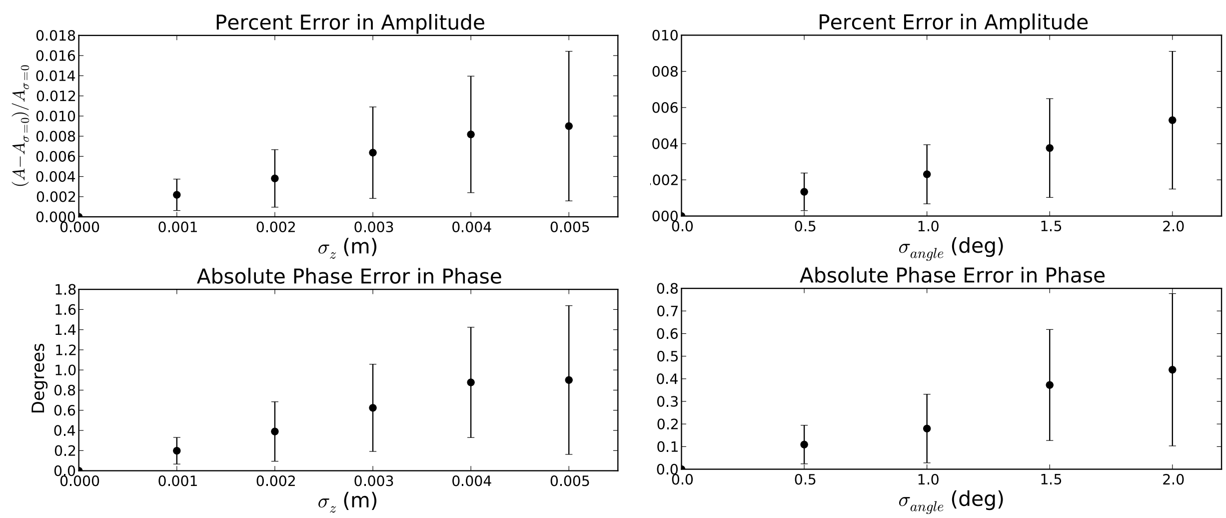

Figure 2.8: Monte Carlo error analysis of a single toroidal mode identification, applying Gaussian noise to probe vertical position (left) and cross sectional angle (right) of each sensor in a synthetic diagnostic set corresponding to the magnetic probe array on the LFS mid-plane. In each case, the error is described as a percent error in the identified mode amplitude (top) and absolute error in the identified phase (bottom).

It should be noted that the case depicted in Fig. 2.8 does not include many of the experimental error sources expected in the true measurement. Imperfect vacuum compensation, for example, may pollute a plasma response measurement with fields from the nonaxisymmetric coils. Unphysical noise and nonlinear integrator drifts may also contribute to the error in a fit and require smoothing or regular baselining in order to isolate the signal of interest. The largest source of error, however, is assumed to be pickup from the large axisymmetric background. Misalignment of sensors in a toroidal array leads to errors $\sigma_{b_\parallel} \sim \sigma_\ell dB_\parallel/d\ell$, where ℓ is the length coordinate along the measurement surface poloidal cross section and $B_\parallel$ is the zeroth order axisymmetric field ($\sim 10^{-1}T$). The result is imperfect removal of the axisymmetric field when hardware differencing sensors in the same toroidal array. This residual axisymmetric pickup has been quantified experimentally as $\sim 10^{-4}$T near the HFS mid-plane, and is compensated for in the hardware differencing as well as hardware toroidal field compensation [61]. Note that this pickup would result in significantly higher errors if the sensors were oriented in the toroidal direction, for which the axisymmetric field is on the order of a Tesla. This is a major motivation for their orientation in the poloidal plane (perpendicular to the toroidal direction).

Note the an estimation of the error for the Maxwell torque (2.9) is immediately available from the uncertainty of the Fourier amplitudes. For demonstration, the torque from a perturbation characterized by a single toroidal mode and single poloidal wavenumber,

$$T_{EM} \sim \frac{k_\phi}{k_\parallel} b_\parallel b_\perp \sim \Delta\phi_\ell \left|b_\parallel\right| |b_\perp| \cos \Delta\phi_{\parallel,\perp}, \tag{2.10}$$

can be expected to have an uncertainty given by,

$$\frac{\sigma^2_{T_{EM}}}{T^2_{EM}} = \frac{\sigma^2_{b_\parallel}}{\left|b_\parallel\right|^2} + \frac{\sigma^2_{b_\perp}}{|b_\perp|^2} + \frac{\sigma^2_{\Delta\phi_\ell}}{\Delta\phi_\ell} + \sigma^2_{\Delta\phi_{\parallel,\perp}} \tan^2 \Delta\phi_{\parallel,\perp}. \tag{2.11}$$

Here, σ is the standard variation of the measured amplitudes $\left|b_\parallel\right|, |b_\perp|$, parallel field phase change across the poloidal extent of the surface $\Delta\phi_\ell$ and phase shift between the parallel and perpendicular components $\Delta\phi_{\parallel,\perp}$. From this, we see that the relative error in the Maxwell torque is dominated by uncertainties in the phase when either the surface in question is short compared to the poloidal wavelength or the phase shift between components is near $\pm 90°$. Assuming maximum torque ($\Delta\phi_{\|,\perp} \sim 0$) and structures similar to those seen in IPEC plasma response modeling ($\left|b_{\|}\right|, |b_{\perp}| \sim 10^{-4} - 10^{-3}$T and $\Delta\phi_{\ell} \sim 360°$ on the LFS), the percent error in the Maxwell torque is approximately the L^2 -norm of the percent error in the two component amplitudes. This would be less than 1.5% in the ideal case projected by the Monte Carlo analysis, and less than 15% even with 10% uncertainty in the experimentally measured amplitudes.

2.3 Overview of Installed Diagnostics

This section reviews the details of the nonaxisymmetric diagnostic suite installed in DIII-D. This is not meant to be a comprehensive description of the DIII-D magnetic sensors. A more complete review of the fabrication, installation, and instrumentation of the upgrade sensors can be found in Ref. [61], while the pre-existing magnetics (including many axisymmetric sensors not discussed here) at detailed in Ref. [62]. The subset of nonaxisymmetric diagnostics described here provides the necessary background for the two component measurements available to meet the electromagnetic stress tensor requirements. The distribution of the sensors is presented as both the culmination of the modeling/synthetic optimization efforts of the previous section and the infrastructure on which the data analysis techniques developed in the following section are built. The sensors detailed here form the foundation for all the magnetic measurements and analysis presented hereafter.

Two fundamental classes of magnetic sensor were added in the upgrade to study non-axisymmetric fields in DIII-D: saddle loops and magnetic probes. Saddle loops are inductive sensors consisting of coaxial cable wrapped N turns around a surface of area A of a axisymmetric surface. The time-integrated voltage across the wire leads (V_l) gives the magnetic flux through the surrounded surface $\int V_{lead} dt = NAB_{\perp}$ [86, 62, 61]. Also fundamentally inductive, magnetic probes are small coils of coaxial cable mounted with their axis of symmetry parallel to the vessel wall. The integrated voltage across the leads for a probe with N turns and cross sectional area A measures the flux NAB_p, where B_p is the component of the field aligned with the coil axis. The majority of these are mounted in the poloidal plane, perpendicular to the toroidal direction.

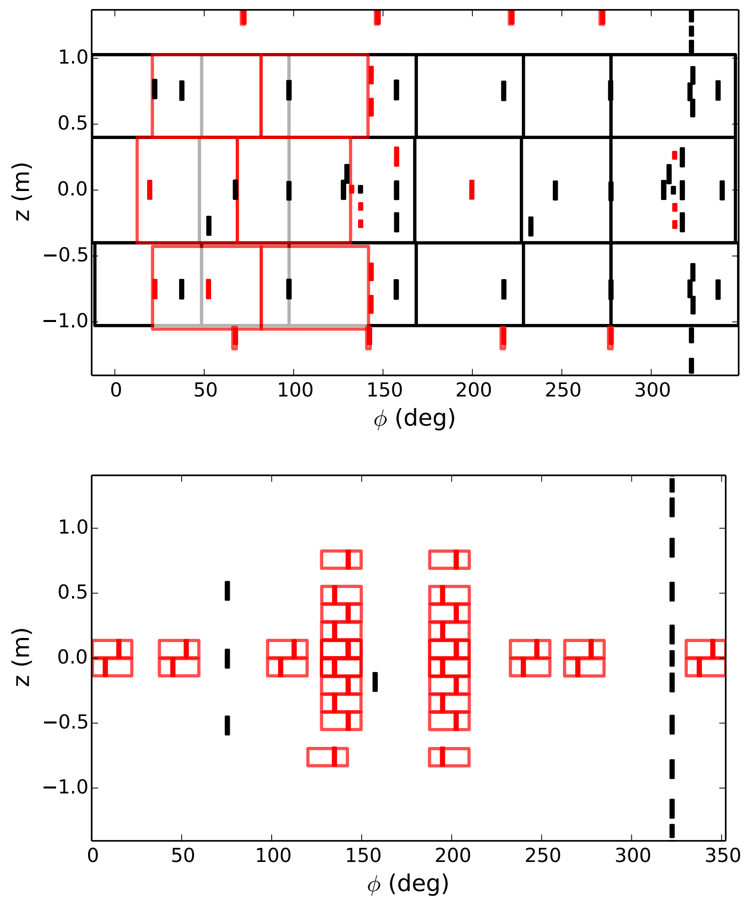

Figure 2.9: The final design of the DIII-D magnetics upgrade, including 101 new sensors (red), added in addition to the 125 internal sensors previously used for nonaxisymmetric measurements (black). Internal saddle loops (open boxes) and magnetic probes (solid rectangles) are shown on the LFS (top) and HFS (bottom) of the vessel.

2.3.1 Fabrication

Both saddle loops and magnetic probes have been installed and reliably collecting data in DIII-D for many years, and much of the same fundamental material and sensor design specifications were used in the magnetics upgrade [81, 62, 61]. Each sensor used high temperature coaxial cable consisting of a 316 stainless steal sheath, MgO ceramic insulator, and a copper center conductor similar to that used in established sensors. Twisted leads were run directly to the vacuum feedthrough couplings, replicating the robust design already in use to avoid any potential for internal disconnections.

Upgrade magnetic probes are of the same racetrack cross section design as pre-upgrade probes [62]. The typical upgrade probe has roughly ~ 90 turns and a cross sectional area of $\sim 7\text{cm}^2$, although slightly smaller cross sections were used on the HFS. Significant deviations from this rule include a number of "double wide" sensors ($NA \sim 2100\text{cm}^2$) used in the LFS arrays and smaller unintegrated sensors ($NA \sim 120-240\text{cm}^2$) not used for the low frequency long wavelength measurements discussed here.

The upgrade saddle loops include large, single-turn sensors as well as new multiple-turn designs. Sensors of the larger single-turn design, consistent with previous internal-saddle loops, were added to the three existing toroidal arrays near the Low Field Side (LFS) mid-plane. Space constraints closer to the diverters motivated a compact, multiple-turn design mounted in tandem with the magnetic probes in the new arrays at the top and bottom of the LFS. The HFS saddle loops were built by pressing cable into channels in pre-fabricated stainless steal frames and mounting the entire loop. This enabled alignment with $\sim 0.16\text{mm}$ precision, to avoid pickup from the large axisymmetric fields in this region.

2.3.2 Distribution

The complete set of in-vessel magnetic diagnostics used for the detection and characterization of nonaxisymmetric modes in DIII-D is shown in Fig. 2.9. The orientation and extent of these sensors were heavily influenced by predictive modeling and the synthetic analysis described in Section 2.2. The final distribution of sensors is naturally divided into two distinct regions: the LFS on the outboard wall of the vessel and HFS comprised primarily of the vertical center stack wall.

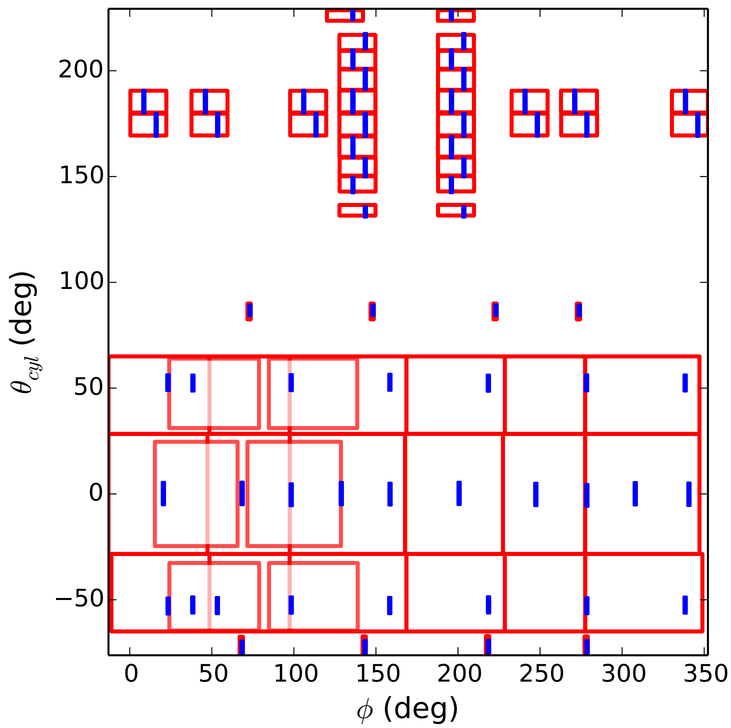

Figure 2.10: The saddle loops (red) and magnetic probes (blue) with hardware differencing to remove axisymmetric contributions, shown on the "unwrapped" surface of the vacuum vessel wall characterized by the machine toroidal (ϕ) and poloidal (θ) angle. This represents the primary subset of diagnostics for detection and characterization of slow and stationary nonaxisymmetric fields.

For the purposes of this work, the LFS distribution consists of 5 toroidal arrays of both saddle loops and magnetic probes. The additional sensors pertain to the additional physics objectives: poloidal arrays for equilibrium reconstruction and characterization of rotating modes and high density arrays for the detection of short wavelength Edge Localized Modes (ELMs). The toroidal arrays are distributed poloidally at $\theta \sim -72°, -53°, 0°, 53°, 87°$, where the angle is defined by $\theta \equiv \tan^{-1}(z/(R - R_0))$ for the point at height z from the mid-plane and radius R from the central axis. The machine major radius, $R_0 = 1.69$, is used as the toroidal axis. Sensors in these arrays are distinct in that each digitized signal is first hardware differenced with the integrated voltage of another sensor in the same array, eliminating the axisymmetric contributions. These differenced arrays provide the most accurate description of quasi-stationary nonaxisymmetric modes, and are thus central to the electromagnetic torque analysis.

Differenced pairs are crucial for the detection of quasi-stationary modes on the HFS as well. Here, only the upgrade installations have been hardware differenced with poloidally aligned pairs. Pairs in the two toroidal arrays, as in the LFS arrays, are chosen asymmetrically to prevent aliasing of any single toroidal mode and optimized for the simultaneous detection of three toroidal modes $1 \leq n \leq 3$. Matched sensors in the twin poloidal arrays at 139 and 199 degrees toroidally are also differenced. The complete collection of differenced arrays is shown in Fig. 2.10. The capabilities of this subset of diagnostics are addressed in detail in the following sections.

2.4 Data Analysis Techniques

In parallel with the hardware commissioning and initial data collection, a new 3D magnetics package was built within the General Atomics Python analysis packages. This analysis package, pyMagnetics, was built specifically for the analysis of electromagnetic torque. It thus focuses on long wavelength ($0 \leq n \leq 3$, $-12 \leq m \leq 12$) and low frequency ($\omega \leq 1$kHz) analysis: exactly the regime that the design of the magnetics upgrade differenced pairs was optimized for. This section will focus on the data analysis developed to obtain the electromagnetic stress by accurately mapping the perturbed field to the vessel surface.

2.4.1 Vacuum Compensation

The compensation of magnetic sensors for vacuum coupling with tokamak operation and error field correction coils is necessary to isolate the perturbed plasma response to toroidal asymmetries. Although the Maxwell stress tensor requires information from both the plasma and the external field, it is not always possible to accurately fit both sources from the magnetics data.

In DIII-D, the primary example of this is the torque applied by internal saddle loop coil arrays (I-coils). Mounted above and below the mid-plane on the vessel wall, these coils have very little coupling to the poloidal field magnetic probes by design. Fig. 2.4 shows that the sensors are designed to include minimal contributions from the vacuum fields despite the dominance of those fields across the majority of the vacuum vessel surface. The result is that the structure of the combined (applied and plasma response) fields is not directly obtainable from the measurements. The electromagnetic torque can be obtained, however, by eliminating all vacuum contributions through compensation and mapping the plasma response field to the coil contours to directly measure a $\mathbf{J} \times \mathbf{B}$ force on the coil. To this end, and for the increased physics understanding that comes from clear identification of the plasma fields, multiple compensation techniques have been developed at DIII-D and employed in the pyMagnetics analysis package.

The most basic method of vacuum compensation is to subtract a direct (DC) coupling. This direct coupling is calculated annually at DIII-D for the central solenoid (E) coil, 18 independent field shaping (F) coils, and the error field correction (I and C) coil arrays. This process employs square wave pulses of the individual coils in vacuum. The pulse is longer than any relevant eddy current decay time, and the ratio of the resulting sensor signal flattop to the coil current is recorded in a compensation matrix. Vacuum compensation is then employed throughout the year using the relation,

$$S_i^c = S_i^r - \sum_j C_{ij} I_j, \tag{2.12}$$

where S_i^c is the compensated signal of the ith sensor, S_i^r is the raw signal, C_{ij} is the coupling matrix and I_j is the current in the jth coil. Annual measurement of the compensation matrix corrects for displacements, deformations, and tilts of the coils and sensors during the maintenance periods.

Often, the error field correction coils are used with waveforms varying in time on scales the order of or smaller than the wall time of DIII-D. In these situations, eddy currents in the surrounding structures introduce time dependent (AC) amplitude and phase modifications in the coupling between a sensor and field coil. Two methods for the calculation of these AC couplings, response function and transfer function compensation, are discussed here.

2.4.1.1 Response Function Compensation

The AC coupling between sensors and time varying currents in magnetic coils can be obtained from the dynamic behavior during the DC calibration pulses. After the magnetics upgrade in 2013, and for the first time in DIII-D, a linear response function technique was used to describe the time dependent relationship between each sensors and each of the coils in the DC compensation matrix. The response function, through a convolution integral, defines a system output from the time history of its input,

$$S_i(t) = \int_{t_0}^{t} d\tau R_{ij}(t-\tau)\frac{dI_j}{d\tau} = R(0)I(t) + \int_{0}^{t-t_0} d\tau R'(\tau)I(t-\tau). \tag{2.13}$$

In this case, the coil current I_j and the sensor signal S_i are the coupled system's input and output respectively. A unique response function, $R_{ij}(\tau)$, exists for each sensor-coil coupling.

The strength of the response function technique is in its simplicity and its generality. The definition Eq. (2.13) is taken directly from signal theory and is not dependent on the geometry of a tokamak, the details of its coils, or the design of its sensors. Physically, the phase and amplitude of the sensor response depends on the exponential decay of eddy currents in nearby conducting structures. The most physical approach is to attempt to solve for the decay time constants using an inverse Laplace transform of the signal. When multiple decay constants are involved, however, this approach is infamously confounding and mathematically ill posed [87, 88]. The response function, by assuming a linear system with zero initial conditions and zero-point equilibrium, captures all of the physics in a superposition of after effects represented in the second term in the rightmost side of Eq. (2.13).

The simplest calculation of a system's response function can be obtained from an impulse response. The reaction of a dynamical system to a step function input is identically the response function,

$$S_i(t) = \int_{t_0}^{t} d\tau R_{ij}(t-\tau)\delta(\tau - t_s) = R(t - t_s). \tag{2.14}$$

This is, theoretically, exactly what the sensor signals are measuring during the transient phases at the beginning and end of the step function currents applied for DC coupling measurements.

In practice, the coils take some non-zero time to turn on and to return to zero current. The response function then becomes a least squares solution to the linear equation,

$$\boldsymbol{\mathcal{I}}\mathbf{R} = \mathbf{S}, \tag{2.15}$$

solved for k data points sampled at the regular time step dt such that $\mathbf{S}_k = S(t_k)$, $\mathbf{R}_k = R(t_0 - t_k)$, and $\boldsymbol{\mathcal{I}}$ is a lower triangular matrix $\boldsymbol{\mathcal{I}}_{ij} = dtI'(t_i - t_j)$ with $I'(t < 0) = 0$. The fall time of the coil current can be very sharp in practice ($\sim 1ms$), allowing for Eq. (2.15) to be readily solved for massive amounts of data using sparse matrix techniques. The result is an accurate response function describing the AC coupling between a sensor and coil from the DC limit to an upper frequency limited by the fall time of the coil current. In DIII-D, this limit corresponds to the frequency at which the radial field necessary for electromagnetic torque measurements is shielded by the graphite tiles located between the plasma and the sensors [11]. Above this frequency, standard time domain Fourier analysis techniques [62] or more intricate system identification techniques [89] can produce a more accurate description of rotating modes from even a limited set of poloidal field probes than the time-independent spacial fitting techniques for which the magnetics upgrade was designed. These techniques are not applicable to the direct measurement of electromagnetic torque, and are thus beyond the scope of this discussion.

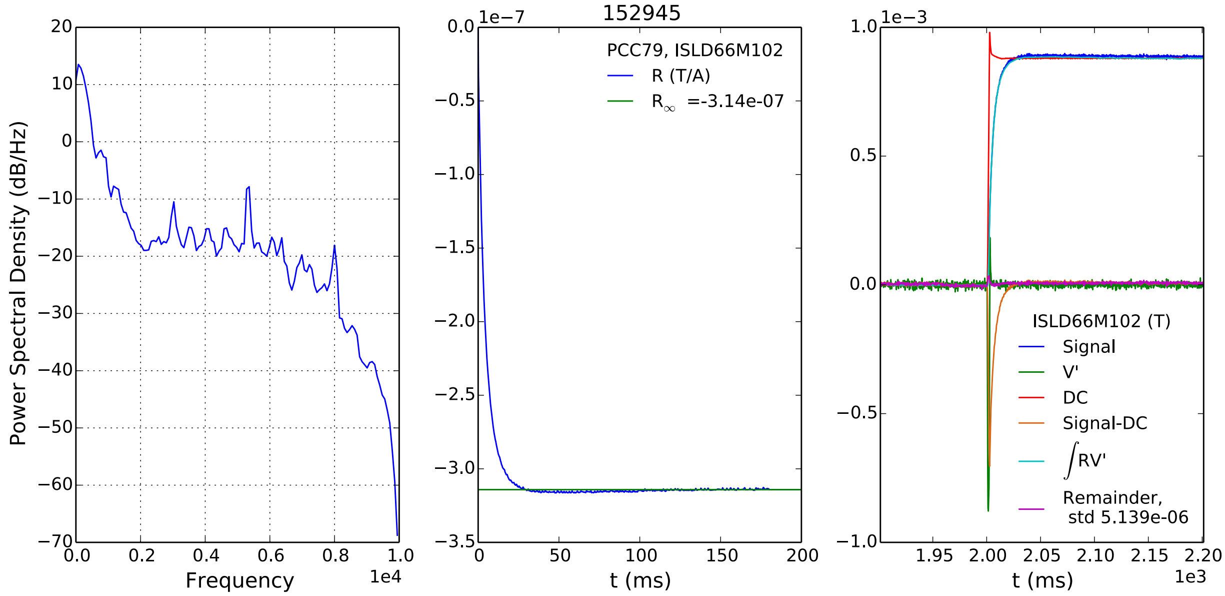

Figure 2.11: A representative response function (middle) fit for the coupling between a single C-coil (PCC79) and midplane saddle loop sensor (ISLD66M102) calculated using 2013 vacuum calibration shot 152945. The power spectrum of the time derivative of the coil current (left) indicates the applicability range in frequency space. The asymptotic limit of the response function (middle) shown in green estimates the DC coupling. Accuracy of the fit is examined through a self consistency check (right) showing the original step function signal, time derivative of the coil voltage V', DC compensation proportional, DC compensated remainder, AC compensation calculated using Eq. (2.13), and the remainder after AC compensation.

An example response function calculation is shown in Fig. 2.11, and a benchmark of it's accuracy is presented in Fig. 2.12. The fit process immediately gives the direct DC coupling between the sensor and coil, which compares favorably to the standard C matrix value (-9.97×10^{-7}). The power spectrum of the current derivative (flat for a perfect step function) indicates the regime of accuracy for a given fit. The maximum frequency above 0 dB/Hz is 0.466kHz, which suggests a more limited range of accuracy than the order of magnitude estimate of $\lesssim$ 1kHz in the discussion above. The application to a completely new waveform in Fig. 2.12 shows the response function is able to accurately describe the amplitude and phase of the sensor coupling over the applicable range of frequencies. Comparison with the direct coupling compensation gives a sense of the importance of this AC compensation. When

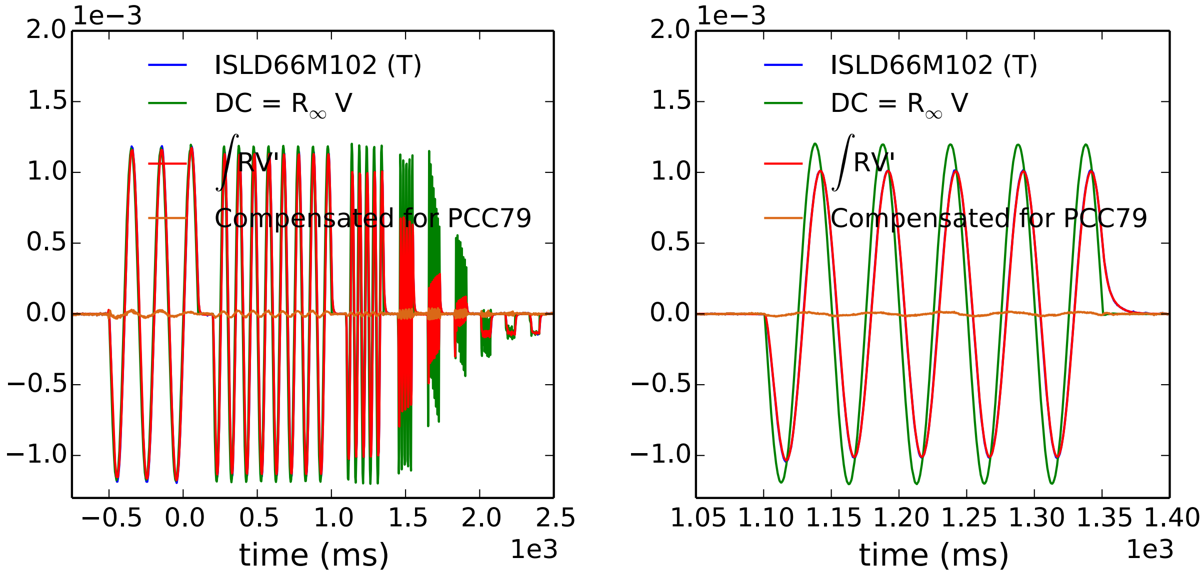

Figure 2.12: Application of response function compensation for a midplane saddle loop (ISLD66M102) during vacuum shot 153236 with C-coil frequency chirping (left). The signals shown include the original baselined signal (blue), the DC compensation (green), the response function compensation (green), and the residual signal after response function compensation is subtracted (orange). Signal values are given in Tesla. A close up of the 20Hz window (right) shows the DC compensation fails to capture the amplitude and phase shift, which would have lead to a false signal of $\sim$ 1G in this case.

studying synchronous plasma response signals on the order of $\sim$ 1G, the vacuum coupling at 20Hz would be a preventatively large source of error in the fitting of mode structures without accurate AC compensation.

2.4.1.2 Transfer Function Compensation

The transfer function compensation technique uses a frequency space description of the coil-sensor system by fitting the response over a wide range of controlled frequency inputs. As with the response function analysis, we assume a Linear Time-Independent (LTI) system with zero initial conditions and zero-point equilibrium. For a continuous time input, the transfer function $H(s)$ is the linear mapping of the Laplace transformed input to Laplace transformed output,

$$\mathcal{L}[S_i(t)] = H(s)\mathcal{L}[I_j(t)]. \tag{2.16}$$

Here $s = \alpha + i\omega$ is in general the complex exponent variable of the Laplace transform. When taken as purely imaginary for a LTI, the Laplace transform reduces to a Fourier transform,

$$\mathcal{L}[f(t)] \equiv \int_{-\infty}^{\infty} f(t)e^{-st}dt = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt. \tag{2.17}$$

Note that in the LTI a sinusoidally driven complex coil current input $I_j(t) = |I_j| \times \exp\left[i\omega t - \arg\left(I_j\right)\right]$ produces a complex response $S_i(t) = |S_i| \exp\left[i\omega t - \arg\left(S_i\right)\right]$ at the driven frequency. In this case, the transfer function can be considered a description of the amplitude and phase lag of the signal output,

$$|H(i\omega)| = \frac{|S_i|}{|I_j|}, \tag{2.18}$$

$$\arg\left(H(i\omega)\right) = \arg\left(S_i\right) - \arg\left(I_j\right). \tag{2.19}$$

Historically, this frequency space description of the signals has been fit for DIII-D after sampling a wide range of input frequencies in vacuum during devoted calibration shots [90]. The Fourier analyzed signals are fit to a rational function described by N zeros and M poles,

$$H(i\omega) = \frac{\sum_{n=1}^{N}\left(i\omega - z_n\right)}{\sum_{m=1}^{M}\left(i\omega - p_m\right)}. \tag{2.20}$$

In practice, a low number of zeros and poles are enough to form a good fit to the amplitude and phase lag for a coil-sensor pair as functions of frequency ($M, N \lesssim 4$). For discrete time series data, the compensated signal can then be described using the functional form,

$$S_i^c(t_k) = S_i^r(t_k) - \left[\sum_{n=1}^{N} a_n I_j(t_{k-n+1}) - \sum_{m=1}^{M} b_m S_i^r(t_{k-m})\right], \tag{2.21}$$

where a_n and b_n are the discrete time domain coefficients obtained from a bilinear transform of the H numerator and denominator polynomial coefficients.

The benefit of this technique is that compensation requires knowledge of only a few prior time steps, as opposed to the full history required by the response function or continuous time series definitions. Real time compensation is thus possible for applications such as RWM feedback. Offline, storage of the compensation requires only $M+N$ coefficients per coil-sensor pair. These benefits have historically been enough to motivate annual vacuum calibration days with devoted frequency scans of the I and C coils for transfer function fitting [90].

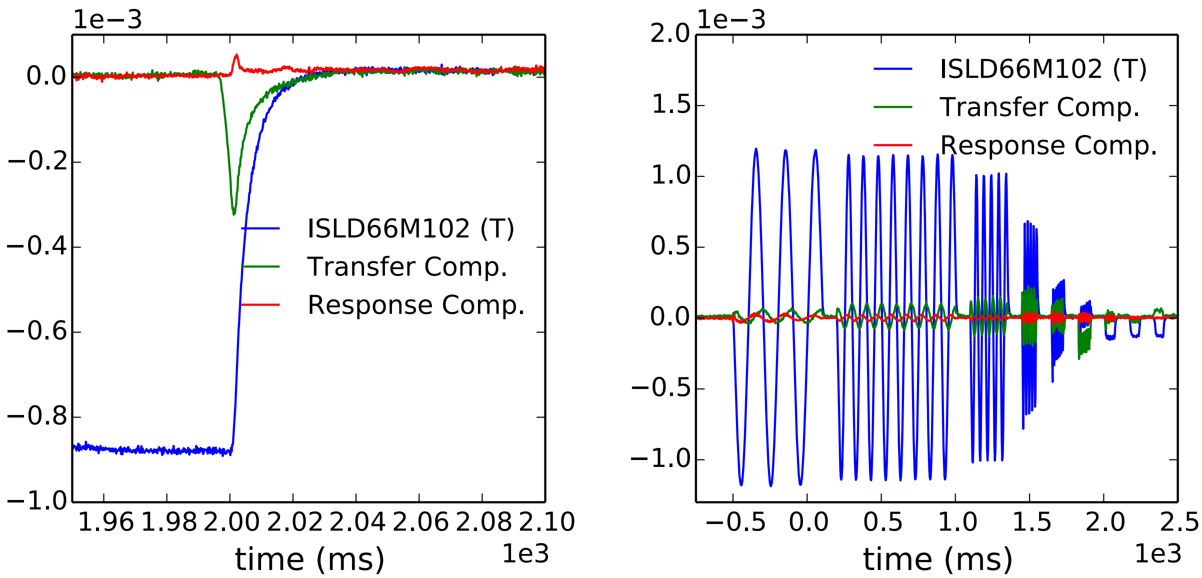

Figure 2.13: The signal measured using midplane saddle loop ISLD66M102 (blue), the transfer function compensated signal (green), and the response function compensated signal (red) calculated for two vacuum calibration shots. The step function in shot 152945 (left) was used in fitting the response function applied in both cases, while the AC chirping shot 153236 (right) was used to fit the transfer function applied in both. All signals are in Tesla.

The use of these calibrations is evolving. The magnetics upgrade and inclusion of transfer function compensation routines in the pyMagnetics package has motivated the computation of many new coil-sensor pair couplings since 2013. The python package routines also provide, for the first time, a chance to cross compare the accuracy of the transfer function technique to AC compensation using a completely independent data set: the response function compensation from the square wave pulse calibrations. A comparison of the two techniques for a representative sensor-coil pair is shown in Fig. 2.13. Although the response function appears to perform better for this case, both reduce the vacuum coupling by an order of magnitude for the lower frequency perturbations. The response function calibration requires no devoted machine time beyond the standard DC coupling shots and the sparse least squares fitting can be easily automated over a time window around the sharp step in coil current. The response function can thus become available for use in magnetics analysis before the transfer functions, which currently require some human control of pole and zero number/location for the thousands of coil-sensor couplings. The transfer function compensation, once available, is less computationally demanding and directly comparable to historical results.

2.4.2 Mode Identification and Fitting

The structure of nonaxisymmetric fields that the magnetics upgrade was designed to study are complicated by deviations of the experimental plasma and vessel geometries from simple cylindrical geometries. Even in the case of a tearing mode, which has a integer poloidal and integer toroidal harmonic (m, n) in magnetic coordinates, the corresponding cylindrical (θ, ϕ) mode signature in the sensor measurements may become significantly distorted by the toroidicity, plasma shaping, and poloidally dependent proximity of the measurement due to non-conformal vessel geometry. In the linear, ideal MHD regime IPEC has shown that the plasma is most sensitive to external fields that overlap with the kink coupled poloidal spectrum. The corresponding plasma response contains a range of poloidal mode numbers for each toroidal harmonic.

This section presents the techniques used in the pyMagnetics analysis package to identify structures in the multichannel magnetics data, and to map these structures to the real space geometry of the DIII-D vessel.

2.4.2.1 Multichannel MHD Mode Identification

The pyMagnetics analysis of the magnetics upgrade signals is designed to be consistent with the stated physics goals of the hardware design. The physics this diagnostic set was designed to address includes the dynamics of mode locking, the response to applied nonaxisymmetric fields and the electromagnetic torque. Each of these goals requires identification and analysis of slowly rotating or quasi-stationary plasma modes. Standard time domain Fourier analysis techniques are thus ill suited for this analysis. Nor would they represent significant improvement of the available mode identification using established poloidal and toroidal arrays [62]. Singular Value Decomposition (SVD) of large sets of magnetics signals for the purpose of structure identification has been successful in describing rotating MHD in JET and DIII-D with limited sets of magnetic sensors in the past [91, 92]. The SVD analysis techniques extend well to the quasi-stationary limit and benefit, in that limit, strongly from increased spatial resolution. This subsection extends SVD analysis to the magnetics upgrade, enabling the desired plasma response to be identified and the desired physics to be addressed.

The SVD of multichannel magnetics arrays determines an eigenbasis based on the direction of maximum coherence in the data. It makes no assumptions about the spacial or temporal structure of the coherence. The SVD identifies coherent basis vectors in the sensor-space, which may correspond to multi-modal structures when mapped to cylindrical coordinates. For each sensor-basis vector, there is a corresponding principle component: the projection of the data on the time eigenvectors of the data time covariance. These are simply the time evolution of the signal structure. The squares of the singular values are the eigenvalues of a given structure and measure the energy of the mode. The data is thus organized into coherent structures, ranked by their respective energy content.

Consider the rectangular $P \times T$ data matrix composed of T synchronized time samples of data from P sensors,

$$\mathbf{D}_{ij} = \frac{1}{\sqrt{TP}} S_i(t_j), \tag{2.22}$$

where the normalization $1/\sqrt{TP}$ is introduced for convenience. The matrices $\mathbf{DD}^T$ and $\mathbf{D}^T\mathbf{D}$ are real, symmetric matrices with identical eigenvalues that form a complete orthonormal basis. These are the normalized sensor-covariance and time-covariance of the data respectively. The covariance of the ith and jth probe is the time average of their product. Similarly, the covariance of ith and jth time sample is the average of their product over all sensors. The normalized covariance matrices are thus defined as,

$$\left(DD^T\right)_{ij} = \frac{1}{P} \times \frac{1}{T} \sum_{k=1}^{T} S_i\left(t_k\right) S_j\left(t_k\right), \tag{2.23}$$

$$\left(D^T D\right)_{ij} = \frac{1}{T} \times \frac{1}{P} \sum_{k=1}^{P} S_k\left(t_i\right) S_k\left(t_j\right). \tag{2.24}$$

The SVD of the data matrix is the factorization,

$$\mathbf{D} = \mathbf{USV}^T, \tag{2.25}$$

where $\mathbf{U}$ is a $P \times P$ orthonormal matrix, $\mathbf{S}$ is a $P \times T$ rectangular diagonal matrix, and V^T is a $T \times T$ orthonormal matrix. The diagonal elements $S_{ij} = s_i \delta_{ij}$ are positive, real values termed the singular values of $\mathbf{D}$. The column vectors $\mathbf{U}_i$ are the normalized eigenvectors of $\mathbf{DD}^T$ referred to here as the sensor-basis vectors and more generally termed the left-singular vectors. The column vectors $\mathbf{V}_i$ are the normalized eigenvectors of $\mathbf{D}^T\mathbf{D}$ referred to here as the time vectors and more generally termed the right-singular vectors. The SVD separates a dynamical system into spatial structures ($\mathbf{U}_i$) and their temporal evolution ($\mathbf{V}_i$), and ranks them by their coherence or energy content (s_i).

A major advantage of the SVD analysis is a reduction in the data used for analysis. Many plasma phenomena of interest are dominated by a single dominant structure in the plasma [17, 52, 84]. In this case, only the top sensor-basis and time vectors may be of significance. In practice, structures with a cumulative energy content of under ~2-5% are typically ignored. In addition to data reduction, this serves as an effective filter for incoherent noise reduction.

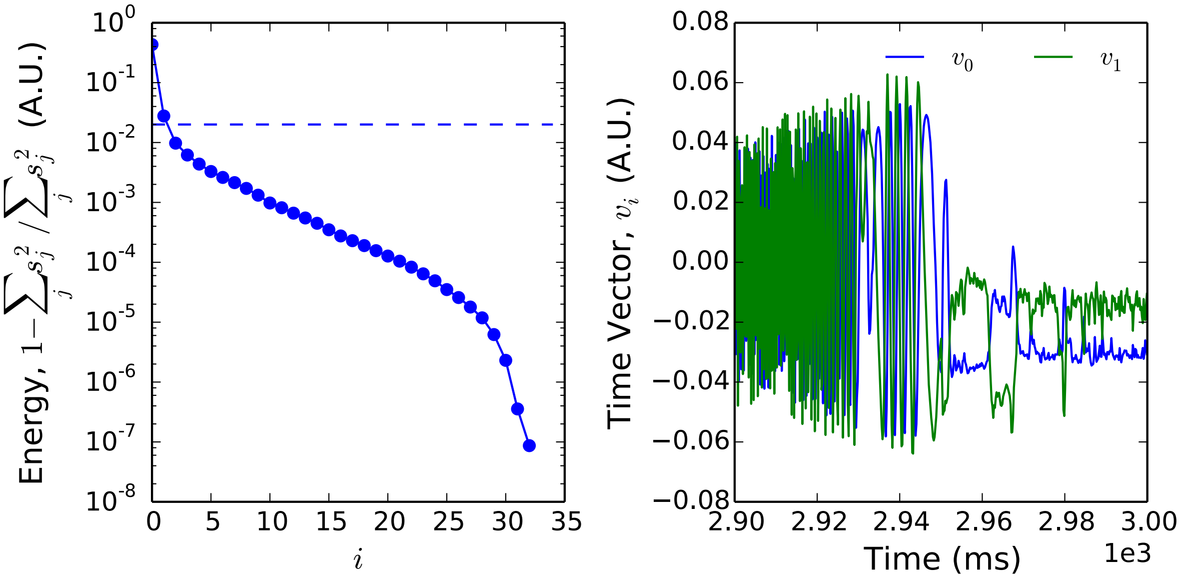

Figure 2.14: The residual of the cumulative SVD eigenstructure energy content (left) defined using the sum over the square of the first i eigenvalues, and the first two time-vectors (right). The first two eigenvectors contain nearly 98% of the total energy (indicated by a dashed line in the left, energy plot). The SVD results use data from the 5 LFS parallel field sensor arrays during a 100ms window surrounding the locking of a large $n = 1$ tearing mode near 2950ms in DIII-D plasma shot 154551.

Separation of the temporal evolution by SVD allows efficient analysis of dynamical plasma phenomena. In the case of a pure sinusoidal oscillation at a constant frequency, the SVD technique reduces to a discrete Fourier transform in time [91, 92]. The physical phenomena of interest, however, often include coupled dynamics that are captured by the SVD as deviations from pure sinusoidal oscillations. The electromagnetic entrainment of islands results in modulations about an applied error field frequency [83], and similar modulations might be produced by coupled tearing modes on distinct flux surfaces or multiple MHD modes rotating at similar frequencies. Of special interest to the analysis of the magnetics upgrade is the ability of the time vectors to capture stationary or quasi-stationary structures. Fig. 2.14 shows the LFS parallel field sensor SVD mode energy content and time vectors of a large tearing mode as it rotates, slows, and locks to the vessel wall. The first two eigenvectors contain over 98% of the energy in this case, and the remaining structures are considered

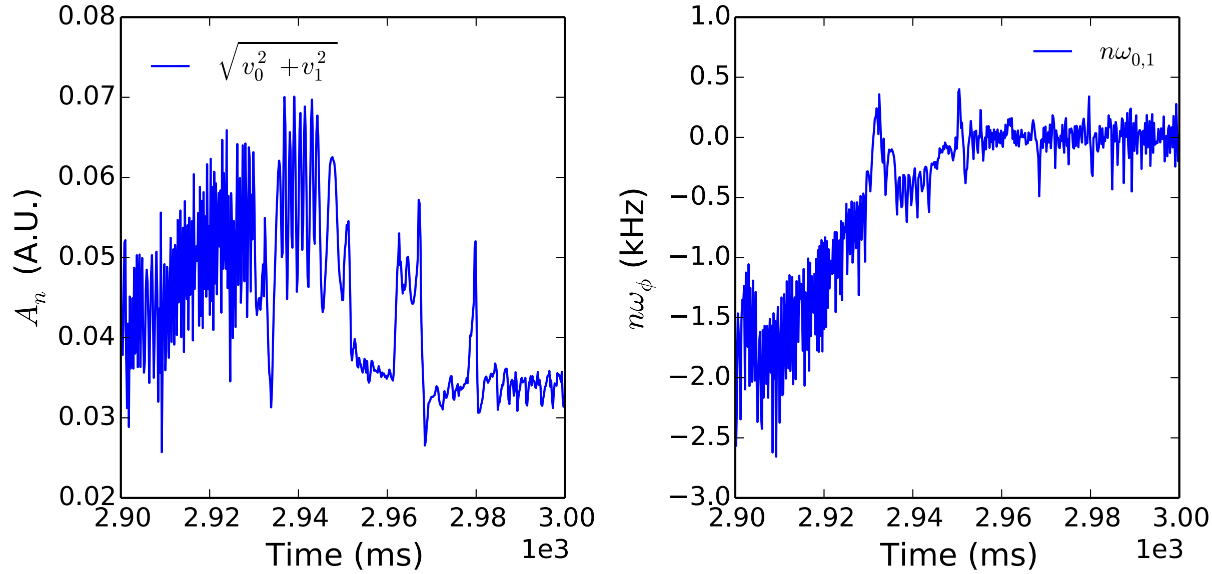

Figure 2.15: The SVD mode amplitude (left) and frequency (right) assuming the two dominant eigenmode time-vectors from Fig. 2.14 are the real and imaginary components of the same toroidal mode, $v_0 + iv_1 = A_n e^{in\phi}$.

incoherent noise. In this case, a single island structure dominates the data, and the two dominant time eigenvectors can be combined as Fourier coefficients to estimate an amplitude and phase of the mode shown in Fig. 2.15.

2.4.2.2 Harmonic Identification

Separation of the spatial structures by SVD greatly reduces the number of spatial modes, enabling advanced spatial identification techniques that would be prohibitively expensive computationally to do for each time step. The SVD itself identifies structures in a sensor-basis. Individual point measurements could then be mapped directly to real space, as has been done previously for measurements of rotating modes using poloidal arrays of magnetic probes [92]. The sensor basis vectors can thus efficiently capture complex poloidal structures from phenomena such as saw-teeth, the plasma kink response to applied fields, and entrained islands. Hardware design and the physics of interest make this direct approach opaque however. Harmonic identification is necessary to interpret differenced signals as well as area averaged saddle loop measurements. It also enables interpolation over the vessel surface for maxwell stress calculations. Each eigenmode of the SVD sensor-basis is thus mapped, using least-squares techniques, to real-space basis functions for increased physics understanding. The SVD greatly reduces the computation cost of this mapping, as low energy eigenmodes are filtered and the mapping to a real-space basis need only be done once per eigenmode to capture the entire time evolution of the measurement.

Given a $P \times N$ basis matrix $\mathbf{B}$ composited of N real-space basis functions evaluated for P sensor locations, the weighting coefficient vector $\mathbf{C}$ for a SVD sensor-basis vector $\mathbf{U}_i$ are found by solving the linear equation,

$$\mathbf{B} \cdot \mathbf{C} = \mathbf{U}_i, \tag{2.26}$$

using a pseudo-inverse,

$$\mathbf{C} = \left(\mathbf{B}^T \mathbf{B}\right) \mathbf{B}^T \cdot \mathbf{U}_i. \tag{2.27}$$

This is similar to the least-squares fitting technique used for detection of nonrotating modes in DIII-D prior to the magnetics upgrade, which fit a $N \times T$ coefficient matrix using the raw data matrix in place of the sensor-basis vector [62]. Immediately, we see that the new computation is reduced by a factor of T (typically much greater than any other dimension).

The accuracy of the least squares fit can be improved by conditioning the basis matrix $\mathbf{B}$. A SVD of this matrix decomposes it into eigenmodes in real and sensor space ranked by their mutual coherence. That is, it determines a real-space basis that the sensor set is most sensitive to. The basis can then be manipulated by eliminating eigenvalues below the inverse of a desired condition number c ($s_i < c^{-1} \times s_{max} \to 0$) and re-combining the factorized singular vectors with the reduced singular value matrix. This performs the fit in the most readily measurable space determined by the sensor set. Equivalently, it ignores combinations of the basis functions that would be poorly determined by the sensors. Fig. 2.16 shows that this technique greatly improves confidence in the fit when many poloidal harmonics are needed to describe a coherent structure identified by the data matrix SVD. Fitting the eigenmodes identified in Fig. 2.14 using a a single toroidal mode number $n = 1$ and 6 cylindrical poloidal modes $-6 \leq m \leq -1$ using Eq. (2.28) without conditioning fails to identify clear poloidal harmonics. With conditioning, harmonics are clearly identified and their phase evolution shows the expected rotation followed by locking. The conditioning in this example used a conditioning parameter $c = 10$ over two orders of magnitude lower than the original basis condition number, reducing the rank of the basis matrix from 12 to 9.

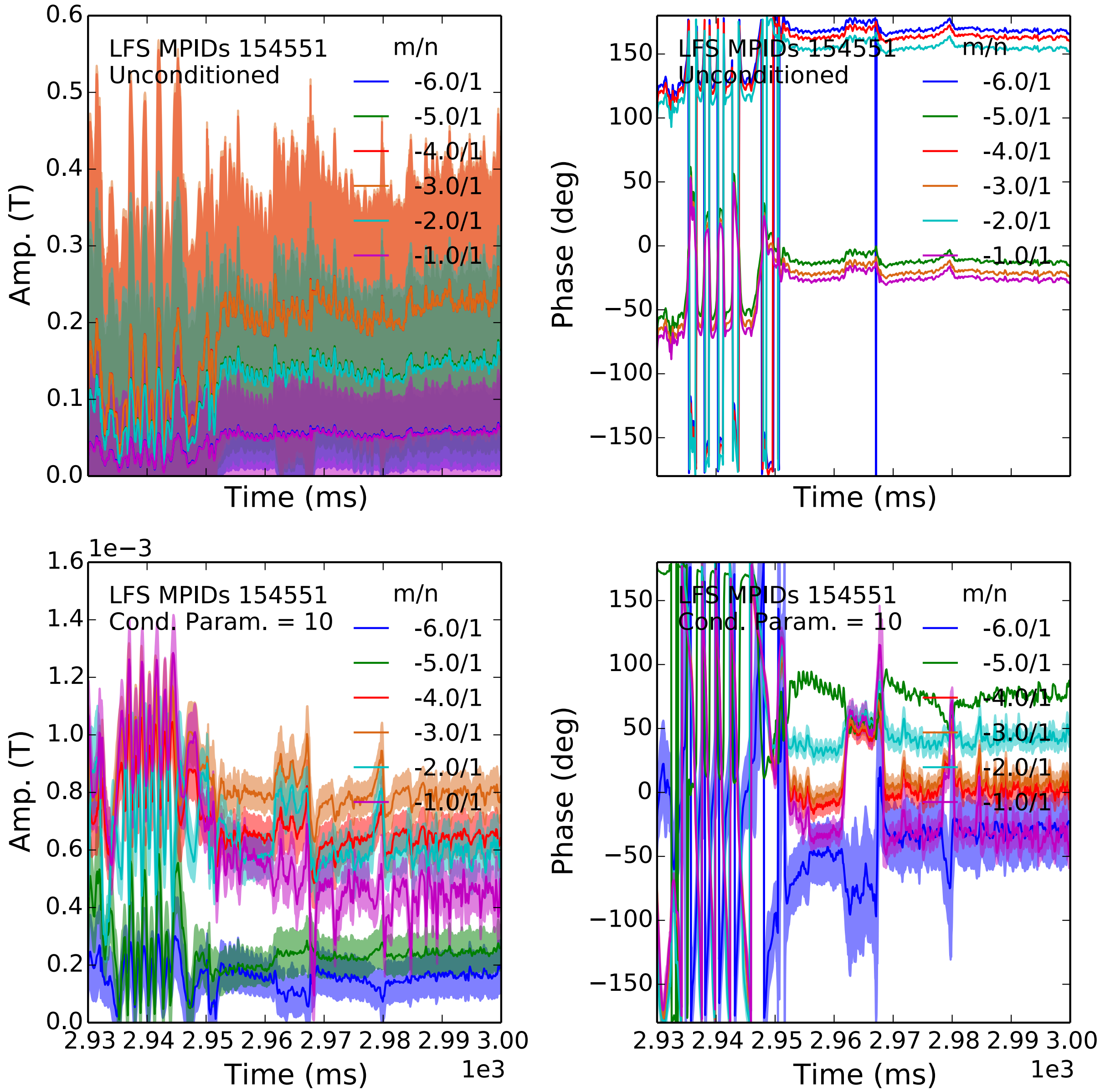

Figure 2.16: Time evolution of the amplitude (left) and phase (right) of the cylindrical Fourier harmonics fit using Eq. (2.28) and the 2 eigenstructures identified in Fig. 2.14. Without conditioning (top), the fit fails to identify clear poloidal harmonics. With conditioning (bottom), harmonics are clearly identified. The basis matrix uses a single toroidal $n = 1$ harmonic, and 6 poloidal harmonics $-6 \leq m \leq -1$. The conditioning used reduces the condition number of the basis matrix to 10, reducing the effective rank from 12 to 9.

The obvious limitation of this technique is that it must retain enough sensor-optimized real-space basis functions to accurately capture the physical phenomena. Fig. 2.17 shows the tradeoff: The condition number of unconditioned basis matrixes quickly becomes unacceptably large when including short wavelength (large $|m|$) modes, while the basis function rank retention with decreased condition number plummets when conditioning to values below ~ 10. Typical values of $c = 10 - 10^2$ retain approximately 10 orthonormal sensor-optimal basis vectors for identification of LFS structures with a single toroidal mode number. This is typically sufficient for the low wavelength phenomena of interest to this diagnostic suite.

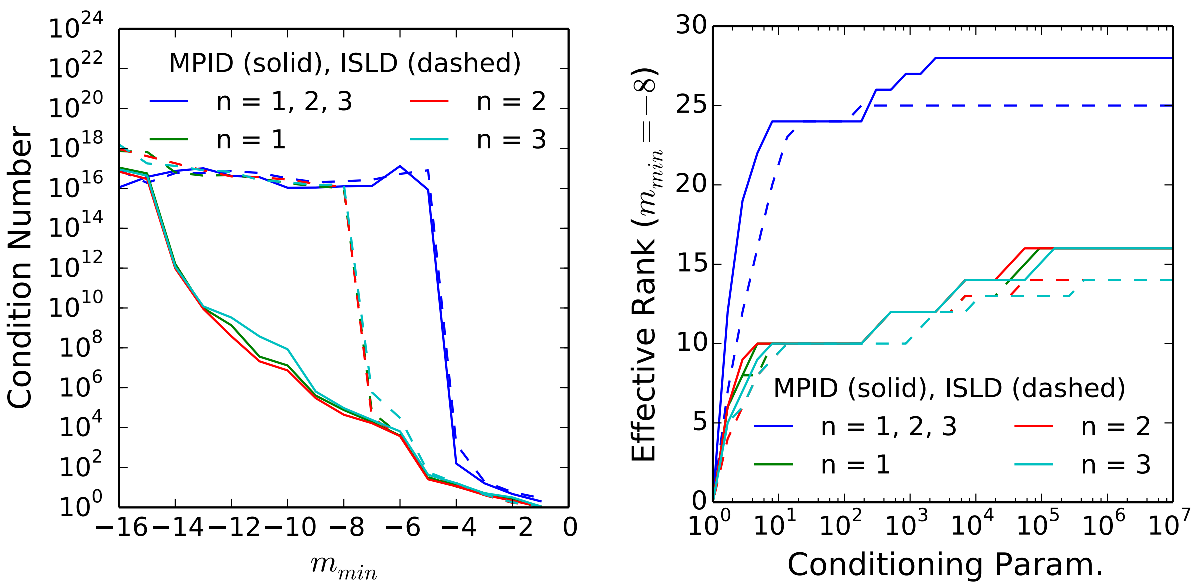

Figure 2.17: The condition numbers (left) of 2D cylindrical basis function matrices using the 5 LFS differenced parallel field sensor (MPID, solid) and internal saddle loop (ISLD, dashed) arrays (left) and all integer m such that $m_{min} \leq m < 0$ are plotted as functions of m_{min}. The condition number of the basis containing toroidal harmonics $1 \leq n \leq 3$ is shown in blue, while the condition number of basis matrices with only the individual $n = 1$ (green), $n = 2$ (red), and $n = 3$ (cyan) toroidal harmonics are also shown. The effective rank of the matrices using $m_{min} = -8$ is shown (right) as a function of the forced condition number eigenvalue cutoff.

2.4.2.3 Two Dimensional Geometric Basis Functions

Toroidicity, as well as plasma and vessel elongation and shaping, implies that the cylindrical poloidal harmonic m is not a good quantum number in DIII-D. Poloidal harmonics are not eigenvectors for observable quantities such as energy and momentum, and measured quantities such as the magnetic field at the vessel are often a mixture of many m modes due to the lack of symmetry. The significant deviation of the DIII-D vessel from a cylindrical approximation is shown in Fig. 2.18. The majority of the magnetic sensors in DIII-D are mounted on the LFS surfaces, which are shown to more closely approximate a spherical cross-section than a large aspect ratio cylindrical cross-section. The center stack, a vertical surface, is poorly described by either cross-sectional geometry and the most natural geometric parameters are the height and toroidal angle (z, ϕ) for HFS sensors. For these reasons, it should not be taken a priori that a cylindrical basis is the best basis for the fitting and interpolation described above. In this section we compare two geometric basis functions: Fourier decomposition in the cylindrical angles and spherical harmonics.

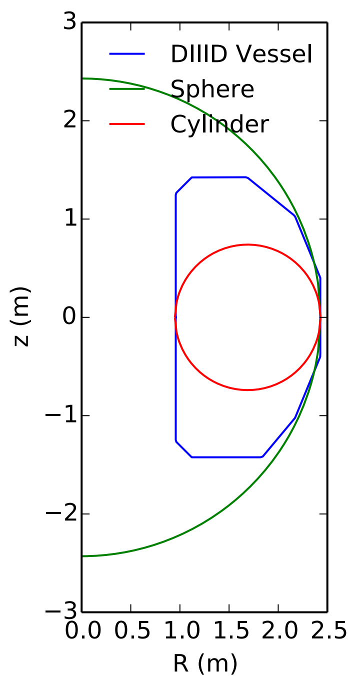

Figure 2.18: A poloidal cross-section of the DIII-D vacuum vessel wall, with cylindrical and spherical approximations that intersect the wall at the LFS midplane. Note the LFS is more closely matched by a spherical approximation then a cylindrical one, while the center-stack is poorly matched by either.

Fourier decomposition in the poloidal angle $\theta = \tan^{-1}(z \ / (R - R_0))$ and azimuthal angle ϕ uses basis matrix elements,

$$B_{ij} = \frac{1}{2}\left\{\frac{1}{\Delta\phi_i\Delta\theta_i}\int_{\phi_i-\Delta\phi_i/2}^{\phi_i+\Delta\phi_i/2} d\phi \int_{\theta_i-\Delta\theta_i/2}^{\theta_i+\Delta\theta_i/2} d\theta \exp\left[i\left(n_j\phi - m_j\theta\right)\right]\right., \quad (2.28)$$
$$\left. - \frac{1}{\Delta\phi_i^p\Delta\theta_i^p}\int_{\phi_i^p-\Delta\phi_i^p/2}^{\phi_i^p+\Delta\phi_i^p/2} d\phi \int_{\theta_i^p-\Delta\theta_i^p/2}^{\theta_i^p+\Delta\theta_i^p/2} d\theta \exp\left[i\left(n_j\phi - m_j\theta\right)\right]\right\},$$

where the (θ_i, ϕ_i) is the center of the ith sensor, $(\Delta\theta_i, \Delta\phi_i)$ is the poloidal and toroidal extent of the ith sensor, the similar quantities with a superscript p are the center and extent of the sensor's differenced pair, and (n_j, m_j) are the toroidal and poloidal harmonics of the jth mode. The toroidal modes of interest have been discussed in the context of hardware design, and are $n \leq 4$ in DIII-D. The poloidal modes of interest are the natural modes of the plasma (tearing, kink, etc.), which tend to have the equilibrium helicity [52]. For operation with the standard toroidal field direction $(-\mathbf{e}_\phi)$ and plasma current $(+\mathbf{e}_\phi)$, this corresponds to the negative poloidal harmonics. Note that the basis matrix describes the sensor signals, which are an average over the extent of the sensor.

The spherical harmonic basis matrix,

$$B_{ij} = \frac{1}{2}\left\{\frac{1}{\Delta\phi_i\Delta\Theta_i}\int_{\phi_i-\Delta\phi_i/2}^{\phi_i+\Delta\phi_i/2} d\phi \int_{\Theta_i-\Delta\Theta_i/2}^{\Theta_i+\Delta\Theta_i/2} d\Theta e^{in_j\phi} P_{m_j}^{n_j}\left(\cos\Theta\right)\right. \quad (2.29)$$
$$\left. -\frac{1}{\Delta\phi_i\Delta\Theta_i}\int_{\phi_i^p-\Delta\phi_i^p/2}^{\phi_i^p+\Delta\phi_i^p/2} d\phi \int_{\Theta_i^p-\Delta\Theta_i^p/2}^{\Theta_i^p+\Delta\Theta_i^p/2} d\Theta e^{in_j\phi} P_{m_j}^{n_j}\left(\cos\Theta\right)\right\}, \quad (2.30)$$

relies on associated Legendre polynomial $P_m^n\left(\cos\Theta\right)$ decomposition to describe the behavior in polar angle $\Theta = \cos^{-1}\left(z/\rho\right)$. Here z is again the vertical height and ρ is the distance from the origin at the center of the tokamak. These are orthogonal on the sphere defined by a radius $\rho = \text{const}$. Imposing regularity at the poles of this sphere in the solution of Laplace's equation forces integer polar harmonics such that $m_j \geq |n_j|$. The polar spacing between the closest toroidal arrays on the LFS is $\sim 12°$. This suggests an appropriate upper bound on the polar mode number $m \leq 8$, comparable to the bound given by the Nyqvist condition in the cylindrical Fourier basis.

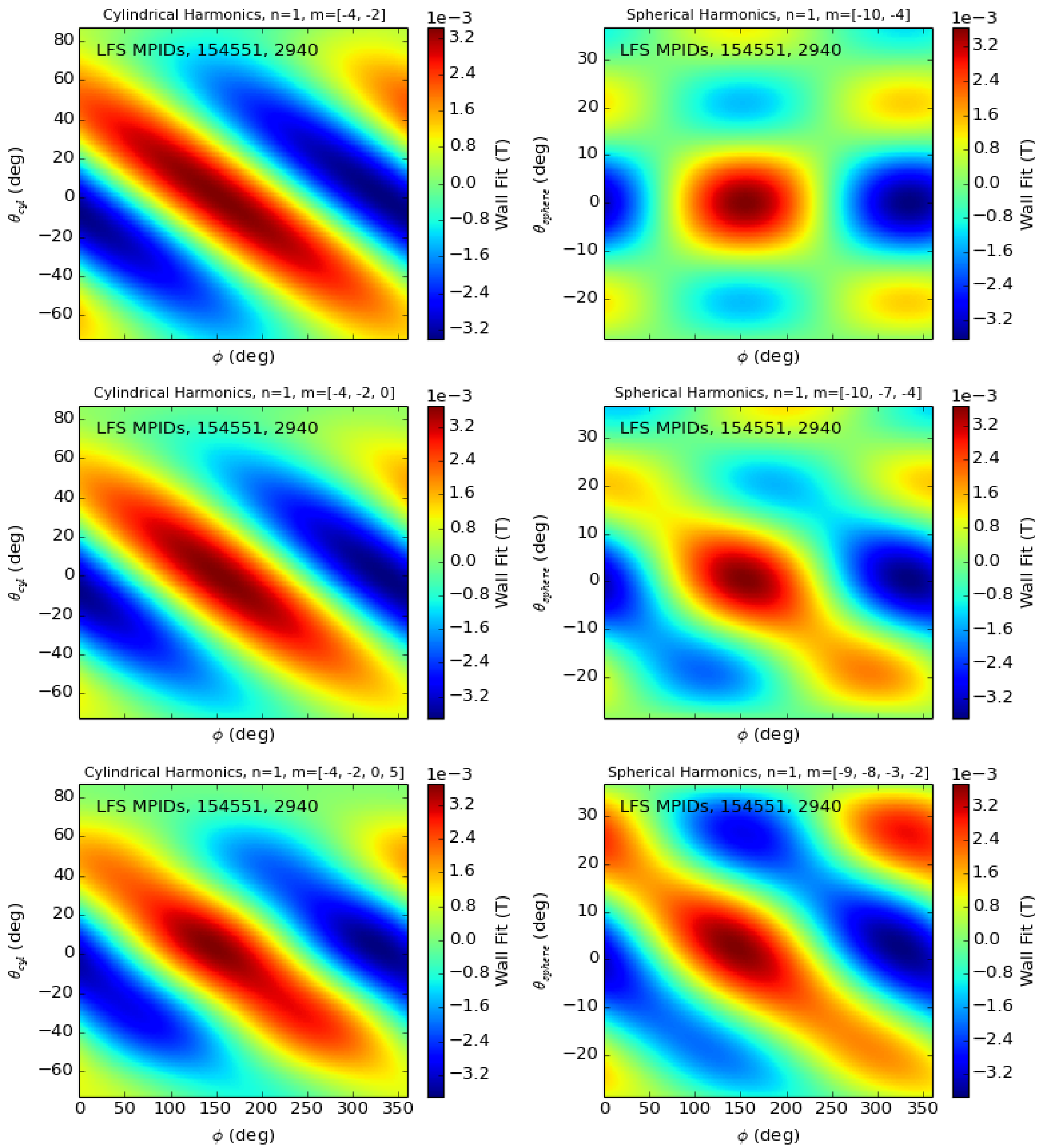

Figure 2.19: Contour plots of the plasma magnetic field at the vessel wall reconstructed using the 5 LFS magnetic probe differenced arrays (MPIDs) and plotted across machine toroidal (ϕ) and poloidal (θ) angle. The reconstructions on the left use cylindrical basis functions including a single $n = 1$ toroidal harmonic and 2 (top), 3 (middle) or 4 (bottom) poloidal harmonics. The reconstructions on the right use spherical basis functions, again with a single toroidal and 2, 3, or 4 (top to bottom) poloidal mode numbers. The reconstructions are of a saturated and locked tearing mode (discharge 154551, time 2940 ms).

Spherical harmonics have the advantage that the polar harmonic is a good quantum number on a surface that closely resembles that of the DIII-D LFS wall. Sensitivity to this geometric matching is in large part determined by the radial solution to Laplace's equation, ρ^{-m}, for modes from the plasma. Using the radius $\rho = 2.43$m used in Fig. 2.18 and a maximum deviation of the R+2 array $\Delta\rho = 0.2$m gives an approximate error $\sigma_B/B \sim m\Delta\rho/\rho \sim 0.1m$. A similar cylindrical harmonic analysis would give errors on the order of unity ($\sim m\Delta r/r$), while the toroidicity alone couples the poloidal harmonics and introduces additional errors associated with the ability to measure only a severely truncated spectrum.

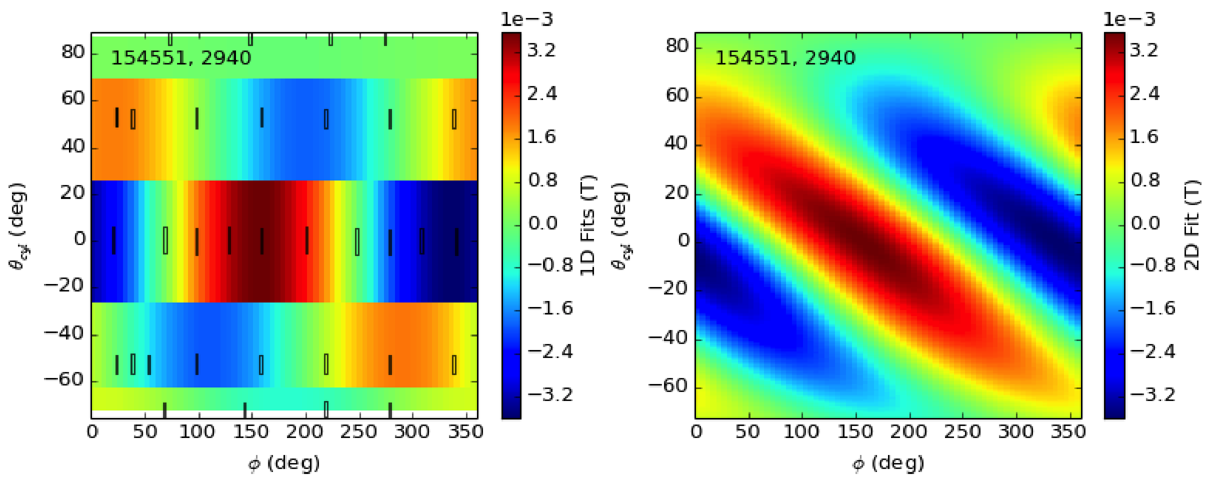

Figure 2.20: Contours of the same stationary plasma mode field shown in Fig. 2.19, reconstructed as individual toroidal fits at each array (left) and smoothly extended to the full surface using a third order spline (right).

The sampled magnetic field is reconstructed using the cylindrical and spherical basis matrices in Fig. 2.19. In each case, the 5 toroidal arrays of differenced magnetic probe arrays are used and amplitudes of prescribed mode spectrum are fit according to Eq. (2.27) after conditioning the basis matrix ($c = 10^2$). The signals are from a large $n = 1$ tearing mode at the $q = 2$ surface in a beam heated plasma. This strong, single-mode source is expected to produce a smooth structure at the vessel wall. Thus, the full 2D fits may be compared to the polynomial spline interpolation of the $n = 1$ amplitude and phase from each individual array shown in Fig. 2.20. The strength of the cylindrical Fourier decomposition is shown by the reasonable agreement with the interpolated mode using few poloidal harmonics. The spherical harmonics begin to reproduce a smooth structure only when more (4) poloidal harmonics are included. For this reason, and for the clear connection to standard theoretical models, the remainder of this work uses exclusively cylindrical basis functions.

Chapter 3

Measured Electromagnetic Torque

Two distinct methods of measuring electromagnetic (EM) torque are used in this thesis. The first is the Maxwell stress tensor technique, described in Section 2.1,

$$T_{EM} = \int dAR\frac{1}{\mu_0}B_\phi B_\perp. \tag{3.1}$$

This technique is applicable in the low m quasi-stationary limits discussed, and used when large plasma modes exchange momentum with eddy currents in the vessel structure.

The second approach is unique to the geometry of the I-coils on DIII-D, which lie on the interior measurement surface. These coils produce sharp, high m fields in the surface as shown in Fig. 2.4 and cannot be described consistently by the measurements. When the interaction of interest is between nonaxisymmetric fields produced by the I-coils and the plasma response, the EM torque can be calculated directly using,

$$T_{EM} = \int d\ell R\left(\mathbf{J}\times\mathbf{B}\right)_\phi. \tag{3.2}$$

Here the integral is taken along the contour parameterized by length ℓ for each individual coil on the LFS surface. The only contribution to the toroidal torque is from the plasma response field $b_\perp$. This requires precise vacuum compensation of the large coupling between the internal saddle loop sensors and the I-coils, but reduces the number of components necessary for a direct torque measurement.

Both calculations rely on the extensive analysis techniques detailed in the previous chapter. The computational steps of any EM torque calculation are largely similar, and are summarized as follows:

1. Hardware integrated voltage measurements are taken from inductive sensors oriented normal or parallel to the vessel surface but perpendicular to the toroidal field.
2. Unwanted pick-up is removed by zeroing the signals and applying vacuum compensation.
3. The eigenstructure and temporal dynamics of each component are separated by SVD analysis, and the uncorrelated signals removed.
4. The eigenstructure is decomposed into Fourier harmonics using conditioned least squares fitting techniques.
5. The EM torque is obtained by:
    a) Integrating the measured Maxwell stress according to Eq. (2.4).
    b) Interpolating the plasma response field to the EFC coils, and integrating the torque according to Eq. (3.2).

The calculations of the EM torque described here are a unique tool for the exploration of plasma rotation in the presence of nonaxisymmetric fields. Peak torques from large tearing modes with toroidal mode number $n = 1$ have been previously measured using the Maxwell stress tensor [11], but never before has the stress from multimodal plasma structures been mapped to the vessel surface with the accuracy of the upgrade and its devoted analysis. This allows the exploration of detailed dynamics in time as well as contributions from nonresonant spectra. The measurements are model-independent, passive, and robust. This chapter presents a range of examples, showcasing the wide range of plasma parameters, spacial scales and time scales for which the EM torque can be measured.

The example measurements of EM torque in DIII-D are organized as follows. Section 3.1 shows the dynamic locking of a large $n = 1$ plasma mode. The measurement exhibits the characteristic inaccuracy due to shielding above rotation frequencies $\omega_\varphi \approx 0.5$kHz, oscillates due to the interaction with a nonaxisymmetric error field at low ω_φ, and matches a simple axisymmetric torque balance when the plasma is stationary. The direct I-coil coupling technique is then demonstrated in Section 3.2, showing qualitative agreement with tearing mode model torque estimates and enabling EFC estimations consistent with standard multi-shot optimization techniques. Finally, a perturbative EFC torque optimization is presented and compared to alternative optimization metrics in Section 3.3. Although a single mode model is used to fit the EFC optimum, no such assumption is used in the direct measurement of the data and no assumption is made as to the resonant or nonresonant nature of the torque on the plasma.

3.1 $n = 1$ Mode Locking

This section presents an example of electromagnetic torque measurements using the Maxwell stress tensor to study the slowing and locking of large plasma modes. Here, the momentum is exchanged between the plasma and the surrounding structures in which eddy currents are induced by the rotating plasma mode. The characteristics of the measurement are thus dictated by the details of the conducting structures.

The EM torque measured using the Maxwell stress technique is shown for a large tearing mode that locks to the lab frame in Fig. 3.1. The large tearing mode was identified using SVD analysis (Fig. 2.14 and Fig. 2.16), and interpolated across the LFS wall using the cylindrical basis (Fig. 2.19). The HFS contribution is ignored in this case, justified by much lower amplitude combined with the R^2 weighting in Eq. (2.4). The final result shows the $n = 1$

EM torque exchange between the mode and the vessel wall as the mode slows and locks.

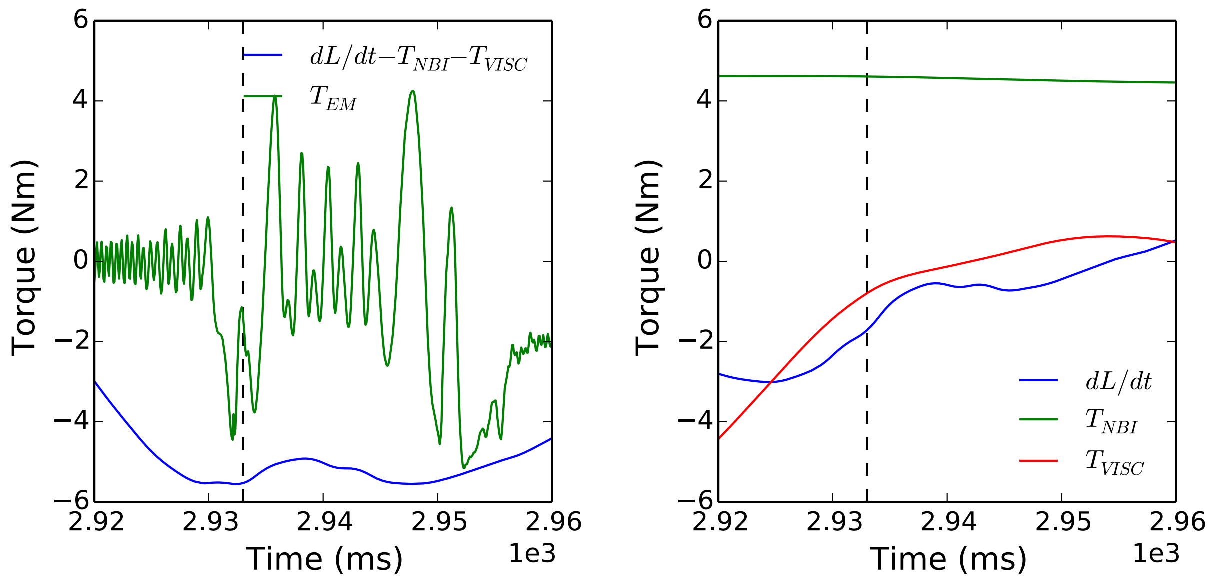

Figure 3.1: Left: The integral EM torque (green) and axisymmetric torque balance (blue) terms measured during DIII-D shot 154551 as a large $n = 1$ plasma mode slows and locks to the vessel wall. Right: the individual axisymmetric terms in the torque balance. A vertical dashed line marks a large EM torque transient corresponding to the temporary penetration of the perpendicular field shown in Fig. 3.2.

The maxwell stress measurement is compared in the figure to axisymmetric measurements of terms in a simplified torque balance,

$$T_{EM} = \frac{dL_\varphi}{dt} - T_{VISC} - T_{NBI}. \tag{3.3}$$

The first and second term, dL_φ/dt and $T_{VISC} = -L_\varphi/\tau_\varphi$, are estimated from axisymmetric density and rotation measurements. The final term T_{NBI}, is estimated using beam powers and purveyances and does not include effects such as prompt ion losses that may reduce the injected torque in the presence of large nonaxisymmetric perturbations. The corresponding values given in Fig. 3.1 have been subjected to low-pass filter with cutoff frequency $1/\tau_\varphi$ to remove transient behavior such as beam pulsing. The momentum confinement time $\tau_\varphi \approx$ 270ms is calculated at the beginning of the time window shown assuming $T_{EM} = 0$ prior to the growth of the mode, and assumed constant throughout the locking. The EM torque is not filtered, in order to better show the dynamic nature of the measurements.

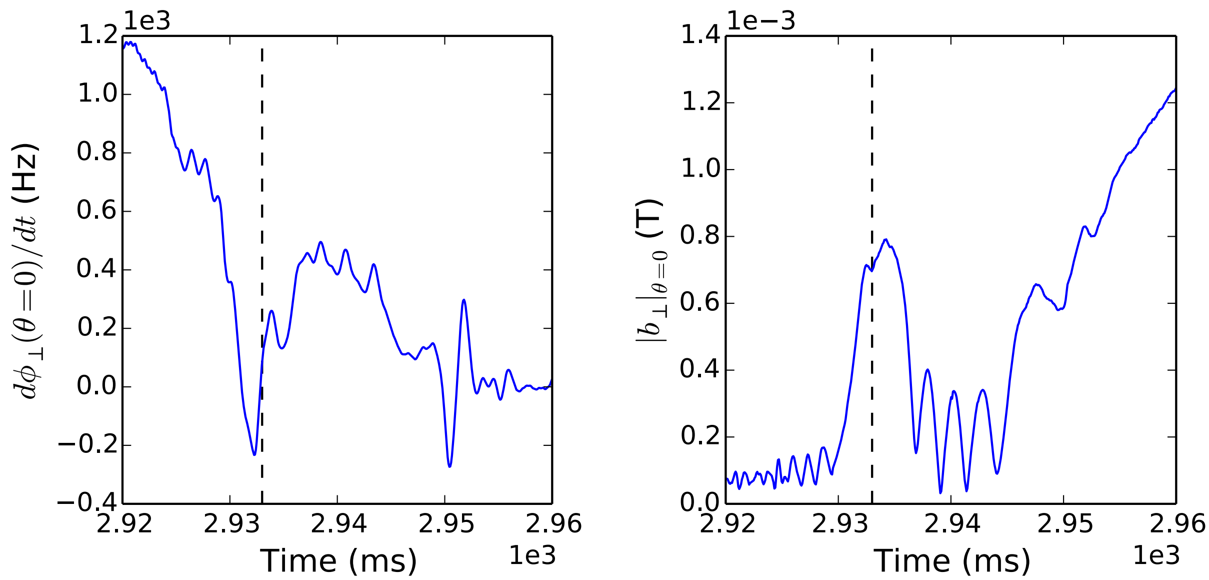

Figure 3.2: The $n = 1$ plasma mode toroidal rotation frequency measured by the LFS midplane array of parallel field sensors (left) and amplitude measured by LFS midplane array of saddle loops during shot 154551. Note that the dashed line correlated with a peak EM torque in Fig. 3.1 corresponds to a transient dip in the rotation and peak in the radial field.

Several pervasive characteristics of Maxwell stress measurements in DIII-D are apparent in the dynamics of this mode locking. It is known that the protective carbon tiles mounted on the inner vessel wall act as a second conducting wall with effective time constant $\tau_w \approx$ 0.14ms, shielding the sensors from fields perpendicular to the wall at high frequencies [11]. As expected, the torque measured behind these tiles is negligible compared to the axisymmetric terms at frequencies above $\sim$ 1kHz. Fig. 3.2 shows the rotation frequency of the plasma mode and the amplitude of the measured field perpendicular to the vessel wall. Large perpendicular fields are not measured until the rotation is on the order of 100Hz or less. At higher rotation, little EM torque is measured and the majority of the electromagnetic momentum exchange is between the plasma and the tiles. The mode locks at 2935ms into the discharge, and the measured EM torque jumps to the appropriate value to balance the axisymmetric terms. The plasma mode then briefly unlocks and rotates around the torus at approximately 0.4kHz. Oscillations in the measured torque during this time indicate the presence of an intrinsic error field [93] and the peak value is partially suppressed as some portion of the momentum exchange is with the tiles. After 2952ms the plasma locks again, and the measured Maxwell stress again agrees with the torque balance. The induced eddy currents in the wall decay with time constant $\tau_w \approx$ 4ms, and the electromagnetic momentum exchange transitions to dominantly an exchange with an unmeasured intrinsic error field. The measured torque thus decays after locking.

These large tearing mode structures were measurable at and near the mid-plane prior to the magnetics upgrade. As shown Fig. 2.19, the field structures have well defined wavefronts that can be effectively described by a single poloidal and single toroidal wavenumber. The peak EM torque measured at the time of locking is shown to agree with the approximate axisymmetric torque balance for a wide variety of plasmas and injected torques in Ref. [11]. The upgrade allows mode detailed descriptions of the fields, however, including multiple field structures and/or sources. This has been put to use in a number of error field correction (EFC) applications post-upgrade that are the subject of the remaining sections.

3.2 Tearing Mode Entrainment

The entrainment and steering of magnetic islands using the DIII-D EFC coils enables alignment of islands for study by toroidally localized diagnostics, suppression by electron cyclotron current drive [94, 82], and estimation of optimal intrinsic error field correction [83]. The last of these capabilities has been performed using detailed tearing mode and plasma response models [83], that can be benchmarked against the model-independent measurement of EM torques in DIII-D. This section presents this comparison for one such case, showing good agreement between the modeled and measured EM torque.

Large I-coil fields in the example discharge, 126623, induce and entrain a $m/n = 2/1$

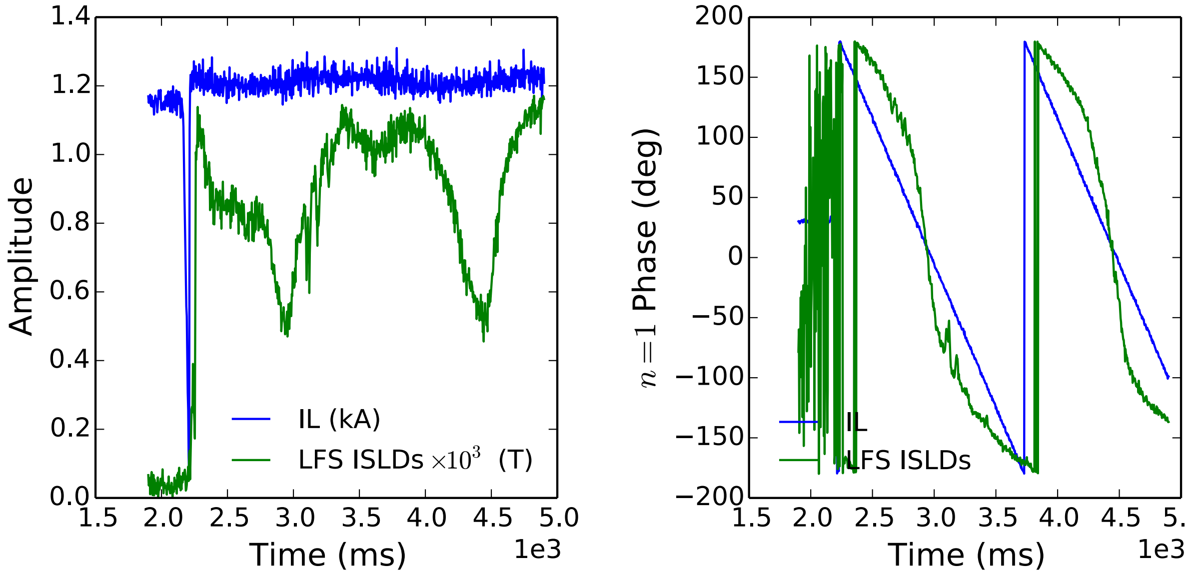

Figure 3.3: The amplitude (left) and phase (right) of the applied I-coil current and midplane plasma response measured perpendicular to the inner vessel wall during the saturation and entrainment of a $m/n = 2/1$ tearing mode in DIII-D shot 126623.

tearing mode as shown in Fig. 3.3. The discharge is a lower single-null plasma with toroidal field on axis $B_T = 1.6$T, plasma current $I_p = 1.0$MA, normalized beta $\beta_N = 1.2$, and safety factor at the surface encompassing 95% of the poloidal flux $q_{95} = 4.5$. The coils are turned on with 1kA in both the upper (IU) and lower (IL) coil sets, and relative phasing $\phi_{UL} = \phi_{IU} - \phi_{IL} = 240°$. The induced plasma mode is allowed to saturate (at $\sim 10^{-3}$T) and lock the plasma to the stationary frame. The saturated, locked mode is then rotated in the co-current toroidal direction by rotating the I-coil field with a toroidal frequency $f_\phi = 0.67$Hz. This is slow compared to the momentum confinement time estimated from the viscous torque balance prior to the coil on time $1/\tau_\varphi \approx T_{NBI}/L_\varphi = 4.16$Hz, and the island is expected to remain in a simplified torque balance,

$$T_{EM} = T_{EM}(coil) + T_{EM}(EF) = 0. \tag{3.4}$$

Here the time derivative of the angular momentum, viscous torque, and neutral beam torque are assumed negligible in the quasi-stationary plasma. These terms are shown in Fig. 3.4, supporting this approximation during times of peak EM torque magnitude. The torque from interactions between plasma fields and eddy currents induced in the vessel wall is also ignored as the fields freely penetrate the vessel wall at the low frequency of the rotation ($\omega_\varphi \tau_w \sim 10^{-3}$). The EM torque balance thus consists of two distinct components: the torque exchange between the plasma and I-coils $T_{EM}(coil)$ and the torque exchanged between the plasma and the intrinsic EF sources $T_{EM}(EF)$. The first can be directly measured using the magnetic sensors.

The direct measurement of the toroidal torque applied by the I-coils in discharge 126223 is shown in Fig. 3.4. The torque is calculated using Eq. (3.2), the known I-coil currents and geometries, and the full 2D structure of the plasma response measured on the interior vessel wall. Also included in Fig. 3.4 is the torque calculated using the missing bootstrap current and total field on the $q = 2$ flux surface [83], each requiring detailed modeling (using ONETWO and IPEC respectively). Note there is a factor of 2 applied to the model result, accounting for a unknown discrepancy in the modeled and measured torque amplitude. The measured and modeled torque agree well qualitatively however, oscillating a-symmetrically as the I-coil applied field is rotated around toroidally.

The oscillatory toroidal dependence of the applied EM torque is due to an unaccounted for intrinsic EF. The torque from the intrinsic EF cannot be directly measured, as the vacuum field is removed in the baselining process required to null the axisymmetric pickup and integrator drifts accumulated since the beginning of the discharge. The value, however, can be estimated from the simplified torque balance Eq. (3.4). The implicit assumption is that the radial field of the plasma mode is always in phase with the total (applied and intrinsic) EF. The structure of the intrinsic EF is assumed constant, the structure of the applied field is held constant, and the saturated plasma response structure is assumed constant in the single mode model. These assumptions allow the intrinsic EF to be expressed in terms of an equivalent correction coil current defined by amplitude and phase $\mathbf{I}_0 = I_0 e^{i\phi_0}$. The total EF

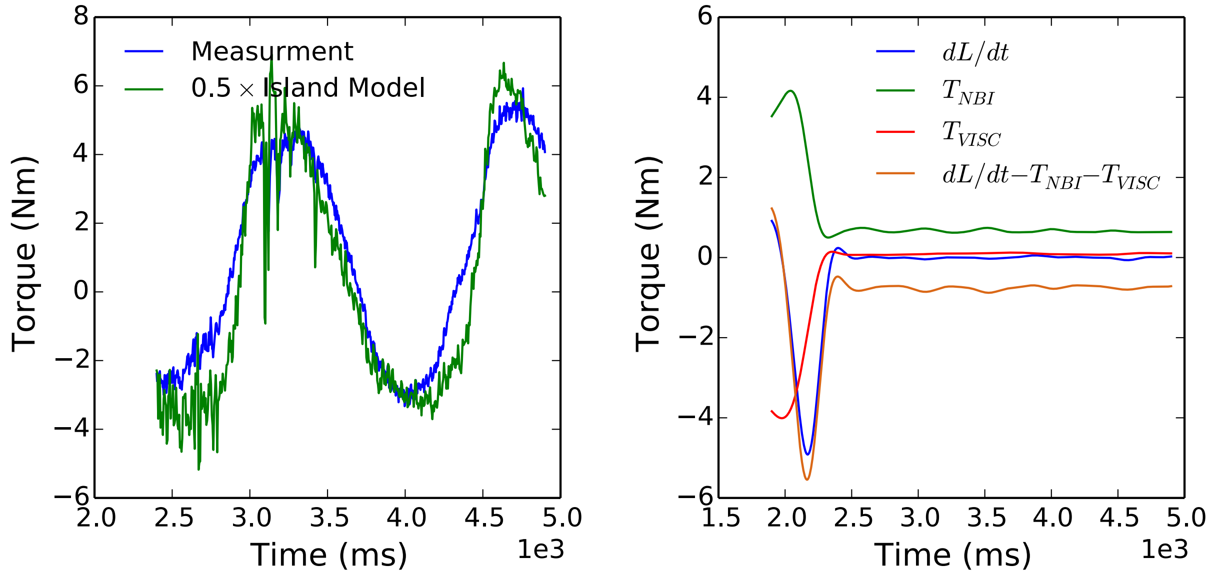

Figure 3.4: Left: The EM torque measured during the entrainment of a saturated tearing mode (blue) and normalized prediction (green) using a detailed island model and plasma response modeling. Right: The axisymmetric torque balance terms during this time, filtered to remove high frequency contributions above $1/\tau_\varphi = 4.16$Hz.

can then be expressed in the lower I-coil space, $\mathbf{I}_T = \mathbf{I} + \mathbf{I}_0$.

The EFC current $\mathbf{I}_0$ can be estimated from Eq. (3.4) by noting that the plasma response remains in phase with the total EF [83],

$$\angle \mathbf{b}_\perp = \angle \mathbf{I}_T, \tag{3.5}$$

or from the an estimate of the individual EM torque contributions [93],

$$T_{EM} = K_{EM} \mathbf{b}_\perp \times |\mathbf{I}_{IC} - \mathbf{I}_0| \,, \tag{3.6}$$

$$T_{EM}(coil) = -K_{EM} b_\perp I_0 \sin(\phi_\perp - \phi_0) \,, \tag{3.7}$$

where $\mathbf{I} = Ie^{-i\phi}$ is the complex representation of the applied $n = 1$ I-coil waveform, $-\mathbf{I}_0$ is the effective EF in this complex I-coil phase space, $\mathbf{b}_\perp = b_\perp e^{i\phi_\perp}$ is the measured plasma response, and K_{EM} is a factor dependent on the geometric coupling between the I-coil and plasma mode. Here the $\angle$ operator returns the angle of a complex vector (i.e. $\angle \mathbf{b}_\perp = \phi_\perp$). The optimal EF correction, $\mathbf{I}_0$, can be found by fitting Eq. (3.5) or Eq. (3.7) during the entrainment. These two fits are shown in Fig. 3.5 for measurements at the LFS mid-plane. The two fits agree on the EFC phase to within 6° of one another but have a discrepancy of 140A in the magnitude of the EFC. This is considered good agreement, comparable or lower than the typical discrepancies between the magnetic entrainment technique and alternate techniques [83]. It is necessary to note that this good agreement is not found everywhere across the poloidal cross section of the vessel however, suggesting that multimodal effects may play a role in the measured torque. It is also possible that an additional torque proportional to $b_\perp^2$ would reduce the discrepancy [93, 84]. This is beyond the scope of this example however, which is designed to show the simplicity and directness of the model free measurement.

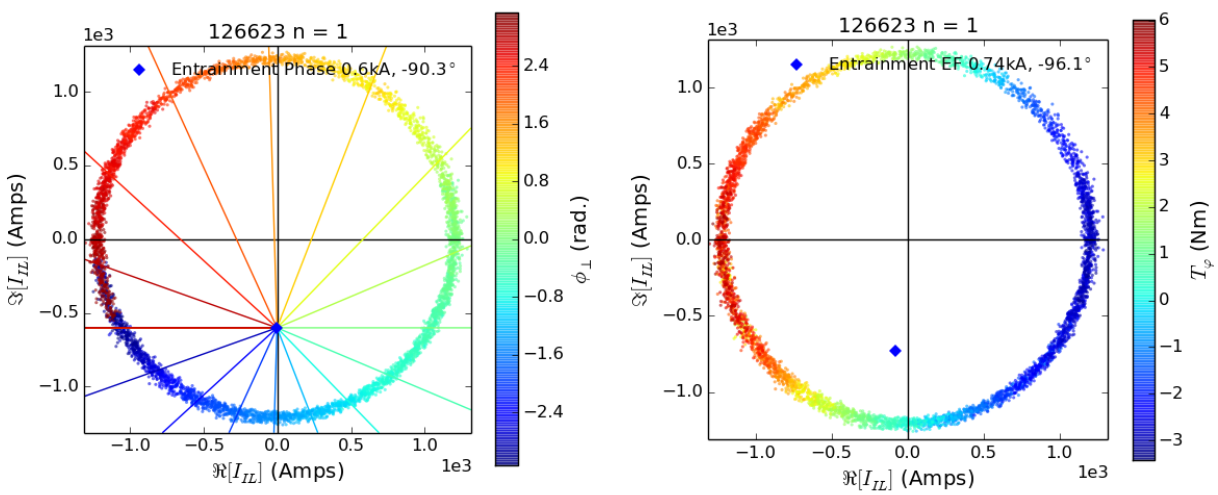

Figure 3.5: Left: The measured phase of the plasma field perpendicular to the LFS vessel midplane is indicated by by color and plotted on the trajectory of the complex I-coil current $\mathbf{I}_{IL}$. The optimal EFC (blue diamond) is determined by a least-squares fit to Eq. (3.5), and solid contours of constant total EF phase, $\angle \mathbf{I}_T$, are shown for the fit. Right: The measured EM torque plotted on the rotating trajectory of the complex coil current. The optimal EFC (blue diamond) is determined by a least-squares fit to Eq. (3.7). Both plots use data from between 1900 and 4900ms in shot 126623. In each case, the x an y axes are the real and imaginary components of the lower I-coil current rotated by 120 degrees to translate to an effective mid-plane phase for consistency with Ref. [83].

3.3 Perturbative EFC Optimization

This section presents the application of the I-coil EM torque measurement technique to perturbed, stable equilibria in which there are no macroscopic changes in the plasma equilibrium such as the island penetration and saturation observed in the previous sections. The EM torque measured during a scan of the applied I-coil EF still depends on the total EF in the perturbative limit, and can again be used to optimize the applied EFC.

In a I-coil phase space scan the relative amplitude and phasing of the two sets of I-coils are fixed, once again reducing the variations to a two dimensional vector space $\mathbf{I} = Ie^{in\phi}$. This two dimensional space is scanned by varying the amplitude and toroidal phase of the fixed I-coil waveform. If the scan is done slowly compared to the momentum confinement time, the plasma remains in a stable torque balance determined by the axisymmetric terms as well as a integral electromagnetic torque determined by the total (applied and intrinsic) EF. These stable scans are used to directly probe the plasma response to applied fields.

Historically, axisymmetric figures of merit such as the normalized beta β_N and angular momentum L_φ as well as the midplane nonaxisymmetric plasma response have been optimized using these scans to determine the intrinsic EF [54, 95, 29]. Fig. 3.6 shows the evolution of these parameters as even parity I-coils are slowly scanned for a typical H-mode EFC optimization discharge. Also shown, the measured EM torque follows a simple torque balance estimated using Eq. (3.3). Note that each of the axisymmetric torque balance terms shown have been filtered to remove noise at frequencies above $1/\tau_\varphi$, where the momentum confinement time $\tau_\varphi = 71$ms is estimated from the approximate axisymmetric torque balance at 2.1 seconds. The applied EM torque T_{EM}, angular momentum $L_\varphi \propto T_{VISC}$, and normalized beta β_N are all observed to oscillate in unison as the applied field alternately compensates and compounds the intrinsic error field.

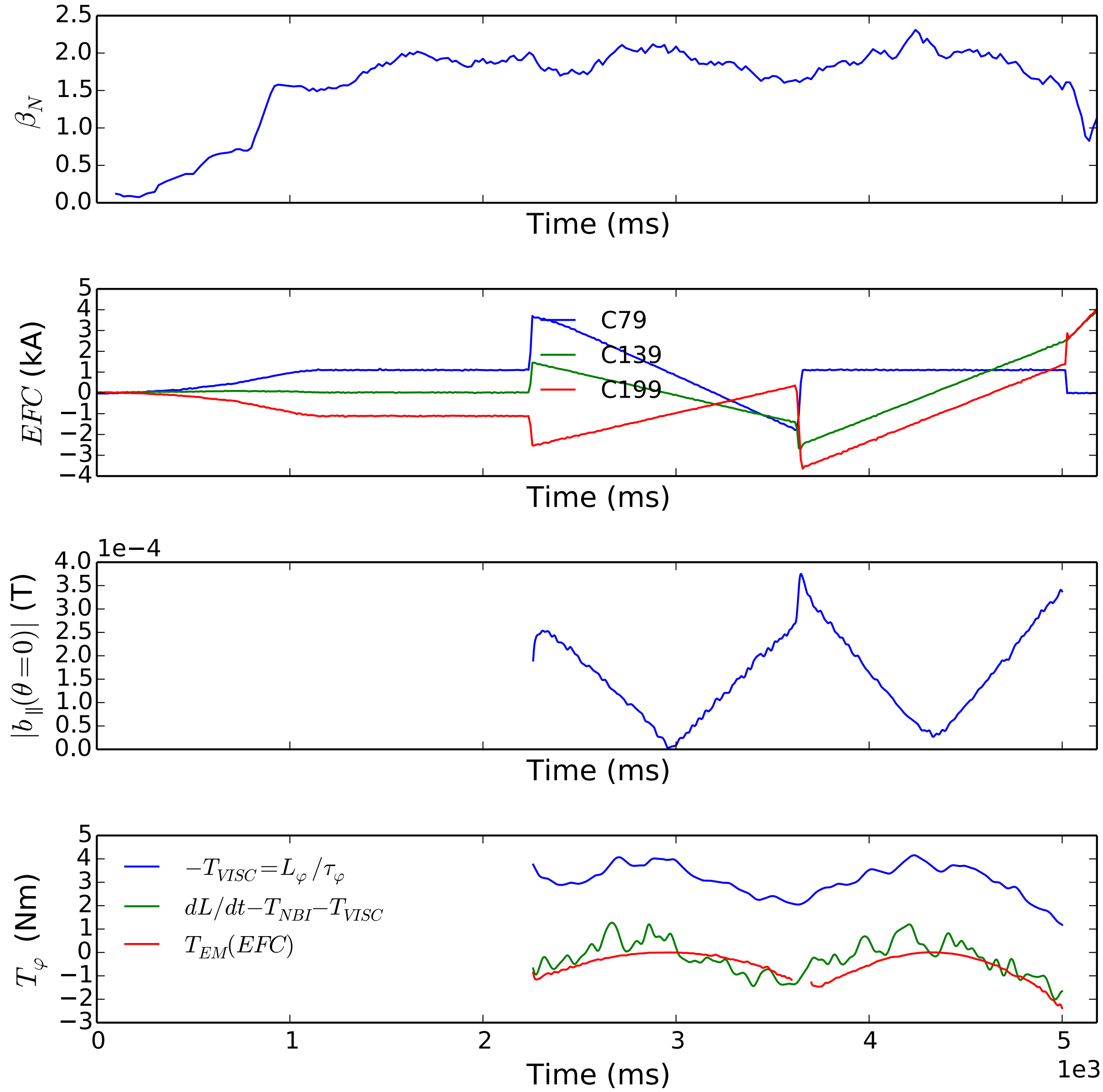

Figure 3.6: A perturbative EFC scan in discharge 152797: The high confinement mode enables access to $\beta_N \approx 2.0$ (top), the C-coil currents are swept in a scan of the $n = 1$ EFC waveform phase and amplitude beginning at 2250ms (second from top), the MPID measured midplane plasma response amplitude evolves in response to the total EF (second from bottom), as do the perturbed terms of the torque balance (bottom) before the discharge is eventually terminated after 5 seconds.

The direct measurement of the EM torque provides continuous measurements of the integral torque applied by the coils during the phase space scan. The total EM torque, however, is in general a non-linear function of the applied and intrinsic nonaxisymmetric fields. As before, the assumption that the plasma response is dominated by the coupling of the total field to a "dominant mode" [49, 45, 14, 83, 29] allows the intrinsic EF to be expressed as single vector in the coil space $-\mathbf{I}_0$. The EM torque then becomes a function of the difference between the applied and optimal EFC current $T_\varphi\left(\mathbf{I}-\mathbf{I}_0\right)$.

Fig. 3.7 shows the optimal EFC modeled using $T_\varphi = \mathcal{A}\left|\mathbf{I}-\mathbf{I}_0\right|^2 + \mathcal{B}$ gives a result similar to the standard L_φ optimization as well as optimization of the right hand side of Eq. (3.3) using a similar functional dependence. The optimal EFC currents ($\mathbf{I}_0$) are shown in the complex phase space, while a reduction to the real one dimensional variable $|\mathbf{I}-\mathbf{I}_0|$ confirms the quadratic dependence. The result clearly shows two distinct quadratic branches. These correspond to the two crosses of the "X" scanned in the complex coil space. The discrepancy between them can be taken as an estimate of the error associated with the technique. Note that the other figures of merit, such as the axisymmetric torque terms shown in Fig. 3.6 have noisy oscillations of a similar order ($\sim$ 1Nm), resulting in a similar error estimate in the quadratic fit. The standard deviations of the optimal EFC amplitude and phase are 0.12kA and 6.4° respectively comparing the full fit and fits to the individual branches of the "X" scan.

The EM torque measurements presented here do not use any physics model for the plasma response or torque distribution within the plasma. The EFC model has been presented and used to good effect with minimal understanding of the underlying physics. The behavior of the plasma is fundamentally different, however, than the behavior during or after large events measured in the previous sections. Here a large saturated island structure in the plasma is not interacting with external fields. Instead a perturbed plasma response is stably increased or decreased as the total EF slowly increases or decreases. The EFC model shows the perturbative EM torque is a second order quantity, consistent with a square of the perturbed fields. A fundamental understanding of the second order EM torque inside a perturbed equilibrium is needed to justify this model of the integral torque and thus confirm

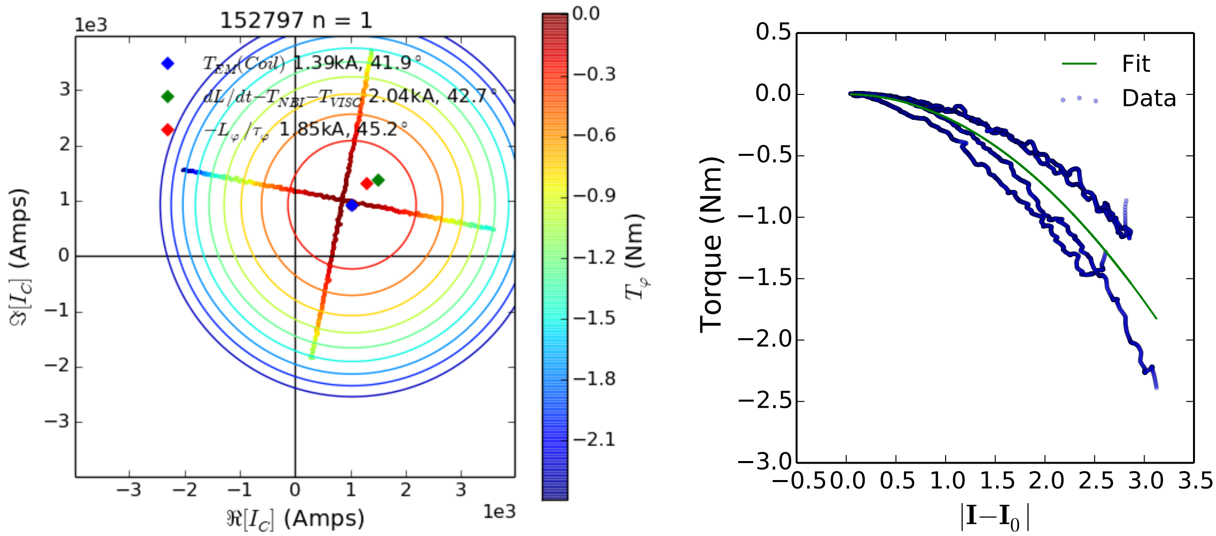

Figure 3.7: Left: The EM torque measured between 2250ms and 5000ms in discharge 152797 is indicated by color and plotted on the "X-scan" trajectory of the complex C-coil current $\mathbf{I}_C$. Estimates of the EFC optimum (diamonds) are given by least-squares fits of the EM torque (blue), axisymmetric torque balance terms (green) and viscous torque (red) to a quadratic amplitude of the total EF and offset. Solid lines show the contours of constant torque for the EM torque fit. Right: The measured EM torque (blue circles) plotted against the amplitude of the total EF, and the least-squares quadratic fit (green).

validity of the EM torque measurements in this perturbative regime. Although resonant perturbations in intrinsically tearing stable plasmas can induce local EM torques with quadratic dependence on the EF [16], the following chapters present a theoretical model including all harmonics and their nonlinear coupling. The model is often referred to as neoclassical toroidal viscosity (NTV) and associated with nonresonant fields. The following chapter details how the theory and the corresponding computational model built to study such torques can be used to account for the EM torque induced by the complete spectrum throughout the plasma. Chapter 6 returns to experimental measurements of the torque, showing that the model agrees well with the experimental measurements in this example as well as many others.

Chapter 4

The Perturbed Equilibrium Nonambipolar Transport Code

It has long been known that magnetic toroidal asymmetries in tokamaks as small as $\delta B/B \approx 10^{-4}$ can cause toroidal flow damping [22, 69, 17, 96, 97]. The responsible electromagnetic torques are often split into "resonant" torque on island structures and "nonresonant" torque due to neoclassical toroidal viscosity (NTV). Resonant braking received immediate attention as a common cause of disruptions and has been regularly observed, modeled, and compensated in tokamaks [69, 17, 83, 93]. Recently, NTV has received attention as a potential method of rotation and stability control using applied nonaxisymmetric perturbations to alternately decrease or increase the plasma rotation [10, 98, 28]. The Perturbed Equilibrium Nonambipolar Transport (PENT) code was developed to accurately model the NTV torque that results from intrinsic and applied nonaxisymmetric fields in experiment, providing a physical intuition to the observed phenomena and predictive optimization of applied torque.

When magnetic field strength cannot be expressed as a function of a flux surface label ψ and distance along a field line l, such that $B = B(\psi, l)$, the action $J = \oint dl M v_\parallel$ of a particle with mass M and velocity parallel to the field lines $v_\parallel$ is not conserved on flux surfaces and the particle experiences radial drifts across flux surfaces [99]. This radial transport depends on the particle species and is not intrinsically ambipolar (the electron and ion flux may differ by a factor other than the charge ratio, causing a divergence of electric charge density). The nonambipolar transport produces radial currents that in turn produce $\mathbf{J} \times \mathbf{B}$ toroidal torque. This torque from nonambipolar transport in tokamaks is often called neoclassical toroidal viscous (NTV) torque, and has been observed to be the dominant influence on rotation in the presence of the nonresonant nonaxisymmetric perturbations [97, 10, 100].

Theoretical predictions of the NTV torque, however, are nontrivial due to detailed phase space structure that is dependent on different particle orbits, precessions, and collisions. A number of theories have been developed in limited regimes to simplify or ignore one or more such dependencies. The effects of passing and trapped particles are studied separately and various trapped particle effects are also studied separately in many different collisionality (ν) regimes including the $1/\nu$ regime, ν regime, $\sqrt{\nu}$ regime, and superbanana-plateau regime, or combined using approximate connection formulae [24].

In recognition of the practical importance of overlapping between the regimes, a combined NTV model has been developed using a simple Krook collision operator but including all bounce and precession orbits [23]. The combined theory treats a wide range of kinetic regimes simultaneously, automatically incorporating all plasma particles throughout energy and real space. This model is fully implemented in PENT [56], in conjunction with the Ideal Perturbed Equilibrium Code (IPEC) [44, 46], for general aspect ratio and shaped plasmas.

This chapter is organized as follows. Section Section 4.1 outlines the combined NTV theory, including a complete treatment of pitch angle dependencies in the bounce frequency, precession frequency and perturbed action. Explicit connections to the nonambipolar transport of particles and the electromagnetic torque are made in Section 4.2. The limited validity of previous large aspect ratio and deeply trapped particle approximations is then discussed in contrast to PENT's general geometric treatment in Section 4.3. Finally, the wide variety of kinetic resonances captured by PENT are detailed in Sec. 4.4.2. Here again, the PENT computation is contrasted with reduced approximations across multiple kinetic regimes and shown to recover the expected limiting behavior.

4.1 Neoclassical Toroidal Viscosity (NTV)

In this section we derive an analytical form of the toroidal torque that arises from neoclassical nonambipolar transport, often referred to as NTV torque, valid across the kinetic regimes of interest in modern tokamaks. The results reproduce those previously developed in a reduced large aspect ratio limit [23, 60], but are given here in their most general form. The derivation thus serves an important purpose in orienting the reader to the geometry dependent quantities and their ultimate role in the NTV torque. The structure of the derivation will be as follows: First, a general form for toroidal torque in magnetic coordinates is expanded using an anisotropic perturbed pressure tensor and linear expansion of the nonaxisymmetric equilibrium quantities. Next, the torque is written in its phase space representation and closed using the first order linearized drift-kinetic equation. The zeroth order drift-kinetic solution, a Maxwellian distribution function, is used in the final step to connect to familiar offset rotation physics [10, 9].

The general relation, given by Boozer [101], for toroidal torque in a closed magnetic confinement device is,

$$T_\varphi = \int d^3x \left(\frac{\partial \mathbf{x}}{\partial \varphi} \cdot \nabla \cdot \mathbf{\Pi} \right) = -\frac{1}{2} \sum_{ij} \int d^3x \frac{\partial g_{ij}}{\partial \varphi} \Pi^{ij}, \tag{4.1}$$

where g is the coordinate metric tensor and $\mathbf{\Pi}$ is the anisotropic pressure tensor with equilibrium $\mathbf{\Pi} = 0$ on magnetic surfaces. We now introduce a simple tensor pressure, in which the parallel and perpendicular pressures are distinct $\mathbf{\Pi} = \left(\delta p_\parallel - \delta p_\perp\right) \hat{\mathbf{b}}\hat{\mathbf{b}} + \delta p_\perp \boldsymbol{I}$ [102, 101]. Here $\delta p_\parallel$ and $\delta p_\perp$ are the perturbed pressure parallel to and perpendicular to the field, $\hat{\mathbf{b}}$ is the unit vector in the direction of the field, and $\mathbf{I}$ is the identity matrix.

We define our field in Clebsch representation with magnetic coordinates $(\psi, \vartheta, \alpha)$ such that $\mathbf{B} = \nabla\alpha \times \nabla\psi$ where ψ is the poloidal flux and the poloidal angle is defined as relating the toroidal angle α to the standard toroidal angle $\phi = \alpha + q(\psi)\vartheta$ with the safety factor q [103, 104]. This gives,

$$\hat{b}\hat{b}^{ij} = \left(\frac{\frac{\partial \mathbf{x}}{\partial \vartheta} \cdot \frac{\partial x^i}{\partial \mathbf{x}}}{\mathcal{J}B}\right)\left(\frac{\frac{\partial \mathbf{x}}{\partial \vartheta} \cdot \frac{\partial x^j}{\partial \mathbf{x}}}{\mathcal{J}B}\right) = \frac{\delta^i_\vartheta \delta^j_\vartheta}{(\mathcal{J}B)^2},$$

and,

$$g_{\vartheta\vartheta} = \frac{\partial \mathbf{x}}{\partial \vartheta} \cdot \frac{\partial \mathbf{x}}{\partial \vartheta} = (\mathcal{J}B)^2,$$

where we have introduced the Jacobian $\mathcal{J}^{-1} = \nabla\psi \times \nabla\vartheta \cdot \nabla\alpha$. The torque equation is thus expanded to,

$$T_\varphi = -\frac{1}{2}\sum_{ij}\int d^3x \frac{\left(\delta p_\parallel - \delta p_\perp\right)}{(\mathcal{J}B)^2}\frac{\partial}{\partial \varphi}(\mathcal{J}B)^2 + \delta p_\perp I^{ij}\frac{\partial g_{ij}}{\partial \varphi}, \tag{4.2}$$

which can be simplified noting the identities $I^{ij} = g^{ij}$ and $\sum_{ij} g^{ij}\partial g_{ij}/\partial\varphi = (2/\mathcal{J})\partial\mathcal{J}/\partial\varphi$,

$$T_\varphi = -\int d^3x \left(\delta p_\parallel - \delta p_\perp\right)\frac{1}{B}\frac{\partial B}{\partial \varphi} + \delta p_\parallel \frac{1}{\mathcal{J}}\frac{\partial \mathcal{J}}{\partial \varphi}. \tag{4.3}$$

The Jacobian and magnetic field are assumed axisymmetric with small nonaxisymmetric perturbations. To first order, these perturbed quantities are the change in flux surface arc-length $\mathcal{J} = \mathcal{J}(\psi, \vartheta)(1 + \nabla\cdot\boldsymbol{\xi})$ and the Lagrangian perturbed field $\mathbf{B} = \mathbf{B}(\psi, \vartheta) + \delta\mathbf{B}(\psi, \vartheta, \varphi)$ [60]. This gives,

$$T_\varphi \quad = -\int d^3x \quad \left[\left(\delta p_\parallel - \delta p_\perp\right)\frac{1}{B}\frac{\partial \delta B}{\partial \varphi} + \delta p_\parallel \frac{\partial}{\partial \varphi}(\nabla\cdot\boldsymbol{\xi})\right], \tag{4.4}$$

where the unperturbed quantities are all the zeroth order axisymmetric values and the integrand -a second order quantity- represents the toroidal torque in a nonaxisymmetric perturbed equilibrium.

Much of the important physics associated with NTV torque is a result of kinetic effects, thus far absent from this derivation. Inserting the phase space integral definitions of each perturbed pressure, $\delta p_\parallel = \int d^3v M v_\parallel^2 f_1$ and $\delta p_\perp = \int d^3v M v_\perp^2 f_1/2$ where f_1 is the perturbed distribution function, into Eq. (4.4) and converting to energy coordinates ($E = Mv^2/2, \mu = Mv_\perp^2/2B$) we arrive at a more insightful form of the toroidal torque,

$$
\begin{aligned}
T_\varphi \quad = \quad & -\frac{4\pi}{M^2}\int d^3x d\mu dE f_1 \frac{\mathcal{J}BM}{v_\parallel}\left[(2E-3\mu B)\frac{\partial}{\partial\varphi}\frac{\delta B}{B}\right. \\
& \left. + (2E-2\mu B)\frac{\partial}{\partial\varphi}(\nabla\cdot\boldsymbol{\xi})\right]. \qquad (4.5)
\end{aligned}
$$

Note that the second term in the bracket above corresponds to the arc-length change in the perturbed equilibrium. This term is often ignored in large aspect ratio NTV approximations that use $\mu B \approx E$ for trapped particles. It can contribute comparable amounts of torque in experimental geometries, however, and is explicitly kept in the PENT code for this reason [56, 29].

4.1.1 Drift Kinetic Equation for the Perturbed Distribution Function

At this point, it is necessary to solve for the perturbed distribution function. An appropriate model for the ideal perturbed equilibrium is the bounce averaged, first order drift-kinetic equation [105]. Explicitly, the drift-kinetic equation,

$$
\mathbf{v}_\parallel\cdot\nabla\delta f_s + \mathbf{v}_{ds}^{\alpha}\frac{\partial\delta f_s}{\partial\alpha} - C\delta f_s = -\mathbf{v}_{ds}^{\psi}\frac{\partial f_s}{\partial\psi}, \qquad (4.6)
$$

is used with the Clebsch representation and Krook collision operator. The bounce averaged equation can be written as,

$$\left\langle -2\pi i(\ell - \sigma nq)\frac{\partial h}{\partial \vartheta} v_{\|}\frac{\mathbf{B}}{B}\cdot\nabla\vartheta - 2\pi i n\mathbf{v}_{ds}\cdot\nabla\alpha + \nu_s \right\rangle \delta f_{\ell s} = \left\langle -\mathcal{P}_{\ell}{}^{-1}\mathbf{v}_{ds}\cdot\nabla\psi\frac{\partial f_s}{\partial\psi}\right\rangle, \tag{4.7}$$

where the subscript s refers to the particle species, the distribution function has been expanded as an axisymmetric f and perturbed δf components, $\mathbf{v}_d$ is the drift velocity, and a simple Krook collision operator $Cf_s = -\nu_s f_s$ is assumed. Note that a more detailed operator, such as a pitch angle scattering operator, may have a quantitative impact in the (relatively) higher collisionality NTV regimes but would require a differential drift-kinetic solver or further separation into limiting regimes rather than the regime-independent analytic solution obtained in the combined theory. Although the Krook operator is not momentum-conserving, any unphysical nonambipolar transport that it may induce independent of nonaxisymmetric effects is ignored in this perturbative formalism. The many and overlapping regimes in tokamak plasmas are treated by sacrificing the accuracy of the collision operator, with the validity of this approach demonstrated by ability of the final solution to match particle code and experimental results.

The angular brackets in Eq. (4.7) refer to the bounce average defined by $\langle A\rangle_b \equiv \oint\left(Adl/v_{\|}\right)/\oint\left(dl/v_{\|}\right) = \frac{\omega_b}{2\pi}\oint d\vartheta A\mathcal{J}B/v_{\|}$ with the bounce frequency $\omega_b \equiv 2\pi/\oint d\vartheta\ \mathcal{J}B/v_{\|}$. Note that the parallel electric field term for electrons has not been included in Eq. (4.6) since the Ware pinch effect is annihilated by the subsequent bounce averaging. The bounce average isolates the nonaxisymmetry driven perturbation to the distribution [106], and is well defined only for those species with closed orbits such that,

$$\delta f = \delta f(\mathbf{v},\psi)e^{-i2\pi n\alpha}e^{-i2\pi(\ell-\sigma nq)h(\mathbf{v},\vartheta)} = \delta f_{\ell}\mathcal{P}_{\ell}. \tag{4.8}$$

Here, σ is 1 for passing and 0 for trapped particles, $h(\mathbf{v},\vartheta) = \left(\int_0^{\vartheta} d\theta \mathcal{J} B v_d^{\alpha}/v_{\parallel}\right) / \oint d\theta \mathcal{J} B v_d^{\alpha}/v_{\parallel}$ is the α-phase of a particle as a function of ϑ, n is the toroidal mode number of the nonaxisymmetric perturbations, $\ell - \sigma n q$ is the number of toroidal transits after which the particle returns to an original point in space, and the phase-factor $\mathcal{P}_{\ell} = e^{-i2\pi(\ell-\sigma n q)h}$ contains all the ϑ dependence. In writing Eq. (4.7) we have assumed decoupling between ℓ-species, which can be strictly proved in the circular large aspect ratio limit with $\omega_E \gg \omega_D$ (see Sec. 4.1.1.1).

The drift velocity in Eq. (4.7) is the sum of gradient, electric, and curvature drifts,

$$\begin{aligned}\mathbf{v}_d &= \frac{\mathbf{B}}{eB^2} \times \left(\mu \nabla B + e\nabla\Phi + M v_{\parallel}^2 \frac{[(\mathbf{B}\cdot\nabla)\mathbf{B}]}{B^2}\right) &(4.9)\\ &= \frac{v_{\parallel}}{B}\nabla\times\left(\rho_{\parallel}\mathbf{B}\right) - \frac{v_{\parallel}}{B^3}\rho_{\parallel}\left[\mathbf{B}\cdot(\nabla\times\mathbf{B})\right]\mathbf{B}, &(4.10)\end{aligned}$$

where e is the charge of the particle species, Φ is the electric potential, and we have defined $\rho_{\parallel} \equiv M v_{\parallel}/eB$. In writing Eq. (4.10) from (4.9) we used the conservation of total energy ($\mathcal{E}$), $\nabla\mathcal{E} = 0 = M v_{\parallel}\nabla v_{\parallel} + \mu\nabla B + e\nabla\Phi$, and a few vector formulae.

Inserting Eqs. (4.8) and (4.10) into Eq. (4.6), we obtain,

$$\left[-i\omega_b(\ell-\sigma n q) - in(\omega_E+\omega_D) + \nu_s\right]\delta f_{\ell s} = \frac{\omega_b}{2\pi e}\frac{\partial \delta J_{-\ell}}{\partial\alpha}\frac{\partial f_{s0}}{\partial\psi}. \tag{4.11}$$

Here we have defined the electric and magnetic precession frequencies as,

$$\omega_E \equiv -2\pi\frac{\partial\Phi}{\partial\psi}, \tag{4.12}$$

and,

$$\omega_D \equiv 2\pi\left\langle\left(E-\frac{3}{2}\mu B\right)\frac{2}{eB}\frac{\partial B}{\partial\psi} + (E-\mu B)\frac{2}{e\mathcal{J}}\frac{\partial\mathcal{J}}{\partial\psi}\right\rangle_b, \tag{4.13}$$

as well as the variation in the action,

$$\begin{aligned}
\frac{\partial \delta J_{\pm\ell}}{\partial \alpha} &\equiv M \int d\vartheta \mathcal{P}_{\ell}^{\pm 1} \delta \left(v_{\|} \mathcal{J} B \right), && (4.14)\\
&= \oint d\vartheta \frac{\mathcal{J} B}{v_{\|}} \mathcal{P}_{\ell}^{\pm 1} \left[(2E - 3\mu B) \frac{1}{B} \frac{\partial \delta B}{\partial \alpha} + \right.\\
&\quad \left. (2E - 2\mu B) \frac{1}{\mathcal{J}} \frac{\partial}{\partial \alpha} \nabla \cdot \boldsymbol{\xi}_{\perp} \right]. && (4.15)
\end{aligned}$$

Again, the bounce-averaging process has eliminated the $\partial/\partial\vartheta$ terms from $v_{di}^{\alpha,\psi}$ normally associated with axisymmetric neoclassical transport (see Section B.1 for detailed derivations). It should be noted that by doing so, we make the momentum transport solution inherently a nonlocal quantity.

The solution (4.11) is valid for both trapped and passing particles, and contains contributions from curvature drifts and the perturbed arc-length in ω_D and δJ_ℓ often ignored in large aspect ratio approximations [23, 107]. The dependencies on aspect ratio and shaping enter the torque calculation through these terms.

4.1.1.1 A Note on Bounce Harmonic Coupling

The analytic solution of the drift kinetic equation used here has implicitly decoupled each (ℓ, σ) species of particle. This allows the phase factor to be moved to the right hand side of Eq. (4.7). It is necessary to completely decouple each combination of rotation and drift direction to completely remove phase-dependence from the bounce integrands on the left hand side and recover the standard frequencies $\omega_b, \omega_D, \omega_E, \nu_i$ in the resonance operator. This provides the physical intuition that is important in a semi-analytical approach such as the one taken here. It also dramatically reduces the computational demand. The separability, however, relies on the orthogonality of phase factors

$$\int_{\vartheta_{b-}}^{\vartheta_{b+}} d\vartheta e^{-2\pi i \tilde{\ell}_1 h} e^{2\pi i \tilde{\ell}_2 h} = \delta_{\tilde{\ell}_1 \tilde{\ell}_2}, \qquad (4.16)$$

where δ is the Kronecker delta. Equation (4.16) is only applicable if $h(\mathbf{v},\vartheta) = \vartheta/\vartheta_t$ for $\vartheta_t = \vartheta_{b+} - \vartheta_{b-}$and $\tilde{\ell}$ is an integer, which in turn requires,

- $\omega_E \gg \omega_D(\vartheta)$ such that $h(\mathbf{v},\vartheta) \approx \frac{1}{2\pi}\omega_b \int_0^{\vartheta} d\theta \mathcal{J} B/v_\parallel$
  - This is also necessary to recover strict orthogonality in the v_{di}^{α} term by making the drift independent of ϑ, allowing it to be pulled out of the integral, and leaving only ω_E times the integral (4.16).
- The large aspect ratio cylindrical approximation $B = B_0(1-\epsilon\cos\vartheta)$, such that $h(\vartheta) \sim K(k, \pi\vartheta/2\vartheta_t)/4K(k) \sim \vartheta/\vartheta_t$ as in [23]. Here K is the complete integral of the first kind and $k = \sqrt{(1-\Lambda+\epsilon\Lambda)\,/\,(2\epsilon\Lambda)}$.
  - This approximation is also necessary to have $\partial h/\partial\vartheta$ independent of ϑ and recover the integral (4.16) in the first term of Eq. (4.7).
- Passing particles only contribute at rational surfaces for the toroidal mode, $nq = m$. For passing particles, $\tilde{\ell} = \ell \pm nq$ and non-integer values would result in coupling. Away from the rational surface, however, passing particles experience phase-mixing and do not actually complete closed orbits for which the bounce average has physical meaning. Thus, these particles have negligible contributions to the nonambipolar cross field transport.

In general geometry, Eq. (4.16) is not true as both h can have a wide variety of ϑ dependencies. The simple integration of phase factors cannot be recovered in the first place since each of the $v_\parallel^{\vartheta}$ and v_d^{α} terms in Eq. (4.7) also have ϑ dependence and cannot be passed through the bounce integral operator. Extensive numerical benchmarking, however, has confirmed that these terms are negligible with the shaping of interest [25, 57], and the decoupled analytic solution of the drift kinetic equation will be used throughout the rest of this work.

4.1.2 Torque from Thermal Species

The bounce averaged torque is given by inserting the nonaxisymmetry driven perturbed distribution (4.11) in the phase space integral form of the toroidal torque (4.5),

$$T_\varphi = \frac{1}{eM^2}\int d\psi d\alpha d\mu dE \mathcal{R}_\ell \left|\frac{\partial \delta J_\ell}{\partial \alpha}\right|^2 \frac{\partial f_{s0}}{\partial \psi}. \tag{4.17}$$

Here, the resonant operator $\mathcal{R}_\ell = \omega_b / \left\{ i\left[\left(\ell - \sigma n q\right)\omega_b - n\left(\omega_E + \omega_D\right)\right] - \nu_s \right\}$ contains important bounce harmonic resonance physics [23, 60]. The well known offset rotation is explicitly included in the resonant operator when introducing the zeroth order drift-kinetic solution Maxwellian,

$$f_s = \frac{N}{\left(2\pi T/M\right)^{3/2}} \exp\left(-\frac{\mathcal{E} - e\Phi}{T}\right), \tag{4.18}$$

where N is the species density, T is the species temperature and the density, temperature and electric potential are all functions of the flux variable ψ only. The derivative of this distribution with respect to ψ is taken at constant total energy,

$$\left.\frac{\partial f_s}{\partial \psi}\right|_{\mathcal{E}} = -\frac{e f_s}{2\pi T}\left[\omega_E + \omega_{*N} + \omega_{*T}\left(\frac{E}{T} - \frac{3}{2}\right)\right]. \tag{4.19}$$

Here, we have used the familiar diamagnetic frequencies $\omega_{*N} = -2\pi T N'/eN$ and $\omega_{*T} = -2\pi T'/e$ where the prime denotes a derivative with respect to ψ.

Placing Eq. (4.19) in Eq. (4.17), assuming perturbations of the form $\delta A \sim e^{in\alpha}$, and making a final conversion to normalized energy coordinates ($x = E/T, \Lambda = \mu B_0/E$) gives,

$$T_\varphi = -\frac{n^2}{\sqrt{\pi}}\frac{R_0}{B_0}\int d\psi NT \int d\Lambda \bar{\omega}_b \left|\delta \bar{J}_\ell\right|^2 \int dx \mathcal{R}_{T\ell}, \tag{4.20}$$

with,

$$\mathcal{R}_{T\ell} = \frac{\left[\omega_\varphi + \omega_{*T}\left(x - \frac{5}{2}\right)\right] x^{5/2} e^{-x}}{i\left[\left(\ell - \sigma n q\right)\omega_b + n\left(\omega_E + \omega_D\right)\right] - \nu_s}. \tag{4.21}$$

Here, B_0 refers to the magnitude of the magnetic field on axis and both $\bar{\omega}_b = \omega_b R_0/\sqrt{2xT/M}$ and $\delta\bar{J}_\ell^2 = \delta J_\ell^2/2xTMR_0^2$ are unit-less quantities. Note that this expression is valid for a single Maxwellian particle species, and should be summed over species if more than one significantly contributes to the torque. The toroidal rotation frequency has been included in Eq. (4.21) using radial force balance neglecting poloidal rotation $\omega_\varphi = \omega_E + \omega_{*T} + \omega_{*N}$ [58, 9] in order to explicitly show that the resonant operator -and ultimately the torque on a given magnetic surface- is proportional to the rotation with an offset value.

The result is left in its complex form since the imaginary component of the torque has been shown to be proportional to the perturbed kinetic energy, $2n\delta W_k$ [60]. The calculations at the heart of PENT, therefore, is physically equivalent to those done by stability codes MARS-K [59, 108], MISK [109, 110], and MISHKA [111] in the case of static, marginally stable resistive wall modes (see Chapter 5).

The final form given by Eqs. (4.20) and (4.21) represents an expression for the toroidal torque and perturbed kinetic energy driven by nonaxisymmetric perturbations to an axisymmetric equilibrium valid across kinetic regimes. Although it uses a simplified collision operator, it contains important bounce harmonic resonance physics that can dominate the torque in experimental devices. Here, in its complete form, the resonance operator is a function of both energy and pitch (x, Λ), such that geometric shaping or low aspect ratio effects may influence the NTV through sensitive resonance conditions. Capturing these kinetic resonances in the true geometric shaping of modern experiments is an important computational task, and is the subject of Section 4.3 and Section 4.4. Before these details are introduced, Section 4.2 provides physical intuition explicitly relating the torque and neoclassical nonambipolar transport.

4.2 Nonambipolar Transport as NTV

The torque derived in the previous section is the electromagnetic torque produced from nonambipolar transport across field lines and is thus inescapably entwined with transport physics. This section provides a new derivation of the NTV torque that makes these connections explicit, while providing a clear physical picture for the process of momentum exchange between the plasma and external world.

4.2.1 Nonambipolar Particle Transport

It has been asserted that symmetry breaking causes nonambipolar transport. Explicitly, the transport across flux surfaces is the product of the number of particles in the perturbed distribution and the cross-field drift velocity,

$$\Gamma_{NA} \quad = \quad \int d^3x d^3v v_d^{\psi_n} \delta f \tag{4.22}$$

$$\Gamma_{NA} \quad = \sum_\ell \quad \frac{2\pi}{M^2} 2\pi \int d\psi d\alpha dE d\mu \delta f_\ell \int d\vartheta \frac{\mathcal{J}B}{v_\parallel} \mathcal{P}_\ell \mathbf{v}_d \cdot \nabla\psi / \psi'. \tag{4.23}$$

Here we expanded the velocity-space volume element $\int d^3v = \int dE d\mu 2\pi B/M^2 v_\parallel$, factored the perturbed distribution as before $\delta f = \sum_\ell \delta f_\ell \mathcal{P}_\ell$, and converted to the Clebsch coordinate toroidal angle. The derivative of the poloidal flux with respect to a normalized poloidal flux, $\psi' = d\psi/d\psi_n$, is a constant of the volume and can be pulled out of the integrals.

The poloidal integral is identical to the one on the right hand side of the drift kinetic equation (4.7). Comparing Eq. (4.7) and Eq. (4.11) we have,

$$\langle -\mathcal{P}_\ell \mathbf{v}_{di} \cdot \nabla\psi \rangle = \frac{\omega_b}{2\pi e} \int d\vartheta \frac{\mathcal{J}B}{v_\parallel} \mathcal{P}_\ell v_d^\psi = \frac{\omega_b}{2\pi e} \frac{\partial}{\partial\alpha} \delta J_\ell. \tag{4.24}$$

Inserting into the transport equation (4.23),

$$\Gamma_{NA} = -\frac{2\pi}{e\psi'}\sum_{\ell}\frac{2\pi}{M^2}\int d\psi d\alpha dE d\mu \delta f_{\ell}\frac{\partial}{\partial\alpha}\delta J_{\ell}. \tag{4.25}$$

Comparing to Eq. (4.5), in which the poloidal integration also reduces to a bounce averaged variation in the action, we obtain the simple relation,

$$\Gamma_{NA} = \frac{2\pi}{e\psi'}T_{\varphi}. \tag{4.26}$$

That is, the NTV torque is proportional to the nonambipolar transport across field lines. The usual neoclassical transport drives are found in the denominator of the resonance operator. The terms proportional to ω_E, ω_{*N} and ω_{*T} are the potential, density and temperature gradient driven transport respectively.

4.2.2 Electromagnetic Torque from Nonambipolar Transport

An alternate derivation of the NTV torque now presents itself. Starting from the assumption that the torque is inherently an electromagnetic torque caused by currents from nonambipolar transport interacting with equilibrium fields, we write,

$$\begin{aligned} T_{\varphi} &= \int d^3x d^3v e\delta f \mathbf{v}_d \times \mathbf{B}\cdot\nabla\psi \\ T_{\varphi} &= \int d^3x d^3v e\delta f v_d^{\psi} B^{\vartheta}. \end{aligned} \tag{4.27}$$

We use the covariant field in straight field line magnetic coordinates,

$$2\pi\mathbf{B} = q\nabla\psi\times\nabla\vartheta + \nabla\varphi\times\nabla\psi, \tag{4.28}$$

and convert to a normalized poloidal flux to arrive at the familiar form

$$T_\varphi = \frac{e\psi'}{2\pi}\int d^3x d^3v v_d^{\psi_n}\delta f. \tag{4.29}$$

Comparing this electromagnetic torque to the nonambipolar transport definition (4.22), and through Eq. (4.26) to the toroidal viscosity torque, we see that it is identically the NTV torque derived in Section 4.1.

We arrive at the conclusion that the torque due to the neoclassical toroidal viscosity is the toroidal component of the second order electromagnetic torque created by the nonambipolar transport across flux surfaces. The fact that it is proportional to this transport by a simple factor corresponding to the poloidal field is then demystified.

This new derivation of the NTV provides important insights into the physical nature of the torque. This derivation was made simple by the extensive hindsight provided by the derivation from Section 4.1, but did not require it. A complete derivation could work directly from Eq. (4.29), solve the drift kinetic equation (4.6), and arrive at the final form given in Eq. (4.20). Deriving the torque in such a way makes the explicit assertion that the force on the plasma is electromagnetic in nature, and thus that momentum is exchanged with the external world in the form of electromagnetic fields.

The nature of the momentum exchange between plasma and the surrounding tokamak in the presence of resonant fields has been the subject of a extensive field of study with many publications [17, 69, 11, 2, 83, 84]. The ultimate nature of the momentum exchange due to nonresonant fields, however, has remained opaque. Some publications claimed ballistic momentum exchange due to particle losses across the separatrix contained the momentum imparted by NTV torques [70] while others left the subject unaddressed. This limited the experimental community to active measurement techniques such as changes in the neutral beam torque or Charge Exchange Recombination (CER) spectroscopy for rotation profile measurements. The electromagnetic nature of the NTV torque derived here implies that the integral torque can be measured using the Maxwell stress tensor technique detailed in Chapter 2, which requires no additional perturbation of the plasma and is insensitive to the basic plasma parameters.

4.3 Pitch Angle Dependencies and Geometric Effects

The PENT code has been developed to calculate the NTV torque from Eqs. (4.20) and (4.21), without any geometric simplifications in the bounce frequencies, precession frequencies or perturbed action. The combined theory presented in Ref. [23] used circular large aspect ratio (CLAR) approximations and further removed pitch (Λ) dependence of bounce and magnetic precession frequencies,

$$\omega_b \approx \tfrac{1}{2}\sqrt{\tfrac{\epsilon x \Lambda}{2}}\omega_t \tfrac{\pi}{K(k)} \approx \frac{\pi}{4}\sqrt{\frac{\epsilon}{2}}\omega_t\sqrt{x}, \tag{4.30}$$

$$\omega_D \approx \frac{q^3\Lambda}{\epsilon}x\left[\frac{E(k)}{K(k)} - \frac{1}{2}\right] \approx \frac{q^3\omega_t^2}{4\epsilon\omega_g}x. \tag{4.31}$$

Here ϵ is the inverse aspect ratio, ω_t is the transit frequency, ω_g is the gyro-frequency, K and E are the complete elliptic integrals of the first and second kind respectively and their argument is $k = \sqrt{(1-\Lambda+\epsilon\Lambda)\,/\,(2\epsilon\Lambda)}$. The reduced large aspect ratio (RLAR) approximations on the far right correspond approximately to the Λ-averaged circular larger-aspect-ratio approximations for trapped particles that they follow [23]. In practice, these were designed to separate the computationally demanding double integration in (x, k) space of the resonant operator assuming $\omega_E \gg \omega_D$ in most applications. The PENT code, in contrast, solves the full bounce averaged expressions for both frequencies throughout the phase space populated by passing and trapped particles.

The CLAR and RLAR approximations deteriorate as the aspect ratio approaches unity or shaping is introduced into the plasma equilibrium. For demonstration, the NTV torque from trapped ions as calculated using the full and RLAR formalisms in an aspect ratio scan of Solov'ev equilibria is presented here. The Solov'ev equilibria are analytic solutions of the Grad-Shafranov equation for which the current and pressure profiles are fully defined by the elongation κ, safety factor on axis q_0, field on axis B_0, major radius R_0 and inverse aspect ratio $\epsilon_a = a/R_0$ where a is the minor radius [112, 113]. The equilibria used here have common parameters $\kappa = 1$, $q_0 = 2.2$, $B_0 = 1$T, $R_0 = 1$m, and are henceforth referred to solely by their inverse aspect ratio. Fig. 4.1 shows flux surfaces for three such equilibria. Note that the Solov'ev solution is defined for $\epsilon_a < 0.5$, as shaping at low aspect ratio weights the boundary towards the $R = 0$ vertical axis.

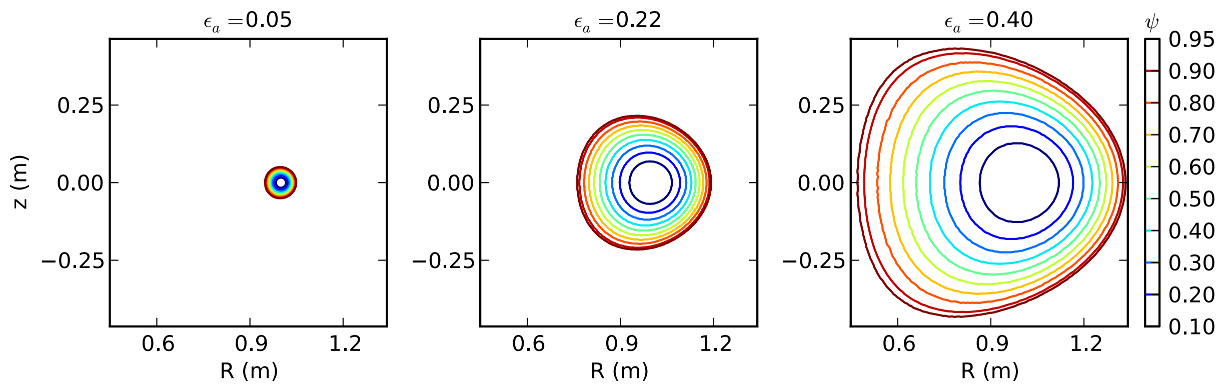

Figure 4.1: Solov'ev equilibria with $\kappa = 1$, $q_0 = 2.2$, $B_0 = 1T$ and $R_0 = 1m$. The equilibria have limited shaping, and approach the cylindrical limit for $\epsilon_a \to 0$.

The ratio of the NTV torque calculated using the full calculation in PENT to the RLAR approximation across a scan of the inverse aspect ratio is presented in Fig. 4.2. In each case, the Solov'ev equilibrium is perturbed in IPEC using an applied 1 Gauss non-resonant $m/n = 2/1$ mode on the outermost flux surface where m and n are the poloidal and toroidal harmonics respectively. The electron and deuterium ion density and temperature profiles, as well as the electric precession profiles are set constant as $n(\psi) = n_0(1 - 0.7\psi)$, $P(\psi) = P_0(1-\psi)$ and $\omega_E = \omega_{E0}(1-\psi)$ with on-axis values $n_0 = 1.12 \times 10^{-20}\text{m}^{-3}$, $P_0 = 2.33 \times 10^4\text{Pa}$, and $\omega_{E0} = 0.63\text{kHz}$. The collisionality in this case is near the $1/\nu$ collisionality regime [24]. As the inverse aspect ratio approaches 0, the equilibria become cylindrical and the RLAR agrees well with the PENT calculation. At larger ϵ_a, however, both the large aspect ratio and the circular approximations begin to break and the RLAR torque significantly underestimates the total torque predicted by the combined theory without geometric simplification.

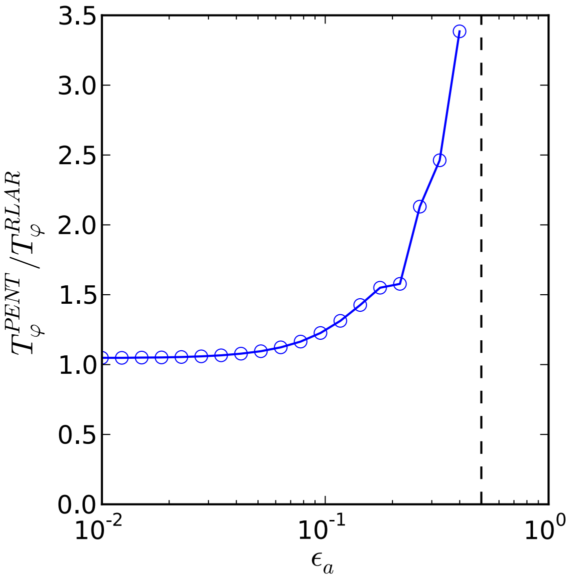

Figure 4.2: The ratio of NTV torque as calculated by PENT and the RLAR approximation approaches unity as the inverse aspect ratio ϵ_a approaches 0, but becomes large ϵ_a approaches its maximum value of 0.5.

A closer look at the simplifications (4.30) and (4.31) in the Solov'ev aspect ratio scan provides insight into the physics missed in the RLAR approximation. Fig. 4.3 shows the bounce and precession frequencies in the RLAR and circular large aspect ratio approximations compared to the full bounce-averaged values for all trapped particles on the $\psi_n = 0.66$ surface for representative large and low aspect ratio Solov'ev equilibria. Note that the RLAR approximation ignores strong Λ-dependences. In a large aspect ratio case such as $\epsilon_a = 0.05$, this reduction can be valid for ω_b in average although a subtle interplay between the disappearing ω_b and potentially large $|\delta J_\ell|^2$ for weakly trapped particles is not accounted for. As ϵ_a increases, ω_b calculated with the full geometry can differ largely from the large aspect ratio and thus further from the RLAR approximation. The deviation at higher ϵ_a is also severe for ω_D. The RLAR approximation ignores the sign change between the deeply and marginally trapped species and thus will be unable to capture a limiting behavior for $\omega_E \to \omega_D$, and it is shown that even the circular large aspect ratio approximation can become significantly incorrect for ϵ_a approaching its maximum. The next section looks in more detail at how retaining the Λ dependence of these profiles can play an important role in describing particle resonances precisely.

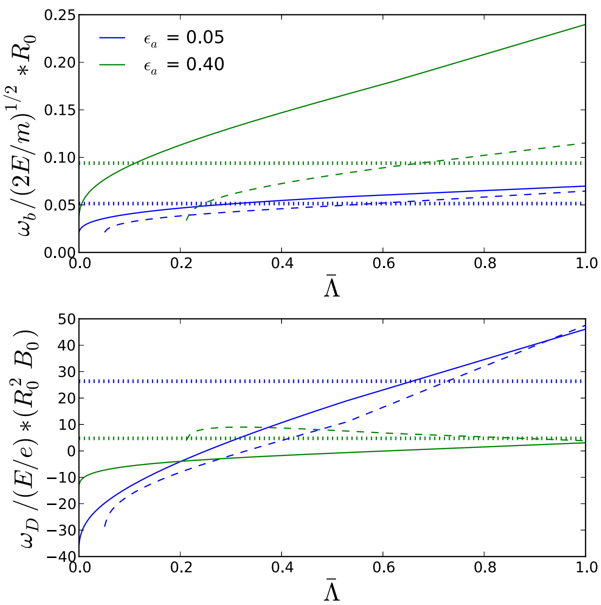

Figure 4.3: The pitch angle profiles of the bounce (top) and magnetic precession (bottom) frequencies in PENT (solid), the CLAR approximation (dashed) and the RLAR approximation (dotted) for trapped ions on the $\psi_n = 0.66$ surface of two Solov'ev equilibria. The normalized axis $\bar{\Lambda} = (\Lambda - B_0/B_{max})/(B_0/B_{min} - B_0/B_{max})$ increases from 0 at the trapped-passing boundary to 1 at the deeply trapped limit.

4.4 Kinetic Resonances

Bounce-harmonic resonant (BHR) particles, satisfying $(\ell - \sigma n q)\,\omega_b + n\,(\omega_E + \omega_D) \approx 0$, have been shown to dominate the nonambipolar transport and resulting toroidal torque in kinetic regimes relevant to present day and future tokamaks [23, 27, 53]. This is the case in the limit that the collisionality is comparable to or less than the precession and bounce frequencies off resonance. These conditions are common in modern tokamaks, and thus deserves explicit attention here.

Typical frequencies from the resonance condition, can be seen in Tab. 4.1. The values are taken from near the edge ($\psi_n = 0.9$) where perturbations from external asymmetries are expected to be large. The ratio $(\omega_E + \omega_D)/\nu_i$ serves as a proxy for the sensitivity to precession resonant ($\ell = 0$) particles. Bounce frequencies tend to be relatively high, and thus $\ell \neq 0$ bounce harmonic resonances are expected to become important at higher rotation (see Sec. 5.5.1).

Plasma Scenario	ν_i	$\omega_E + \omega_D$	ω_b	$\nu_i / (\omega_E + \omega_D)$
DIII-D Ohmic	1×10^3	2×10^5	5×10^4	$\mathcal{O}(10^{-2})$
DIII-D H-mode	2×10^3	5×10^4	5×10^4	$\mathcal{O}(10^{-1})$
DIII-D QH-mode	2×10^2	1×10^5	3×10^4	$\mathcal{O}(10^{-3})$
ITER	5×10^2	7×10^3	1×10^4	$\mathcal{O}(10^{-1})$

Table 4.1: Typical frequencies (rad/s) from the denominator of the resonant operator (4.21) evaluated for ions at $\psi_n = 0.9$ for a range of scenarios. Precession frequencies and bounce frequencies are pitch angle averaged circular large aspect ratio approximations. Low n resonances can be expected to dominate the NTV when $\nu_i / (\omega_E + \omega_D) \lesssim \mathcal{O}(1)$.

An important property of the combined NTV theory is its smooth connection of kinetic regimes as defined by the ratios of these frequencies. These regimes are defined by limiting behavior from dominant resonances in energy space, the fundamentals of which will be discussed in this section. Further discussion of the final NTV torque as well as comparisons between multiple models can be found in Chapter 5.

4.4.1 Energy Resonances

Many of the fundamental NTV characteristics are readily apparent in the innermost energy integral from Eq. (4.20). The rotation dependence, for example, can easily be seen traced through multiple regimes. Fig. 4.4 shows rotation scans of the energy integral characterized by $n\omega_E/\nu_s$. Each case shows the torque linearly increases with the rotation in the ν regime at $|n\omega_E/\nu_s| \ll 1$, but actually decreases with rotation in the and $\nu - \sqrt{\nu}$ regime at high rotation $|n\omega_E/\nu_s| \gg 1$. Integrals that include magnetic precession drifts and/or bounce harmonic particles have extended regions of positive correlation with rotation, and exhibit sharp peaking at the transition between regimes. This behavior is directly translated to the final NTV torque rotation dependence.

Sharp resonances in the normalized energy, $x = E/T$, can dominate the resonance operator and ultimately the NTV on a given flux surface. Fig. 4.4 shows these sharp resonances. In this case the resonances dominate the integrand when present, and their presence is responsible for the resonance observed in the integral rotation scan. The value $\nu_s/\left(\omega_E + \omega_D\right)$ is small for the resonant cases, and comparable to the values in Tab. 4.1 for a number of experimental scenarios.

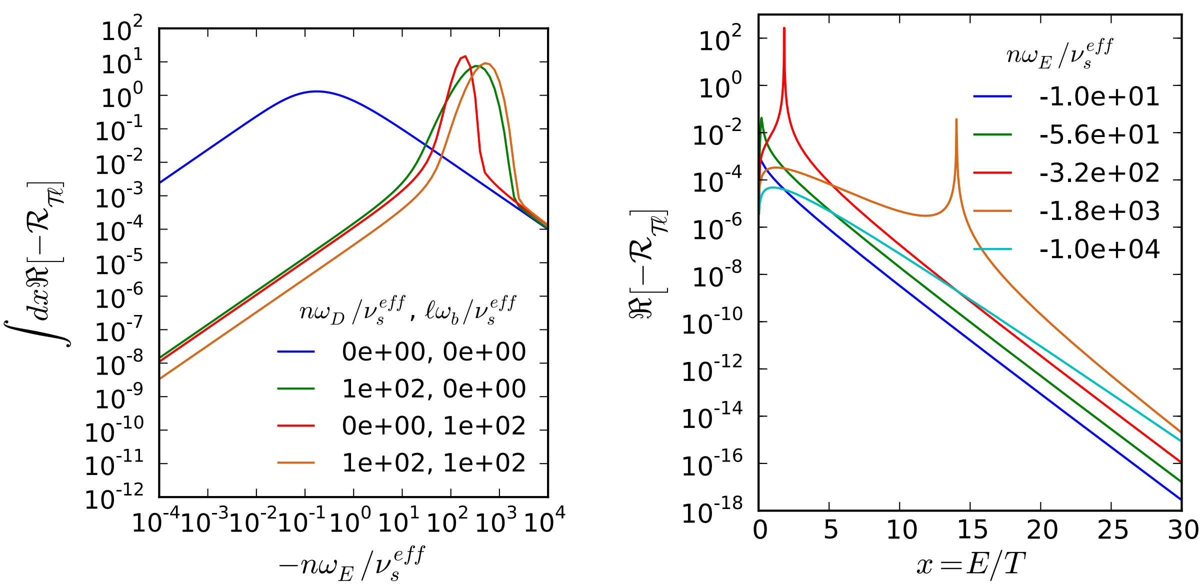

Figure 4.4: Rotation scans of the energy integral (left) show the transition between regimes of positive and negative correlation influenced by the inclusion of magnetic precession and/or bounce harmonics. Sample energy integrands (right) show sharp energy resonances responsible for the peak near $n\omega_E/\nu_s \approx -7 \times 10^2$ in the final (orange) integral curve to the left.

The location of energy resonances is easily obtained by factoring out the energy dependence from each frequency in the resonance condition,

$$x n \hat{\omega}_D + \sqrt{x}\left(\ell - \sigma n q\right)\hat{\omega}_b + n\omega_E = 0, \tag{4.32}$$

where $\hat{\omega}_D = \omega_D/x$ and $\hat{\omega}_b = \omega_b/\sqrt{x}$ are independent of x. This is a quadratic equation for $x^{1/2}$ with the solution,

$$x^{1/2} = -\frac{1}{2}\frac{(\ell - n\sigma)}{n}\frac{\hat{\omega}_b}{\hat{\omega}_D} \pm \frac{1}{2}\sqrt{\frac{(\ell - n\sigma)^2}{n^2}\left(\frac{\hat{\omega}_b}{\hat{\omega}_D}\right)^2 - 4\frac{\omega_E}{\hat{\omega}_D}}, \tag{4.33}$$

giving in turn,

$$x = \frac{1}{2}\frac{(\ell - n\sigma)^2}{n^2}\left(\frac{\hat{\omega}_b}{\hat{\omega}_D}\right)^2 - \frac{\omega_E}{\hat{\omega}_D} \pm \frac{1}{2}\frac{(\ell - n\sigma)}{n}\frac{\hat{\omega}_b}{\hat{\omega}_D}\sqrt{\frac{(\ell - n\sigma)^2}{n^2}\left(\frac{\hat{\omega}_b}{\hat{\omega}_D}\right)^2 - 4\frac{\omega_E}{\hat{\omega}_D}}, \tag{4.34}$$

if both $x^{1/2}$ solutions are positive. Negative solutions for Eq. (4.33), are not valid solutions on the positive real energy axis integration path. Note that there may be 0, 1, or 2 resonances on the positive real axis dependent on the relative values of ω_E and $\hat{\omega}_b$. If only precession resonant trapped particles are considered ($\ell, \sigma = 0$), for example, there is one resonance $x = -\omega_E/\hat{\omega}_D$ if $\omega_E/\hat{\omega}_D < 0$. In the case that there are no positive real resonances, the resonance operator is peaked at $x = 3/2, 5/2, 4$ for the ω_D, ω_E, and ν_s dominated cases respectively (here we have used a $x^{-3/2}$ energy dependence for the collisionality as in Ref. [23]). Including bounce harmonic resonances, it is possible for the dominant contribution to the torque to come from species with higher energy and lower collisionality. This effectively changes the relevant NTV regime for a given surface, further motivating the use of a combined NTV formalism.

In the strict limit of zero collisionality, the solutions of Eq. (4.34) are true poles. The bounce harmonic resonant particles correspond to the energies for which these poles are on the real axes. The torque is then due to the residues, which are possible to calculate analytically. Rewriting the energy integral in Eq. (4.20) in a simplified form,

$$\begin{aligned} I_x &= \int_0^\infty dx \frac{-if(x)}{x + bx^{1/2} + c} = \int_0^\infty dz \frac{-i2zf(z)}{z^2 + bz + c} \\ I_x &= \int dz \frac{-i2zf(z)}{\left(z + \frac{1}{2}b - \frac{1}{2}\sqrt{b^2 - 4c}\right)\left(z + \frac{1}{2}b + \frac{1}{2}\sqrt{b^2 - 4c}\right)} \end{aligned} \tag{4.35}$$

we find the pole contribution to the torque,

$$
\begin{aligned}
Re(I_x) &= \pi i H(b^2-4c)\left\{\left[\frac{-i2zf(z)}{\left(z+\frac{1}{2}b+\frac{1}{2}\sqrt{b^2-4c}\right)}\right]_{-\frac{1}{2}b+\frac{1}{2}\sqrt{b^2-4c}}\right. \\
&\left. +H(c)\left[\frac{-i2zf(z)}{\left(z+\frac{1}{2}b-\frac{1}{2}\sqrt{b^2-4c}\right)}\right]_{-\frac{1}{2}b-\frac{1}{2}\sqrt{b^2-4c}}\right\}.
\end{aligned}
\tag{4.36}
$$

Here we have defined the variable $z=\sqrt{x}$ and energy independent normalized frequencies $b=(\ell-n\sigma)\,\hat{\omega}_b/n\hat{\omega}_D$ and $c=\omega_E/\hat{\omega}_D$. The first Heaviside function, H, isolates cases for which the poles are on the real axis and the second Heaviside function isolates only those on the positive axis. Note that the case $b=c=0$ is not taken into account in Eq. (4.36), but would give no torque. See Appendix D for details on how this is treated computationally in PENT.

4.4.2 Pitch Angle Resonances and the Superbanana Plateau

The precision of the ω_b and ω_D representations in two dimensional energy space (x,Λ) becomes critical when $\nu_s/n\,(\omega_E+\omega_D)\to 0$. In this limit the integration of the resonant operator becomes stiff and essentially produces a δ function for BHR particles.

One such regime, the so-called Superbanana Plateau (SBP) [107, 114, 98, 115, 116], is expected when ω_E is as small as ω_D and $\nu_s\to 0<|\omega_D|$. An analytic expression for SBP NTV characterized by precession resonances ($\ell=0$) has been derived in Ref. [107] using circular large aspect ratio approximations for ω_b, ω_D, and δJ as well as a Krook collision operator (known to be accurate in this regime [117, 118]).

The analytical SBP approximation requires an effective collisionality less than the precession frequency away from the kinetic resonance, whereas an effective collision frequency much larger than the precession frequency results in $1/\nu$ regime behavior [119]. Here, the effective collisionality will be identified by the normalized parameter $\nu_*=\nu_i Rq/(\epsilon^{3/2}v_{th})$, where $v_{th} = \sqrt{2T/M}$ is the ion species thermal velocity.

Fig. 4.5 shows the torque as calculated in PENT and the RLAR approximation as a function of ν_* across the SBP and $1/\nu$ regimes for a perturbed Solov'ev equilibrium consistent with those in Section 4.3 and $\epsilon_a = 0.3$. The torque is calculated over the entire plasma. The PENT profiles peak near the rational surface at $\psi_n \approx 0.66$, and $\nu_*(\psi_n = 0.66)$ is used for the axis in Fig. 4.5. Only the precession resonances ($\ell = 0$ species) are taken into account for consistency with [107], and small $\omega_E \sim \mathcal{O}(\omega_D)$ is chosen to eventually reach SBP limits and to compare with the analytic SBP calculation. One can see that the RLAR and PENT torque calculations agree in the high-collisionality limit, but the RLAR calculation fails to capture the plateau behavior expected at low collisionality. On the other hand, it is also shown that the PENT plateau torque is a factor of ~ 4 lower than the SBP calculation. In order to compare these semi-analytic results with a more fundamental computation, the drift-kinetic δf guiding-center particle code, POCA [27, 116], is used to calculate the NTV torque in the same Solov'ev targets. POCA is a local code that includes a momentum conserving pitch-angle scattering collision operator, general geometry, and finite orbit effects by considering annuli in ψ with finite width. The PENT results have closer agreement with the POCA results than the analytic theory does, indicating that the phase space structure of the resonance condition contains important details beyond the circular large aspect ratio approximations.

The superbanana plateau captured in these PENT computations is a result of BHR particles dominating the torque with the resonance near the precession frequency zero crossing in Λ. The analytic derivation of the SBP torque relies on this resonance in Λ, approximating the real part of the resonant operator (4.21) as a delta function in pitch angle in the limit $n(\omega_D + \omega_E) \gg \nu_i$ [107]. It is emphasized in Ref. [107] that the existence of the plateau is a consequence of treating this singularity in the resonance operator independent of geometric details. Therefore, it is not surprising that the RLAR approximation failed to capture SBP behavior [114] when this singularity in Λ is not included. The analytic SBP calculation predicts the plateau behavior, but can be incorrect quantitatively in general equilibria due to

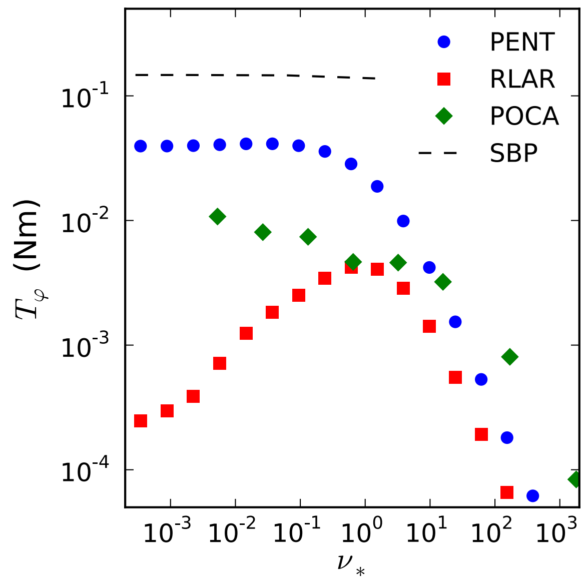

Figure 4.5: NTV torque across SBP, $1/\nu$, and intermediate regimes as computed in PENT with full pitch angle dependent ω_b, ω_D, and δJ (blue circles) compared to the RLAR approximation (red squares) shows the PENT calculation recovers the collisionality independence of the superbanana plateau. A SBP limit circular large aspect ratio calculation (dashed line) and POCA δf particle simulation (green diamonds) are also shown.

the sensitive geometric dependency of ω_D.

It should be noted that global particle simulations by the FORTEC-3D [120] code did not show the SBP limiting behavior [114]. FORTEC-3D includes a full Fokker-Planck collisional operator with both energy and pitch-angle scattering rather than the pitch-angle scattering employed by POCA or Krook operator used by the semi-analytic theories in this limit. It is possible that particles can be de-correlated quickly from the sharp SBP resonant conditions by the Fokker-Planck operator or that global scale orbit widths at extremely low collisionality introduce a new phase mixing not captured in the local codes or theory. This is beyond the scope of this work but is clearly another important subject that a global simulation can test in the future.

Fig. 4.6 shows the pitch angle dependence of the combined theory NTV torque in PENT smoothly transition from the isotropic $1/\nu$ regime toward the delta function resonance behavior at the BHR condition near the precession null in the SBP limit. The amplitude of the delta function like resonance in the SBP depends on its location in Λ, which is highly dependent on the details of ω_D. Through changes in the resonant pitch condition, the curvature drift emphasized in Section 4.1 has a large effect on the ultimate value of the plateau torque calculated in PENT. This, combined with the generalization of the perturbed action to include changes in the arc-length, brings the combined NTV torque plateau down and into better agreement with the more general and computationally expensive POCA results.

This type of detailed comparison between the PENT code, analytic approximations, and independent models is important for numerical validation as well as increased physics understanding. Before presenting experimental applications, it is thus important to compare and contrast the PENT code with other hybrid kinetic MHD modes in perturbed nonaxisymmetry.

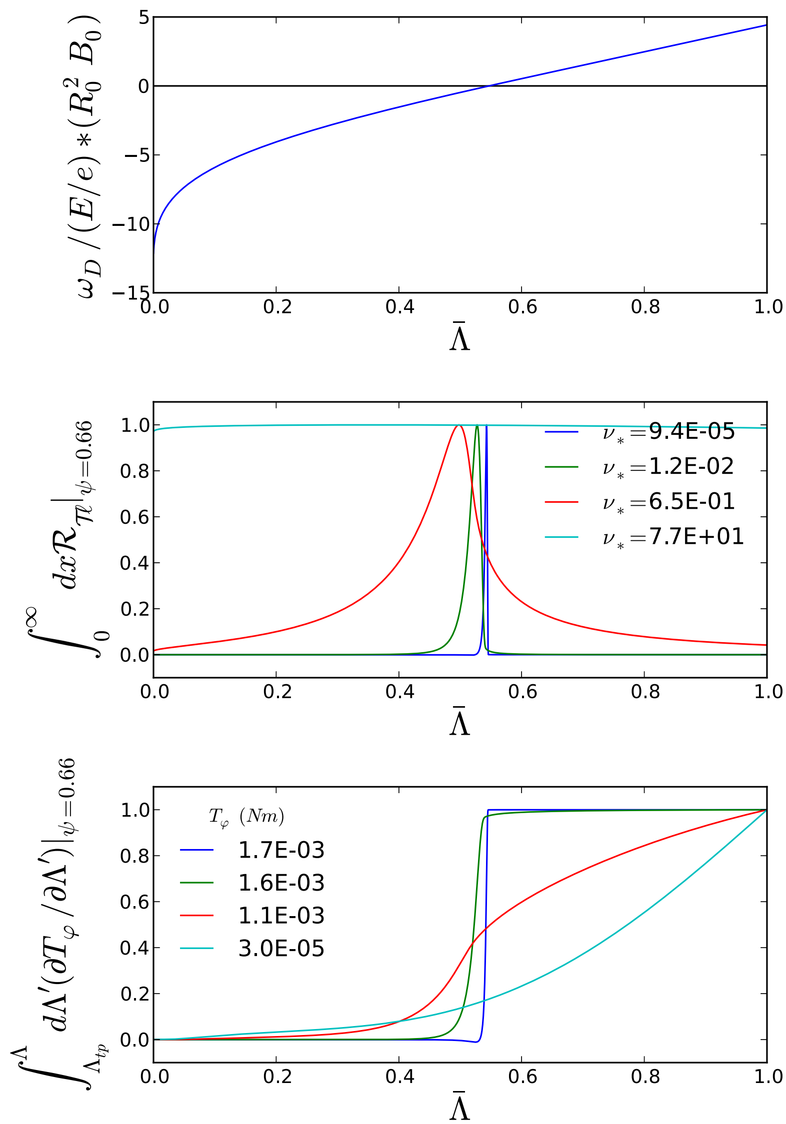

Figure 4.6: The magnetic precession (top), normalized energy integrated resonant operator (middle) and normalized cumulative integral torque (bottom) on the $\psi_n = 0.66$ surface for 4 collisionalities spanning the SBP and $1/\nu$ regimes. The torque is 1.7×10^{-3}, 1.6×10^{-3}, 1.1×10^{-3}, and 3×10^{-5} Nm for ν_* values of 9.4×10^{-5} (dark blue), 1.2×10^{-2} (green), 6.5×10^{-1} (red), and 7.7×10^{1} (light blue) respectively. Sensitivity to precession resonances in Λ becomes more pronounced as the collisionality is reduced. The Λ resonance clearly dominates the NTV torque in the low collisionality limit.

Chapter 5

Numerical Benchmarking of PENT

The PENT code computes the toroidal torque as defined by the combined NTV model detailed in Chapter 4. In this chapter, the PENT code will be given context by comparison and benchmarking with independent nonaxisymmetric tokamak modeling efforts. These benchmarks provide assurance that model captures the physics of interest. This, in turn, allows confidence in the quantitative comparisons between the torque modeled by PENT and the torque measured in experiment presented in the following Chapter 6. The necessary generalizations of a cross-model benchmark will only temporarily delay the direct application of PENT to experiment.

Much of the established modeling effort for perturbed equilibria, and as a consequence much of this benchmarking process, focuses on the perturbed kinetic energy. The motivation can be seen clearly in practical application of tokamak plasmas for fusion energy, which requires a relatively high ratio of plasma stored energy to magnetic field energy for efficient energy production. This ratio is commonly characterized by an operational parameter normalized beta, $\beta_N = aB\beta/I_p$. Here a in the plasma minor radius (m), B is the is the field (T) and I_p is the plasma current (MA), and β is the ratio of plasma pressure to magnetic pressure. At high β_N - often near 3.5 - tokamak plasmas become susceptible to large scale MHD instabilities that grow on the fast Alfvénic time scales, disrupting the plasma current and destroying confinement. The presence of a close fitting wall requires perturbed fields to penetrate the conducting structure, limiting the growth rate of instabilities to the much slower eddy current decay time of the wall. The stability of the resulting mode, known as a Resistive Wall Mode (RWM), places a limit on β_N and has thus motivates extensive theoretical modeling in an effort to understand predict these stability limits.

Of particular interest here, and in RWM theory, is the kinetic energy principle [121]. Rewriting the force balance equation in terms of changes in the potential MHD and kinetic energies (δW, δW_k), the energy principle states that if any small displacement $\xi_\perp$ from equilibrium causes the potential energy to decrease then the plasma is unstable and the corresponding mode will grow exponentially.

Like PENT, many hybrid kinetic MHD models assume a fixed displacement $\xi_\perp$ unchanged by the kinetic effects (torques in the force balance) that are assumed to be small compared to the fluid plasma equilibrium ($\delta W_k \ll \delta W_p$). In this small perturbation limit, the RWM dispersion relation can be written in terms of the perturbed energies [122, 58, 59],

$$(\gamma + i\omega_r)\,\tau_w = -\frac{\delta W_\infty + \delta W_k}{\delta W_b + \delta W_k}. \tag{5.1}$$

The mode growth rate γ and toroidal rotation ω_r have been normalized by the wall current decay time and natural time scale of the mode τ_w. Here δW_∞ is the sum of the plasma fluid and vacuum perturbed energies with no wall, and δW_b is the plasma fluid and vacuum perturbed energies with an ideal wall at location b. These terms are real and do not, by themselves, contribute to the toroidal rotation. The perturbed kinetic energy contribution, δW_k, is responsible for the complex dispersion relation including a perturbed toroidal mode frequency. The kinetic energy contribution is given by,

$$\delta W_k = -\frac{1}{2}\int d^3x\,(\xi_\perp^* \cdot \nabla \cdot \mathbf{\Pi}), \tag{5.2}$$

where $\xi_\perp$ is the perturbed displacement perpendicular to the field and $\mathbf{\Pi}$ is the same anisotropic perturbed pressure tensor found in Eq. (4.1). In fact, the similarities between Eqs. (4.1) and (5.2) result in a simple equivalency principle [60, 25], $T_\varphi = i2n\delta W_k$, valid in the same small perturbation limit as the dispersion relation (5.1). The physical interpretation of this equivalency, in the context of RWM studies, is that the torque from nonambipolar transport caused by the RWM perturbation determines the resulting toroidal rotation frequency of the mode. Conversely, the imaginary component (divided by $2n$) of the NTV torque calculated by PENT from the perturbed fields is the kinetic energy transferred from the RWM.

The kinetic energy equivalency principle allows direct benchmarking of PENT with codes developed explicitly for tokamak stability studies. This chapter consists of comparisons with two such codes, MISK [58] and MARS-K [59]. Benchmarks using two analytical Solov'ev equilibria and a projected ITER scenario show good agreement between the codes. Section 5.1 provides detailed descriptions of these equilibria. The kinetic frequencies relevant to the resonance operator at the core of each calculation are shown to agree among the codes and are compared to simple analytical limits in Section 5.2. Section 5.3 and Section 5.4 then give comparisons between the energy integral and perturbed action, serving as both detailed benchmarks and an important motivators for the new NTV physics included in PENT. Finally, parametric scans are used in Section 5.5 to extend the comparison of the final torque and kinetic energy through multiple analytic regimes. All three codes show clear transitions between regimes and agreement is maintained through a range of computationally challenging kinetic resonances.

5.1 Benchmark Equilibria

The benchmarking of PENT, MISK, and MARS-K used three equilibria. The first is a simple analytical Solov'ev equilibrium with no rational surfaces, large aspect ratio, and nearly circular flux surfaces. This equilibrium was used in a previous stability code benchmarking effort [59], and provides a clear comparison with analytic T_φ and δW_k circular large aspect ratio approximations. The second is an Solov'ev equilibrium with two $n = 1$ rational surfaces, smaller aspect ratio and vertically elongated flux surfaces. This provides important insight into the importance of the perturbed equilibrium's treatment of rational surfaces as well as some general geometry considerations. The final equilibrium, a projected ITER equilibrium, demonstrates the applicability of the models to present and future experiments as well as illustrating common levels of deviation from simple circular large aspect ratio approximations.

5.1.1 Solov'ev Equilibria

Solov'ev equilibria [112, 113] are a set of analytical solutions to the Grad-Shafranov equation. In these solutions, the current and pressure profiles are fully defined by the elongation κ, safety factor on axis q_0, field on axis B_0, major radius R_0 and inverse aspect ratio $\epsilon_a = a/R_0$ where a is the minor radius. In the DCON [46, 123] ideal MHD stability code used with PENT, these profiles are specified as follows,

$$F(\psi_n) = R_0, \tag{5.3}$$

$$\mu_0 P(\psi_n) = \frac{2a^2}{(2q_0R_0)^2}(\kappa^2+1)(1-\psi_n), \tag{5.4}$$

where the poloidal current profile is represented by $F = RB_\phi$, the field on axis is normalized to unity and the normalized poloidal flux ψ_n is 0 on axis and 1 at the plasma boundary. The flux surfaces are described geometrically in the (R, z) plane by,

$$\psi_n = \frac{\kappa a^2}{2q_0} - \frac{\kappa}{2q_0R_0^2}\left[\left(\frac{Rz}{\kappa}\right)^2 + \frac{1}{4}\left(R^2 - R_0^2\right)^2\right], \tag{5.5}$$

and the plasma volume is constant as a function of the normalized flux.

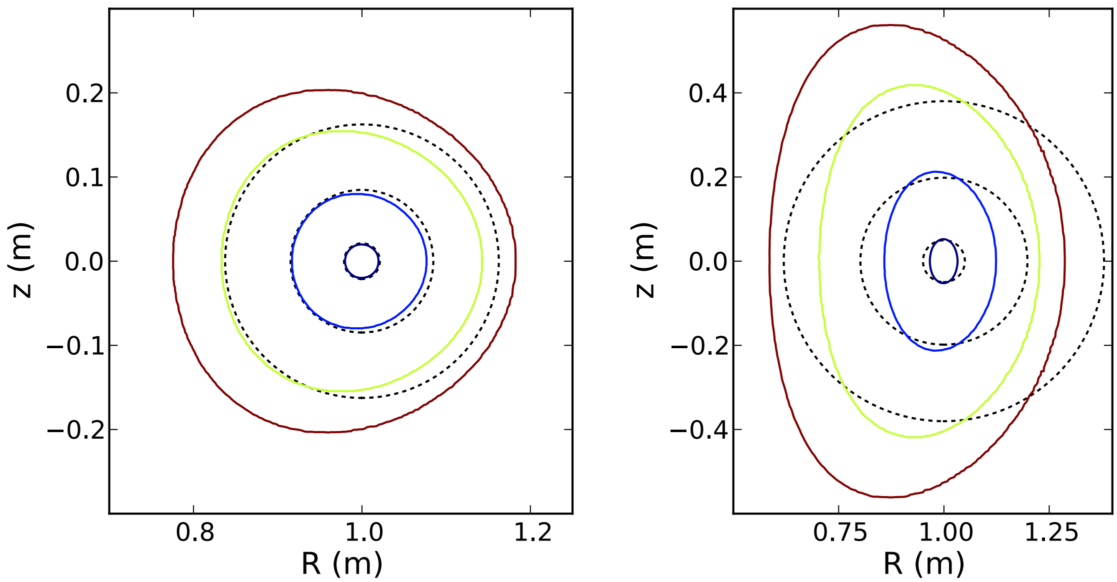

Figure 5.1: Poloidal cross sections of the Solov'ev 1 (left) and Solov'ev 3 (right) equilibria flux surfaces. The surfaces shown correspond to ψ_n values 0.10 (black), 0.16 (blue), 0.58 (green) and 1.0 (red). Also shown for each internal flux surface, are circular cross-sections centered on axis with radii $r = \epsilon_a \sqrt{\psi_n}$ (black dotted).

Although not explicitly in DCON, the q profile for Solov'ev equilibria is analytically soluble,

$$q(\psi_n) = q_0 \frac{2}{\pi} \frac{\sqrt{1 + 2\epsilon_r}}{1 - 4\epsilon_r^2} E(k), \tag{5.6}$$

where $\epsilon_r \equiv (R - R_0)/R_0$ is the local inverse aspect ratio on the LFS mid-plane, E is the complete elliptic integral of the second kind and $k \equiv \sqrt{4\epsilon_r / (1 + 2\epsilon_r)}$.

Two Solov'ev equilibria are used in this chapter. The first, labeled "Solov'ev 1" is the near-circular large aspect ratio case without any $n = 1$ rational surfaces in the plasma. This equilibrium is described by Eqs. (5.3-5.6), with the defining parameters $q_0 = 1.2$, $R_0 = 1$, $a = 0.2$, and $\kappa = 1$. The second equilibria used is labeled "Solov'ev 3", is vertically elongated and has two $n = 1$ rational surfaces. The Solov'ev 3 equilibrium is defined in DCON by the parameter set $q_0 = 1.9$, $R_0 = 1$, $a = 0.33$, and $\kappa = 1.6$. Cross sections of the flux surfaces for the two equilibria are shown in Fig. 5.1. Both cases are compared to circular cross section approximations, and show the inward shift of flux surfaces that is characteristic of Solov'ev equilibria. Fig. 5.4 gives their q profiles, which terminate at q_{edge} 1.41 and 3.26 for Solov'ev 1 and 3 respectively.

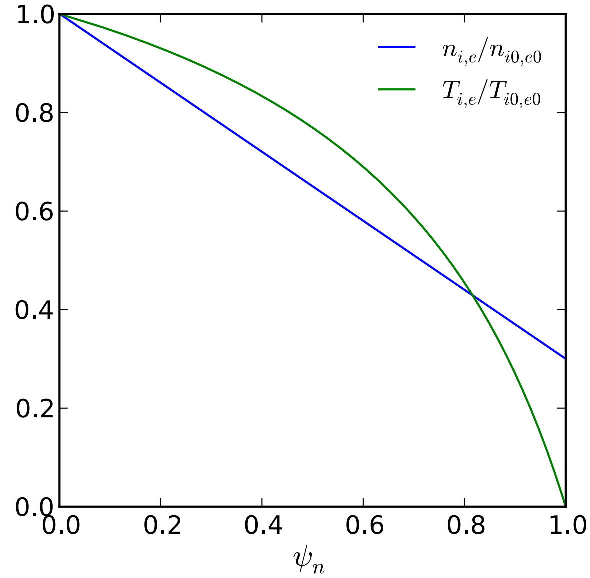

Figure 5.2: The normalized density and temperature profiles used for the Solov'ev 1 and 3 equilibria. The product NT of these gives pressure profiles consistent with (5.4).

For the purposes of this chapter, we assert that the Solov'ev plasmas are entirely comprised of thermal deuterium ions and electrons with equal densities ($n_i = n_e$) and temperatures ($T_i = T_e$). The pressure profiles, determined by Eq. (5.4), have on axis pressures $P_0 = 2.210 \times 10^4$Pa and $P_0 = 4.273 \times 10^4$Pa for the Solov'ev 1 and 3 equilibria respectively. The ratio of the on axis ion gyro frequency to the on axis Alfvén frequency is chosen to be $\omega_{gi0}/\omega_{A0} = 121$ for consistency with earlier work [59]. This value gives $\omega_{A0} = 395.9$krad/s and the unrealistically high density $n_0 = 1.518 \times 10^{21}\mathrm{m}^{-3}$. The density profile is further chosen to be described by $n_{i,e} = n_0(1 - 0.7\psi_n)$, which also determines the temperature profiles $T_{i,e} = (P_0/2n_0)\,(1 - \psi_n)\,/\,(1 - 0.7\psi_n)$. These profiles are shown in Fig. 5.2. Finally, the electric precession profile $\omega_E = \omega_{E0}(1-\psi_n)$ with $\omega_{E0} = 3.959$krad/s is asserted in both cases. This default on axis value corresponds to $\omega_{E0}/\omega_{A0} = 10^{-2}$, and will be scanned over several orders of magnitude in the final section of this chapter.

5.1.2 ITER equilibrium

The ITER equilibrium used in this chapter is the current ITER design for $\beta_N = 2.9$ operation with plasma current $I_p = 9$MA [124]. The equilibrium axis has a major radius $R_0 = 6.2$m and magnetic field $B_0 = 5.3$T. The elongation is comparable to the Solov'ev 3 analytical equilibrium but with much stronger shaping, especially the triangularity near the separatrix at the bottom of the plasma. The ITER shape and safety factor profiles are shown in Fig. 5.3 and Fig. 5.4 respectively. As is clear in the later plot, the ITER equilibrium contains many more $n = 1$ rational surfaces than either of the analytic equilibria.

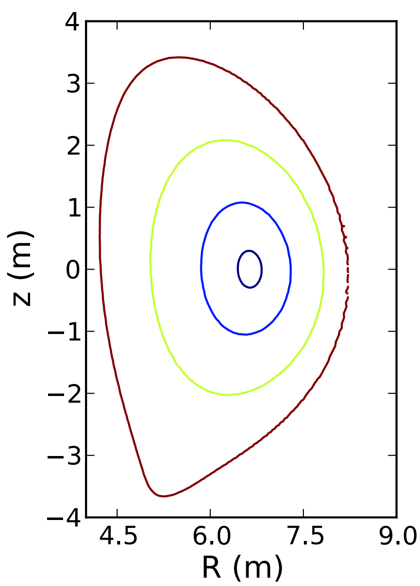

Figure 5.3: Poloidal cross section of the ITER equilibrium flux surfaces. The surfaces shown correspond to ψ_n values 0.10 (black), 0.16 (blue), 0.58 (green) and 1.0 (red). The equilibrium has much stronger shaping than either analytic equilibrium discussed previously.

The kinetic profiles for this equilibrium are determined by the steady-state target, which contains three distinct species: thermal deuterium ions, thermal electrons, and high energy alpha particles. thermal ion and electron densities are again assumed equal, $n_i = n_e$, forsaking quasi-neutrality. For the purposes of this chapter, only thermal species are considered in the calculation of $(T_\varphi, \delta W_k)$. These profiles are shown in Fig. 5.5. The on axis values are $n_0 = 7.22 \times 10^{19}\text{m}^{-3}$, $T_{i0} = 31.32$keV, and $T_{e0} = 34.25$keV. The ratio $\omega_{E0}/\omega_{A0} = 1.62 \times 10^{-2}$,

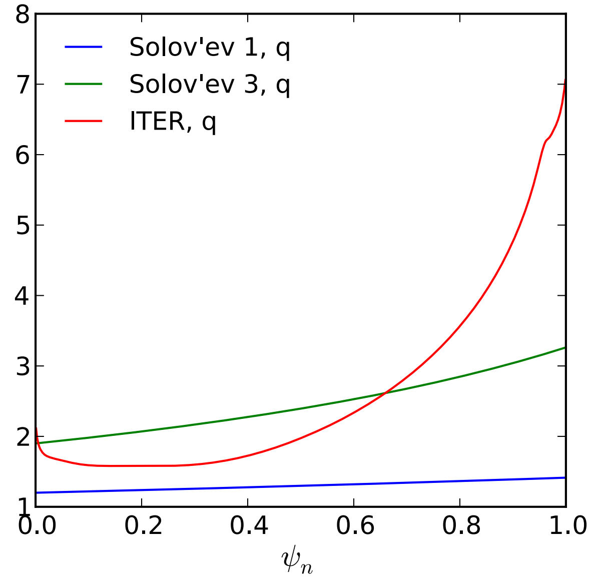

Figure 5.4: Safety factor profile for the three benchmark equilibria. The Solov'ev equilibrium has no $n = 1$ rational surfaces, the Solov'ev 3 equilibrium has 2 and the ITER case contains 7 including two $q = 2$ surfaces. Also significant, is the proximity of ITER's rational surfaces to both the magnetic axis and plasma edge.

is similar to that used for the analytic equilibria and will be similarly scanned in Section 5.5.

5.2 Kinetic Frequencies

At the heart of the torque calculation (4.20) is the resonance operator,

$$\mathcal{R}_{T\ell} = \frac{\left[\omega_\varphi + \omega_{*T}\left(x - \frac{5}{2}\right)\right] x^{5/2} e^{-x}}{i\left[\left(\ell - \sigma n q\right)\omega_b + n\left(\omega_E + \omega_D\right)\right] - \nu_s}, \tag{5.7}$$

which contains a large number of kinetic frequencies. Namely: the bounce frequency ω_b, electric precession frequency ω_E, magnetic precession frequency ω_D, collision frequency ν_s for species s, rotation frequency ω_φ, temperature gradient diamagnetic frequency ω_{*T} and by virtue of the assumed redial force balance the density gradient diamagnetic frequency ω_{*N}. These frequencies all have flux surface dependence and many have energy and pitch angle dependence as well. All, however, are taken to be zeroth order in the perturbed displacement

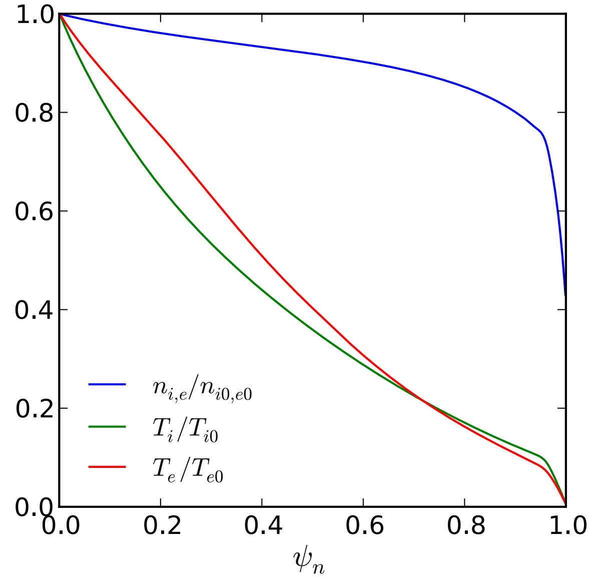

Figure 5.5: The normalized density and temperature profiles for thermal ion and electron species in the ITER equilibrium. Note that quasi-neutrality is broken by the presence of an alpha particle population (not shown). On axis values are $n_{i0,e0} = 7.22 \times 10^{19}\mathrm{m}^{-3}$, $T_{i0} = 31.32\mathrm{keV}$, and $T_{e0} = 34.25\mathrm{keV}$.

and can be calculated directly from the axisymmetric equilibria given in Section 5.1.

5.2.1 Rotation frequencies

The resonance operator (4.21) has implicit in its denominator the assumption of radial force balance with negligible poloidal rotation. This relates the rotation, diamagnetic and electric precession frequencies $\omega_\varphi = \omega_E + \omega_{*T} + \omega_{*N}$, which are all flux function quantities. The electric precession is taken as a fundamental input in PENT. The diamagnetic frequencies,

$$\omega_{*N} = -2\pi \frac{T}{eN}\frac{\partial N}{\partial \psi}, \tag{5.8}$$

$$\omega_{*T} = -2\pi \frac{1}{e}\frac{\partial T}{\partial \psi}, \tag{5.9}$$

are computed using cubic splines of the density and temperature profiles.

The Solov'ev rotation profiles and its three constituents are shown together in Fig. 5.6

for the nominal values of ω_{E0} given in Section 5.1. Note that the components $(\omega_E, \omega_{*N}, \omega_{*T})$ are on the same order for much of the plasma. An increase or reduction of ω_{E0} by an order of magnitude would result in rotation proportional to the electric precession ($\omega_\varphi \approx \omega_E$) or independent of it through almost the entire plasma. The one exception is the edge, where the temperature gradient drive will always dominate the rotation in these profiles.

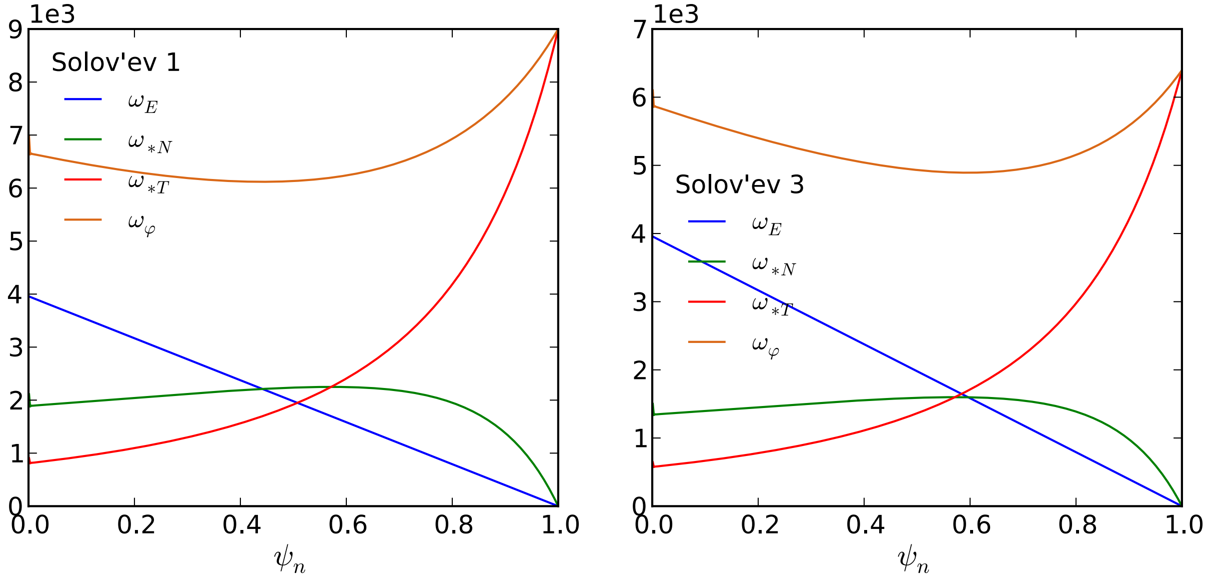

Figure 5.6: Solov'ev 1 (left) and 3 (right) rotation profiles, diamagnetic frequencies and electric precession. The electric precession is explicitly input into PENT, while the others are calculated from radial force balance and the ion density and temperature profiles.

The ITER equilibrium has a smooth and monotonic rotation profile. It steadily decreases from a large value on axis to zero at the edge, with the only deviation being a small steeper portion at the pedestal edge. The density, temperature, and electric potentials are all changing rapidly in this region and their corresponding gradient frequencies can be seen to spike. Unlike the Solov'ev equilibria, the ITER profiles contain a zero crossing of ω_E. This is a likely place for large kinetic resonance effects, as it represents a significant decrease in the resonant operator denominator with low collisionality. The electric precession is again comparable to the diamagnetic frequency, and scaling will result in similar behaviors as discussed for the Solov'ev cases.

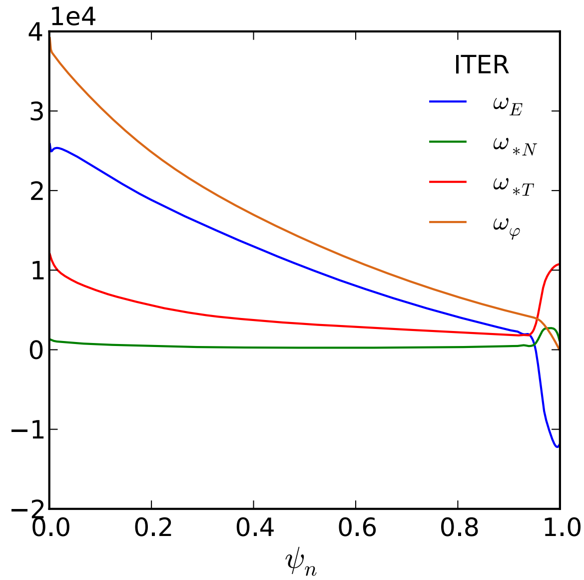

Figure 5.7: ITER rotation profile, diamagnetic frequencies and electric precession.

5.2.2 Bounce Frequency

The bounce frequency of a particle is defined as,

$$\omega_b \equiv \frac{2\pi}{\oint d\vartheta \mathcal{J} B / v_\parallel}, \tag{5.10}$$

where $\mathcal{J}$ is the Jacobian, B is the strength of magnetic field and $v_\parallel$ is the particle velocity parallel to the magnetic field. This definition is valid in the limit of zero-width orbits for which the motion can be completely described on a single flux surface parameterized by the length ℓ such that $d\ell \equiv d\vartheta \mathcal{J} B$.

The bounce frequency can be solved in the circular large aspect ratio (CLAR) limit,

$$\omega_b^{CLAR} = \begin{cases} \frac{1}{2}\sqrt{\frac{\epsilon x \Lambda}{2}}\omega_t \frac{\pi}{K(k)}, & \text{trapped} \\ \frac{1}{2}\sqrt{x\left(1-\Lambda+\epsilon\Lambda\right)}\omega_t \frac{\pi}{K(1/k)}, & \text{passing} \end{cases} \tag{5.11}$$

where the energy space variables $x = E/T$, $\Lambda = \mu B/B_0$ and $k^2 = \left(1-\Lambda+\epsilon\Lambda\right)/\left(2\epsilon\Lambda\right)$ are consistent with Chapter 4. Here, K is again the complete elliptic integral of the first kind and $\omega_t \equiv \sqrt{2T/m}/qR_0$ is the transit frequency.

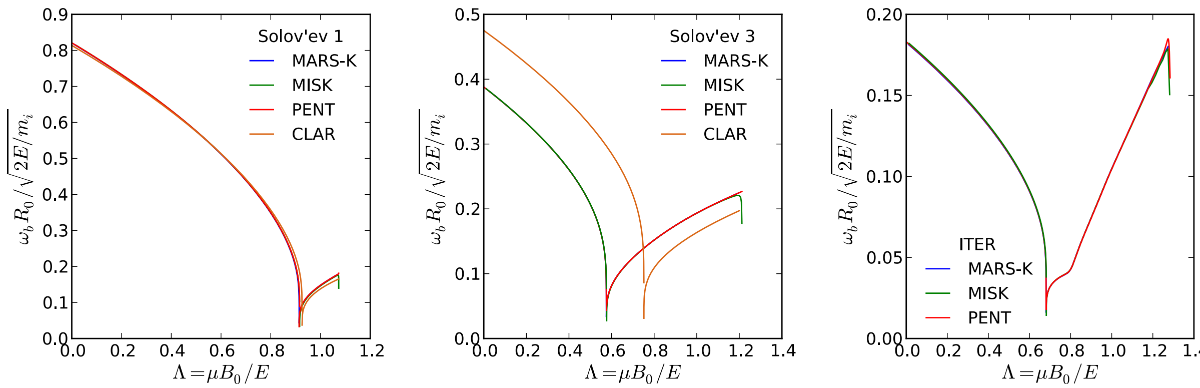

Figure 5.8: The bounce frequency pitch angle profile at the $\psi_n = 0.16$ surface of the Solov'ev 1 equilibrium (left), $\psi_n = 1$ surface of the Solov'ev 3 equilibrium (middle), and $\psi_n = 0.98$ surface of the ITER (right) as calculated by MARS-K, MISK, and PENT. Also included, is the cylindrical approximations for the two Solov'ev cases ($\epsilon_r \approx 0.08$ and $\epsilon_r \approx 0.33$ respectively).

The three codes, PENT, MISK, and MARS-K all must calculate the bounce frequency across energy and pitch space. Each code has a unique method of handling the integrable singularity at the turning points where $v_\parallel = 0$ (See Appendix D for PENT). A comparison between the three methods, and the CLAR limit for reference, is crucial to gain confidence in the basic bounce integration techniques of the codes as well as a necessary building block in the resonant operator that must be confirmed consistent before making further comparisons. Fig. 5.8 shows the bounce frequency as calculated by all three codes and the CLAR reference at ψ_n (ϵ_r) 0.160 (0.08), 1.0 (0.33) and 0.982 (0.322) in the Solov'ev 1, Solov'ev 3 and ITER equilibria respectively. All three codes have good agreement throughout the parameter space. The CLAR limit agrees well with the core Solov'ev 1 results as expected, but begins to significantly overestimate the trapped-passing boundary in Λ for the elongated plasmas at lower (although still large) aspect ratio.

The bouncy frequency can be analytically solved in the deeply trapped ($k \to 0$) limit for Solov'ev equilibria [125]. The analytic expression,

$$\omega_b = \omega_t \sqrt{x} \frac{qR_0}{q_0} \frac{\left[\frac{\epsilon}{2} \frac{F^2}{(1+2\epsilon)} + \epsilon \left(\frac{\kappa\epsilon}{q_0} \right)^2 + \epsilon^2 \frac{(1-\kappa^2)}{2q_0^2} (1+2\epsilon) \right]^{1/2}}{\frac{F^2}{1+2\epsilon} + \left(\frac{\kappa\epsilon}{q_0} \right)^2}, \tag{5.12}$$

is compared to the three codes in Fig. 5.9. This formula is explicitly for Solov'ev equilibria rather than a CLAR approximation to the equilibria and thus agrees well with the general geometry computational results throughout the radial profile of both cases.

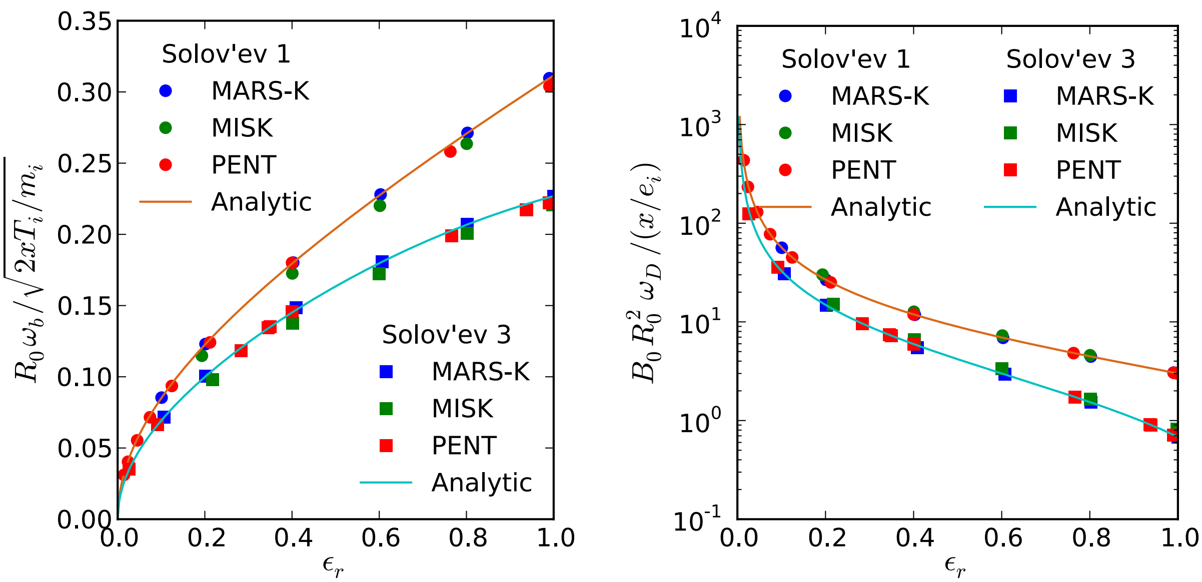

Figure 5.9: The deeply trapped limit bounce frequency (left) and magnetic precession frequency (right) as computed by MARS-K, MISK and PENT across the Solov'ev 1 and Solov'ev 3 equilibria compared with the analytic solutions.

5.2.3 Magnetic Precession Frequency

The magnetic precession frequency calculated in PENT was defined in Chapter 4's derivation of the NTV torque as,

$$\omega_D \equiv 2\pi \left\langle \left(E - \frac{3}{2}\mu B \right) \frac{2}{eB} \frac{\partial B}{\partial \psi} + (E - \mu B) \frac{2}{e\mathcal{J}} \frac{\partial \mathcal{J}}{\partial \psi} \right\rangle_b, \tag{5.13}$$

where the angular brackets denote the bounce average defined by $\langle A \rangle_b \equiv \oint \left(A dl / v_\parallel \right) / \oint \left(dl / v_\parallel \right)$ $= \frac{\omega_b}{2\pi} \oint d\vartheta A \mathcal{J} B / v_\parallel$. This form is different from the one published in Ref. [57], which comes from the somewhat different derivation in the stability codes originating explicitly from the RWM δW_k. The two forms are equivalent mathematically, but serve as a reminder of the numerical differences in the codes originating from their different applicational emphasis. The closed integral bounce average again has an integrable singularity at the bounce points ($v_\parallel = 0$), and again it is treated with different numerical methods in the three codes.

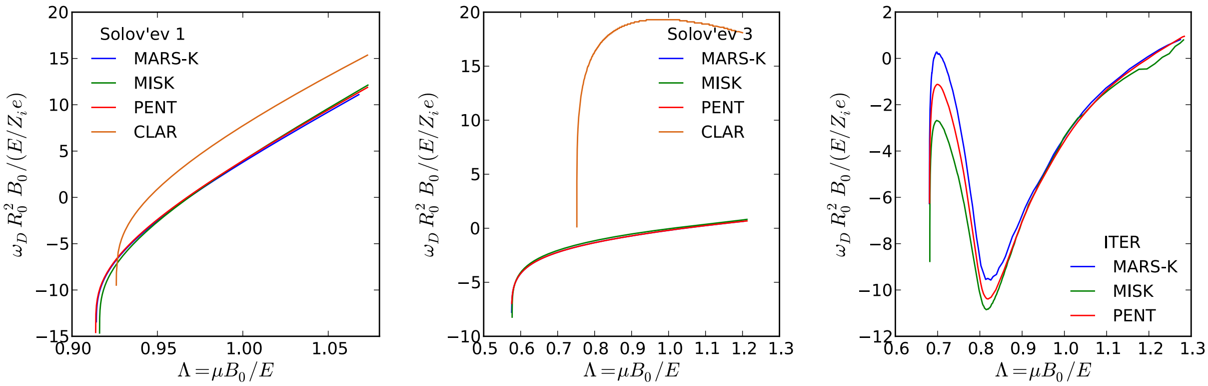

Figure 5.10: The trapped particle magnetic precession frequency pitch angle profile at the $\psi_n = 0.16$ surface of the Solov'ev 1 equilibrium (left), $\psi_n = 1$ surface of the Solov'ev 3 equilibrium (middle), and $\psi_n = 0.98$ surface of the ITER (right) as calculated by MARS-K, MISK, and PENT. Also included, is the CLAR approximations for the two Solov'ev cases ($\epsilon_r \approx 0.08$ and $\epsilon_r \approx 0.33$ respectively).

The magnetic precession frequency in the CLAR limit [59],

$$\omega_D = x\Lambda \frac{q^3}{\epsilon} \frac{\omega_t^2}{\omega_g} \left[(2s+1) \frac{E(k)}{K(k)} + 2s \left(k^2 - 1 \right) - \frac{1}{2} \right], \tag{5.14}$$

is directly proportional to the aspect ratio and ignores the curvature drift altogether. The true precession can be expected to be more sensitive to low aspect ratio effects and general geometric shaping effects than the bounce frequency, which had a weaker aspect ratio dependence $\left(\epsilon^{1/2} \right)$. Here, $s \equiv (r/q) dq/dr$ is the magnetic shear.

Again, an analytic solution exists in the deeply trapped limit for Solov'ev equilibria

[125],

$$\omega_D = \frac{xT}{e}\frac{q_0}{\epsilon R_0 \kappa}\frac{\left[\frac{F^2}{(1+2\epsilon)^2} - \frac{\kappa^2\epsilon}{q_0^2}\right]^{1/2}}{\frac{F^2}{1+2\epsilon} + \left(\frac{\kappa\epsilon}{q_0}\right)^2}. \tag{5.15}$$

Here e is the charge of the particle species and T its temperature. Like the CLAR limit, this value has a strong dependance on the inverse aspect ratio ϵ. The radial profile of deeply trapped limit magnetic precession for each code is shown to agree well to this analytic solution for both Solov'ev equilibria in Fig. 5.9. This is in contrast with comparisons to the CLAR limit in Fig. 5.10, where the lower aspect ratio cases show large deviations from the approximation. All codes are in good quantitative agreement, however, giving confidence in their general treatment of the surface geometry dependent drifts.

5.2.4 Collision Frequency

The combined NTV theory derived in Chapter 4 and implemented in PENT assumes, in general, a modified Krook collision operator with energy but no pitch angle dependence. The effective operator for trapped ℓ harmonic particles is [23],

$$\nu_s = x^{-3/2}\nu_s^{eff}\left[1 + \left(\frac{\ell}{2}\right)^2\right], \tag{5.16}$$

The effective collisionality above is the standard energy independent Krook operator, with aspect ratio dependence for trapped particles $\nu_s^{eff} = \sum_b \nu_{sb}/2\epsilon$, that compensates for the anisotropic energy distribution of that population [105].

A number of reduced operators are also available in the PENT code. Of primary importance are the energy independent $\nu_s = \nu_s^{eff}$ operator, and collisionless $\nu_s = 0$ cases. Both cases allow analytic simplifications of the resonant operator energy integration historically used in MARS-K [126], although all three codes now include energy dependent capabilities.

For the Solov'ev and ITER equilibria results presented here, the collisionality was explicitly set to zero. This simplifies our benchmark in eliminating any possible discrepancy in

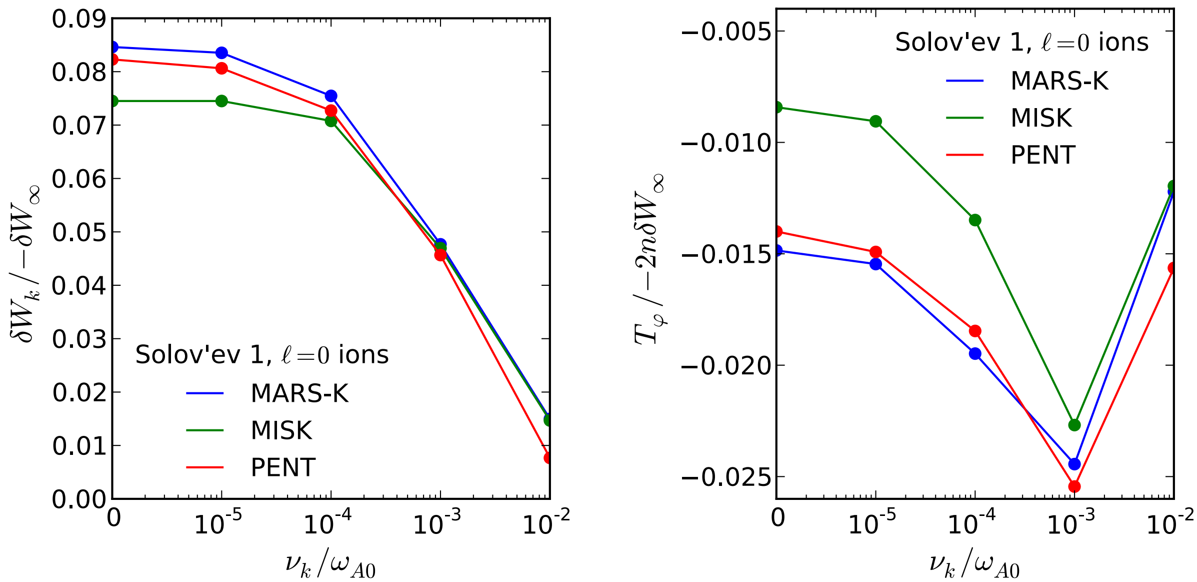

Figure 5.11: Convergence of the kinetic perturbed energy (left) and toroidal torque (right) in the case of a constant collisionality limiting to zero. Thermal ions with $\ell = 0$ bounce harmonic are considered.

the collisionality. It challenges the numerical integration techniques, however, by introducing poles on the energy axis at kinetic resonances. The approach taken by PENT to handle these poles is to integrate over a complex contour in x (see Appendix D). A simplified numerical handling, however, may be used to more clearly demonstrate the sensitivity of the numerical integrations to these singularities. Fig. 5.11 demonstrates the convergence of each code to a collisionless limit using a scan of small, constant collisionality. This figure shows that each code accurately handles the numerical integration challenges inherent to the zero collisionality limit. A detailed analysis of the quantitative agreement between the codes in this limit will be the subject of the following sections.

5.3 Energy Integrated Resonance Operator

The innermost integral in the expression for NTV torque in a perturbed equilibrium is independent of the perturbation. Instead, it contains the resonant operator,

$$I_x(\psi_n, \Lambda) = \int_0^\infty dx \mathcal{R}_{T\ell}, \tag{5.17}$$

which in turn is comprised of the frequencies described above for the unperturbed, axisymmetric equilibrium. An important difference in the development of PENT and the stability codes MISK and MARS-K is that the derivation of the kinetic stability caries the mode rotation frequency (ω_r) and growth rate (γ). The stability resonance operator is [57],

$$\mathcal{R}_{S\ell} = \frac{\left[n\omega_\varphi + n\omega_{*T}\left(x - \frac{5}{2}\right) - \omega_r - i\gamma\right] x^{5/2} e^{-x}}{\left[\left(\ell - \sigma nq\right)\omega_b + n\left(\omega_E + \omega_D\right)\right] - i\nu - \omega_r - i\gamma}, \tag{5.18}$$

and reduces to the torque resonance operator $\mathcal{R}_{T\ell}$ in the limit $\omega_r, \gamma \to 0$. This limit is precisely the regime of validity for the perturbative approach discussed in the introduction of this chapter, allowing direct comparison of the stability and torque codes under the equivalency principle [60]. For the purposes of this benchmark, the mode rotation and growth rate are explicitly taken to be zero ($\omega_r, \gamma = 0$).

Comparisons of energy integral pitch angle profiles from MARS-K, MISK, and PENT are shown in Fig. 5.12, Fig. 5.13 and Fig. 5.14 for the Solov'ev 1, Solov'ev 3 and ITER equilibria. The surfaces used in the bounce and magnetic precession frequency figures are used in the Solov'ev 1 and ITER cases, while the Solov'ev 3 energy integrals are compared further in to the equilibrium. Only select ℓ harmonics for the trapped thermal ions are shown to demonstrate the quantitative agreement between codes. The figures demonstrate the decreasing contribution of increasing $|\ell|$ harmonics, although the codes include many ℓ harmonics for accuracy when calculating the final T_φ and δW_k. They also show the tendency of the resonant operator to preferably weight the weakly trapped particle contribution. This must be combined with preferential weighting of the deeply trapped particles in the perturbed action (see Section 5.4), but is a significant indication that the exact calculation of the trapped-passing boundary and detailed pitch angle dependencies are important in these cases.

An important historical difference in the development of the three codes under consideration is that MARS-K originally relied entirely on analytic solutions of the energy integral [126]. Analytical solutions are possible when the collisionality is taken to be independent of energy and only the precession resonance ($\ell = 0$) is considered. Additional bounce harmonic resonances ($\ell \neq 0$) can be solved in the limit of no magnetic precession ($\omega_D \ll \omega_b$). For compatibility with this method, the strict assertion $\omega_D = 0$ for all $\ell \neq 0$ was included in these benchmarks. During the coarse of the benchmarking process, however, MARS-K developed and implemented numerical integration of I_x. Despite not being used in this benchmark, all three code have the capability to handle both energy dependent collisionality and bounce harmonics with magnetic precession.

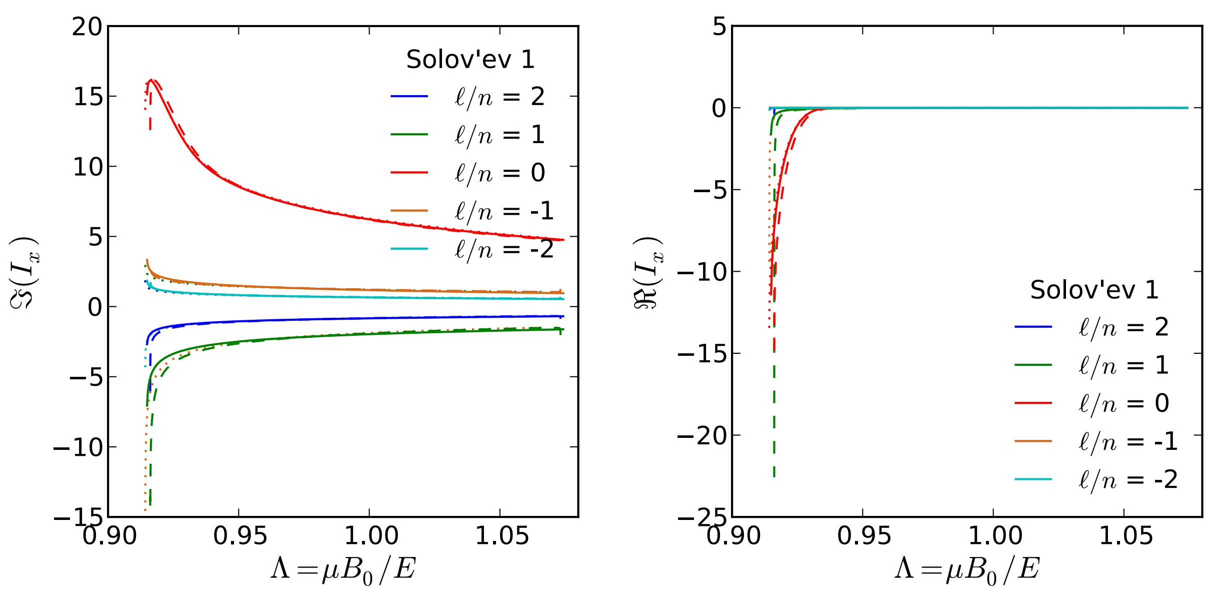

Figure 5.12: Pitch angle profiles of the energy integral I_x for trapped thermal ions on the $\psi_n = 0.16$ surface of the Solov'ev 1 equilibrium. Computations of the integral are in quantitative agreement for MARS-K (dotted), MISK (dashed), and PENT (solid).

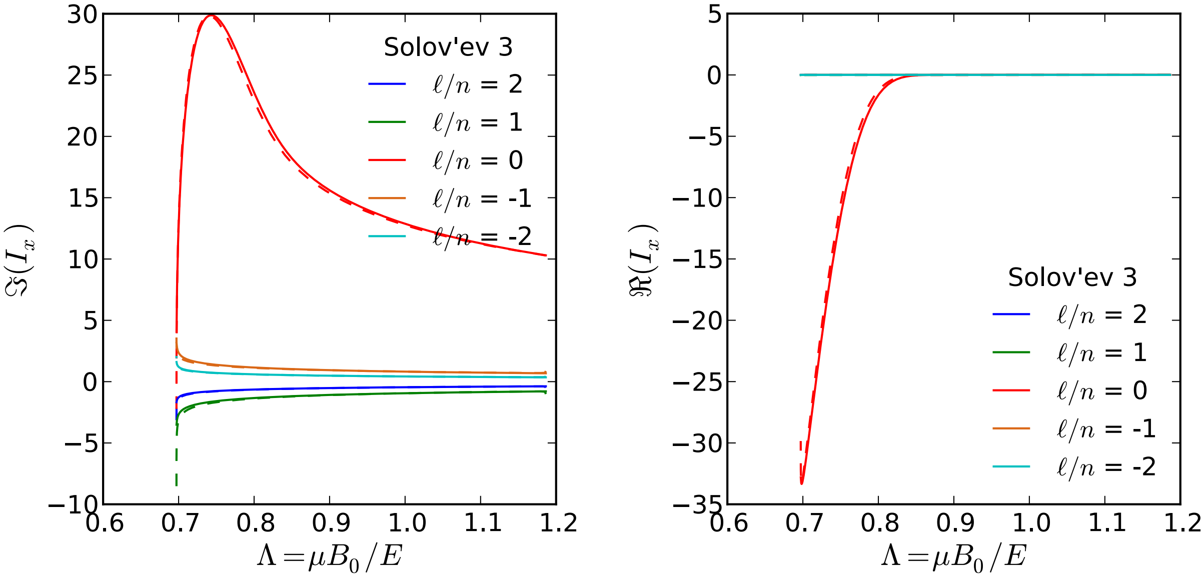

Figure 5.13: Pitch angle profiles of the energy integral I_x for trapped thermal ions on the $\psi_n = 0.585$ ($q = 2.5$) surface of the Solov'ev 3 equilibrium. Computations of the integral are in quantitative agreement for MARS-K (dotted), MISK (dashed), and PENT (solid).

5.4 Perturbed Action

The pitch angle profile of the torque and kinetic energy contains the perturbed action δJ_ℓ, calculated from a nonaxisymmetric perturbed equilibrium. This is the first introduction of perturbed quantities in the benchmark calculations thus far, and the perturbed equilibria must be examined in detail before continuing to the final comparison of action, torque and energy.

5.4.1 An Approximate RWM Eigenfunction

Stability analysis using the RWM dispersion relation (5.1) was a primary goal of this benchmarking. As such, the nonaxisymmetric perturbation of interest was the fluid RWM eigenfunction. MARS-K directly calculates this eigenfunction. MISK and PENT, however, approximate the RWM using the marginally stable ideal kink eigenfunction with an ideal wall. The kinetic codes rely on calculations of these eigenfunctions from PEST [76] and DCON+IPEC

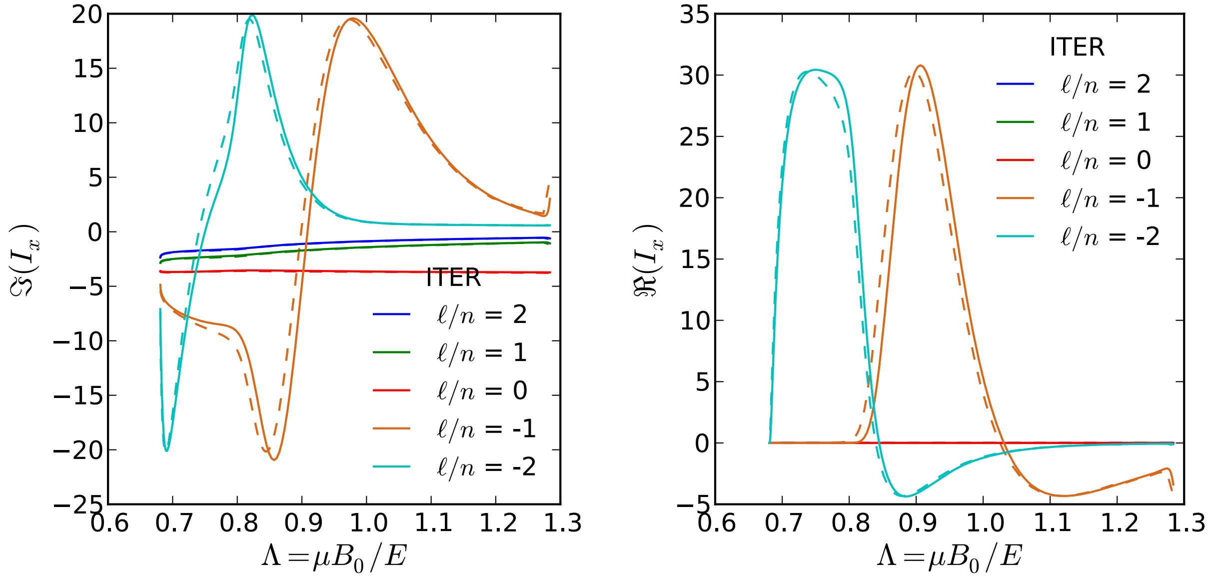

Figure 5.14: Pitch angle profiles of the energy integral I_x for trapped thermal ions on the $\psi_n = 0.982$ surface of the ITER equilibrium. Computations of the integral are in quantitative agreement for MISK (dashed), and PENT (solid).

[46, 113, 44] respectively. In both cases, an ideal wall position is moved progressively inward until the marginal point is found. This approach has been found to be a good approximation to the fluid RWM [59], and has been compared to experiment [67]. Validation in all three equilibria used here is shown in Fig. 5.15, which compares the poloidal Fourier harmonics of the normal displacement of the marginally stable ideal kink from PEST and DCON to the fluid RWM displacement from MARS-K. The Fourier harmonics are decomposed in PEST coordinates, which have well separated resonant and nonresonant harmonics.

One important spectral difference in the codes is that both Solov'ev eigenfunctions have a toroidal mode number of $n = -1$ in MARS-K due to geometrical conventions in that code. All other calculations, including all PENT calculations, are toroidal mode number $n = 1$.

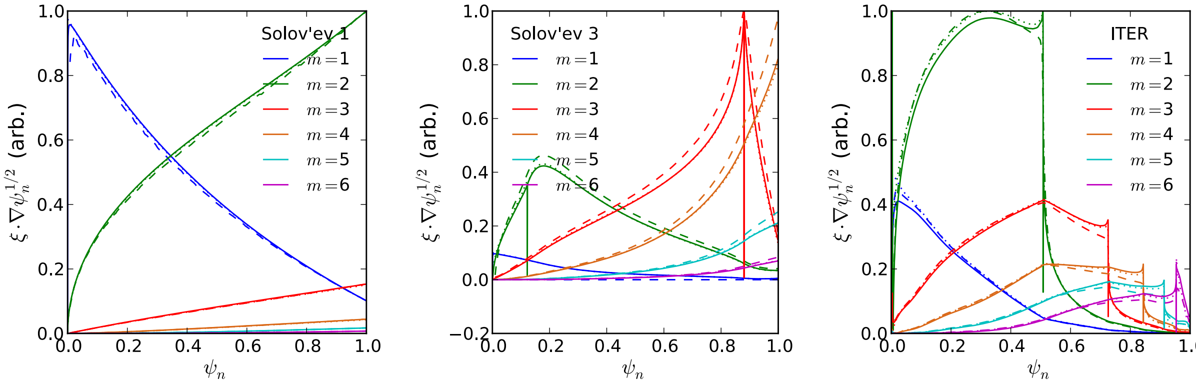

Figure 5.15: Poloidal Fourier harmonics of the normal displacement used with the Solov'ev 1 (left), Solov'ev 3 (middle), and ITER (right) equilibria. The marginally stable ideal kink mode with an ideal wall is computed by DCON (solid) and PEST (dashed) PENT and MISK respectively. A fluid RWM eigenfunction is calculated by and used in MARS-K (dotted). All result are decomposed in the PEST coordinate system.

5.4.2 Pitch Angle Profile of Perturbed Action

The toroidal torque T_φ and kinetic energy δW_k have a quadratic dependence on the perturbed action derived in Chapter 4,

$$\delta J_{\pm\ell}\left(\psi_n,\Lambda,x\right) \equiv -2inxT\left\langle \mathcal{P}_\ell^{\pm1}\left[\left(1-3\Lambda\frac{B}{B_0}\right)\frac{\delta B}{B}+\left(1-\Lambda\frac{B}{B_0}\right)\frac{1}{\mathcal{J}}\nabla\cdot\boldsymbol{\xi}_\perp\right]\right\rangle_b, \tag{5.19}$$

which contains the first order perturbed equilibrium quantities. This quantity is renormalized to become a perturbed Hamiltonian and written using $\boldsymbol{\kappa}\cdot\boldsymbol{\xi}_\perp$ in the stability codes [59, 57]. Here the curvature $\boldsymbol{\kappa}=\hat{\mathbf{b}}\cdot\nabla\hat{\mathbf{b}}$, and the first order quantities are related by,

$$\frac{\delta B}{B}=-\left(\boldsymbol{\kappa}\cdot\boldsymbol{\xi}_\perp+\nabla\cdot\boldsymbol{\xi}_\perp\right). \tag{5.20}$$

The perturbed action is a function of position, energy and pitch. The normalization used in PENT, $\left|\delta\bar{J}_\ell\right|^2=\left|\delta J_\ell\right|^2/2xTMR_0^2$, and similar normalizations in the stability codes remove the energy dependence, reducing the dependence to one of pitch angle only for a given flux surface. Fig. 5.16 shows good agreement in the pitch angle profiles of the perturbed action term used in PENT and MISK for surfaces in the Solov'ev equilibria. Note that MARS-K includes bounce harmonic coupling in the quadratic action, and is thus not included in these figures. The close agreement between MARS-K and PENT results presented in Section 5.5 suggest that this harmonic coupling is negligible, as was assumed in the combined theory derivation.

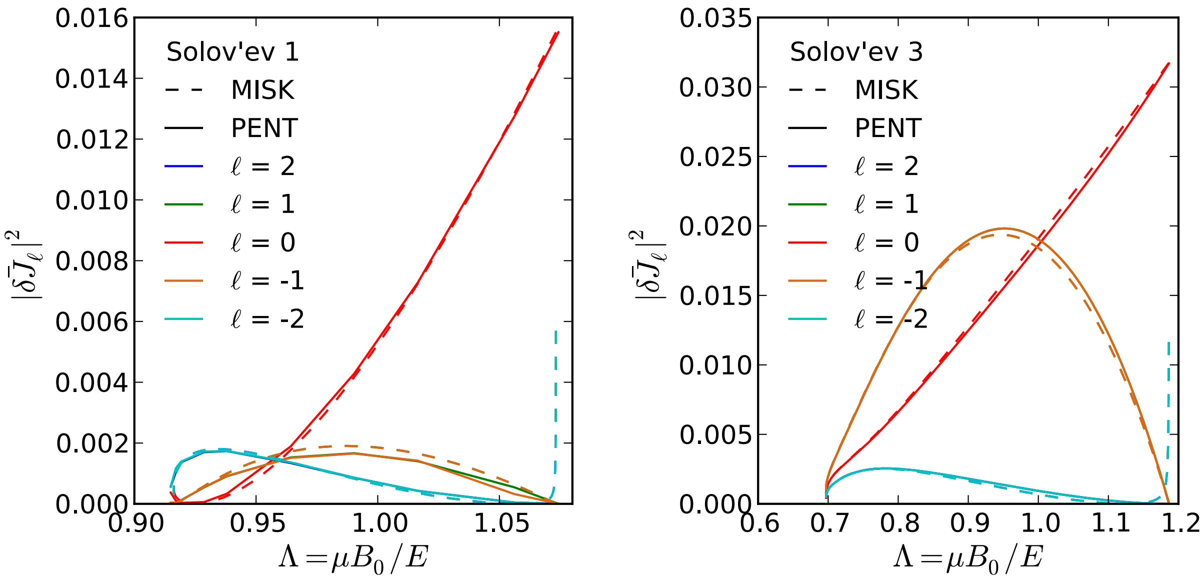

Figure 5.16: The trapped particle perturbed action pitch angle profile at the $\psi_n = 0.16$ surface of the Solov'ev 1 equilibrium (left) and $\psi_n = 0.585$ surface of the Solov'ev 3 equilibrium (right) as calculated by MISK (dashed) and PENT (solid). Distinct bounce harmonic components are not calculated in MARS-K.

Similar to the previous bounce averages, the bounce orbit integrand of the perturbed action is integrally singular. In addition, the perpendicular displacement is singular at rational rational surfaces due to Alfvén resonances in the ideal perturbed equilibria.

5.4.2.1 Sensitivity to Singular Behavior

A major difference in the perturbed equilibria used by PENT and the stability codes is the regularization of singular displacements in IPEC. The Alfvén resonance singularities in the ideal displacement calculated by DCON and IPEC are non-integrable [58, 45]. The ideal

MHD physics used to calculated this displacement, however, is known to break down near the rational surface. IPEC thus asserts a regular condition over a small region comparable to an ion gyro-radius around the rational surface as a ad-hock imitation of resistive layer physics otherwise not captured (see Appendix D and references therein). The result is a smoothly connected perpendicular displacement across rational surfaces.

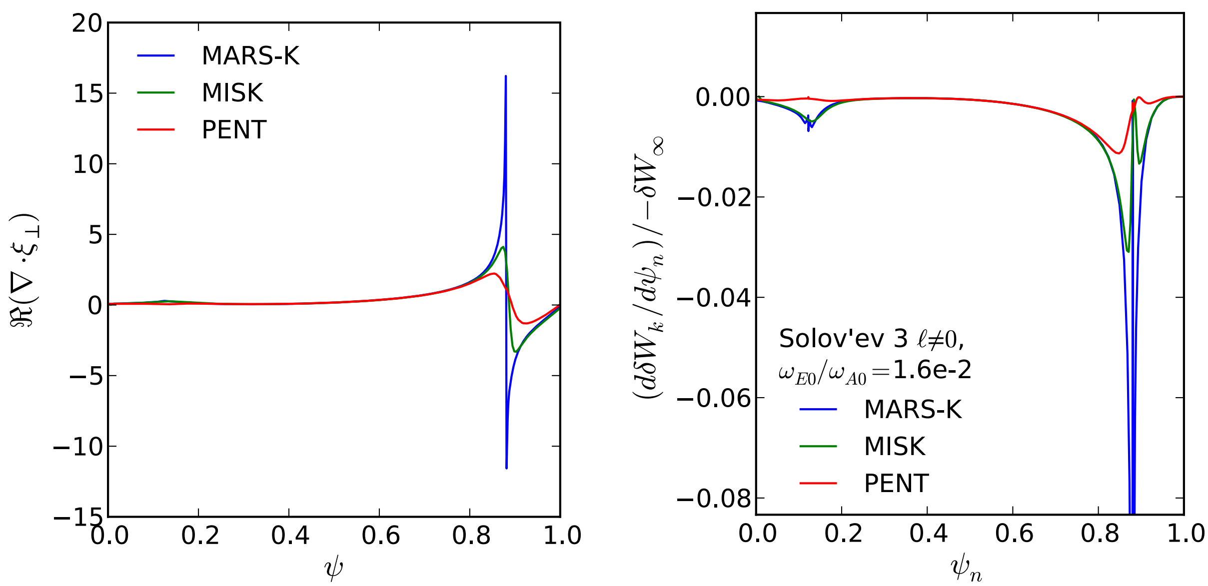

Figure 5.17: Divergence of the perpendicular displacement (left) and kinetic energy (right) in the perturbed Solov'ev 3 equilibrium computed in MARS-K, MISK, and PENT. The $\nabla \cdot \boldsymbol{\xi}_\perp$ profiles are taken along a line extending vertically from the magnetic axis to the upper plasma edge, and have been normalized to intersect at the $q = 2.5$ surface ($\psi_n = 0.585$).

Fig. 5.17 compares the first order quantity $\nabla \cdot \boldsymbol{\xi}_\perp$ and the ultimate torque computed in MARS-K, MISK and PENT for the Solov'ev 3 equilibrium. The sharp, singular behavior of the un-regularized displacements (especially in MARS-K) is a significant contributions to the perturbed action in these codes and thus their ultimate torque. The inclusion of these regions may account for an historical tendency of perturbative MARS-K calculations to predict larger kinetic effects than self-consistent MARS-K calculations [108]. The effect of these singularities is immediately apparent in the ψ_n profiles of T_φ, and results in quantitative differences even in this simple example. Good agreement, even qualitatively, in the ITER equilibria (with many rational surfaces) was only obtained after the profiles where truncated around the rational surfaces in the stability codes [57].

5.5 Regime Validation using Parametric Scans

Given a perturbed equilibrium and kinetic profiles $N(\psi)$, $T(\psi)$, and $\omega_E(\psi)$, PENT calculates the NTV torque and perturbed kinetic energy δW_k. The torque for thermal species was derived in Chapter 4,

$$T_\varphi = -\frac{n^2}{\sqrt{\pi}}\frac{R_0}{B_0}\int d\psi NT \int d\Lambda \bar{\omega}_b \left|\delta \bar{J}_\ell\right|^2 \int dx \mathcal{R}_{T\ell}. \tag{5.21}$$

The stability codes calculate expressions equivalent to Eq. (5.21) divided by $2ni$, corresponding to δW_k. This result has not assumed any ordering of the fundamental frequencies ω_b, ω_D, ω_E, ω_{*N}, ω_{*T}, and ν. It is thus valid across a wide range of kinetic regimes defined by relative orderings of these and the corresponding behavior of the resonant operator. It has already been seen in the previous chapter, however, that these regimes include resonant physics resulting in computationally challenging properties.

This section compares results from PENT to the stability codes MISK and MARS-K across a wide variety of kinetic regimes. The work examines in detail the contributions of multiple particle species, comparing the trapped, passing, precession resonant, bounce harmonic resonant, electron and ion species in the collisionless limit where kinetic resonances are expected to be the dominant physics.

5.5.1 Scan of the Electric Precession

The three way benchmark between MARS-K, MISK, and PENT was motivated originally by RWM stability physics. Accordingly, the ultimate result of this section will be a comparison of the RWM growth rate and rotation over a wide range of plasma rotation.

5.5.1.1 Fluid Growth Rates

The dispersion relation (5.1) contains both a kinetic contribution δW_k and fluid energies δW_∞ and δW_b. Each fluid term is, in turn, composed of a plasma fluid, surface, and vacuum term. PENT uses DCON to calculate the fluid term from the approximate RWM function described in Sec. 5.4.1, which is coupled with the VACUUM code to include the vacuum terms [113]. The various stability codes do not all utilize standard units, and the relevant physics will be presented in unit-less ratios throughout this section.

All cases considered have conformal walls with locations b/a of 1.15, 1.10, and 1.50 for the Solov'ev 1, Solov'ev 3, and ITER equilibria respectively. The ratio of the fluid terms corresponds to the fluid growth rate normalized by the wall time, $\gamma_f \tau_w = -\delta W_\infty / \delta W_b$, and this quantity is given for each code and each equilibrium in Tab. 5.1. Note that the large fluid growth rate in MARS-K for the Solov'ev 3 case comes from relatively low δW_b, while the larger PENT (DCON) value for ITER is a result of relatively low δW_∞. The former is expected to effect the final normalized RWM dispersion relation, while the later is not due to the ubiquitous normalization of δW_∞.

	MARS-K	MISK	PENT
Solov'ev 1	0.842	0.891	0.893
Solov'ev 3	0.546	0.428	0.432
ITER	1.466	1.477	1.168

Table 5.1: The normalized fluid growth rate $\gamma_f \tau_w$ used in the RWM dispersion relation for MARS-K, MISK, and PENT. MISK and PENT use values from PEST and DCON respectively.

5.5.1.2 Kinetic Perturbed Energy and NTV Torque

The normalized kinetic energy terms $\delta W_k / -\delta W_\infty$ from each code were calculated over a wide range of rotation values, manipulated by self similar scaling of the prescribed electric precession profiles given in Section 5.1. The result of this scan are shown in Fig. 5.18 - Fig. 5.23.

Note that the large electron contribution in these figures is an artifact of the collisionless limit used. Electrons and ions have identical or comparable density and temperature profiles in each case, resulting in similar pressure, rotation frequencies and neoclassical offsets for the two species. The only remaining dependence on the mass is embedded in ω_b, and results in a bounce harmonic peak in the $\ell \neq 0$ electron contribution (not shown) roughly two order of magnitude higher in ω_E than the corresponding ion peak. Collisionality typically introduces a large disparity between ion and electron contributions.

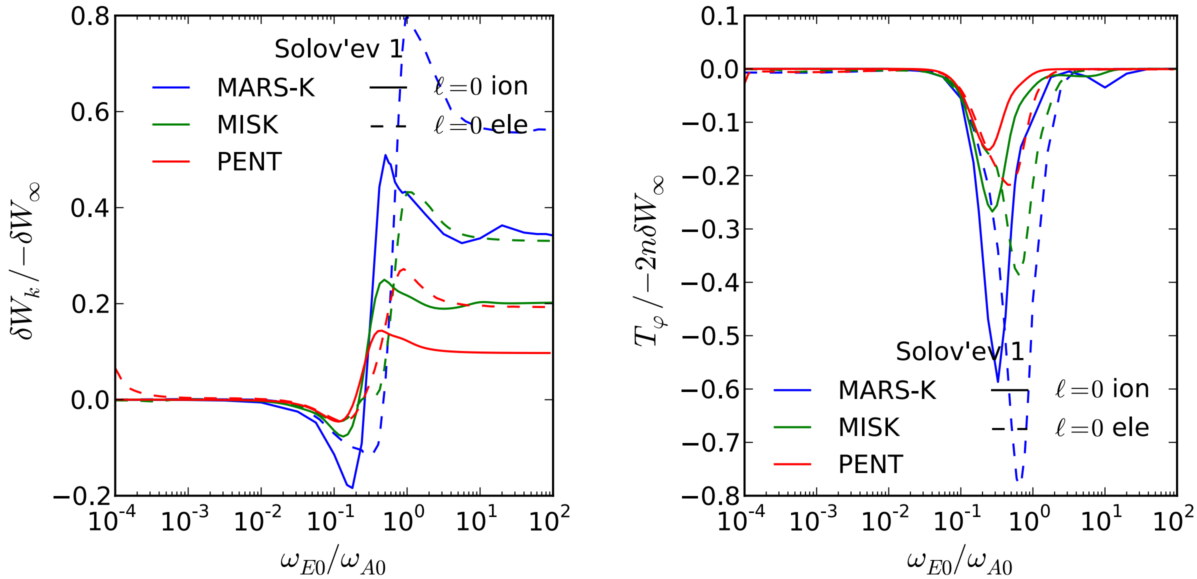

Figure 5.18: Solov'ev 1 RWM kinetic energy and NTV torque across a scan of electric precession frequency for $\ell = 0$ trapped thermal ions and electrons as calculated by MARS-K, MISK, and PENT.

Many of the differences in the results of this scan can be traced to the treatment of the rational surfaces discussed in Sec. 5.4.2.1. The first evidence of this is that the scan using the Solov'ev 1 equilibrium, which contains no rational surfaces, shows impressive quantitative agreement between the three codes. The Solov'ev 3 scan is reported without any explicit treatment of the rational surfaces in MARS-K or MISK. The quantitative differences between the three codes at high rotation are a consequence of the sensitivity of bounce harmonic and circulating particles to the singularity at the rational surfaces. This can be seen in

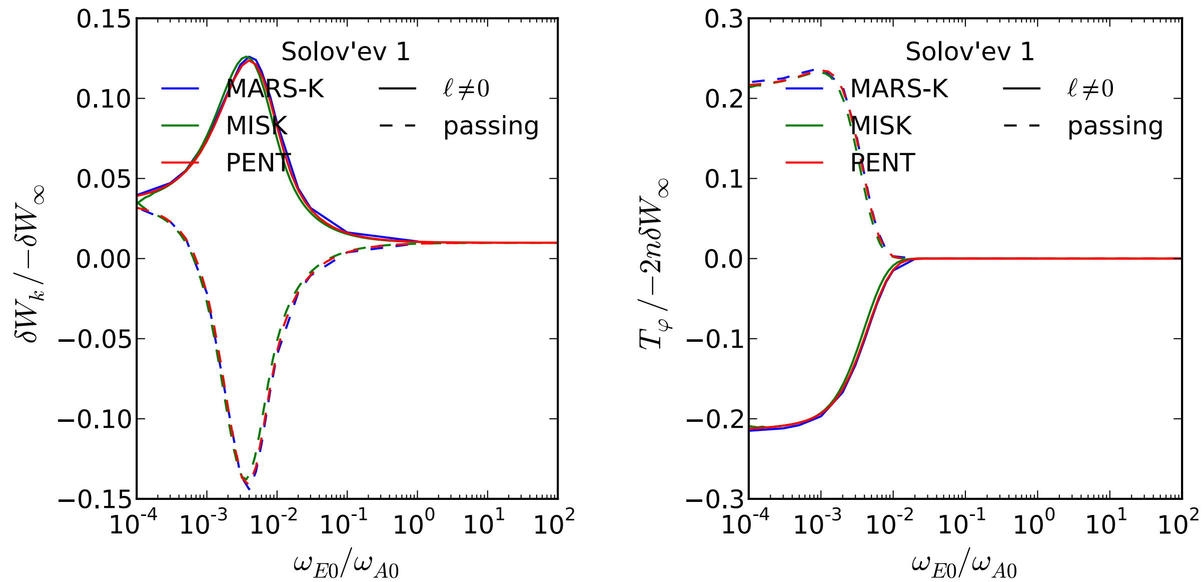

Figure 5.19: Solov'ev 1 RWM kinetic energy and NTV torque across a scan of electric precession frequency for trapped and circulating $\ell \neq 0$ thermal ions as calculated by MARS-K, MISK, and PENT.

Fig. 5.24, which shows the radial profile of the precession resonance ($\ell = 0$) and bounce resonance ($\ell \neq 0$) contributions to the perturbed kinetic energy at the highest rotation. The precession resonant contribution is in close quantitative agreement for all three codes. It shows only a thin resonance near the rational surface, which does not significantly contribute to the integrated profile. The integral result of the bounce harmonic contribution, however, is dominated by the region near the $q = 3$ rational surface. Here, we see evidence of the unregularized displacements resulting in unphysical spikes in the second order quantities at the rational surfaces. These resonances are even larger at the many resonant surfaces within the ITER equilibrium, and the level of agreement seen in the precession scan was only obtained after imposing a layer width of $\Delta q = 0.1$ around each rational surface inside of which the MARS-K and MISK profile contributions were ignored.

A number of common features are evident in the electric precession scans for each equilibria. First, the precession resonance ($\ell = 0$) becomes large at low ω_E. The exact location of the peak precession resonance is different for δW_k and T_φ due to different ω_E

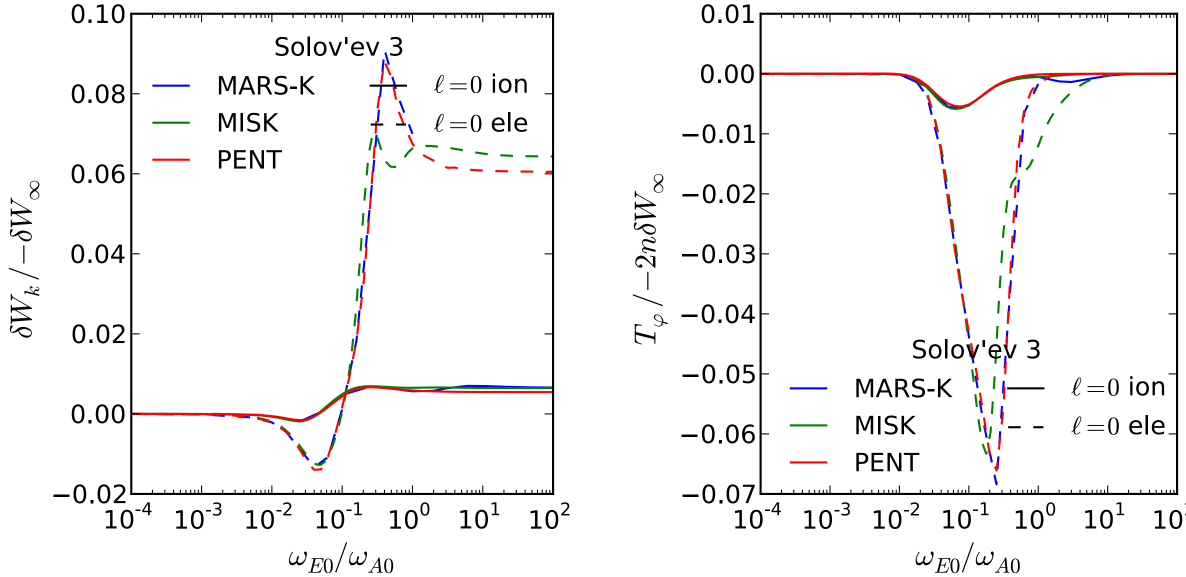

Figure 5.20: Solov'ev 3 RWM kinetic energy and NTV torque across a scan of electric precession frequency for $\ell = 0$ trapped thermal ions and electrons as calculated by MARS-K, MISK, and PENT.

dependencies in the numerator of the real and imaginary parts of $\mathcal{R}_{T\ell}$. The behavior of each is further dependent on the relative values of the diamagnetic drifts, as evidenced by the similar behavior for the Solov'ev cases and qualitatively different behavior of the ITER case at low ω_E. The bounce harmonic resonances ($\ell \neq 0$) are evidenced by NTV peaks in the relatively large ω_E range between $0.1\omega_{A0}$ and ω_{A0}. These appear to have a significant effect on passing particles as well as trapped particles. Finally, in the extreme limit at the maximum ω_E all results limit to a constant δW_k value and zero NTV torque. This is the Chew-Goldberger-Low (CGL) limit.

5.5.1.3 The CGL Limit

The numerical models in Fig. 5.18 - Fig. 5.23 approach an asymptotic limit at large ω_E known as the CGL high frequency limit. The perturbed energy can be solved directly in this limit

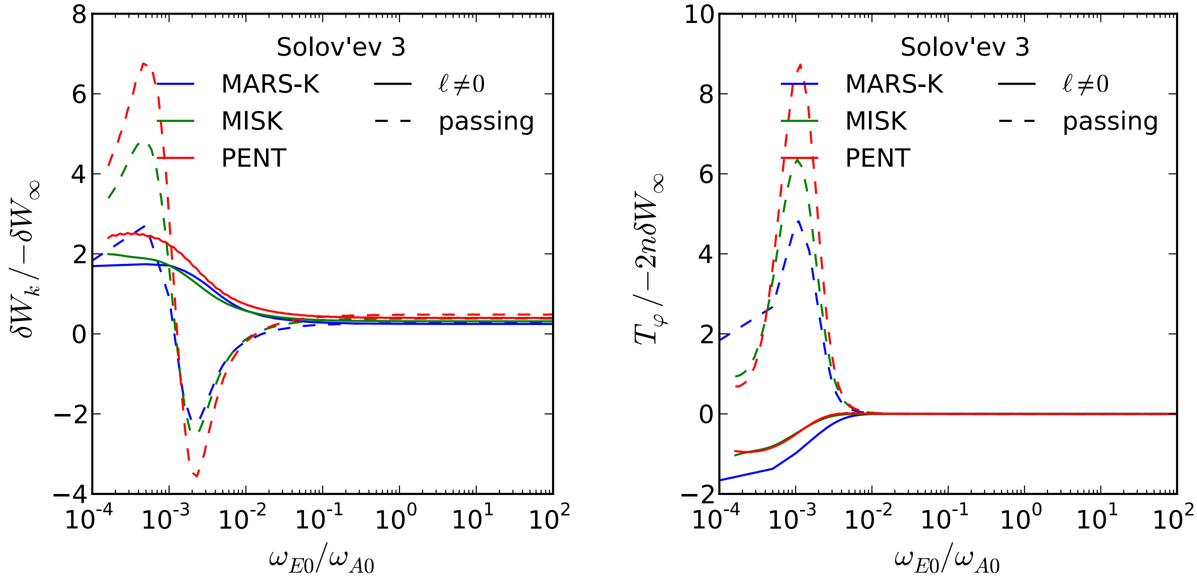

Figure 5.21: Solov'ev 3 RWM kinetic energy and NTV torque across a scan of electric precession frequency for trapped and circulating $\ell \neq 0$ thermal ions as calculated by MARS-K, MISK, and PENT.

using CGL perturbed pressures [127],

$$\delta W_{CGL} = \frac{1}{2}\int \frac{5}{3}p\left|\nabla\cdot\boldsymbol{\xi}_\perp\right|^2 dV + \frac{1}{2}\int \frac{1}{3}p\left|\nabla\cdot\boldsymbol{\xi}_\perp + 3\boldsymbol{\kappa}\cdot\boldsymbol{\xi}_\perp\right|^2 dV. \tag{5.22}$$

Here p is the total (ion and electron) scalar pressure.

This limit is an useful tool for benchmarking the perturbed action contribution to the full solution because it effectively removes the kinetic resonances from the energy integral. In the high ω_E limit, the energy integral approaches a constant,

$$I_x(\omega_E \gg \omega_b, \nu) \rightarrow \int_0^\infty x^{5/2}e^{-x}dx = \frac{15\sqrt{\pi}}{8}. \tag{5.23}$$

The PENT code has the ability to optionally substitute this explicit energy integral for the full numerical integration. This, combined with the standard numeric result for large ω_E and the direct CGL solution (5.22) provides three separate calculations of the CGL limit.

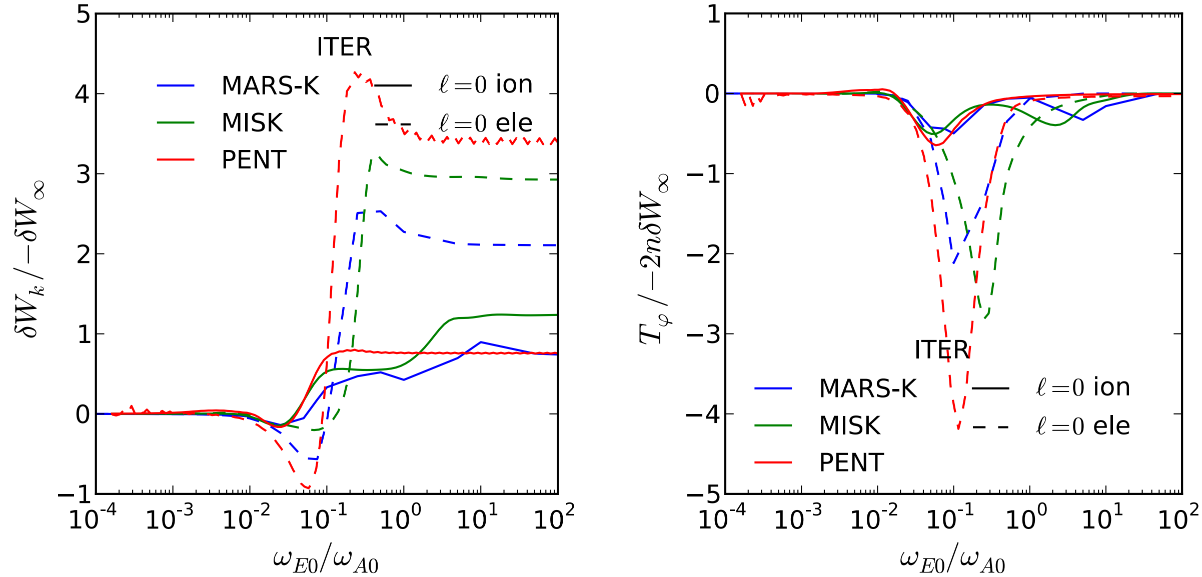

Figure 5.22: ITER RWM kinetic energy and NTV torque across a scan of electric precession frequency for $\ell = 0$ trapped thermal ions and electrons as calculated by MARS-K, MISK, and PENT.

These values are compared between MARS-K, MISK, and PENT in Tab. 5.2. The table clearly shows a large discrepancy in the Solov'ev 3 values due to the treatment of the singular surfaces, which are integrated over without any special truncation in the MARS-K and MISK calculations. The singular contributions have been removed from these two codes using a sharp Δq cut-off when integrating over the ITER profile. The discrepancy between PENT and the other codes in this case are a result of the lower $|\delta W_\infty|$ in DCON.

5.5.1.4 RWM Growth Rate and Rotation

The dispersion relation (5.1) can be solved for the RWM growth rate and rotation calculated by each code over the scan in electric precession. The complex perturbed energy used in this calculation must include all the separate contributions discussed above, including passing, trapped, ion, and electron contributions. The results are shown in Fig. 5.25 - Fig. 5.27. The intermediate rotations are the most unstable in each case. Often, there are two distinct regions destabilized by the precession resonant trapped particles (relatively low rotation)

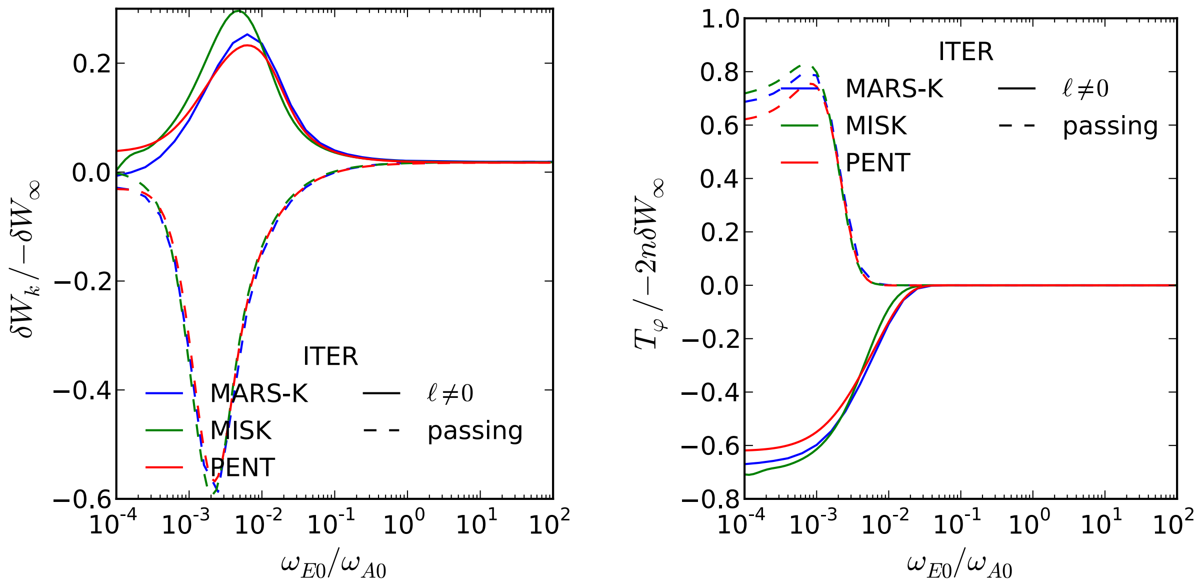

Figure 5.23: ITER RWM kinetic energy and NTV torque across a scan of electric precession frequency for trapped and circulating $\ell \neq 0$ thermal ions as calculated by MARS-K, MISK, and PENT.

and circulating particles (slightly higher rotation). These regions align roughly with regions of the fastest mode rotation. Note that the displacement of the MARS-K Solov'ev 1 growth rate as compared to the other codes is a result of the fluid energies not the kinetic calculations. A similar displacement exists in the Solov'ev 3 scan. This second case, however, also includes differences between the three codes in the kinetic contribution due to circulation and bounce harmonic ($\ell \neq 0$) ions at high values of the electric precession.

Kinetic effects are of differing levels of importance for the stability of the three equilibria. In the Solov'ev 1 equilibrium, the growth rate is never changed by more than $\sim 25\%$ from the fluid result and the rotation of the mode stays just a fraction of the wall time throughout the scan over many orders of magnitude in precession frequency. The changes in the growth rate due to kinetic resonances become comparable to the fluid growth rate itself in the Solov'ev 3 case, although the mode rotation remains small. The most dramatic deviations from fluid growth rates appear in the ITER case at the low and high precession frequency limits. The precession resonant ($\ell = 0$) trapped particles at low ω_{E0}/ω_{A0} and the circulating combined

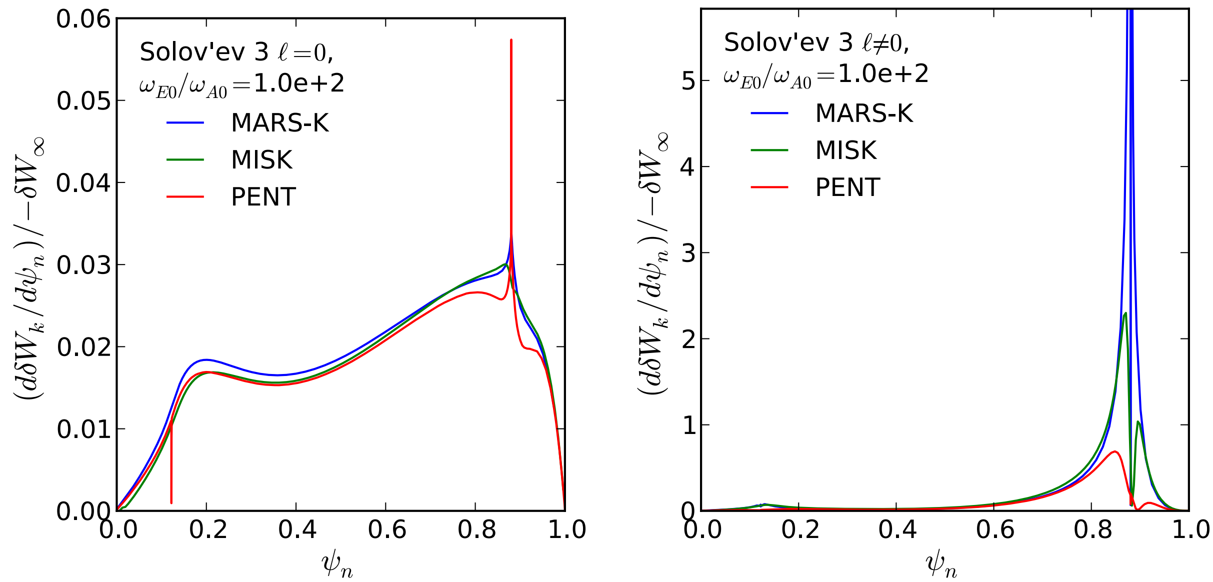

Figure 5.24: Radial profiles of the perturbed kinetic energy contribution from the precession resonant (left) and bounce resonant (right) ions in CGL limit for the Solov'ev 3 equilibrium as calculated by MARS-K, MISK, and PENT.

with bounce resonant particles at high ω_{E0}/ω_{A0} are enough to kinetically stabilize the RWM. The mode rotation frequency near the marginal stability points can become larger, surpassing the inverse wall time.

The conclusion of this parametric scan is that MARS-K, MISK, and PENT agree on the physical mechanisms that may lead to kinetic stabilization at either high or low rotation in tokamaks. There is rough quantitative agreement between the codes, the sensitivities of which have been addressed. Qualitative behavior, showing regions of high and low stability are in good agreement between the codes. The consistent result of instability at intermediate rotation surrounded by kinetic stabilization at low and high rotation agrees with experimental observations [110]. It should be noted that the results presented here are subject to the approximations outlined above (most notably zero collisionality and the absence of magnetic precession for bounce harmonic particles). While not representative of the full stability predictions for ITER, this benchmark lends the needed confidence to such predictions made by any one code in the future.

	MARS-K	MISK	PENT
Solov'ev 1	1.57×10^{-1}	1.54×10^{-1}	1.55×10^{-1}
		1.54×10^{-1}	1.52×10^{-1}
	1.57×10^{-1}	1.56×10^{-1}	1.47×10^{-1}
Solov'ev 3	1.98×10^{0}	1.09×10^{0}	6.32×10^{-1}
		1.07×10^{0}	6.20×10^{-1}
	1.84×10^{0}	1.01×10^{0}	6.36×10^{-1}
ITER	6.11×10^{0}	6.72×10^{0}	9.88×10^{0}
		6.67×10^{0}	1.01×10^{1}
	6.55×10^{0}	6.53×10^{0}	1.07×10^{1}

Table 5.2: The CGL limit $\delta W_k / -\delta W_\infty$ as calculated using the full numerical integration with $\omega_{E0}/\omega_{A0} = 100$ (top), the analytic energy integral $I_x = 15\sqrt{\pi}/8$ (middle), and the direct solution using the CGL pressures (bottom).

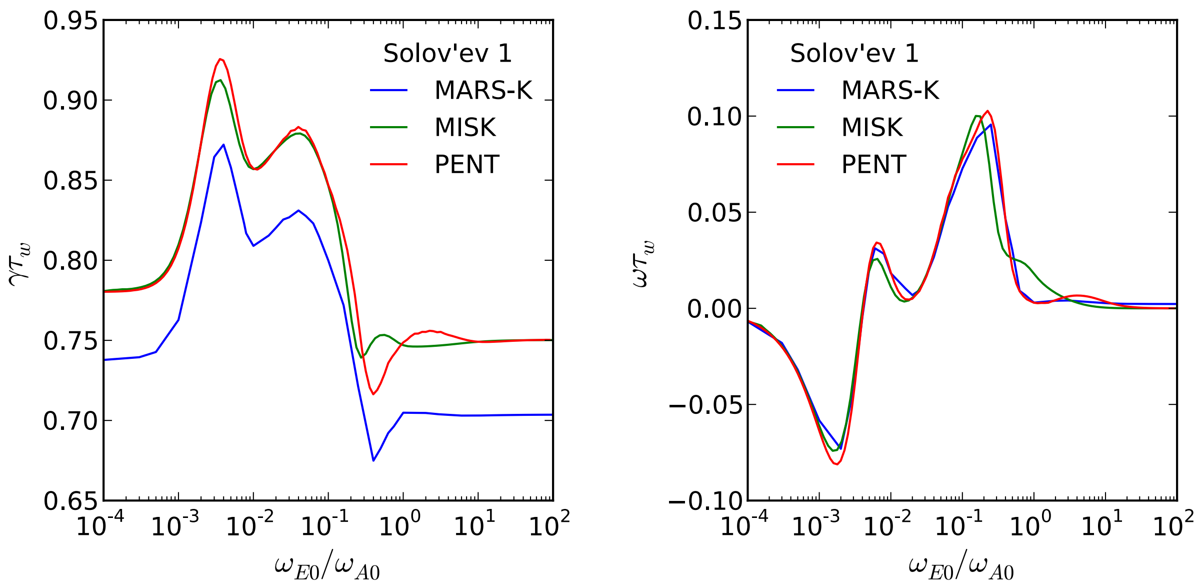

Figure 5.25: Solov'ev 1 RWM normalized growth rate and rotation across a scan of electric precession frequency as calculated by MARS-K, MISK, and PENT.

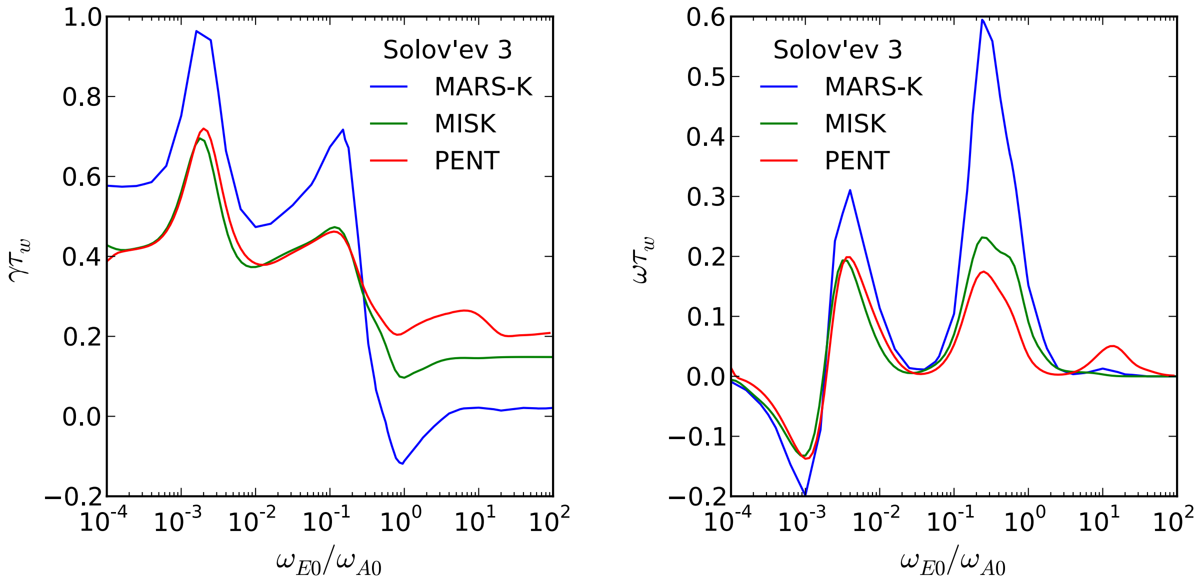

Figure 5.26: Solov'ev 3 RWM normalized growth rate and rotation across a scan of electric precession frequency as calculated by MARS-K, MISK, and PENT.

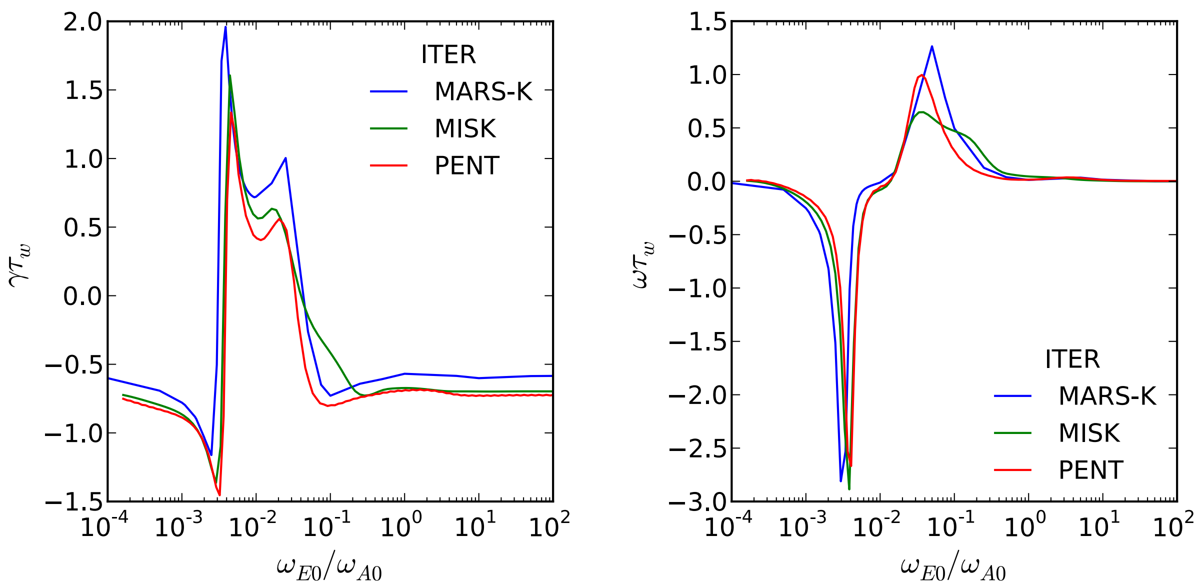

Figure 5.27: ITER RWM normalized growth rate and rotation across a scan of electric precession frequency as calculated by MARS-K, MISK, and PENT.

Chapter 6

A Single Mode Model for NTV Torque

While the previous chapter presented an extensive benchmark of the PENT model and the physics captured therein, this chapter presents a reduction of the variables used to control the application of NTV in practice. The two are not, however, incongruent. The PENT model is applied here with confidence that it is fully capturing the geometric, resonant, and nonlinear physics of the generalized combined NTV theory. Only through this knowledge can the task of reducing the interplay of these many effects to a practical tool for the application of torque be rigorously performed. Just such a reduction is what is presented here, a practical single mode model that is shown to describe the nonlinear coupling between the applied field and the applied torque in experiments.

Neoclassical toroidal viscosity, as modeled by PENT, is rigorously a nonlinear phenomenon. The NTV torque from one nonaxisymmetric field source cannot be summed together with the NTV torque from a second nonaxisymmetric field source to find the torque caused by the two sources together. This has many important consequences for the practical application of nonaxisymmetric fields for rotation control. Without further simplification, the correction of intrinsic nonaxisymmetries would require a perfect matching of the source spectrum. This would be impractical, and the infinite possibilities of source spectra would render finite error field correction (EFC) coil arrays useless. The direct application of desired torque using applied fields would contain similar difficulties, as every combination of fields and plasma equilibria would need to be modeled independently in order to gain predictive control capabilities.

The sensitivity of the plasma response to a single "dominant mode" [45, 75] spectrum can reduce the NTV spectral dependence to a single parameter quantifying the extent to which the external field overlaps with the dominant mode spectrum. This reduces the application of NTV to a single mode model, enabling efficient feed forward predictions of the integral torque resultant from any combination of external field sources.

This chapter is organized as follows. The dominant plasma mode, overlap parameter, and effective coupling of NTV torque to a single mode are introduced in Section 6.1. The power of the single mode model is then demonstrated by feed forward EFC applications in Section 6.2. These applications represent a extremely effective reduction of the NTV model in many DIII-D experimental scenarios. The residual effects are addressed in Section 6.3, motivating future improvements to the model.

6.1 NTV Coupling to a Dominant Plasma Response

Modeling in PENT and experiments in the DIII-D tokamak show that the integral NTV torque can be dominated by the coupling of applied fields to a single spectrum. This spectrum, deemed the dominant mode for the NTV torque, is similar to the dominant mode for the resonant plasma response found in the Ideal Perturbed Equilibrium Code (IPEC) but not identical. This section presents the experimental verification of the coupling predicted in PENT, the validation of which allows powerful linearization of the NTV torque applied by external coils.

6.1.1 Experimental Configuration

The coupling of the NTV torque to a single dominant mode was measured in a Quiescent H-mode (QH-Mode) scenario in the DIII-D tokamak. The QH-mode is characterized by the high confinement of H-mode, an absence of edge localized modes (ELMs), and the presence of a benign edge harmonic oscillation (EHO) [128, 129]. The QH-mode is dependent on rotation shear in the edge but independent of its sign, allowing for either co-current or counter-current rotation [130, 115, 28]. Counter-current rotation was chosen for this work, taking advantage of the observation that nonaxisymmetric fields drive QH-Mode plasmas in DIII-D toward a counter-current neoclassical offset rotation [115, 28]. This clearly distinguishes the NTV momentum source from possibly competing terms in the momentum balance such as resonant braking or prompt fast ion loss, which are both momentum sinks independent of the sign of the rotation.

Neutral beam injection (NBI) torque is used to produce strong initial rotation in the diamagnetic, co-current direction. When the amount of NBI is reduced such that the angular momentum evolves on a time scale that is slow compared to the momentum confinement time τ_ϕ, the plasma remains in torque balance. In an axisymmetric plasma,

$$\frac{dL_\phi}{dt} \approx 0 = T_{NBI}^{sym} + T_{INT} - L_\phi/\tau_\phi. \tag{6.1}$$

The injected torque T_{NBI}^{sym} balances the intrinsic T_{INT} and viscous torque $T_{VISC} = -L_\phi/\tau_\phi$. In the presence of nonaxisymmetric fields, additional terms corresponding to the NTV torque T_{NTV}, prompt ion loss, and resonant braking may appear in the right hand side of Eq. (6.1). Assuming that the NTV is the dominant term, the evolution of a QH-Mode during a slow

NBI ramp in the presence of applied nonaxisymmetric fields is described by,

$$\frac{dL_\phi}{dt} \approx 0 = T_{NTV} + T_{NBI} + T_{INT} - L_\phi/\tau_\phi,$$
$$T_{NTV}\left(L_\phi\right) = -\left[T_{NBI}\left(L_\phi\right) - T_{NBI}^{sym}\left(L_\phi\right)\right]. \tag{6.2}$$

Here, we have assumed the momentum confinement time and intrinsic torque are unchanged by the small nonaxisymmetry.

Equation (6.2) describes the NTV torque as the difference in NBI torque needed to obtain a given angular momentum with and without nonaxisymmetric fields. This is determined experimentally by comparing successive discharges with applied nonaxisymmetries to a reference scenario with no applied nonaxisymmetric fields. A comparison of two such discharges is given in Fig. 6.1. Large amounts of NBI torque are used to produce fast rotation and β_N of approximately 2. At 2250ms the NBI torque is slowly ramped toward zero, the rotation evolves in torque balance until reaching a bifurcation, characterized by a sudden locking to the tokamak frame, a disruption, and the termination of the discharge. The presence of nonaxisymmetric fields, indicated here as the amplitude of currents in the lower I-coil array, introduces a NTV torque that accelerates the plasma and extends the discharge.

The DIII-D tokamak is outfitted with 3 nonaxisymmetric coil arrays of 6 discrete coils each (see Fig. 2.2). This enables application of clean toroidal harmonics of $n \leq 3$, and rotation of the $n \leq 2$ harmonics. For the purposes described here, the C-coil array (external to the vessel) is used for correction of the intrinsic $n \leq 2$ error field. The upper I-coil (IU) and lower I-coil (IL) arrays (internal to the vessel) are used to apply the maximum additional $n = 2$ perturbation, corresponding to a peak current amplitude of 4.3kA as shown in Fig. 6.1. The poloidal spectrum of the applied nonaxisymmetric field is thus reduced to a function of the relative toroidal phase of currents in the two I-coil arrays.

Fig. 6.2 shows the variation in poloidal spectra applied to the plasma surface obtained by adjusting the phasing between the IL and IU arrays, $(\phi_{IL} - \phi_{IU})$. Experimentally, these

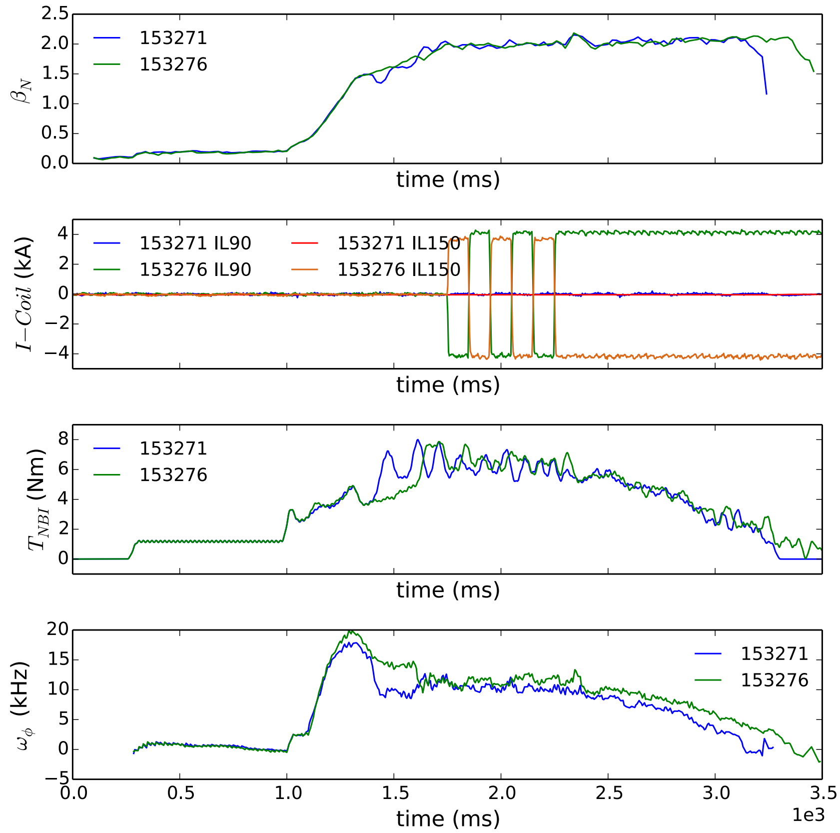

Figure 6.1: Evolution of a baseline QH-mode shot (153271) with no applied I-coil field and a phase scan shot (153276) with 4.3 kA $n = 2$ fields with the lower array (IL) oriented at 330°. In each case the β_N (top) evolves to approximately 2. The I-coil currents (second from top) are turned on, flipped 180° 5 times for diagnostic proposes, and then held constant from 2250ms as the neutral beam torque (second from bottom) is ramped toward 0. Shots with applied I-coil fields maintain faster rotation (bottom) as the injected torque is removed.

spectra were obtained by changes to the IL phase ϕ_{IL} with a constant IU phase $\phi_{IU} = 90°$. There is a sharp peak at $m = -1$ to 1 in all cases. The secondary peak in the positive m spectra (right-handed pitch, like the field lines) shifts from $m = 8$ to 16. Also shown, is the IPEC calculated "dominant mode" spectrum, which is discussed in detail in Sec. 6.1.3 and peaks at $m = 15$ in this equilibrium. The lower I-coil phases 30° (blue) and 90° (green) couple best to the dominant mode, while the cases with higher I-coil phases 150° (red) and 210° (orange) have nulls in key dominant mode harmonics.

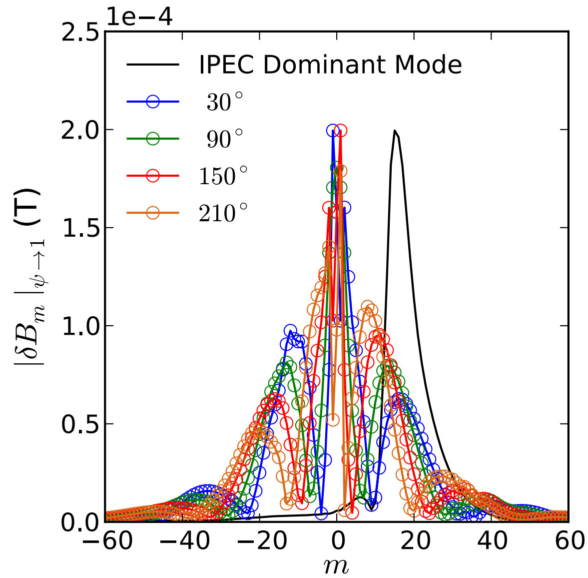

Figure 6.2: The IPEC dominant mode and applied poloidal spectra in PEST coordinates on the plasma surface for various $n = 2$ I-coil phasings. Both upper and lower I-coils have 4.3kA $n = 2$ current waveforms, and the upper I-coil phase is fixed at 90°.

6.1.2 Coupling Model

The PENT code was used to model the dependence of the NTV torque in these QH-mode plasmas on the applied poloidal spectrum. Equilibria properties used in the model were taken from the middle of the torque ramp, corresponding to a period of significant NTV torque in the presence of those nonaxisymmetric spectra for which large torques were observed. The QH-mode kinetic profiles during this period were characterized by strong rotation, with electric precession frequencies approaching the typical bounce frequency of trapped particles and much larger than the magnetic precession. The large aspect ratio approximations of these frequencies, and the effective Krook collision operator used in PENT are shown in Fig. 6.3. The kinetic precession resonance term is much larger than the collisionality, placing the majority of the plasma in the $\nu - \sqrt{\nu}$ regime. The integral torque, also shown in Fig. 6.3, is dominated by large NTV in the edge region. This region contains the highest perturbed fields and largest displacements in the perturbed equilibrium. The sharp spikes in the NTV, however, are due mainly to $\ell = \pm 1$ bounce harmonic resonances. This emphasizes the importance of these resonances, as well as the inaccuracy of the RLAR approximations that show no edge resonance for these species.

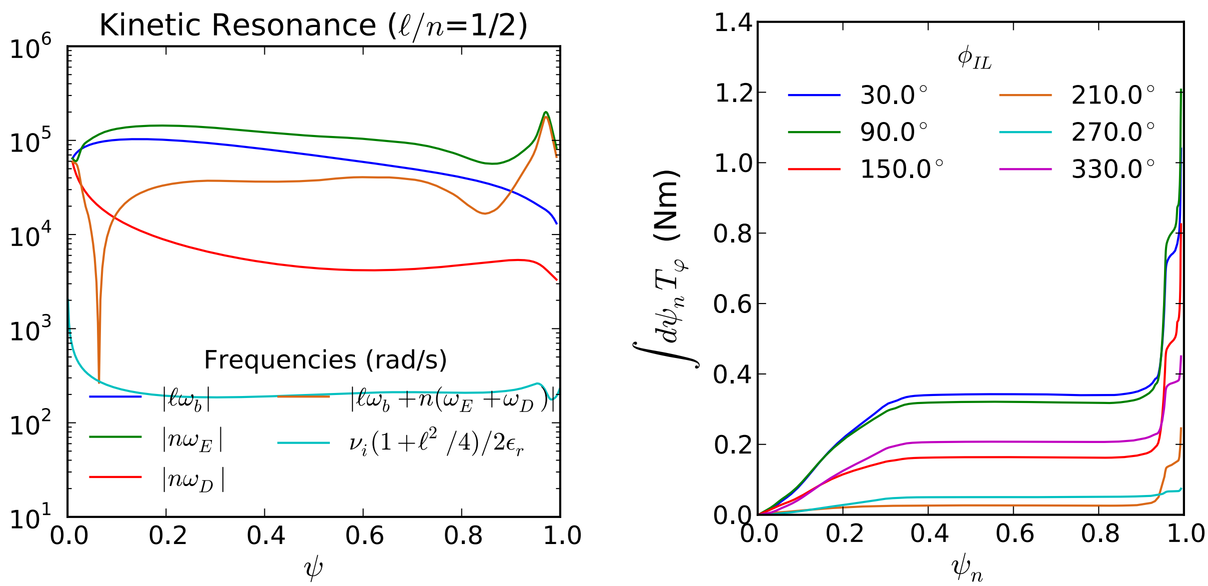

Figure 6.3: Left: The reduced large aspect ratio approximations of the bounce frequency (blue), electric precession (green), magnetic precession (red), and sum kinetic resonance term (orange) compared to the collisionality (cyan) profile in normalized flux for shot 153276. Right: The modeled torque density profile calculated in this equilibrium by PENT for 6 phases of the lower I-coil ϕ_{IL}.

The integral torque modeled by the PENT code has a sinusoidal dependence on the phasing between IU and IL arrays, shown in Fig. 6.4. The phasing is described here in terms of the IL phase only, fixing $\phi_{IU} = 90°$ as done in the experiment. Constant 4.3kA $n = 2$ currents are distributed to both arrays for each phasing. The quality of the fit is clear from Fig. 6.4, which includes a shaded band corresponding to the uncertainty of the fit function. Immediately, the sinusoidal dependence shows that maximum $n = 2$ torque can be obtained in these plasmas using 78° phasing, while the spectrum produced by 258° phasing has minimal coupling to the torque. The difference in extrema shows the importance of the poloidal spectra, as the applied torque can vary by an order of magnitude.

The one dimensional phasing scan can be mapped to a contour in the two dimensional phase space of the complex lower I-coil current. As the scan was done at constant IL am-

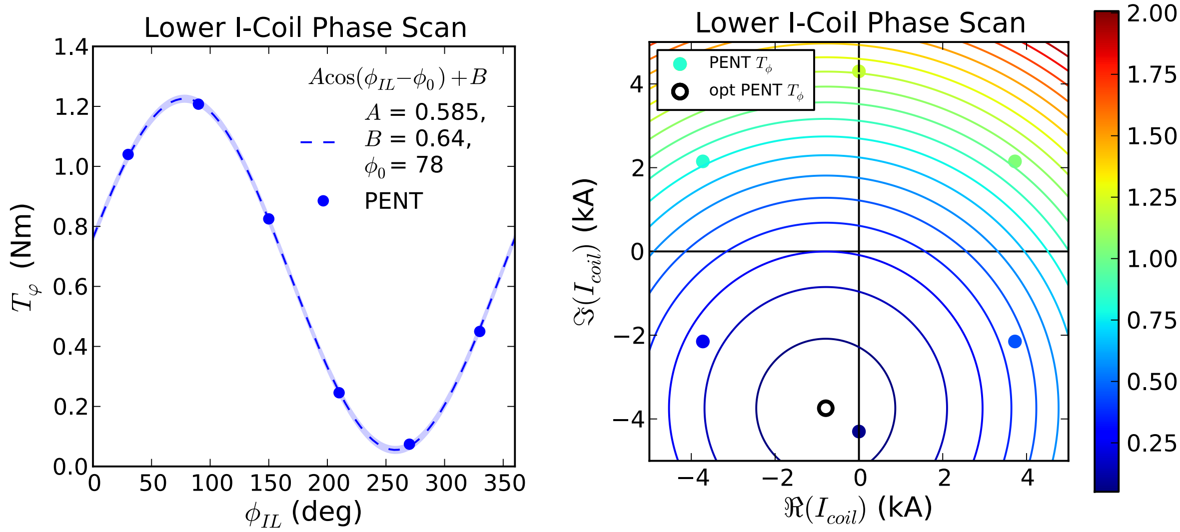

Figure 6.4: Left: The PENT code modeled torque (circles) fit a sinusoid (dashed line and shaded error) with respect to phasing. These models include poloidal mode coupling, general geometric shaping and aspect ration, intermediate regimes and bounce harmonic resonances Right: The modeled torques, indicated by color, fit to circular contours (solid lines) in the complex coil current phase space defined by the real and imaginary components of the applied $n = 2$ lower I-coil waveform. The center of these contours is indicated as a black circle. Note that the constant upper I-coil field is a hidden variable in these figures.

plitude, the modeled points lie on a circular contour in this phase space and the sinusoidal dependence of the torque in ϕ_{IL} corresponds to the intersection of this contour with circular contours of constant torque quadratically approaching their centroid extremum.

Fig. 6.4 shows IL phase space contours of the torque described by,

$$T_\varphi = \mathcal{A}\left|\mathbf{I}-\mathbf{I}_0\right|^2 + \mathcal{B} \tag{6.3}$$

The scan defined by a constant amplitude I and varied phase ϕ is described by the family of phase space vectors, $\mathbf{I} = Ie^{in\phi}$, while the extremum is given by constant amplitude and phase, $\mathbf{I}_0 = I_0 e^{in\phi_0}$. Inserting these into Eq. (6.3), the dependence is reduced to the simple sinusoid,

$$T_\varphi = Ae^{in(\phi-\phi_0)} + B, \tag{6.4}$$

with amplitude $A = -2\mathcal{A}II_0$, phase $n\phi_0$ and offset $B = \mathcal{A}(I^2 + I_0^2) + \mathcal{B}$ also shown.

6.1.3 Relation to Resonant Coupling

IPEC calculates the linear plasma response to external nonaxisymmetric fields, allowing superposition of the results. Extensive sensitivity studies have been performed using the IPEC model for $n = 1$ resonant locking thresholds [50, 75, 29]. Although the ideal model does not capture the field penetration physics, the resonant component suppressed by shielding currents prior to penetration is readily obtainable. The linear coupling between resonant components and spectra applied to a control surface (taken to be the plasma surface) in IPEC can be expressed by a simple matrix equation,

$$\mathbf{\Phi}_r = \mathbf{C} \cdot \mathbf{\Phi}_x, \tag{6.5}$$

where $\mathbf{\Phi}_r$ is a r-row vector containing the area normalized resonant field shielded by surface currents at each of the r resonant surfaces in the plasma, $\mathbf{\Phi}_x$ is a m-row vector representing the m applied spectral components on the control surface given by Eq. (6.6), and elements of the coupling matrix $\mathbf{C}_{i,j}$ are the linear coefficients relating poloidal spectrum to the resultant resonant components. Here, the resonant components contains both the applied vacuum field and the plasma response.

Fourier decomposition of the area weighted spectrum is defined as,

$$\Phi_{x,m,n} = \frac{1}{A} \oint_S d\varphi d\vartheta \mathcal{J} \left|\nabla\psi\right| \left(\delta\mathbf{B}_x \cdot \hat{n}\right) e^{i(m\vartheta - n\varphi)}, \tag{6.6}$$

where S is a flux surface with area A and unit normal $\hat{n}$, $\delta\mathbf{B}_x$ is the externally applied field. The spectral components $\Phi_{x,m,n}$ have units of flux and are independent of the choice of magnetic coordinates $(\psi, \vartheta, \varphi)$ corresponding to the Jacobian $\mathcal{J}$.

Singular value decomposition of the coupling matrix $\mathbf{C}$ gives eigenvectors $\mathbf{\Phi}_i$ and eigenvalues λ_i of $\mathbf{C}^\dagger\mathbf{C}$, ranked by the energy content of the resonant response to the applied spectrum. The first eigenvalue is regularly an order of magnitude above the next highest [13, 45], indicating the dominance of the corresponding eigenspectrum $\mathbf{\Phi}_1$. The magnitude of the resonant response is then well defined by the extent to which an applied spectrum overlaps with the dominant mode. That is, the resonant response is proportional to the overlap parameter,

$$\mathcal{C}_i \equiv \sqrt{\int d\varphi' \left[\int da \left(\delta\mathbf{B}_x \cdot \hat{n}\right)\left(\delta\mathbf{B}_i \cdot \hat{n}\right)\right]^2} = \left|\mathbf{\Phi}_x \cdot \mathbf{\Phi}_i\right| . \tag{6.7}$$

Note that the same parameter is sometime used as a proxy for coupling to the plasma Kink mode, a global MHD instability. This approximation is valid in the limit of high β (the ratio of plasma pressure to magnetic pressure) and strong shaping, in which the plasma response is dominated by the least stable (Kink) mode. In this limit, the plasma response is large and dominates over the applied vacuum perturbation. Extensive work has been done showing that the overlap is an accurate metric for error field penetration thresholds and extending the technique to future devices error field correction [75, 13, 45, 54].

The validity of the single mode model is confirmed for the QH-mode plasmas used here, by a ratio of the first eigenvalues $\lambda_1/\lambda_2 = 41.0$. Fig. 6.5 shows the expected sinusoidal dependence on the phasing and corresponding mapping of the dominant mode to the phase space of the lower I-coil array. The phase dependence agrees with that of the NTV extremum. The extremum I-coil current amplitude is slightly larger. However, the quadratic nature of the torque in coil-space has produced a shallow and broad minimum near the single mode optima. Calculated in PENT, the difference between the NTV at the two optima is 1% of the phase scan sinusoidal amplitude.

The quality of the overlap as a metric parameterizing the nonresonant torque is apparent in the quality of a one dimensional fit across the phasing scan shown in Fig. 6.6. the nonlinear torque is well correlated with the square of the linear metric in these QH-Mode plasmas.

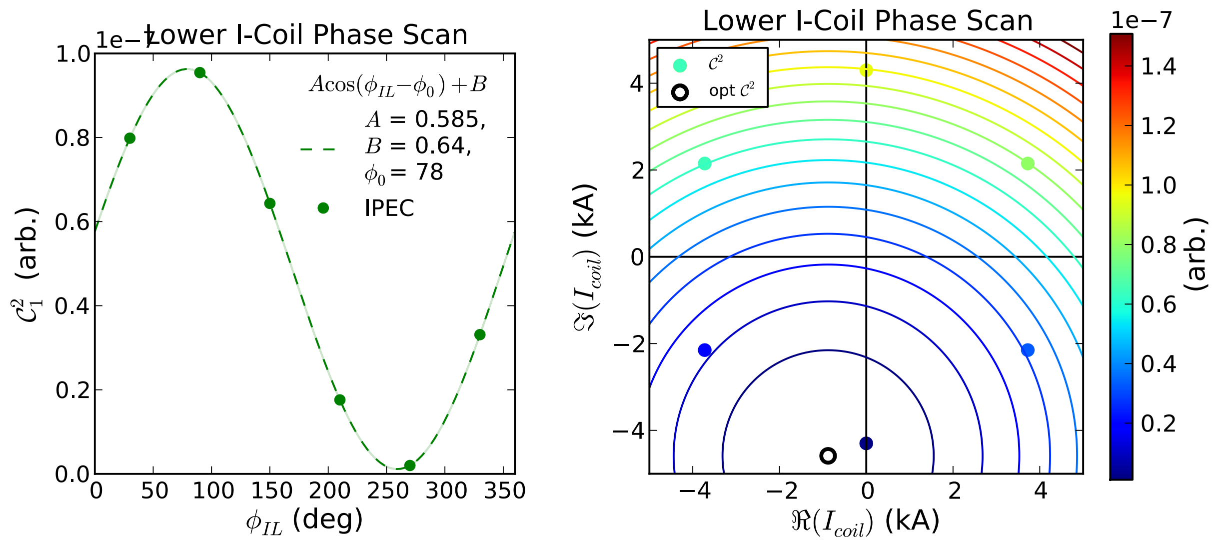

Figure 6.5: Left: The IPEC overlap (green circles) fit to a sinusoid (dashed line and shaded error) with respect to phasing. Left: The overlap, indicated by color, is described by circular contours in the lower I-coil phase space. The center of these contours (black circle) corresponds to the minimum coupling of the applied field to the IPEC dominant mode.

This is expected, as the high β_N results in perturbations dominated by the plasma response. The plasma response being proportional to the overlap, the torque is then proportional to the square of the overlap through a simplification of the variation in the action.

6.1.4 Experimental Measurement of the Coupling

The predicted NTV dependence on the poloidal field spectrum was confirmed by the I-coil phasing scan done in DIII-D. Fig. 6.7 shows a typical angular momentum L_φ profile evolution for the experimental discharges. Consistent with the PENT prediction, the applied fields prop up edge L_φ while the decreasing NBI torque brings down the core rotation. The reference shot profiles, in contrast, scale more regularly and self-similarly in time. This confirms the NTV torque profile is concentrated in the plasma edge. The concentration of the profile also enables the reduction of the NTV rotation dependence to a single variable, the edge rotation. Edge rotation, $\omega_\varphi(\psi_n = 0.8)$, is thus a good space in which to compare torque trajectories between multiple discharges. In this experiment, the rotation is obtained from the velocity of

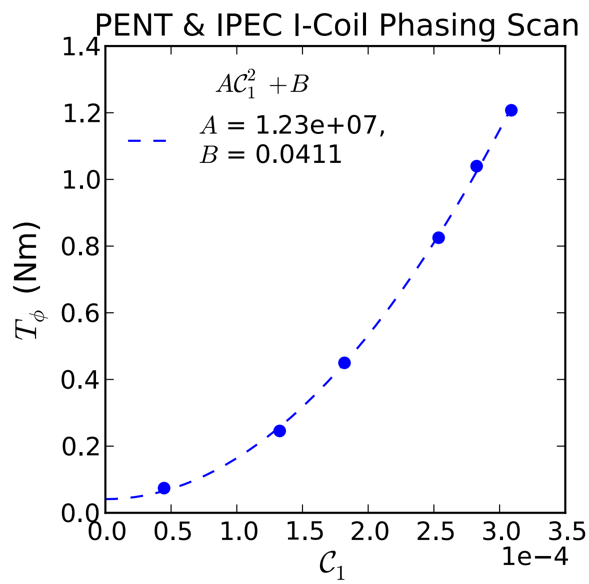

Figure 6.6: The nonlinear torque calculated using the PENT code (circles) fit to a quadratic dependence (dashed line) on the IPEC overlap metric for the six point phasing scan of the I-coils. metric.

the minority impurity C^{6+} measured by charge exchange recombination (CER) spectroscopy [131] assumed to be proportional to that of the majority plasma ion species D^+.

The difference between the injected neutral beam torque required for equilibrium at a given rotation with and without applied nonaxisymmetric fields gives a direct measurement of the integral NTV torque according to Eq. (6.2). Fig. 6.8 shows the evolution of the NTV torque measured in this way as a function of the CER edge rotation. Each curve is a third degree polynomial fit to points corresponding to the reconstructed profile times shown in Fig. 6.7, and the shaded error bars indicate propagation of errors determined by the fit accuracy. The temporal evolution of individual points is indicated in the reference shot, giving a sense of the scatter associated with the measurements.

The phasing dependence of the measured NTV torque is shown in Fig. 6.9. The torque is taken from the difference in NBI torques at the edge rotation $\omega_\varphi(0.8) = 3 \times 10^4$krad/s, corresponding to the time of the kinetic EFIT equilibrium and profiles used in the PENT model. The NTV measurements show a clear sinusoidal dependence on the poloidal spectrum,

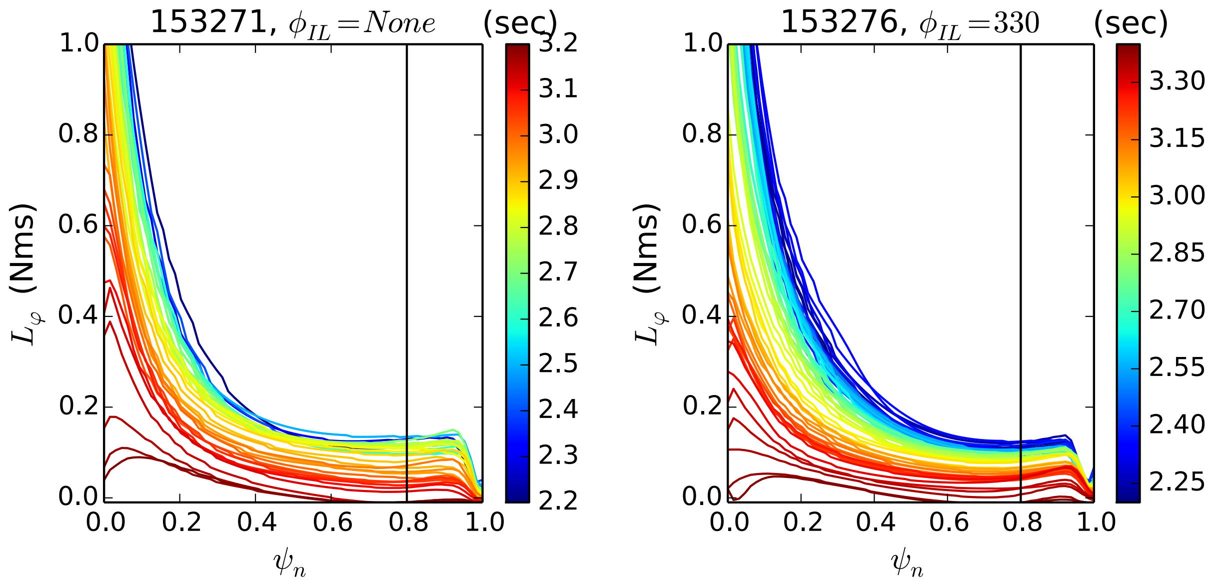

Figure 6.7: The angular momentum profile evolution of reference shot 153271 (left) and shot 153276 with applied I-coil fields (right). The profiles in normalized flux have been calculated separately at many times, indicated by color, during the NBI torque ramp-downs in each shot. The edge location $\psi_n = 0.8$ is indicated by a vertical line, as it is used to define the NTV rotation space dependence.

in good agreement with the PENT model amplitude and phase. Both the full nonlinear PENT predication and the linear IPEC dominant mode prediction of the maximum NTV coupling are within the error of the fit phase $\phi_0 = 65° \pm 18°$. The corresponding single-mode minima are clustered in the lower I-coil phase space. Fig. 6.10 shows the single mode phase space map produced using the experimental measurement, as well as the minima fit using the PENT prediction and dominant mode proxy. Both model minima lie within a shallow bowl of the phase space map fit to the experimental data. The map extrapolates the NTV at these other optima to be 0.11Nm and 0.18Nm respectively, each differing from the 0.07Nm experimental minimum by less than 8% of the 1.32Nm peak in the sinusoidal fit to the experimental torque. This difference between the extrapolated NTV at each model minimum is within the error bar on any given torque measurement ($\sim$ 0.4Nm), making either a sufficient choice for minimal rotation drive within the accuracy of this experiment.

An unanticipated observation from the experiment was a general decrease in the NTV

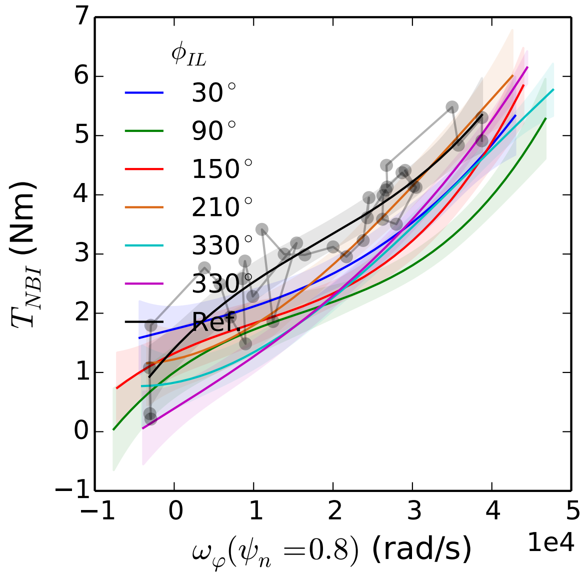

Figure 6.8: The neutral beam torque trajectories in rotation space of a reference discharge with no applied I-coil fields and discharges with 4.3kA $n = 2$ I-coil waveforms defined by the lower I-coil phase ϕ_{IL}. Trajectories (solid lines) are third order polynomial fits, with error bars indicated by the shaded regions of similar color. Individual measurements are shown connected in temporal order for the reference discharge (grey points and line), and are representative of the experimental scatter.

torque as the edge rotation approaches zero. Previous results in DIII-D QH-mode plasmas have observed a NTV peak between the neoclassical offset and zero rotation [100, 98], and bifurcation of the torque balance was expected at maximum measured NTV. A possible reason for this discrepancy is the assumption in Eq. (6.2) that NTV is the dominant torque introduced by the application of nonaxisymmetric fields. The applied error field would be expected to interact strongly with islands open on the rational surfaces, the existence of which depends on resistive physics not captured in the perturbations calculated by IPEC. Nonaxisymmetries may also induce prompt beam ion loss not accounted for in the axisymmetric approximation of the injected torque [132, 133]. Both of these effects damp rotation. They would appear in the analysis above as rightward shifts of the $T_{NBI}(\omega_\varphi)$ curves, and thus a reduction of the difference between any given case and the reference. The positive difference measured for the majority of the rotation space confirms that the NTV, the only

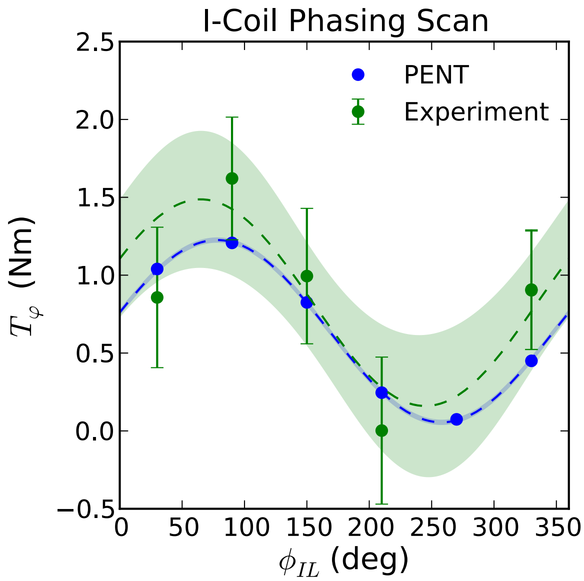

Figure 6.9: The PENT prediction of the torque (blue) compared to the experimental values (green) as a function of lower I-coil phase. The dashed lines and shaded regions of similar color indicate least-squares sinusoidal fits and their error.

nonaxisymmetry term driving rotation in the co-current direction, is the largest term in the torque balance but does not demand that the others be negligible. The additional physics that appears near zero rotation is beyond the scope of this analysis. Concentrating on the period in the discharge in which NTV is dominant, the experimental data confirms the validity of both the PENT model and it's single mode coupling dependence.

6.1.5 Significance for Predictive Capability

The circular IL phase space torque contours given by Eq. (6.3) with negligible offset specify that the torque is determined by the coupling to a single poloidal spectrum, minimized by the relative combination of I-coils corresponding to the contour centroid $\mathbf{I}_0$. This is a single mode model, and represents a major simplification of the analysis necessary to compute the magnetic torque. The NTV torque is, from first principles, nonlinear with respect to the magnetic spectrum due to poloidal mode coupling that occurs in the squared variation of the action. The torque from multiple sources cannot be linearly summed, and complete profile computations must be performed for every new combination of sources. The single mode model makes it possible to map the torque throughout the phase space of multiple sources using a finite number of computations. For the experiment described here, a minimum of 4 independent IL current distributions are enough to solve for the variables $(\mathcal{A}, \mathcal{B}, I_0, \phi_0)$ and map the torque across the entire space of possible IL-IU current combinations.

This extension, from modeling singular points in an infinite space to efficiently mapping the entire space, empowers NTV models to execute feedforward optimizations of the non-axisymmetric field sources for desired physics applications such as error field correction and rotation control. These applications are the subject of the following section.

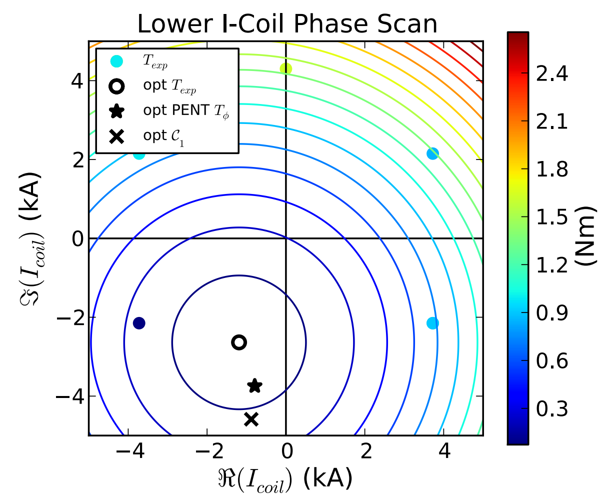

Figure 6.10: The phase space map determined from a single mode model fit to experimental measurements of the NTV torque. The torque is indicated by color for the five experimental points in the lower I-coil phase space. Solid lines indicate contours of constant torque determined by a least-squares fit to Eq. (6.3). The fit minimum (black circle) is similar to minima from single mode fits of the torque predictions from PENT (black star) and IPEC dominant mode overlap (black X).

6.2 Feed Forward NTV Error Field Correction

The previous section introduced the reduction of NTV application to a single mode model in the context of optimal rotation drive perturbing an originally axisymmetric equilibrium rotating in the ion diamagnetic direction. It is more common, however, that the DIII-D tokamak is operated with plasmas rotating opposite the diamagnetic direction. Any intrinsic nonaxisymmetric perturbations in the machine then cause rotation braking along with the associated deterioration of the stability and confinement. The field sources of interest are thus the intrinsic error field and the EFC coil arrays. The optimal operating point is defined as the combination that minimizes the NTV torque. This section will apply the single mode model to these sources and validate the result by comparing the predicted and experimental optima.

The reduction of NTV to a single mode model enables the extension of well established linear EFC optimization techniques [72, 44, 55, 13, 54, 52] to regimes where the resonant physics is no longer the dominant concern. The results presented here use the NTV single mode model to efficiently map the induced torque as a function of multiple nonaxisymmetric field sources in regimes where the NTV is expected to be the dominant effect of the error field. In DIII-D, this includes fast rotating H-mode plasmas and spectra with toroidal mode number greater than 1.

6.2.1 $n = 1$ Error Field Correction

The correction of intrinsic, as well as proxy, error fields with toroidal mode number $n = 1$ has been extensively studied in the DIII-D tokamak [54, 52, 14]. It was not until recently, however, that the importance of the poloidal spectrum in the single mode model was directly tested [29]. These experiments tested the single mode model in both ohmic and H-mode plasmas, finding good agreement between theory and the experimental optimum in each case. The ohmic scenarios explicitly studied resonant physics, using the locked mode density threshold "compass scans" [134, 54, 14]. The H-mode scenarios, however, were fit using the single mode model established in Section 6.1.

The H-mode plasmas in this work were ITER-like, lower single null plasmas with high rotation in the plasma current direction [29]. These scenarios operated with $\beta_N \approx 2.1$. The injected and intrinsic torques, $T_{NBI} \approx 3.5$Nm and $T_{INT} \approx 1.3$Nm, drove rotation levels above 100krad/s in the core in the unperturbed equilibria. The plasma current of 1.2MA and toroidal field of -1.8T resulted in left handed plasmas with $q_{95} \approx 4.1$. The resulting neoclassical offset is in the negative (diamagnetic) direction, causing the NTV to be observed as rotation damping.

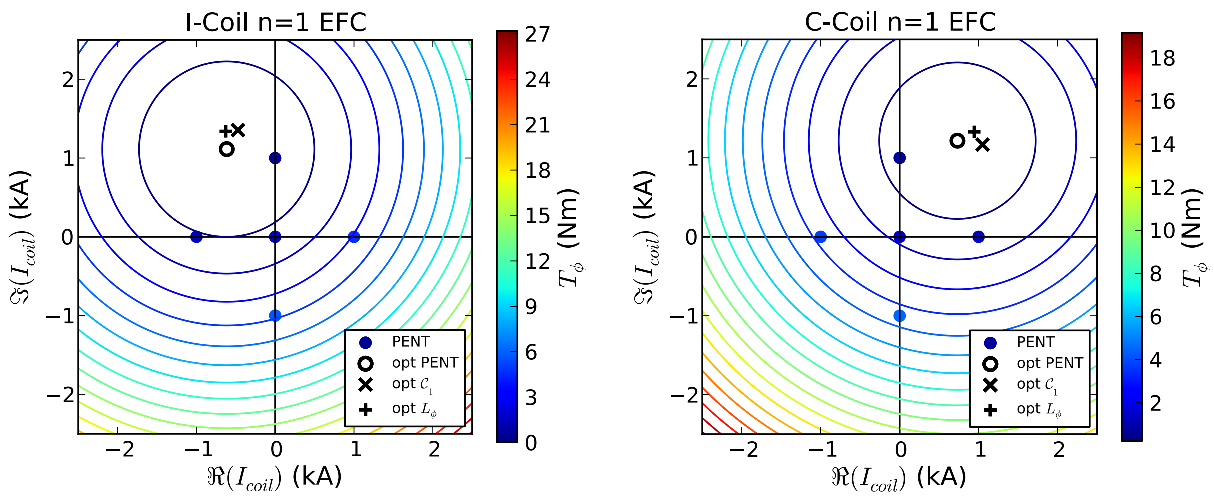

Figure 6.11: Phase space contours of constant torque determined by a least-squares fit to Eq. (6.3) of the torque calculated by PENT (colored circles) for 5 $n = 1$ I-coil EFC currents (left) and 5 $n = 1$ C-coil EFC currents (right). The fit minima (black circle) are similar to minima from single mode fits of the IPEC dominant mode overlap (black X) and measured angular momentum (black +). The intrinsic $n = 1$ error field is implicit in these calculations, applied in each perturbed equilibrium calculation.

The $n = 1$ intrinsic error field in DIII-D has been well diagnosed by in-vessel measurements of the vacuum field due to poloidal field coil tilts and shifts [135, 136]. In IPEC, the intrinsic EF spectrum on a plasma control surface can be input from the SURFMN vacuum model of these asymmetries [137].

In the experiment, correction of this intrinsic error field is defined as the minimization of the angular momentum reduction attributed to the nonaxisymmetric fields. The optimum in amplitude and toroidal phase was found separately for multiple EFC coil sets. The external C-coil array was used to produce a predominantly low m spectrum at the control surface, nearly symmetric in its handedness. The internal I-coils were hardwired with 240° degree phasing, which was predicted to have strong coupling to the IPEC dominant mode for this scenario. For each of these cases, the available phase space was scanned by slowly varying the amplitude and toroidal phase of the applied fields. The measured angular momentum was then fit to a single mode model,

$$L_\phi = \mathfrak{A}\left(\mathbf{I} - \mathbf{I}_0\right)^2 + \mathfrak{B}. \tag{6.8}$$

Here the amplitude and angle of the phase space vector $\mathbf{I}$ defines the current and toroidal phase of the applied EFC coil current distribution.

I-Coil n=1 EFC			C-Coil n=1 EFC		
Metric	$\lvert\mathbf{I}_0\rvert$ (kA)	$\angle\mathbf{I}_0$ (deg)	Metric	$\lvert\mathbf{I}_0\rvert$ (kA)	$\angle\mathbf{I}_0$ (deg)
L_ϕ	1.67	113	L_ϕ	1.89	47
T_φ	1.27	119	T_φ	1.42	59
$\mathcal{C}^2$	1.15	115	$\mathcal{C}^2$	1.26	51

Table 6.1: Comparison of the EFC optima as calculated by single mode model phase space fits to the angular momentum, the NTV torque, and the square of the IPEC dominant mode overlap.

The form of the angular momentum given by Eq. (6.8) can be derived from the torque balance,

$$\frac{dL_\varphi}{dt} = T_{VISC} + T_{EF} + T_{NBI} + T_{INT}, \tag{6.9}$$

perturbed about a slowly evolving equilibrium with constant beam and intrinsic torque,

$$0 = -\delta L_\varphi/\tau_\varphi + T_{EF}. \tag{6.10}$$

Here we have approximated the viscous torque as the angular momentum divided by a (constant) momentum confinement time $T_{VISC} = -L_\phi/\tau_\phi$ and defined the angular momentum in terms of a perturbation about the unperturbed value $L_\phi = L_{\phi,0} + \delta L_\phi$. Coupling to a single dominant mode in the toroidal phase space used here is conceptually identical to the coupling discussed in the previous section. Inserting the model expression for NTV torque (6.3) in as the error field induced torque T_{EF} gives,

$$L_\varphi = -\tau_\varphi \mathcal{A} \left(\mathbf{I} - \mathbf{I}_0\right)^2 - \tau_\phi \mathcal{B} + L_{\varphi,0}, \tag{6.11}$$

which is Eq. (6.8) with the sensitivity $\mathfrak{A} = -\tau_\varphi \mathcal{A}$ and offset $\mathfrak{B} = -\tau_\varphi \mathcal{B} + L_{\varphi,0}$.

Coil Set	τ_φ (s)	$L_{\varphi,0}$ (Nms)	$-\mathfrak{A}/\tau_\varphi$ (Nm/kA^2)	$\mathcal{A}$ (Nm/kA^2)	$-\mathfrak{A}/\left(\mathcal{A}\tau_\varphi\right)$
I-coil	0.040	0.347	0.88	1.19	0.74
C-coil	0.033	0.291	0.57	0.78	0.73

Table 6.2: Comparison of the sensitivities in the NTV and angular momentum phase space models. In each case, the axisymmetric momentum and momentum confinement time is estimated as $L_{\varphi,0} \approx \mathfrak{B}$ and $\tau_\varphi \approx L_{\varphi,0}/T_{NBI}$ respectively. This is valid in the limit $T_{NTV}(\mathbf{I}_0) \ll T_{NBI}$, and effectively sets the torque offset determined from the angular momentum to zero.

The NTV and dominant mode overlap EFC optima are compared to the experimentally optimized values in Tab. 6.1 and Fig. 6.11. The optima predicted using both model metrics agree well with the experimental optimum in angular momentum. Note that all C-coil optima also agree with the direct EM torque measurement EFC fit in Fig. 3.7. The agreement in the optima between the hybrid kinetic MHD calculation and the pure MHD metric is a promising result for future control efforts. As long as the agreement holds, optima can be immediately calculated by IPEC for NTV dominated equilibria. This allows the simple extension of a well known metric, despite the new physics encompassed by these scenarios.

In order to distinguish the NTV physics as the dominant error field effect it is necessary to examine not only the EFC optimum, but the sensitivity and offset as well. These are given in Tab. 6.2, which shows that the NTV model over predicts the sensitivity to the non-axisymmetric fields by roughly 25%. The opposite, an under prediction, would be expected if resonant damping was also significant in these scenarios. It is thus assumed that the NTV is the dominant rotation damping torque. Note that 25% is historically a very good agreement for the model [45, 23]. For a sense of the discrepancy in the predicted and measured NTV strengths, the PENT model I-coil fit predicts 1.81Nm with no EFC while the conversion of the angular momentum fit to torque predicts 2.46Nm. This is in contrast with the 8.76Nm neutral beam torque. Note that the sensitivity and offset of the overlap metric phase space dependence have different units. They do not directly correspond directly to a torque, as there is no net torque on any flux surface in a ideal MHD equilibrium.

The NTV torque contours in Fig. 6.11 correspond to the complete phase space mapping available after only 4 coil combinations are modeled using PENT. For each applied spectrum, a fifth case, corresponding to zero current in the EFC coils, is used to provide an error estimate for the fit. Recall that the intrinsic error field is treated as a constant, implicitly present throughout the applied EFC phase space such that there is a nonzero torque at the origin. This phase space fit represents a infinite reduction of the map between the applied current and the resultant NTV torque for a given coil set.

6.2.2 n=2 Error Field Correction

The validity of the single mode model was tested for $n = 2$ toroidal harmonic fields in 2014. The equilibrium plasma used was similar to that used in the H-mode $n = 1$ EFC experiment above. Single mode model optimization of the angular momentum yielded EFC currents on the order of 800kA in even parity I-coils, roughly half of the corresponding $n = 1$ EFC current. The source of the $n = 2$ intrinsic error field in DIII-D is not well known, however, preventing feed forward predictions of the optimal EFC. To test the computational models, a large proxy error field was applied using the C-coils and then corrected using the I-coil arrays.

The EFC optimization can be reduced to a two dimensional space despite the use of three coil arrays and multiple toroidal harmonics in this experiment. The intrinsic $n = 1$ EF was corrected with the C-coils using feed forward values similar to those shown in Fig. 6.11 and residual effects assumed to be negligible. In addition to this correction, a proxy EF was created by an additional 4kA of $n = 2$ current was driven in the C-coils. This large current overwhelms the intrinsic EF of the machine. The I-coil arrays where then used to correct the proxy error field. Hardwiring the two I-coil arrays in even parity reduced the number of free variables in the problem to 2: the amplitude and phase of the $n = 2$ even parity I-coil current.

The error field correction optimum in the $n = 2$ phase space of the I-coil arrays is shown in Fig. 6.12. The single mode model has been used to fit PENT modeled torques, the square of the IPEC overlap, and finally the experimentally measured angular momentum. The experimental optimum plotted is the difference between the optima with and without the C-coil proxy field, making it the correction of the proxy only as are the modeled optima. As in the $n = 1$ case for this high rotation H-mode, both modeling metrics agree well with the experimental optimum. The PENT fit predicts torques 1% and 8% of the no-EFC case for the IPEC and experimental optima respectively. The sensitivity of the PENT mapping, however, is not physical. The large torques predicted at even 1kA from the PENT optimum are larger than the injected neutral beam torque and would be enough to significantly alter the equilibrium (to the point of reversing the rotation). Compared to the quadratic fit to the experimental momentum, the sensitivity is orders of magnitude too large $-\mathfrak{A}/\mathcal{A}\tau_\varphi = 0.02$. Experimentally, the data is observed to fit a linear dependence, $L_\phi = \mathfrak{A}\left|\mathbf{I} - \mathbf{I}_0\right| + \mathfrak{B}$ well, and the angular momentum is reduced by only 30% 3kA away from the experimental optimum. This suggests that additional changes to the equilibrium such as reduction of β_N and density pump-out that are not captured in the perturbative IPEC and PENT calculations are playing a significant role in the magnetic braking away from the optimum.

The 5 phase space points modeled in PENT provide the fit to the 4 free parameters of Eq. (6.3) as well as an estimate of the fit accuracy. The single mode model fitting error is on the order of the numerical accuracy output by PENT, confirming the accuracy of the single mode model in describing the PENT results. Again, we have demonstrated a mapping of the NTV torque to the entire space available to an experimental coil set using only a small

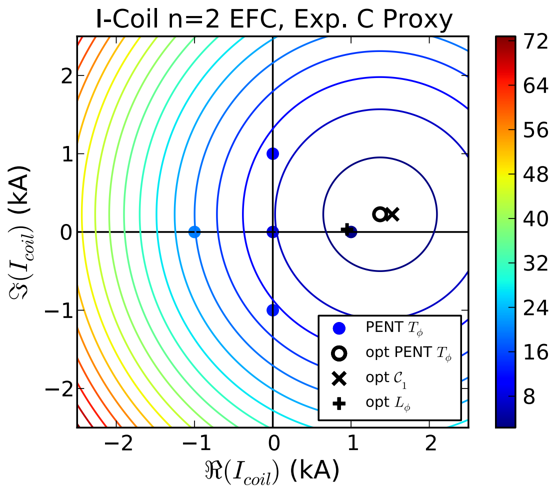

Figure 6.12: Phase space contours of constant torque (in Nm) determined by a least-squares fit to Eq. (6.3) of the torque calculated by PENT (colored circles) for 5 $n = 2$ I-coil EFC currents. The fit minimum (black circles) is similar to minima from single mode fits of the IPEC dominant mode overlap (black X) and measured angular momentum (black +). An additional field from a 4kA $n = 2$ C-coil waveform is applied in each perturbed equilibrium calculation.

number of computations. The agreement of the mapping with experimental data confirms that the relevant physics is captured for the regimes of interest.

6.2.3 TBM Error Field Correction

The extension of the single mode model's applicability to all DIII-D magnetic braking experiments, resonant or nonresonant, enables straight forward optimization in scenarios where multiple effects are expected to be important. Test Blanket Module (TBM) experiments performed in DIII-D provide a good example of such a complicated EFC scenario. The DIII-D TBM mockup, shown in Fig. 6.13, is a local coil designed to simulate the perturbations caused by discrete ferromagnetic blanket modules that will be installed in ITER [54, 95, 133, 138]. It's localized nature means that it produces a wide spectrum of toroidal harmonics at the plasma surface. The application to ITER incentivizes both experiments with low beam torque and experiments at high β_N. In the former, $n = 1$ resonant braking

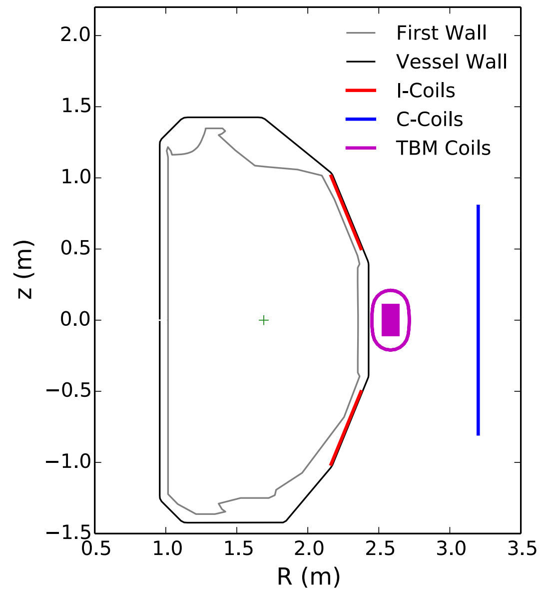

Figure 6.13: Schematic cross section of the DIII-D first wall (grey), vessel (black), I-coils (red), C-coils (blue) and TBM coils (purple). The TBM horizontal racetrack (oval) creates a toroidal field and the vertical solenoid (rectangle) creates a poloidal field. Unlike the toroidal EFC arrays, the entire apparatus has a toroidal angular cross section of less than 3°.

and mode locking are the direct concern of EFC. In the later, large amounts of neutral beam power creates a plasma similar to those mentioned above. In these plasmas, the large kink response amplifies nonresonant fields throughout the plasma and the high rotation favors NTV ($T_\varphi \sim \omega_\varphi$) over resonant torques ($T_\varphi \sim \omega_\varphi^{-1}$). For both scenarios, multiple toroidal harmonics make significant contributions to the torque.

The toroidal spectrum of the TBM induced NTV torque in a ITER-like lower single null scenario with $\beta_N \sim 2.1$ is shown in Fig. 6.14. Also shown are three torque spectra corresponding to optimized EFC correction with 6, 9 and 18 separately controlled power circuits. Here each circuit represents a power supply, which can drive current in a predetermined set of I-coils and/or C-coils. Three circuits are needed to give full $n = 1$ or $n = 2$ amplitude and phase control in any given set, while the $n = 3$ is generally controlled using a single circuit (fixed phase). The EFC currents were calculated using a simple C-dimensional fit to the IPEC overlap metric, where C is the number of circuits. Equal weighting was given to the $n = 1$, 2, and 3 overlap fields, while $n > 3$ harmonics were not taken into consideration. Fig. 6.14 shows that the $n = 4$ torque is significantly increased in 2 of the 3 EFC cases, which is an unfortunate compounding of the uncontrolled sidebands.

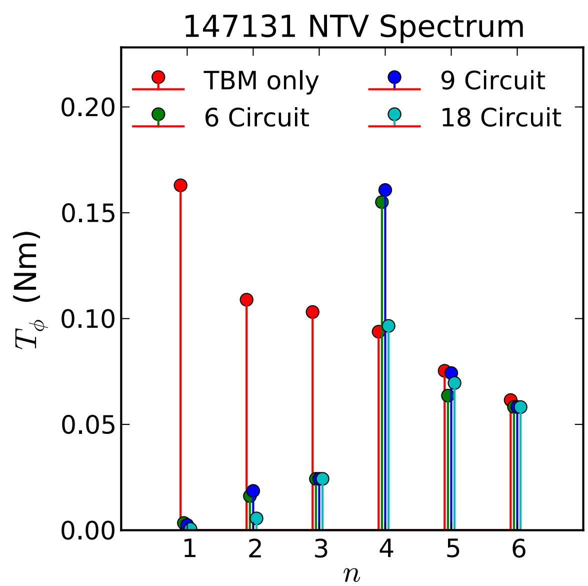

Figure 6.14: The toroidal mode spectrum of the NTV applied by 1kA in the TBM coils calculated by PENT without correction (red), and with with 6 (green), 9 (blue), or 18 (cyan) independent power supplies devoted to optimal correction of the first three toroidal harmonics using the I and C-coils.

This multi-harmonic, multi-circuit feed forward EFC is a demonstration of the power of linear combination in the single mode model. The application of the dominant mode overlap metric to both resonant and nonresonant physics, makes the optimization of the low and high toroidal harmonics a single straight forward process. Although the experimental application of these precise fields was hindered by a ground fault in the TBM coil, the PENT torque predictions agree with previously reported results. From Fig. 6.14, it is estimated that the $n = 1$ TBM torque is roughly 30% of the total TBM torque. This is consistent with the 25% recovery in rotation found with experimentally optimized single circuit $n = 1$ EFC [95]. For this scenario, an additional 20% recovery would be expected with $n = 2$ and only slightly less with $n = 3$ correction. The 18 circuit feed forward correction is thus predicted to recover

70% of the rotation.

6.2.4 Error Field Correction with Sparse Coil Sets

The single mode model allows the application of desired nonaxisymmetric fields to be smoothly transitioned between EFC coil sets. Experiments in DIII-D testing the limits of this application for ELM suppression with sparse EFC arrays, providing a test of the single mode model's practical limits. The experiment simulated the failure of individual coils in an EFC array by pseudo-randomly removing individual I-coils between repeated discharges. The original I-coil currents were a even parity $n = 3$ waveform, and the reduction of the $n = 3$ fields was mitigated by increased C-coil currents. As coils were removed, sidebands were introduced and both the toroidal and poloidal spectra broadened. ELM suppression, however, was maintained.

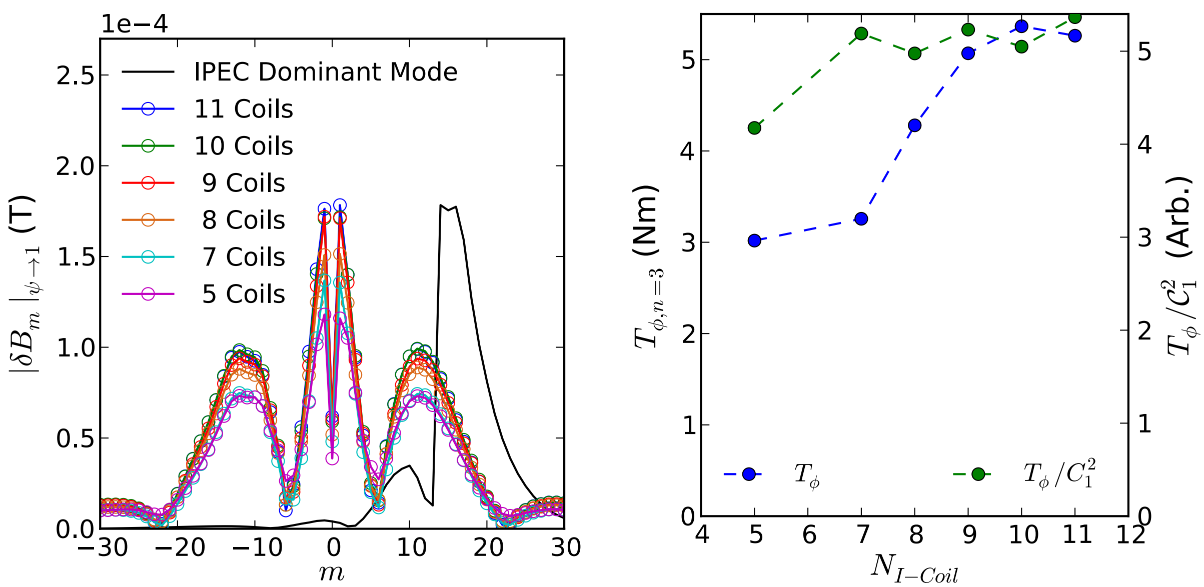

Figure 6.15: Left: The external field poloidal spectra (colored circles) on the IPEC control surface applied by the I-coils during pseudo-random removal of individual I-coils from a even parity $n = 3$ waveform. The normalized IPEC dominant mode spectrum (black) is also shown. Right: The NTV torque applied by each reduced number of coils as calculated by PENT (blue) and torque normalized by the square of the IPEC overlap metric (green).

The IPEC dominant mode metric approximation of the NTV coupling remains valid for I-coils reduced by nearly half the original coil number despite the torque calculated by PENT dropping by nearly 40 percent. Fig. 6.15 shows a figure of merit, $T_\phi/\mathcal{C}_1^2$, for the IPEC dominant mode description of the NTV coupling as the coil set is reduced. It remains constant as coils are removed randomly from the even parity configuration, indicating that the NTV is simply proportional to the square of the overlap throughout this range of shots. The EFC coil currents and torque are changing significantly throughout this range of shots, traversing a multi-dimensional phase space conceptually similar to those shown in previous sections. Torque proportional to the overlap throughout these shots confirms that the NTV can be reduced to a single mode model with a dominant spectrum approximately that of the IPEC dominant mode $\mathbf{\Phi}_{NTV} \approx \mathbf{\Phi}_1$. This proportionality holds until the final case, with fewer than half the original 12 coils active.

6.3 Residual Effects

The single mode model is an inexact reduction of the full nonlinear NTV physics, and residual effects must be present. The applicability of the single mode NTV EFC mapping as well as the IPEC overlap metric approximation has been demonstrated throughout the previous sections. The agreement with experimental optima confirms the residual effects are small compared to the dominant single mode coupling. Agreement is not exact however, as can be seen in the differences in the precise coil current optima using the two metrics. This section will explicitly address the residual NTV induced by applied spectra perpendicular to the dominant mode spectrum.

The residual effect of spectra orthogonal to the dominant mode were directly tested for both $n = 1$ and $n = 2$ EFC poloidal spectrum optimization experiments in DIII-D. There is, in principle, an infinite space orthogonal to the single dominant mode. The experimentally accessible orthogonal space is reduced to a single waveform when utilizing only 2 EFC arrays

(I-coils with fixed phasing and C-coils) as was done in these experiments. The waveform corresponds to the optimal correction of the first array by the second. The EFC arrays were fixed in this relative configuration and operated as a single array, $(I+C)_\perp$, with a spectrum orthogonal to the dominant mode.

The $(I+C)_\perp$ configuration was used to correct the residual EF remaining after the optimal dominant mode correction was applied. The NTV torque calculated by PENT again fit Eq. (6.3) in the $(I+C)_\perp$ phase space. This is a consequence of slight differences in the NTV dominant mode and the IPEC dominant mode, $\mathbf{\Phi}_{NTV} \cdot \mathbf{\Phi}_1 \neq 0$. Coupling to a second dominant eigenmode is ruled out by the accuracy of the previous single mode fits, which included the total NTV calculated using the full experimental spectra. In contrast, the IPEC overlap is fit separately for each of the individual eigenspectra. In the phase space of the $(I+C)_\perp$ configuration the dominant mode coupling is null by definition $\mathcal{C}_1 \equiv 0$, and the RFA response is dominated by the overlap with the second eigenspectrum $\mathcal{C}_2$. Fig. 6.16 shows that the experimentally determined EFC optimum agrees with the PENT calculated torque minimum and not the null of the IPEC second eigenspectrum overlap.

The sensitivity of the torque in the $(I+C)_\perp$ is drastically lower than the sensitivities calculated in either the I-coil or C-coil phase spaces. The $0.07\mathrm{Nm/kA}^2$ sensitivity is 6% and 10% of the values given in Tab. 6.2 for the I-coils and C-coils respectively. This is interpreted as a consequence of the similarity between the NTV single mode and IPEC dominant mode spectra. A large portion of the current in the $(I+C)_\perp$ configuration drives fields perpendicular to the dominant NTV spectrum, with only a small portion effectively contributing to the NTV EFC. The fact that the IPEC overlap null does not accurately describe the experimental optimum confirms that the global kink structure of the IPEC dominant mode response rather than the resonant field itself is the dominant error field effect in these plasmas.

Given a eigenbasis of spectra calculated from the direct coupling to the NTV torque, a similar experiment could be done in the null space of the self consistently calculated dominant NTV inducing spectrum (as apposed to the IPEC overlap approximation). Such an

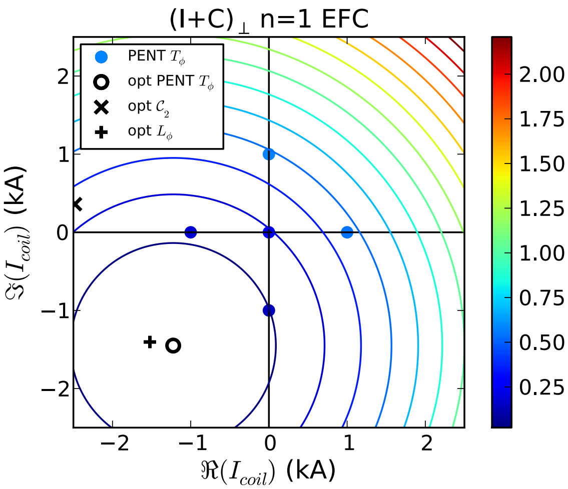

Figure 6.16: Phase space contours of constant torque (in Nm) determined by a least-squares fit to Eq. (6.3) of the torque calculated by PENT (colored circle) for 5 $n = 1$ EFC currents in the phase space of the combined $(I + C)_\perp$ waveform. The fit minimum (black circles) is similar to the minimum of a fit to the measured angular momentum (black +) but far from the minimum of the IPEC second dominant mode overlap (black X). The phase space is parameterized by the C-coil amplitude and phase, and the I-coil currents necessary to maintain orthogonally are implicitly maintained. The intrinsic $n = 1$ EF is also implicitly included in each perturbed equilibrium calculation.

experiment would provide further insight into the validity and limits of the single mode NTV approximation. A calculation of the NTV eigenspectrum is computationally demanding however, due to the nonlinear coupling of poloidal harmonics. This has motivated new theoretical work to include kinetic effects self consistently in the perturbed equilibrium calculations. Doing so is expected to provide more accurate coupling metrics, as well as new insights into the coupling between applied nonaxisymmetric fields and the induced NTV torque.

Chapter 7

Torque in Hybrid Kinetic MHD Perturbed Equilibrium

Thus far in this dissertation, the NTV torque has been solved using ideal perturbed equilibria that do not include the anisotropic kinetic pressure in the equilibrium force balance. The applicability of this limit, $\delta W_k \ll \delta W$ has been discussed in context of both the stable force balance and the RWM dispersion relation and the efficacy of the predictions has been demonstrated in a wide range of experimental applications. This chapter seeks to extend both the accuracy and the applicability of these predictions. The following sections describe initial calculations of self-consistent hybrid drift-kinetic MHD perturbed equilibria. The new equilibrium calculations include kinetic splitting and damping at resonant surfaces, resulting in a finite NTV torque without regularization. The consequences and extension of these results will be the focus of a new effort in predictive modeling of nonaxisymmetric field torque.

The calculation of self consistent kinetic MHD perturbed equilibria is presented as follows. The Direct Criterion of Newcomb (DCON) [47, 46] code's calculation of ideal MHD nonaxisymmetric eigenfunctions is reviewed in Section 7.1. The appropriate kinetic modifications to the energy minimizing Euler-Lagrange equations are then presented in Section 7.2. These solutions are obtained numerically by integrating the PENT code directly into the DCON framework. Initial calculations confirm the predicted kinetic modifications result in splitting and suppression of the ideal MHD resonances near rational surfaces while still converging to the ideal solution when the kinetic terms are reduced. The significance and consequences of these calculations are discussed in Section 7.3, setting the stage for future work in this field.

7.1 Ideal MHD Eigenfunctions

This section reviews the ideal MHD eigenfunction calculations performed by the DCON code [47, 46], which finds the stability of axisymmetric equilibria to nonaxisymmetric perturbations. The code calculates eigenfunction solutions of a high-order Euler-Lagrange system of ordinary differential equations, describing the displacements of flux surfaces that minimize the perturbed energy. Poles of the real Critical Determinant, a generalization of Newcomb's criteria for cylindrical geometry, indicate the presence of internal, ideal MHD instabilities.

The starting assumption of DCON is that the equilibrium in question satisfies the ideal MHD force balance equation,

$$\mathbf{J} \times \mathbf{B} = \nabla P, \tag{7.1}$$

throughout the plasma volume. Here P is the plasma pressure, $\mathbf{B}$ is the magnetic field, and $\mathbf{J} = \nabla \times \mathbf{B}$ is plasma current. Both the field and current are divergence free, and the Grad-Shafranov equation [31, 32] defines simply nested surfaces of constant poloidal flux.

The ideal MHD perturbed energy for such an equilibrium s given by [139],

$$\delta W_{MHD} = \int d\mathbf{x} \left[\delta B^2 + \mathbf{J} \cdot \xi \times \delta \mathbf{B} + (\xi \cdot \nabla P) \nabla \cdot \xi + \gamma P (\nabla \cdot \xi)^2\right], \tag{7.2}$$

where γ is the ratio of specific heats for the equation of state, ξ is the perturbed displacement, and δB is the perturbed field. These perturbed quantities are related by combining the ideal Ohm's law and Faraday's Law,

$$\delta\mathbf{B} = \nabla \times (\xi \times \mathbf{B}), \tag{7.3}$$

which enables an expression of the perturbed energy entirely in terms of the perturbed displacement and equilibrium quantities.

Choosing straight field line Hamada coordinates $(\psi, \vartheta, \varphi)$ and using Eq. (7.3) to eliminate the perturbed field, the perturbed energy becomes,

$$\begin{aligned}
\delta W_{MHD} = \int d\psi d\vartheta d\varphi \Big\{ \Big[& -\nabla\psi \times \nabla\xi^{\alpha} + \left(\mathbf{B}\cdot\nabla\xi^{\psi}\right)\mathbf{e}_{\psi} \\
& - \mathbf{B}\left(\psi_p'\xi^{\psi}\right)'/\psi_p' - q'\psi_p'\xi^{\psi}\mathbf{e}_{\varphi}\Big]^2 \\
& + \left(P'\psi_p''/\psi_p' + J^{\vartheta}q'\psi_p'\right)\xi^{\psi 2} + 2P'\xi^{\psi}\xi^{\psi\prime} \\
& + \xi^{\psi}\mathbf{J}\cdot\nabla\xi^{\alpha} - \xi^{\alpha}\mathbf{J}\cdot\nabla\xi^{\psi} \\
& + \partial\left(P'\xi^{\psi}\xi^{\varphi}\right)/\partial\vartheta + \partial\left(P'\xi^{\psi}\xi^{\varphi}\right)/\partial\varphi + \gamma P\left(\nabla\cdot\xi\right)^2 \Big\}.
\end{aligned} \tag{7.4}$$

Here the angular coordinates are normalized to unit periodicity (standard divided by 2π) such that $\mathcal{J} = 1$, primes denote derivates with respect to the normalized poloidal flux ψ, superscripts denote the contravariant components, and $\xi^{\alpha} = \psi_p'\left(q\xi^{\vartheta} - \xi^{\varphi}\right)$ is effectively the Clebsch angular coordinate component.

The existence of a displacement test function that has negative potential energy δW_{MHD} implies the equilibrium is unstable and the exponential growth of perturbations is to be expected. The job of DCON is thus to minimize Eq. (7.4) with respect to each component of the displacement in order to determine the existence of such a test function while satisfying the boundary conditions. For a perturbed equilibrium calculation, DCON will find plasma eigenfunctions: a set of neighboring perturbed equilibria that are stationary points of the potential energy but, if stable, need to be driven by external forces at the boundary [44, 46].

The final term is the only contribution to the integral energy containing ξ^{ϑ} distinct from the Clebsch component. This term is positive definite, and must be eliminated by minimization of the energy with respect to ξ^{ϑ} for all toroidal mode numbers $n \neq 0$. Thus, incompressible displacement ($\nabla \cdot \xi = 0$) is assumed from this point on. The remaining displacements are expanded in Fourier space,

$$\xi^{\alpha,\psi} = \sum_{n=-\infty}^{\infty} \sum_{m=-\infty}^{\infty} c_{n,m}^{\alpha,\psi} \left(\psi\right) e^{-2\pi i (n\varphi - m\vartheta)}. \tag{7.5}$$

The toroidal mode number n is a good quantum number in tokamaks, and the perturbed energy can be calculated independently for each toroidal mode. The lack of poloidal symmetry, however, results in poloidal mode coupling. In practice, the poloidal spectrum is truncated and the perturbed energy for a single toroidal mode becomes a function of $2 \times M$ coupled Fourier coefficient flux functions. $M = m_{max} - m_{min}$ is the number of poloidal harmonics. Defining the Fourier coefficients as $M-$vectors,

$$\Xi^{\alpha,\psi}\left(\psi\right) \equiv \begin{bmatrix} c_{m_{min}}^{\alpha,\psi} \\ c_{m_{min}+1}^{\alpha,\psi} \\ \vdots \\ c_{m_{max}}^{\alpha,\psi} \end{bmatrix}, \tag{7.6}$$

the potential energy integrand becomes a matrix equation [46, 45],

$$\begin{aligned} \delta W_{MHD} = \int d\psi \Big[& \Xi^{\alpha\dagger}\mathbf{A}\Xi^{\alpha} + \Xi^{\alpha\dagger}\mathbf{B}\Xi^{\psi\prime} + \Xi^{\alpha\dagger}\mathbf{C}\Xi^{\psi} + \Xi^{\psi\prime\dagger}\mathbf{D}\Xi^{\psi\prime} + \Xi^{\psi\prime\dagger}\mathbf{E}\Xi^{\psi} \\ & + \Xi^{\psi\dagger}\mathbf{H}\Xi^{\psi} + \left(\Xi^{\alpha\dagger}\mathbf{B}\Xi^{\psi\prime}\right)^{\dagger} + \left(\Xi^{\alpha\dagger}\mathbf{C}\Xi^{\psi}\right)^{\dagger} + \left(\Xi^{\psi\prime\dagger}\mathbf{E}\Xi^{\psi}\right)^{\dagger} \Big], \end{aligned} \tag{7.7}$$

where the off-diagonal components of the $M \times M$ matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$, $\mathbf{E}$, and $\mathbf{H}$ result in poloidal coupling. Here, daggers denote the Hermitian conjugate and $\mathbf{A}$, $\mathbf{D}$, and $\mathbf{H}$ are self-adjoint. Note also that the displacements satisfy the reality conditions $\Xi_{\alpha,\psi} = \Xi^{*}_{\alpha,\psi}$.

The DCON formalism first finds the stationary point with respect to Ξ^{α} by noting that Ξ^{α} enters Eq. (7.7) only algebraically [46]. Finding extremum with respect to this displacement is equivalent to solving the algebraic equation,

$$\mathbf{A}\Xi^{\alpha} + \mathbf{B}\Xi^{\psi\prime} + \mathbf{C}\Xi^{\psi} = 0, \tag{7.8}$$

which gives,

$$\Xi^{\alpha} = -\mathbf{A}^{-1}\left(\mathbf{B}\Xi^{\psi\prime} + \mathbf{C}\Xi^{\psi}\right). \tag{7.9}$$

Inserting this into (7.7) immediately annihilates the first 3 terms, and gives the reduced form,

$$\delta W_{MHD} \;\; = \int d\psi \;\; \left[\Xi^{\psi\prime\dagger}\mathbf{F}\Xi^{\psi\prime} + \Xi^{\psi\prime\dagger}\mathbf{K}\Xi^{\psi} + \Xi^{\psi\dagger}\mathbf{K}^{\dagger}\Xi^{\psi\prime} + \Xi^{\psi\dagger}\mathbf{G}\Xi^{\psi}\right], \tag{7.10}$$

where,

$$\mathbf{F} \;\; \equiv \;\; \mathbf{D} - \mathbf{B}^{\dagger}\mathbf{A}^{-1}\mathbf{B} = \mathbf{F}^{\dagger}, \tag{7.11}$$

$$\mathbf{G} \;\; \equiv \;\; \mathbf{H} - \mathbf{C}^{\dagger}\mathbf{A}^{-1}\mathbf{C} = \mathbf{G}^{\dagger}, \tag{7.12}$$

$$\mathbf{K} \;\; \equiv \;\; \mathbf{E} - \mathbf{B}^{\dagger}\mathbf{A}^{-1}\mathbf{C} \neq \mathbf{K}^{\dagger}. \tag{7.13}$$

Note that **F** and **G** are Hermitian and **K** is not.

Minimizing the energy respect to the only remaining component of the displacement $\Xi^{\psi}(\psi)$ gives the Euler-Lagrange matrix equation,

$$\left(\mathbf{F}\Xi^{\psi\prime} + \mathbf{K}\Xi^{\psi}\right)' - \left(\mathbf{K}^{\dagger}\Xi^{\psi\prime} + \mathbf{G}\Xi^{\psi}\right) = 0, \tag{7.14}$$

which is a system of M second order ordinary differential equations. It is possible to express this as an equivalent system of first order differential equations,

$$\mathbf{u}' = \mathbf{L}\mathbf{u}, \tag{7.15}$$

where,

$$\mathbf{u} \equiv \begin{bmatrix} \Xi^{\psi} \\ \mathbf{F}\Xi^{\psi\prime} + \mathbf{K}\Xi^{\psi} \end{bmatrix}, \tag{7.16}$$

is a $2M-$vector and,

$$\mathbf{L} = \begin{bmatrix} -\mathbf{F}^{-1}\mathbf{K} & \mathbf{F}^{-1} \\ \mathbf{G} - \mathbf{K}^{\dagger}\mathbf{F}^{-1}\mathbf{K} & \mathbf{K}^{\dagger}\mathbf{F}^{-1} \end{bmatrix}, \tag{7.17}$$

is a complex $2M \times 2M$ matrix.

That the Euler-Lagrange equations be satisfied is a necessary, but not sufficient condition for the corresponding perturbed energy to be a minimum [46, 45]. The additional conditions for a minimum are beyond the scope of this review however, and are not addressed here.

A key insight of the DCON formalism is that the Euler-Lagrange condition reduces the potential energy integral to a surface term only. Inserting Eq. (7.14) in to Eq. (7.10) gives,

$$\delta W_{MHD} = \int d\psi \left[\Xi^{\psi\prime\dagger}\left(\mathbf{F}\Xi^{\psi\prime} + \mathbf{K}\Xi^{\psi}\right) + \Xi^{\psi\dagger}\left(\mathbf{F}\Xi^{\psi\prime} + \mathbf{K}\Xi^{\psi}\right)'\right]. \tag{7.18}$$

Adding the hermitian conjugate of the integrand,

$$\begin{aligned} \delta W_{MHD} \quad &= \tfrac{1}{2}\int d\psi \quad \left[\Xi^{\psi\prime\dagger}\left(\mathbf{F}\Xi^{\psi\prime} + \mathbf{K}\Xi^{\psi}\right) + \Xi^{\psi\dagger}\left(\mathbf{F}\Xi^{\psi\prime} + \mathbf{K}\Xi^{\psi}\right)'\right. \\ &\qquad \left. + \left(\Xi^{\psi\prime\dagger}\mathbf{F} + \Xi^{\psi\dagger}\mathbf{K}^{\dagger}\right)\Xi^{\psi\prime} + \left(\Xi^{\psi\prime\dagger}\mathbf{F} + \Xi^{\psi\dagger}\mathbf{K}^{\dagger}\right)'\Xi^{\psi}\right], \end{aligned} \tag{7.19}$$

gives a symmetric form. The chain rule condenses this form,

$$\delta W_{MHD} = \frac{1}{2}\int_{\psi_1}^{\psi_2} d\psi \left[\Xi^{\psi\dagger}\left(\mathbf{F}\Xi^{\psi\prime} + \mathbf{K}\Xi^{\psi}\right) + \left(\Xi^{\psi\prime\dagger}\mathbf{F} + \Xi^{\psi\dagger}\mathbf{K}^{\dagger}\right)\Xi^{\psi}\right]', \tag{7.20}$$

which needs only the fundamental rule of calculus to give,

$$\delta W_{MHD} = \frac{1}{2}\left[\Xi^{\psi\dagger}\left(\mathbf{F}\Xi^{\psi\prime} + \mathbf{K}\Xi^{\psi}\right) + \left(\Xi^{\psi\prime\dagger}\mathbf{F} + \Xi^{\psi\dagger}\mathbf{K}^{\dagger}\right)\Xi^{\psi}\right]\Big|_{\psi_1}^{\psi_2}, \tag{7.21}$$

$$\delta W_{MHD} = \frac{1}{2}\left[\mathbf{u}_1^{\dagger}\mathbf{u}_2 + \mathbf{u}_2^{\dagger}\mathbf{u}_1\right]\Big|_{\psi_1}^{\psi_2}. \tag{7.22}$$

Equation (7.22) has been expressed in terms of the first order differential system's first M components $\mathbf{u}_1 = \Xi^{\psi}$ and second M components $\mathbf{u}_2 = \mathbf{F}\Xi^{\psi\prime} + \mathbf{K}\Xi^{\psi}$. The DCON code uses an adaptive grid ODE solver [140] to solve for the $2M$ solutions of the first order system of equations (7.15), using the shooting technique initialized with an $2M$ orthogonal set of boundary conditions. The boundary conditions are all imposed on Ξ^{ψ}, with half at the axis and half at the edge. Any unique solution of the Euler-Lagrange equation, can be expressed as a weighted sum of these $2M$solutions,

$$\mathbf{u}(\psi) = \mathbf{U}(\psi)\,\mathbf{c} = \begin{bmatrix} \mathbf{U}_{11}(\psi) & \mathbf{U}_{12}(\psi) \\ \mathbf{U}_{21}(\psi) & \mathbf{U}_{22}(\psi) \end{bmatrix} \begin{bmatrix} \mathbf{c}_1 \\ \mathbf{c}_2 \end{bmatrix}, \tag{7.23}$$

where $\mathbf{U}$ is a $2M \times 2M$ matrix of solutions, $\mathbf{c}$ is an $M-$vector of constant weighting coefficients, and the subscripts refer to $M-$component partitions. Applying the boundary condition $\Xi^{\psi}(0) = 0$ on axis, Eq. (7.23) immediately gives $\mathbf{c}_1 = \Xi^{\psi}(0) = 0$. Throughout the plasma, $\mathbf{u}_1(\psi) = \mathbf{U}_{12}\mathbf{c}_2$ and $\mathbf{u}_2(\psi) = \mathbf{U}_{22}\mathbf{c}_2 = \mathbf{U}_{22}\mathbf{U}_{12}^{-1}\mathbf{u}_1$. We finally arrive at the compact expression,

$$\delta W_{MHD} = \Xi^{\psi\dagger}\mathbf{W}_p\Xi^{\psi}\Big|_{\psi_2}, \tag{7.24}$$

where the plasma matrix $\mathbf{W}_p \equiv \mathbf{U}_{22}\mathbf{U}_{12}^{-1}$ is built from the $2M$ set of $2M$ solution vectors to the Euler-Lagrange Eq. (7.15). Singular value decomposition (SVD) analysis of this plasma matrix forms eigenvectors corresponding to Euler-Lagrange solutions.

The DCON code has generalized this approach to treat free boundary, including the vacuum coupling [123]. The plasma matrix, however, contains the ideal MHD physics that must be modified by kinetic effects to calculate perturbed equilibria including self consistent kinetic pressure effects. With this in hand, the kinetic modifications become relatively strait forward.

7.2 Hybrid Drift-Kinetic MHD Eigenfunctions

This section describes a new calculation, including kinetic modifications to the perturbed energy integrand and calculating the plasma eigenfunctions using the same Euler-Lagrange matrix equation methods developed in the ideal MHD DCON code. Previous chapters have shown that an anisotropic kinetic pressure tensor can result in significant toroidal NTV torque and effect plasma stability through $\delta W_k = T_\varphi/2ni$. These calculations have thus far used perturbed equilibria under the assumed limit $\delta W_k \ll \delta W_{MHD}$. That is, the limit in which the kinetic forces did not significantly alter the perturbed equilibrium. Here, the kinetic physics is self consistently included in the calculation of the plasma eigenfunctions. When coupled to the external drive of an applied field, the result will be a perturbed equilibrium that self consistently includes anisotropic kinetic pressure in the perturbed force balance equation.

We start from the form of the NTV given in Eq. (4.17),

$$T_\varphi = \frac{4\pi^2 n^2}{eM^2}\int d\psi dE d\mu \mathcal{R}_\ell \left|\delta J_\ell\right|^2 \frac{\partial f_0}{\partial \psi}. \tag{7.25}$$

The perturbed action can be written in the $m-$vector form,

$$\delta J = \mathcal{W}_X^T \Xi^{\psi\prime} + \mathcal{W}_Y^T \Xi^{\psi} + \mathcal{W}_Z^T \Xi^{\alpha}, \tag{7.26}$$

where $\mathcal{W}_{X,Y,Z}^{\dagger}$ are row vector operators on the displacement column vectors $\Xi^{\psi\prime,\psi,\alpha}$ (see Appendix C). Isolating the real kinetic energy ($\Re\left(\delta W_k\right) = \Im\left(T_\varphi\right)/2\pi n$) and plugging (7.26) into (7.25) gives,

$$\delta W_k = \frac{2\pi n}{eM^2} \int d\psi dE d\mu \left[\Im\left(\mathcal{R}\right) f' \left(\mathcal{W}_X^T \Xi^{\psi\prime} + \mathcal{W}_Y^T \Xi^{\psi} + \mathcal{W}_Z^T \Xi^{\alpha} \right)^{\dagger} \right.$$
$$\left. \cdot \left(\mathcal{W}_X^T \Xi^{\psi\prime} + \mathcal{W}_Y^T \Xi^{\psi} + \mathcal{W}_Z^T \Xi^{\alpha} \right) \right], \tag{7.27}$$

which has 9 distinct terms,

$$\begin{aligned} \delta W_k \quad = \quad & \frac{2\pi n}{eM^2} \int d\psi dE d\mu \Im\left(\mathcal{R}\right) \frac{\partial f_0}{\partial \psi} \left\{ \Xi^{\alpha\dagger} \mathcal{W}_Z^* \mathcal{W}_Z^T \Xi^{\alpha} \right. \\ & + \Xi^{\alpha\dagger} \mathcal{W}_Z^* \mathcal{W}_X^T \Xi^{\psi\prime} + \left(\Xi^{\alpha\dagger} \mathcal{W}_Z^* \mathcal{W}_X^T \Xi^{\psi\prime} \right)^{\dagger} \\ & + \Xi^{\alpha\dagger} \mathcal{W}_Z^* \mathcal{W}_Y^T \Xi^{\psi} + \left(\Xi^{\alpha\dagger} \mathcal{W}_Z^* \mathcal{W}_Y^T \Xi^{\psi} \right)^{\dagger} \\ & + \Xi^{\psi\prime\dagger} \mathcal{W}_X^* \mathcal{W}_Y^T \Xi^{\psi} + \left(\Xi^{\psi\prime\dagger} \mathcal{W}_X^* \mathcal{W}_Y^T \Xi^{\psi} \right)^{\dagger} \\ & \left. + \Xi^{\psi\prime\dagger} \mathcal{W}_X^* \mathcal{W}_X^T \Xi^{\psi\prime} + \Xi^{\psi\dagger} \mathcal{W}_Y^* \mathcal{W}_Y^T \Xi^{\psi} \right\}. \end{aligned} \tag{7.28}$$

This integrand is explicitly Hermitian, consistent with the DCON Euler-Lagrange formalism, due to the fact that only the imaginary component of the resonance operator is retained for the real energy calculation. Kinetic modifications of the energy principle have been studied previously in order to more accurately predict stability boundaries in the zero-frequency and collisionless limit, for which the resonance operator is explicitly imaginary and the kinetic modifications are thus Hermitian [141, 142, 143, 127, 144, 4]. It has been shown in many situations, including work presented in this dissertation, that the poles of the resonant operator produce finite imaginary terms (non-zero torque) even in this limit. The complex kinetic energy is non-Hermitian in general, and thus the energy principle stability analysis is not applicable.

The self-adjoint property of the energy integrand is essential to determine the stability, but not required to obtain the perturbed force balance equation by minimizing the potential energy [43]. This analysis is limited to finding the eigenfunctions for which the total perturbed energy, $\delta W_{MHD} + \delta W_k$, is stationary with respect to ξ. The calculations are thus limited to solving the perturbed force balance, a necessary but not sufficient condition for stability. The DCON code Euler-Lagrange matrix formalism is used to find these eigenfunction displacements.

The kinetic energy can then be written in a form identical to Eq. (7.7),

$$\delta W_k = \int d\psi \left[\Xi^{\alpha\dagger}\mathbf{A}_k\Xi^{\alpha} + \Xi^{\alpha\dagger}\mathbf{B}_k\Xi^{\psi\prime} + \Xi^{\alpha\dagger}\mathbf{C}_k\Xi^{\psi} + \Xi^{\psi\prime\dagger}\mathbf{D}_k\Xi^{\psi\prime} + \Xi^{\psi\prime\dagger}\mathbf{E}_k\Xi^{\psi} \right.$$
$$\left. + \Xi^{\psi\dagger}\mathbf{H}_k\Xi^{\psi} + \left(\Xi^{\alpha\dagger}\mathbf{B}_k\Xi^{\psi\prime}\right)^{\dagger} + \left(\Xi^{\alpha\dagger}\mathbf{C}_k\Xi^{\psi}\right)^{\dagger} + \left(\Xi^{\psi\prime\dagger}\mathbf{E}_k\Xi^{\psi}\right)^{\dagger} \right], \tag{7.29}$$

where the new kinetic matrices are defined as,

$$\mathbf{A}_k = \frac{2\pi n}{eM^2}\int dEd\mu\Im\left(\mathcal{R}\right)\frac{\partial f_0}{\partial\psi}\mathcal{W}_Z^*\mathcal{W}_Z^T, \tag{7.30}$$
$$\mathbf{B}_k = \frac{2\pi n}{eM^2}\int dEd\mu\Im\left(\mathcal{R}\right)\frac{\partial f_0}{\partial\psi}\mathcal{W}_Z^*\mathcal{W}_X^T, \tag{7.31}$$
$$\mathbf{C}_k = \frac{2\pi n}{eM^2}\int dEd\mu\Im\left(\mathcal{R}\right)\frac{\partial f_0}{\partial\psi}\mathcal{W}_Z^*\mathcal{W}_Y^T, \tag{7.32}$$
$$\mathbf{D}_k = \frac{2\pi n}{eM^2}\int dEd\mu\Im\left(\mathcal{R}\right)\frac{\partial f_0}{\partial\psi}\mathcal{W}_X^*\mathcal{W}_X^T, \tag{7.33}$$
$$\mathbf{E}_k = \frac{2\pi n}{eM^2}\int dEd\mu\Im\left(\mathcal{R}\right)\frac{\partial f_0}{\partial\psi}\mathcal{W}_X^*\mathcal{W}_Y^T, \tag{7.34}$$
$$\mathbf{H}_k = \frac{2\pi n}{eM^2}\int dEd\mu\Im\left(\mathcal{R}\right)\frac{\partial f_0}{\partial\psi}\mathcal{W}_Y^*\mathcal{W}_Y^T. \tag{7.35}$$

Note that the matrices $\mathbf{A}_k$, $\mathbf{C}_k$, and $\mathbf{H}_k$ are Hermitian. These 6 $M \times M$ matrices are calculated by PENT and contain all the non-linear poloidal mode coupling of the calculation. By including PENT directly into the DCON stability code, the self-consistent hybrid kinetic MHD eigenfunctions are found by minimizing the combined energy,

$$\delta W_h = \delta W_{MHD} + \delta W_k. \tag{7.36}$$

The hybrid energy has an identical form as the ideal MHD energy given in (7.7), and the methods developed by DCON to minimize a integrand of this form are applied directly to the new hybrid equation. As discussed above, the computed eigenfunctions are solutions of the perturbed force balance equation but not sufficient to determine the stability of the plasma. Fig. 7.1 shows a comparison between the harmonics of the least stable $n = 1$ eigenfunction calculated by DCON in the ideal MHD limit and calculated including a kinetic contribution. The equilibrium used in this calculation is again the Solov'ev 3 equilibrium presented in Section 5.1. A scaling of the kinetic matrices by k, such that $\delta W_h = \delta W_{MHD} + k\delta W_k$, is used as an approximate scaling of the kinetic pressure by a factor of 10^2 in order to produce an observable difference in the global structure of the eigenfunction harmonics. The similarity between the eigenfunctions, is confirmation that the RWM physics studied in Chapter 5 was in the perturbative limit in which the kinetic effects do not have a large nonlinear effect on the RWM itself. There are, however, significant differences between the eigenfunctions at the two rational surfaces. This is a consequence of kinetic suppression of the ideal MHD resonances, detailed in the following section.

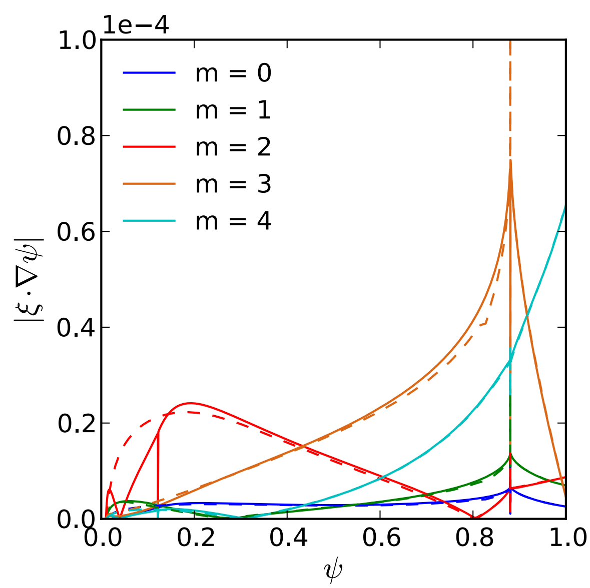

Figure 7.1: The first five poloidal harmonics of the displacement normal to the flux surfaces for the first $n = 1$ ideal MHD DCON eigenfunction (solid) and hybrid drift-kinetic MHD eigenfunction (dashed) for the marginally stable Solov'ev 3 equilibrium. The normal displacements are plotted against normalized poloidal flux.

7.2.1 Resonance Splitting and Suppression

The Euler-Lagrange equation (7.15) becomes singular at the zeros of the matrix $\mathbf{F} = \mathbf{D} - \mathbf{B}^\dagger \mathbf{A}^{-1}\mathbf{B}$, which is inverted in the operator (7.17). To examine the zeros of $\mathbf{F}$, we must write the matrix explicitly. For this purpose it is useful to define the diagonal matrices,

$$\mathbf{M}_{m,m'} = m\delta_{m,m'}, \tag{7.37}$$

$$\mathbf{Q}_{m,m'} = (m - nq)\delta_{m,m'}, \tag{7.38}$$

and geometric matrices,

$$[\mathbf{G}_{ij}]_{m,m'} = \int_0^{2\pi} d\vartheta g_{ij} e^{i(m'-m)\vartheta}. \tag{7.39}$$

The diagonal factors are simply definitional, although the significance of the rational surface is immediately discernible in the fact that at rational surfaces $m - nq = 0$ nulls $\mathbf{Q}$. The geometric matrix is a collection of Fourier coefficients for the standard metric tensor $g_{ij} = \mathbf{e}_i \cdot \mathbf{e}_j$. The fundamental DCON matrices comprising $\mathbf{F}$ can be written in terms of these matrices,

$$\mathbf{A} = (2\pi)^2 \left[n\left(n\mathbf{G}_{22} + \mathbf{G}_{23}\mathbf{M}\right) + \mathbf{M}\left(n\mathbf{G}_{23} + \mathbf{G}_{33}\mathbf{M}\right)\right], \tag{7.40}$$

$$\mathbf{B} = -2\pi i\psi_p' \left[n(\mathbf{G}_{22} + \mathbf{G}_{23}\mathbf{M}) + \mathbf{M}(n\mathbf{G}_{23} + \mathbf{G}_{33}\mathbf{M}) - (n\mathbf{G}_{23} + \mathbf{M}\mathbf{G}_{33})\mathbf{Q}\right], \tag{7.41}$$

$$\mathbf{D} = \psi_p'^2 \left[(\mathbf{G}_{22} + q\mathbf{G}_{23}) + q\left(\mathbf{G}_{23} + q\mathbf{G}_{33}\right)\right]. \tag{7.42}$$

Forming $\mathbf{F}$ from these definitions, there is a cancelation of terms that leaves the operator second-order singular,

$$\mathbf{F} = \mathbf{D} - \mathbf{B}^\dagger \mathbf{A}^{-1}\mathbf{B}, \tag{7.43}$$

$$\mathbf{F} = (\chi'/n)^2 \mathbf{Q}\left[\mathbf{G}_{33} - (n\mathbf{G}_{23} + \mathbf{M}\mathbf{G}_{33})(2\pi)^2\mathbf{A}^{-1}(n\mathbf{G}_{23} + \mathbf{M}\mathbf{G}_{33})\right]\mathbf{Q}. \tag{7.44}$$

The two $\mathbf{Q}$ matrices cause an non-integrable singularity at the rational surfaces. DCON uses an adaptive mesh ODE solver [140] to approach up to a small distance from the surface and crosses the surface enforcing the small asymptotic limit $\xi_{mn} \sim Q^{-1/2-\sqrt{-D_I}}$ for the resonant component of the normal displacement [46, 45]. Here, $Q = m - nq$ and D_I is the Mercier criterion for interchange stability [145]. The physics assumptions relevant for the perturbed equilibrium and consequential NTV torque have been discussed in Sec. 5.4.2.1, which details the need to regularize the ideal MHD solution at the rational surface to obtain a finite estimate of the NTV and kinetic energy. These are nuanced treatments of the rational surfaces necessary due to the inherent singularities in ideal MHD.

In the hybrid energy minimization, the form of the Euler-Lagrange equation remains unchanged and the singularity is determined by the new hybrid matrix,

$$\mathbf{F}_h = \mathbf{D} + \mathbf{D}_k - (\mathbf{B} + \mathbf{B}_k)^{\dagger} (\mathbf{A} + \mathbf{A}_k)^{-1} (\mathbf{B} + \mathbf{B}_k) . \tag{7.45}$$

Using the definitions already available for the ideal matrices, it is possible to recast this equation in the form,

$$\mathbf{F}_h = (\mathbf{Q} - \mathbf{P}^{\dagger})\bar{\mathbf{F}}(\mathbf{Q} - \mathbf{P}) + \mathbf{D}_k - (\psi_p'/2\pi n)^2 \mathbf{A}_k - \mathbf{P}^{\dagger}\bar{\mathbf{F}}\mathbf{P}. \tag{7.46}$$

by defining,

$$\bar{\mathbf{F}} = (\psi_p'/n)^2 \left[\mathbf{G}_{33} - 4\pi^2 i \left(n\mathbf{G}_{23} + \mathbf{M}\mathbf{G}_{33} \right)^{\dagger} (\mathbf{A} + \mathbf{A}_k)^{-1} \left(n\mathbf{G}_{23} + \mathbf{M}\mathbf{G}_{33} \right) \right], \tag{7.47}$$

$$\mathbf{P} = \bar{\mathbf{F}}^{-1}(-2\pi i \psi_p'/n) \left(n\mathbf{G}_{23} + \mathbf{M}\mathbf{G}_{33} \right)^{\dagger} (\mathbf{A} + \mathbf{A}_k)^{-1} \left(\mathbf{B}_k + \frac{i\psi_p'}{2\pi n} \mathbf{A}_k \right). \tag{7.48}$$

This in term can be written formally as,

$$\mathbf{F}_h = (\mathbf{Q} - \mathbf{P} - \mathbf{R}_1)^{\dagger} \, \bar{\mathbf{F}} \, (\mathbf{Q} - \mathbf{P} - \mathbf{R}_2) \, , \tag{7.49}$$

using the matrices $\mathbf{R}_1 = -\bar{\mathbf{F}}^{-1}\mathbf{Q}^{-1}\mathbf{R}_X^{\dagger}\mathbf{Q}$ and $\mathbf{R}_2 = \bar{\mathbf{F}}^{-1}\mathbf{R}_X$ with the solutions of a complex

Riccati matrix equation,

$$\mathbf{Q}\mathbf{R}_X\mathbf{Q}^{-1}\bar{\mathbf{F}}^{-1}\mathbf{R}_X + \mathbf{Q}\mathbf{R}_X\mathbf{Q}^{-1} - \mathbf{P}^\dagger\mathbf{R}_X + \mathbf{D}_K + (\psi_p'/2\pi n)^2\mathbf{A}_k - \mathbf{P}^\dagger\bar{\mathbf{F}}\mathbf{P} = 0.$$

The new form of $\mathbf{F}_h$ shows the singular surfaces are shifted by $\mathbf{P}$ from the rational surfaces and split into two distinct singularities. Depending on the sign and magnitude of these modifications, the singularity may be suppressed altogether by removing any zeros of the modified $\mathbf{F}_h$ for a given harmonic. When present, each zero now represents a lower order singularity. In principle, evaluating $\mathbf{P}$ and examining $\mathbf{F}_h$ near the rational surfaces enables one to find the new singular surfaces exactly. In practice the lower order singularity becomes integrable, and can be handled directly by the adaptive ODE solver. Thus, the hybrid Euler-Lagrange equation can be solved in a single continuous solution from the core to the plasma edge. Fig. 7.2 shows the $m = 3$ harmonic of the least stable $n = 1$ eigenfunction near the $q = 3$ surface of the Solov'ev 3 equilibrium for the ideal calculation and two kinetic calculations. In the ideal calculation, the resonant harmonic of the displacement is explicitly suppressed at the rational surface to avoid non-integrable singularities in the Euler-Lagrange equation. The singularity is suppressed in the hybrid Euler-Lagrange equation including the kinetic terms with the kinetic profiles in Section 5.1, allowing a smooth and continuous solution of the Euler-Lagrange equation across the profile. A scaling of the kinetic contributions, $\delta W_h = \delta W_{MHD} + k\delta W_k$ with $k = 10^2$, shows a split resonance. Again, the solution is integrable and a single adaptive mesh ODE solution is obtained for the entire plasma volume. These hybrid calculations are demonstrations of a self consistent treatment of the resonant surfaces, enabled by the kinetic splitting and damping of the resonances.

7.2.2 Convergence to the Ideal Limit

At low plasma pressures, well below the ideal MHD stability limit, the kinetic contributions to perturbed energy are expected to be negligible. The global structure of the hybrid and

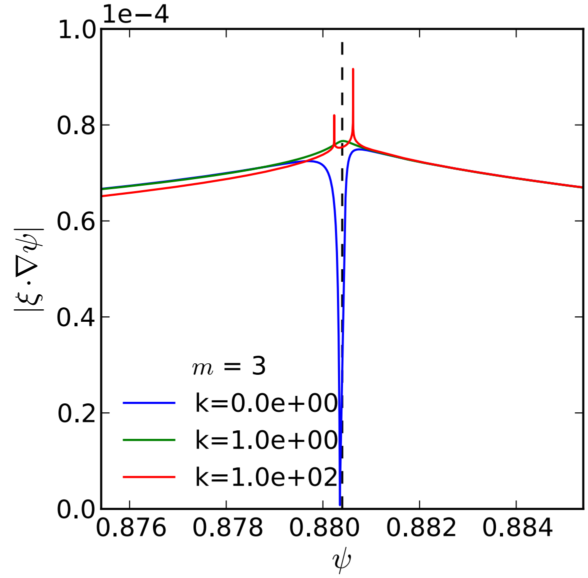

Figure 7.2: Normalized poloidal flux profiles of the $m = 3$ poloidal harmonic for the least stable $n = 1$ eigenmode near the $q = 3$ surface (dashed line) in the Solov'ev 3 equilibrium. The three cases represent the ideal MHD solution (blue), unmodified hybrid solution (green), and a hybrid solution with artificially scaled kinetic contributions (red).

ideal eigenfunctions are thus expected to converge at low pressure, with the vanishingly small shift and/or split of the resonances contributing little to the total perturbed energy. This convergence is demonstrated in Fig. 7.3 using the Solov'ev 3 equilibrium described in Section 5.1. Here again, a kinetic pressure scan has been approximated by a scaling of the kinetic energy matrices such that the calculation effectively minimizes the energy $\delta W = \delta W_{MHD} + k\delta W_k$ where δW_k is determined by the kinetic profiles established in the referenced section. The initial pressure in this example is low, producing a smoothing of the rational surface resonances but little global modification of the eigenfunctions. At higher pressures, the eigenfunctions are distorted on a global scale resulting in increased δW. As expected, the calculations of the kinetic energy and NTV torque of the RWM approximation using IPEC and PENT show dependences similar to that of the hybrid energy. At lower pressures the hybrid result quickly converges to that of the ideal MHD stability calculation. Correspondingly, the self consistent PENT calculations converge to the perturbative solution calculated using the ideal eigenfunctions. Note that the lower limit of this scan was determined by numerical stability limits determined by the return of sharp resonances, and work is on going to form a numerical treatment that can smoothly transition to the fully ideal limit.

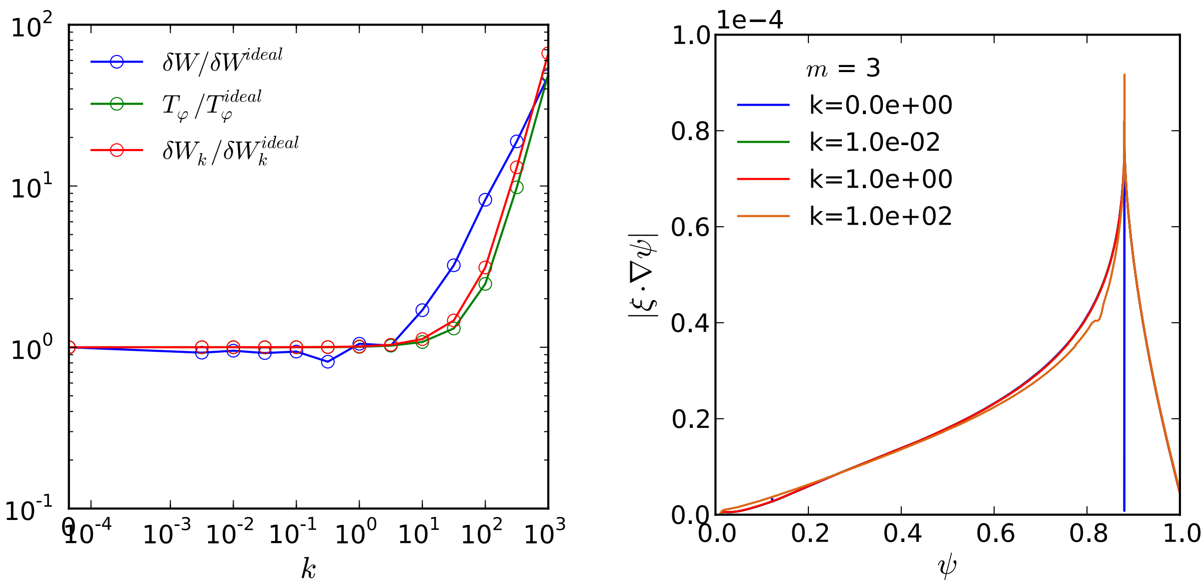

Figure 7.3: Left: The hybrid perturbed energy, NTV torque, and kinetic energy normalized to values calculated using the ideal MHD eigenfunctions as the kinetic effects in the eigenfunction calculation are scaled by the factor k. Right: The $m = 2$ poloidal harmonic of normal displacement plotted as a profile in normalized poloidal flux for sample hybrid eigenfunctions.

7.3 Discussion and Consequences

The calculation of hybrid drift-kinetic MHD eigenfunctions has significant consequences for the calculation of perturbed equilibria and toroidal torques from nonaxisymmetric fields. The splitting and suppression of resonant behavior shown in Sec. 7.2.1 allows a unified treatment of the plasma volume. When the hybrid eigenfunctions are coupled to external forces using the virtual casing method developed in IPEC [44, 45], the new force balance represents a tensor pressure perturbed equilibrium self-consistently including the anisotropic kinetic pressure tensor. Significantly, the torque calculated by PENT for a generalized perturbed equilibrium using the hybrid eigenmodes is now self-consistently finite, and has no need for ad-hoc regularization of the perturbation at the rational surface. This is the first incarnation of a *generalized perturbed equilibrium code*, which is being developed to also include resistive MHD forces in the inner layer of rational surfaces [146].

The hybrid calculation of toroidal torque caused by nonambipolar transport in nonaxisymmetric equilibria represents an important new tool for experimental application. The torque predicted by PENT in generalized perturbed equilibria is not restricted by the assumption $\delta W_k \ll \delta W$. Predictions of the torque can be rigorously extended to important experimental regimes, including NTV dominated plasmas ($T_{NTV} \gg T_{NBI}$) and high plasma pressures near or above the ideal MHD no wall limit. Further, the perturbed equilibrium dominant mode metric will be modified by the kinetic contributions of the DCON eigenmodes. This is expected to further solidify the agreement in the coupling between applied fields and the observed torque. Thus an new dominant mode metric will be available for optimization of the torque applied by nonaxisymmetric fields, both improved in the perturbative regime and extended to new regimes. This will be the focus of continued work on generalized perturbed equilibrium, and a continued collaboration between the Princeton Plasma Physics Laboratory and the DIII-D tokamak at General Atomics.

Chapter 8

Conclusion

This dissertation presents the effects of nonaxisymmetric fields on the rotation of tokamak equilibria. New experimental and computational tools were developed and presented here in order to study these effects in detail. A significant upgrade to the nonaxisymmetric magnetic diagnostics in the DIII-D tokamak enabled detailed mapping of the electromagnetic stress across the tokamak vessel wall. The combined NTV model was re-derived as an equivalent electromagnetic torque, formalizing the NTV torque as an exchange of momentum between the external axisymmetric field source and the plasma through the very electromagnetic fields measured by the new diagnostics. The extension of the combined NTV model to generalized geometry enabled accurate computation of the applied torque in the wide range of experimental geometries and kinetic regimes relevant to DIII-D and future tokamaks. The application of these tools showed that the toroidal torque from nonambipolar transport can be dominated by the plasma response to a single dominant external poloidal spectrum, enabling the reduction of the nonlinear physics to a linear coupling between the applied field and the dominant mode. This single mode model was shown to have extensive applications in the DIII-D tokamak, formally extending previous field optimization techniques to new NTV dominated regimes. Finally, the effects of the kinetic transport induced by nonaxisymmetric fields were self consistently included directly in hybrid kinetic MHD perturbed equilibrium calculations to extend the validity of the models to new high conferment regimes in which the effects of the kinetic response to nonaxisymmetric fields significantly modify the equilibrium. The products and insights of these many efforts continue to be actively used in the design and analysis of tokamak experiments, even as they themselves continue to be improved.

8.1 Summary of Dissertation

The motivation of this work is detailed in Chapter 1, which provides the basic definitions of electromagnetic torque and nonaxisymmetric equilibrium calculations. Here, the fundamental link between nonaxisymmetric fields and their observable consequences is established. The inevitability of small nonaxisymmetries and the potential for the dominance of the resulting torque over other sources of rotation in future devices provides the major motivation for accurate prediction and control of perturbed equilibria.

The direct measurement of electromagnetic (EM) torques provides additional incentive for this study through the observation of significant changes to the plasma consistent with those listed in the motivation. The measurement of nonaxisymmetric fields and their resultant torques is described in Chapter 2 and Chapter 3, with the former detailing the necessary magnetic diagnostics and data analysis techniques while the later provides practical examples. Here the focus is narrowed to the DIII-D tokamak in order to examine measurement techniques and observations in detail. Both resonant an nonresonant plasma response fields and torques are shown to be observed in the DIII-D tokamak when applying nonaxisymmetric fields. The resonant field torque agrees well with single-mode island locking theory. The observation of EM torque in rotating perturbed equilibrium motivates the need for a similarly robust model for the nonresonant physics.

In answer to the above observations, Chapter 4 and Chapter 5 present a semi-analytic model describing the kinetic energy and torque in perturbed tokamak equilibria. The root of the toroidal torque, called Neoclassical Toroidal Viscosity (NTV), is emphasized here to be nonambipolar transport across field lines. A novel derivation of the NTV as electromagnetic torque is presented in Section 4.2, solidifying the connection to the phenomenon observed in Chapter 3. A new computational model, the Perturbed Equilibrium Nonambipolar Transport code (PENT), is introduced and benchmarked in detail. The model is shown to contain new physics inaccessible to previous large aspect ratio approximations, and to agree well with the established kinetic stability codes MSIK and MARS-K in regimes of overlapping applicability. Thus validated, the PENT code coupled with DCON and IPEC provides a powerful tool for the analysis of the direct coupling of external coils to the kinetic stability and torque.

Experimental observation and first principle modeling are brought together in Chapter 6. A relatively simple NTV torque dependence on the applied field from DIII-D error field correction (EFC) coils is predicted and confirmed by experimental measurements. The dominance of a single mode plasma response results in the reduction of the nonlinear NTV physics to a linear single mode model, enabling the extension of a linear metric from resonant torque EFC to NTV dominated perturbed equilibria. The "dominant mode" metric is not fundamental to the NTV torque, however, and the torque can be manipulated in space free of resonant coupling.

Finally, the calculation of self-consistent kinetic-MHD perturbed equilibria is presented in Chapter 7. These calculations will enable more fundamental analysis of the NTV eigenspectrum and the extension of the NTV torque EF optimization to high confinement.

8.2 Opportunities Enabled

The work presented in this thesis provides new understanding of the electromagnetic coupling between applied nonaxisymmetric fields and the plasma, in turn enabling further advancements in the study and application of nonaxisymmetry in tokamaks. The hybrid kinetic-MHD perturbed equilibria presented here are the foundational generation of generalized, self-consistent, free boundary perturbed equilibrium calculation that will be improved further with additional physics such as resistivity in the inner-layer near rational surfaces. The predictions of these perturbed equilibria will be further benchmarked against experimental results, and the new capabilities used to further optimize the application of nonaxisymmetric fields for stability and rotation control. Thus will the study of the nonaxisymmetric equilibria continue to advance as it is has been advanced in this dissertation: in a coupled theoretical and experimental approach.

8.2.1 Modeling Efforts

The hybrid kinetic-MHD calculation is the first step in the formation of a Generalization Perturbed Equilibrium Code (GPEC). The coupling of PENT and DCON gives a new perturbed equilibria, self consistently including the anisotropic kinetic pressure tensor in the perturbed force balance in the presence of nonaxisymmetric fields throughout the plasma. At the rational surfaces, the kinetic effect suppress the ideal MHD singularities. These singularities are also suppressed if resistive MHD is included in the layers near the rational surfaces. A solution to the resistive Direct Criterion of Newcomb is being developed to calculate the generalized MHD stability and eigenfunctions using internal boundary conditions that account for the inner-layer dynamics. The coupling of the resistive eigenfunctions with the external forces applied by nonaxisymmetric error fields will provide relaxed perturbed equilibria, allowing saturated currents at the rational surfaces. Combining the global kinetic pressure tensor modifications with the resistive inner layer formulation is the next step in a fully generalized perturbed equilibrium calculation. The calculation of such eigenfunctions, and their coupling to the external nonaxisymmetric field in a generalized free boundary force balance, is the ultimate goal of GPEC.

8.2.2 Experimental Efforts

The predications of the hybrid perturbed equilibrium calculations will provide important insight into regimes previously unaccessible by the perturbative NTV calculations that must be carefully benchmarked with experiment. The new magnetics upgrade provides an important tool for the validation of the nonaxisymmetric equilibria plasma response calculations in new experiments similar to those originally developed to test the limits of the ideal models. In addition to the structure of the plasma response structure, measurements of the NTV torque across and beyond regimes of the perturbative calculation's applicability will provide a direct evaluation of the self-consistent calculations. Validated NTV predictions can then be used to perform more accurate and more extensive optimization of applied nonaxisymmetric fields for rotation control. Important experimental extensions include the accurate prediction of the optimal error field correction in high confinement discharged with pressures above the ideal no-wall limit, the use of a self-consistent "dominant mode" metric for single mode approximation of the applied NTV, and the multimodal optimization of the applied fields for rotation profile control. These are the crucial steps necessary for an improved understanding and the improved application of external nonaxisymmetric fields as a practical tool for stability and rotation control.

Appendix A

Magnetic Coordinates

Care has been taken to use consistent magnetic coordinates throughout this document. Each derivation has striven to use the optimal form for the task at hand while following as closely as possible the methods used in previous literature for consistency. The various forms and their connections are collected in this appendix for easy review.

The first general form of the field used in [23] is the equation for a general field with "good magnetic surfaces" as discussed in (3-4) of [101],

$$\mathbf{B} = \frac{\partial \mathbf{x}/\partial \varphi + \iota \partial \mathbf{x}/\partial \vartheta}{2\pi \mathcal{J}} = \frac{\nabla \psi_t \times \nabla \vartheta + \iota \nabla \varphi \times \nabla \psi_t}{2\pi}. \tag{A.1}$$

Here ψ_t is the toroidal flux enclosed by the magnetic surface. For a nice visualization of toroidal and poloidal flux definitions, see Fig. 1 in [104].

In this document a Clebsch representation of the field with a normalized flux is used. This form is also prevalent in the Park papers [23, 60], which freely switch between coordinate systems. To show the equivalence of the two forms, we first define the rotational transform

$$\iota \equiv \frac{1}{q} \equiv \frac{d\psi_p}{d\psi_t}. \tag{A.2}$$

Here ψ_p is the poloidal flux. Inserting into Eq. (A.1) and generalizing to a normalized surface label coordinate ψ_N,

$$\begin{aligned}
\mathbf{B} &= \frac{1}{2\pi}\left[q\nabla\psi_p \times \nabla\vartheta + \nabla\varphi \times \nabla\psi_p\right] \\
\mathbf{B} &= \frac{\psi_p'}{2\pi}\left[q\nabla\psi_N \times \nabla\vartheta + \nabla\varphi \times \nabla\psi_N\right]. \qquad \text{(A.3)}
\end{aligned}$$

Here $\psi_p' = d\psi_p/d\psi_N$. Note also that Park has a typo in [60] in which he writes the 2π in the denominator and a normalized flux definition in [23] in which the 2π factor does not appear. The Clebsch representation used int he Park papers is now found by defining the angle $\alpha = q\vartheta - \varphi$, which immediately gives

$$\mathbf{B} = \frac{\psi_p'}{2\pi}\nabla\psi_N \times \nabla\alpha. \qquad \text{(A.4)}$$

In the Clebsch representation we are free to pick a third coordinate. Here, we label it χ. In the corresponding covariant representation,

$$\mathbf{B} = B_\psi\nabla\psi_N + B_\alpha\nabla\alpha + B_\chi\nabla\chi, \qquad \text{(A.5)}$$

we can quickly calculate the new coordinate coefficient by $B_\chi = \mathbf{B}\cdot\nabla\psi_N \times \nabla\alpha\mathcal{J} = \mathcal{J}B^2$.

This thesis has used the Clebsch representation found in most of the Boozer literature [147, 106, 104],

$$\mathbf{B} = \nabla\alpha \times \nabla\psi \qquad \text{(A.6)}$$

where ψ is the poloidal flux, the new toroidal angle is $\alpha = \phi - q\vartheta$ and the angle coordinates have been normalized to unity. This is also the representation ultimately used in PENT.

Note that the Clebsch representation is completely general in describing a toroidal field with good magnetic surfaces. The coordinate system is thus fully described by its Jacobian, $\mathcal{J}^{-1} = \nabla\psi \times \nabla\vartheta \cdot \nabla\varphi$, and any generalized coordinates can be converted to Clebsch form.

The power of the Jacobian has been exploited many times in this thesis in the approach to bounce-integration: a line integral has been done varying poloidal angle according to the line-element definitions in the appendix of [104]

$$d\mathbf{x}|_{i,j} = \frac{\partial \mathbf{x}}{\partial x^k} dx^k \to dl = \frac{\partial \mathbf{x}}{\partial \vartheta} d\vartheta = \mathcal{J} B d\vartheta. \tag{A.7}$$

Thus, the derivations throughout this thesis are not specific to any one coordinate system, but rather make use of a common form while containing the system's unique information within the Jacobian.

Appendix B

The Drift-Kinetic Equation

B.1 Bounce Averaged Quantities

This section provides the derivations of the bounce averaged kinetic gyro-drift-kinetic (GDK) equation used in Chapter 4. Starting from Eq. (4.6),

$$\mathbf{v}_\parallel \cdot \nabla \delta f + \mathbf{v}_d^\alpha \frac{\partial \delta f}{\partial \alpha} - C\delta f = -\mathbf{v}_d^\psi \frac{\partial f}{\partial \psi}, \tag{B.1}$$

we first make use of the separable perturbed distribution (4.8) and Clebsch coordinate field $\mathbf{B}(\psi, \vartheta, \alpha) = \nabla\alpha \times \nabla\psi$ to re-distribute the phase dependence,

$$\left[i \left(-2\pi(\ell - \sigma n q) \frac{\partial h}{\partial \vartheta} \right) v_\parallel \frac{\mathbf{B}}{B} \cdot \nabla\vartheta - i2\pi n \mathbf{v}_d \cdot \nabla\alpha + \nu \right] \delta f_\ell = -\mathcal{P}_\ell{}^{-1} \mathbf{v}_d \cdot \nabla\psi \frac{\partial f}{\partial \psi}, \tag{B.2}$$

and then apply the bounce average $\langle A \rangle_b \equiv \oint \left(A dl / v_\parallel \right) / \oint \left(dl / v_\parallel \right) = \frac{\omega_b}{2\pi} \oint d\vartheta A \mathcal{J} B / v_\parallel$ explicitly to each term. Inserting the phase function $h(\mathbf{v}, \vartheta) = \left(\int_0^\vartheta d\theta \mathcal{J} B v_d^\alpha / v_\parallel \right) / \oint d\theta \mathcal{J} B v_d^\alpha / v_\parallel$, the first term on the left hand side is

$$\begin{aligned}
\left\langle \mathbf{v}_\parallel \cdot \nabla \delta f \right\rangle &= \left\langle i \left(-2\pi(\ell - \sigma n q) \frac{\partial h}{\partial \vartheta} \right) \delta f_\ell v_\parallel \frac{\nabla\alpha \times \nabla\psi}{B} \cdot \nabla\vartheta \right\rangle \\
&= i \frac{\omega_b}{2\pi} \oint d\vartheta \frac{\mathcal{J}B}{v_\parallel} \left[v_\parallel \frac{1}{\mathcal{J}B} \left(-2\pi(\ell - \sigma n q) \frac{\frac{\mathcal{J}B}{v_\parallel} v_d^\alpha}{\oint d\theta \mathcal{J} B v_d^\alpha / v_\parallel} \right) \delta f_\ell \right] \\
&= -i\omega_b \left((\ell - \sigma n q) \frac{\oint d\theta \mathcal{J} B v_d^\alpha / v_\parallel}{\oint d\theta \mathcal{J} B v_d^\alpha / v_\parallel} \right) \delta f_\ell \\
&= -i\omega_b \left(\ell - \sigma n q\right) \delta f_\ell . \qquad \text{(B.3)}
\end{aligned}$$

The second GDK term requires the use of the drift velocity (4.10), which contains the gradient, curvature and electric drifts. Taking the α-contravariant component,

$$\begin{aligned}
\mathbf{v}_d \cdot \nabla\alpha &= \frac{v_\parallel}{B} \nabla \times \left(\rho_\parallel \mathbf{B} \right) \cdot \nabla\alpha - \frac{M v_\parallel^2}{e B^4} \left[\mathbf{B} \cdot (\nabla \times \mathbf{B}) \right] \nabla\psi \times \nabla\alpha \cdot \nabla\alpha \\
&= \frac{v_\parallel}{B} \left[\frac{\partial \left(\rho_\parallel \mathbf{B} \cdot \nabla\alpha \times \nabla\psi \mathcal{J} \right)}{\partial \psi} - \frac{\partial \left(\rho_\parallel \mathbf{B} \cdot \nabla\vartheta \times \nabla\alpha \mathcal{J} \right)}{\partial \vartheta} \right] \nabla\psi \times \nabla\vartheta \cdot \nabla\alpha \\
\mathbf{v}_d \cdot \nabla\alpha &= \frac{v_\parallel}{\mathcal{J}B} \left[\frac{\partial \left(\rho_\parallel \mathcal{J} B^2 \right)}{\partial \psi} - \frac{\partial \left(\rho_\parallel \mathbf{B} \cdot \nabla\vartheta \times \nabla\alpha \mathcal{J} \right)}{\partial \vartheta} \right] . \qquad \text{(B.4)}
\end{aligned}$$

The $\partial/\partial\vartheta$ term is zero after the bounce-average,

$$\begin{aligned}
\frac{\omega_b}{2\pi} \oint d\vartheta \frac{\mathcal{J}B}{v_\parallel} \left[- \frac{v_\parallel}{\mathcal{J}B} \frac{\partial \left(\rho_\parallel \mathbf{B} \cdot \nabla\vartheta \times \nabla\alpha \mathcal{J} \right)}{\partial \vartheta} \right] &= -\frac{\omega_b}{2\pi} \oint d\vartheta \frac{\partial \left(\rho_\parallel \mathbf{B} \cdot \nabla\vartheta \times \nabla\alpha \mathcal{J} \right)}{\partial \vartheta} \\
&= \frac{\omega_b}{2\pi} \left(-\rho_\parallel B_\psi \Big|_{\vartheta_0} + \rho_\parallel B_\psi \Big|_{\vartheta_0} \right) \\
&= 0, \qquad \text{(B.5)}
\end{aligned}$$

so the bounce averaged GDK contains only the term,

$$\langle -i 2\pi n \mathbf{v}_d \cdot \nabla\alpha \delta f_\ell \rangle = i 2\pi n \delta f_\ell \frac{\omega_b}{2\pi} \oint d\vartheta \frac{\mathcal{J}B}{v_\parallel} \frac{v_\parallel}{\mathcal{J}B} \left[\frac{\partial \left(\rho_\parallel \mathcal{J} B^2 \right)}{\partial \psi} \right] . \qquad \text{(B.6)}$$

Converting back into the explicit electric and magnetic drifts using $\nabla\mathcal{E} = 0 = M v_\parallel \nabla v_\parallel +$

$\mu\nabla B + e\nabla\Phi$ yields the final terms,

$$\begin{aligned}
\langle -i2\pi n \mathbf{v}_d \cdot \nabla\alpha \delta f_\ell \rangle_b &= -i2\pi n \delta f_\ell \frac{\omega_b}{2\pi} \oint d\vartheta \frac{\mathcal{J}B}{v_\parallel} \left[\frac{M v_\parallel}{e} \frac{\partial v_\parallel}{\partial \psi} + \frac{M v_\parallel^2}{eB} \left(\frac{\partial B}{\partial \psi} + \frac{B}{\mathcal{J}} \frac{\partial \mathcal{J}}{\partial \psi} \right) \right] \\
&= -in\delta f_\ell \omega_b \oint d\vartheta \frac{\mathcal{J}B}{v_\parallel} \left[-\frac{\partial \Phi}{\partial \psi} - \frac{\mu}{e} \frac{\partial B}{\partial \psi} + \frac{M v_\parallel^2}{eB} \left(\frac{\partial B}{\partial \psi} + \frac{B}{\mathcal{J}} \frac{\partial \mathcal{J}}{\partial \psi} \right) \right] \\
&= i2\pi n \delta f_\ell \frac{\partial \Phi}{\partial \psi} + -in\delta f_\ell \omega_b \oint d\vartheta \frac{\mathcal{J}B}{v_\parallel} \left[\frac{(2E - 3\mu B)}{eB} \frac{\partial B}{\partial \psi} \right. \\
&\quad \left. + (2E - 2\mu B) \frac{1}{e\mathcal{J}} \frac{\partial \mathcal{J}}{\partial \psi} \right] \\
&= -in(\omega_E + \omega_D)\delta f_\ell. \qquad \text{(B.7)}
\end{aligned}$$

Here we have use the definition of the electric (ω_E) and magnetic (ω_D) precession frequencies given by Eqs. (4.12) and (5.13) respectively.

The bounce averaging of the third term on the left-hand-side of the GDK equation is a trivial multiplication by unity. Finally, the bounce averaged right-hand-side is,

$$\begin{aligned}
\left\langle -\mathcal{P}_\ell^{-1} \mathbf{v}_d \cdot \nabla\psi \frac{\partial f}{\partial \psi} \right\rangle_b &= -\frac{\partial f}{\partial \psi} \frac{\omega_b}{2\pi} \oint d\vartheta \frac{\mathcal{J}B}{v_\parallel} \mathcal{P}_\ell^{-1} \frac{v_\parallel}{\mathcal{J}B} \left[\frac{\partial \left(\rho_\parallel \mathbf{B} \cdot \nabla\psi \times \nabla\vartheta \mathcal{J} \right)}{\partial \vartheta} \right. \\
&\quad \left. - \frac{\partial \left(\rho_\parallel \mathbf{B} \cdot \nabla\alpha \times \nabla\psi \mathcal{J} \right)}{\partial \alpha} \right], \qquad \text{(B.8)}
\end{aligned}$$

and again one of the terms is identically zero under the bounce average. These $\partial/\partial\vartheta$ terms are known to contribute to transport as ambipolar neoclassical drifts [106], and can be separated into a separate gyro drift kinetic equation. For now, we concentrate only on the nonaxisymmetric effects left after bounce averaging. We are left with,

$$\begin{aligned}
\left\langle -\mathcal{P}_\ell^{-1} \mathbf{v}_d \cdot \nabla\psi \frac{\partial f}{\partial \psi} \right\rangle_b &= \frac{\partial f}{\partial \psi} \frac{\omega_b}{2\pi} \oint d\vartheta \frac{\mathcal{J}B}{v_\parallel} \mathcal{P}_\ell^{-1} \frac{v_\parallel}{\mathcal{J}B} \frac{\partial}{\partial \alpha} \left(\rho_\parallel \mathcal{J} B^2 \right) \\
&= \frac{\partial f}{\partial \psi} \frac{\omega_b}{2\pi e} \frac{\partial}{\partial \alpha} \oint d\vartheta \mathcal{P}_\ell^{-1} \mathcal{J} B M v_\parallel, \qquad \text{(B.9)}
\end{aligned}$$

which contains the action $J_{\pm\ell} = \oint dl M v_\parallel \mathcal{P}_\ell^{\pm}$ .The zeroth order action is assumed to be axisymmetric. This means the highest order derivative in the toroidal angle is that of the variation of the action

$$\frac{\partial J_\ell}{\partial \alpha} = \frac{\partial \delta J_\ell}{\partial \alpha} + \mathcal{O}(\delta^2). \tag{B.10}$$

It is also possible to expand the integrand in a similar manor as in the $\mathbf{v}_d \cdot \nabla \alpha$ term,

$$\begin{aligned} \left\langle -\mathcal{P}_\ell^{-1} \mathbf{v}_d \cdot \nabla \psi \frac{\partial f}{\partial \psi} \right\rangle_b = \frac{\partial f}{\partial \psi} \frac{\omega_b}{2\pi e} \oint d\vartheta \frac{\mathcal{J} B}{v_\parallel} \mathcal{P}_\ell^{-1} \Bigg[(2E - 3\mu B) \frac{1}{B} \frac{\partial B}{\partial \alpha} \\ + (2E - 2\mu B) \frac{1}{\mathcal{J}} \frac{\partial \mathcal{J}}{\partial \alpha} \Bigg], \end{aligned} \tag{B.11}$$

which can be found in Chapter 4. In this case we note that the derivatives again are at least first order, requiring everything else to be axisymmetric for a first order expression and thus the derivatives can be pulled out of the integral. Doing so, we arrive at the compact expression

$$\left\langle -\mathcal{P}_\ell^{-1} \mathbf{v}_d \cdot \nabla \psi \frac{\partial f}{\partial \psi} \right\rangle_b = \frac{\partial f}{\partial \psi} \frac{\omega_b}{2\pi e} \frac{\partial \delta J_{-\ell}}{\partial \alpha}. \tag{B.12}$$

Replacing each of the bounce averaged expressions into the gyro drift-kinetic equation,

$$\left[-i\omega_b(\ell - nq) - in(\omega_E + \omega_B) + \nu\right] \delta f_\ell = \frac{\omega_b}{2\pi e} \frac{\partial \delta J_{-\ell}}{\partial \alpha} \frac{\partial f}{\partial \psi},$$

we explicitly solve for the first order distribution function,

$$\delta f_\ell = \frac{-\omega_b/(2\pi e)}{i\left[\omega_b\left(\ell - \sigma n q\right) + n\left(\omega_E + \omega_B\right)\right] - \nu} \frac{\partial \delta J_{-\ell}}{\partial \alpha} \frac{\partial f}{\partial \psi}. \tag{B.13}$$

This is the first order solution because it contained α derivatives of the perturbed action. All other bounce averaged quantities are taken as their axisymmetric values.

Appendix C

DCON Matrix Equations

C.1 Perturbed Action

The perturbed action used in Chapter 7 took the form,

$$\delta J = \mathcal{W}_X^T \Xi^{\psi\prime} + \mathcal{W}_Y^T \Xi^{\psi} + \mathcal{W}_Z^T \Xi^{\alpha}, \tag{C.1}$$

where $\mathcal{W}_{X,Y,Z}^{\dagger}$ are $M-$row vector operators on the Clebsch displacement column vectors $\Xi^{\psi\prime,\psi,\alpha}$. Here we derive this form, giving the integral equations used in PENT to calculate the operators and thus the kinetic coefficient matrices used to compute hybrid kinetic MHD eigenfunctions.

The bounce averaged perturbed action as given in Eq. (4.15) is,

$$\delta J_{\pm\ell} = \oint d\vartheta \frac{\mathcal{J}B}{v_\parallel} \mathcal{P}_\ell^{\pm 1} \left[(2E - 3\mu B) \frac{\delta B}{B} + (2E - 2\mu B) \frac{1}{\mathcal{J}} \nabla \cdot \boldsymbol{\xi}_\perp \right]. \tag{C.2}$$

Using the relation,

$$\delta B = -B \left[\nabla \cdot \boldsymbol{\xi}_\perp + \boldsymbol{\kappa} \cdot \boldsymbol{\xi}_\perp \right], \tag{C.3}$$

the action integrand can be written in terms of the Clebsch displacements,

$$\delta J_{\pm\ell} = \oint d\vartheta \frac{\mathcal{J}B}{v_\parallel}\mathcal{P}_\ell^{\pm1}\left[-\left(2E-3\mu B\right)\boldsymbol{\kappa}\cdot\boldsymbol{\xi}_\perp + \mu B\frac{1}{\mathcal{J}}\nabla\cdot\boldsymbol{\xi}_\perp\right]. \tag{C.4}$$

Both the curvature dot product and divergence can be expressed by geometric operators on the DCON displacement $M-$vectors,

$$\mathcal{J}B^2\boldsymbol{\kappa}\cdot\boldsymbol{\xi}_\perp = \left(\mathbf{S}\Xi^\psi + \mathbf{T}\Xi^\alpha\right)^\dagger\mathbf{M}, \tag{C.5}$$

$$\mathcal{J}B^2\nabla\cdot\boldsymbol{\xi}_\perp = \left(\mathbf{X}\Xi^{\psi\prime} + \mathbf{Y}\Xi^\psi + \mathbf{Z}\Xi^\alpha\right)^\dagger\mathbf{M}, \tag{C.6}$$

where the final product with $\mathbf{M}_m = e^{im\vartheta}$ is used as the discrete Fourier sum. Inserting Eqs. (C.5) and (C.6) into Eq. (C.4) reproduces the form in Eq. (C.1) with the operators,

$$\mathcal{W}_X^T = \mathcal{M}^T\mathbf{X}, \tag{C.7}$$

$$\mathcal{W}_Y^T = \mathcal{M}^T\left(3\mathbf{S}+\mathbf{Y}\right) - 2\mathcal{E}^T\mathbf{S}, \tag{C.8}$$

$$\mathcal{W}_Z^T = \mathcal{M}^T\left(3\mathbf{T}+\mathbf{Z}\right) - 2\mathcal{E}^T\mathbf{T}. \tag{C.9}$$

Here the bounce integration has been factored into the energy dependent $M-$vectors,

$$\mathcal{M}_m = \frac{1}{2}\sqrt{\frac{M}{2}}\sum_{\ell\sigma}\oint d\vartheta\frac{\mu}{\sqrt{E-\mu B}}\mathcal{P}_{\ell,\sigma}e^{i(m-nq)\vartheta}, \tag{C.10}$$

$$\mathcal{E}_m = \frac{1}{2}\sqrt{\frac{M}{2}}\sum_{\ell\sigma}\oint d\vartheta\frac{E/B}{\sqrt{E-\mu B}}\mathcal{P}_{\ell,\sigma}e^{i(m-nq)\vartheta}. \tag{C.11}$$

Each component of the matrices formed by outer products of the vectors (C.7)-(C.9) represents a unique m-coupling throughout the two dimensional energy space (E,μ), and thus is integrated individually in PENT in order to obtain the corresponding component of the kinetic coefficient matrices (7.30)-(7.35) for use in the DCON Euler-Lagrange eigenfunction formalism.

Appendix D

Computational Treatment of Integrable Singularities

This section details the computational methods used in the Perturbed Equilibrium Nonambipolar Transport (PENT) code to treat the various integrable singularities internal to a combined NTV theory calculation of the toroidal torque due to nonambipolar transport in the presence of nonaxisymmetric fields.

D.1 Spatial Integration

D.1.1 Treatment of Rational Surfaces

The PENT code was developed in conjunction with the DCON-IPEC suite, relying on IPEC to calculate the nonaxisymmetric perturbations used in the perturbed action (4.15). Although not explicitly part of PENT, it is important to address particular properties of the perturbed solution that have significant consequences for the NTV torque. The linearized perturbed force balance equation (Euler-Lagrange equation for δW_p) has an inherent singularity at rational surfaces with $q = m/n$. IPEC uses ideal constraints to enforce that the resonant component of the normal flux is zero at rational surfaces, which corresponds to complete shielding of magnetic islands by singular currents at these ration surfaces [45]. The corresponding normal displacement is either finite or integrable singular at each rational surface. The tangential displacement, however, is not involved in the force balance or energy minimization and consequently is allowed to have non-integrable singularities at the rational surface. Although self consistent within the Ideal MHD model, these solutions would lead to infinite NTV.

The calculation of NTV torque in PENT is not a self consistent modification of the force balance. It is thus an extension beyond the strict applicability of the ideal perturbed equilibrium calculated by IPEC. Although this is valid within the limits of the perturbative approach outlined in Chapter 5, the unphysical singularities in tangential displacement from the ideal perturbed equilibrium would result in infinite torque if directly extended to the combined NTV model. This can be seen by examining the definition of the Lagrangian displacement,

$$\delta B = \frac{\chi'}{\mathcal{J}B}\left(\delta B_\vartheta + q\delta B_\varphi\right) + \xi^\psi \frac{\partial B}{\partial \psi} + \xi^\vartheta \frac{\partial B}{\partial \vartheta}, \tag{D.1}$$

which is singular at the rationals where ξ^ϑ is singular. This is resolved in the IPEC-PENT coupling by asserting a regularization of the displacement in a narrow region near the rational surface [45]. Defining a characteristic width σ in units of $m - nq$, IPEC displacements are regularized as

$$\xi_{mn}^{\vartheta,\varphi} \rightarrow \xi_{mn}^{\vartheta,\varphi} \frac{\left(m-nq\right)^2}{\left(m-nq\right)^2 + \sigma^2}. \tag{D.2}$$

The appropriate physical width is on the order of the ion gyro-radius, and it has been shown that the integral NTV torque is not sensitive to the value used for a wide range of $\sigma \sim 10^{-3} - 10^{-1}$ in experimental plasmas [45].

D.1.2 Treatment of Turning Points in Bounce Averages

The most fundamental integral of the combined NTV theory, the bounce average, contains integrable singularities. The bounce average is defined in Chapter 4,

$$\langle A \rangle_b \equiv \frac{\oint A dl/v_\parallel}{\oint dl/v_\parallel} = \frac{\omega_b}{2\pi}\oint d\vartheta \frac{A\mathcal{J}B}{v_\parallel}. \tag{D.3}$$

In writing this definition in terms of a integration over poloidal angle the line integration has been done according to the line-element definition from Eq. (A.7), and we have defined,

$$\omega_b \equiv \frac{2\pi}{\oint d\vartheta \mathcal{J}B/v_\parallel}. \tag{D.4}$$

The poloidal integral (D.3) appears in both the basic phase space representation of the torque (4.5) and the drift kinetic equation (4.7) and must be explicitly calculated for three distinct quantities,

$$\frac{2\pi}{\omega_b} = \sqrt{\frac{M}{2xT}}\int_{\vartheta_{b-}}^{\vartheta_{b+}} d\vartheta \frac{(2-\sigma)\mathcal{J}B}{\sqrt{1-\Lambda\frac{B}{B_0}}}, \tag{D.5}$$

$$\frac{\omega_D}{\omega_b} = \sqrt{\frac{MxT}{2}}\int_{\vartheta_{b-}}^{\vartheta_{b+}} d\vartheta \frac{(2-\sigma)\mathcal{J}B}{\sqrt{1-\Lambda\frac{B}{B_0}}}\left[\left(1-\frac{3}{2}\Lambda\frac{B}{B_0}\right)\frac{2}{eB}\frac{\partial B}{\partial\psi} + \left(1-\Lambda\frac{B}{B_0}\right)\frac{2}{e\mathcal{J}}\frac{\partial \mathcal{J}}{\partial\psi}\right], \tag{D.6}$$

$$\begin{aligned}\frac{\delta J_{\pm\ell}}{\omega_b} = \sqrt{\frac{MxT}{2}}\int_{\vartheta_{b-}}^{\vartheta_{b+}} d\vartheta \frac{\left(\mathcal{P}_\ell^{\pm 1}+\sigma\mathcal{P}_\ell^{\mp 1}\right)\mathcal{J}B}{\sqrt{1-\Lambda\frac{B}{B_0}}}&\left[\left(1-\frac{3}{2}\Lambda\frac{B}{B_0}\right)\frac{2}{B}\delta B\right.\\ &\left.+\left(1-\Lambda\frac{B}{B_0}\right)\frac{2}{\mathcal{J}}\nabla\cdot\boldsymbol{\xi}\right].\end{aligned} \tag{D.7}$$

where we have used the normalized energy coordinates ($x = E/T, \Lambda = \mu B_0/E$), and $\sigma = 1(0)$ for passing(trapped) particles consistent with Chapter 4. Here each integration follows the orbit motion from the lower bounce point ϑ_{b-} to upper bounce point ϑ_{b+} (identical to the return leg except for the phase factor in the working thin banana approximation) and each integrand is singular at the bounce points, $B\left(\vartheta_{b\pm}\right) = B_0/\Lambda$.

The bounce point singularities above are integrable. A common practice is to approximate the bounce point contribution using an ad-hock Taylor expansion of the singular denominator while assuming the numerator of the integrand is relatively constant near the bounce point. For the general case with a slowly varying numerator $C(\vartheta)$ the upper bounce point contribution from $\vartheta_{b+} - \epsilon$ for small ϵ is given by,

$$\begin{aligned} \int_{\vartheta_{b+}-\epsilon}^{\vartheta_{b+}} d\vartheta \frac{C(\vartheta)}{\sqrt{1-\Lambda\frac{B}{B_0}}} &\approx \int_{\vartheta_{b+}-\epsilon}^{\vartheta_{b+}} d\vartheta \frac{C(\vartheta_{b+})}{\sqrt{\frac{\Lambda}{B_0}\left.\frac{\partial B}{\partial\vartheta}\right|_{\theta_{b+}}\left(\vartheta_{b+}-\vartheta\right)}} \\ &\approx 2C(\vartheta_{b+})\epsilon^{1/2}\left[\frac{\Lambda}{B_0}\left.\frac{\partial B}{\partial\vartheta}\right|_{\vartheta_{b+}}\right]^{-1/2}, \end{aligned} \tag{D.8}$$

and a similar expansion can be done for the lower bounce point contribution from ϑ_{b-} to $\vartheta_{b-} + \epsilon$.

Looking at the specific cases above, we can write explicitly the denominator of each singular term,

$$C_{\omega_b}\left(\vartheta_{b\pm}\right) = \left(2-\sigma\right)\mathcal{J}B_0/\Lambda, \tag{D.9}$$

$$C_{\omega_D}\left(\vartheta_{b\pm}\right) = -\left(2-\sigma\right)\mathcal{J}\left.\frac{\partial B}{\partial\psi}\right|_{\vartheta_{b\pm}}, \tag{D.10}$$

$$C_{\delta J_\ell}\left(\vartheta_{b\pm}\right) = \mathcal{J}\delta B\cos\left(2\pi\ell h\left(\vartheta_{b\pm}\right)\right). \tag{D.11}$$

Here the phase factor term in the action is real ($\sigma = 0$), as this procedure is only necessary if the particle is trapped. Using these values in Eq. (D.8) and the lower bounce point equivalent allows numerical convergence with the square root of ϵ in the regime of validity given by Eq. (D.13). With this insight, PENT employs a change of variables $z_\pm = \left(\vartheta_{b\pm} - \vartheta\right)^2$ and directly computes,

$$\oint d\vartheta \frac{C(\vartheta)}{\sqrt{1-\Lambda\frac{B(\vartheta)}{B_0}}} = \int_{\vartheta_{b-}^{1/2}}^{0} dz_- \frac{2\sqrt{z_-}C(z_-)}{\sqrt{1-\Lambda\frac{B(z_-)}{B_0}}} + \int_0^{\vartheta_{b+}} d\vartheta \frac{2\sqrt{z_+}C(z_+)}{\sqrt{1-\Lambda\frac{B(z_+)}{B_0}}}. \tag{D.12}$$

This method, by approaching the singularity quadratically, is essentially equivalent to using the taylor method above and approaching linearly in ϑ step size and uses the same assumptions as to the order of the singularity.

This taylor expansion (and direct method) above assumes the angular distance of approach, ϵ, is small,

$$\epsilon \ll \frac{\partial B/\partial\vartheta}{\partial^2 B/\partial\vartheta^2}. \tag{D.13}$$

This condition, however, can fail at the passing-trapped boundary and at the deeply trapped limit where the bounce points are on extrema in $B(\vartheta)$. In practice, poloidal cross-sections of axisymmetric tokamak equilibria flux surfaces are regular, smooth, convex curves and it is safe to assume that $B(\vartheta)$ has a non-zero second derivative everywhere on any given surface. Consequently the only divergent points in energy space are the two boundaries (trapped-passing, and deeply trapped) for which a taylor expansion similar to (D.8) would diverge as the natural logarithm of ϵ. Note that for the circular limit the first derivative of the field approaches zero linearly and the bounce contribution consequently limits to a constant as the bounce point approaches the extrema. PENT separates trapped and passing contributions in practice, explicitly integrating between these boundaries uses a variable of closest approach in Λ to ensure the energy integration approaches but does not lie on the singular point. Convergence is insensitive to this parameter, as particles near the trapped passing boundary contributes negligible torque and the Λ dependence is relatively flat in the deeply trapped limit.

D.2 Energy Space Integration

Sharp resonances in energy space (x, Λ) represent a computational challenge in accurately calculating the NTV integrals (4.20). These resonances are not singular in physical plasmas where there must exist some non-zero collisionality. However, the collisionality may become so low as to asymptotically approach singularity and it is often useful for theoretical understanding to consult collisionless results.

D.2.1 Treatment of Energy Resonances

The explicit location of resonances in energy ($x = E/T$) and the resulting poles contribution to the energy integral are given in Sec. 4.4.1. In practice, however, the resonances still represent a computational challenge. Even when the pole contribution is known, approaching these resonances results in sharp peaks in the integrand requiring high resolution for accurate integration. To gain intuition as to the possible severity of these peaks, we insert Eq. (4.32) into the real part (corresponding to the torque) of the resonance operator (4.21),

$$\Re\left(\mathcal{R}_{T\ell}\right) \propto \frac{\nu x^{5/2} e^{-x}}{\left[xn\hat{\omega}_D + n\omega_E\right]^2 - \nu^2}. \tag{D.14}$$

Here the energy dependence has been explicitly pulled out of the magnetic precession $\hat{\omega}_D = \omega_D/x$, and only precession resonance ($\ell = 0$) trapped particles are considered for clarity. In the limit of zero collisionality we find delta function behavior in x,

$$\Re\left(\mathcal{R}_{T\ell}\right) \propto \frac{\nu_i}{\left[xn\hat{\omega}_D + n\omega_E\right]^2 - \nu^2} \xrightarrow{\nu\to 0} \delta\left(-\frac{\omega_E}{\hat{\omega}_D}\right). \tag{D.15}$$

Noting that the resonance condition is quadratic in x, it is sufficient to evaluate the resonance operator on a cubic grid concentrated at the known resonance locations as long as there is collisionality. Optionally, PENT pre-determines these locations and forms a unique cubic grid for spline representations of the integrand for each energy integral. Otherwise, dynamic integration uses the publicly available LSODE package [140]. The upper bound of the computational energy integration is set by an adjustable parameter, but is typically on the order of 100 (corresponding to a reduction in the denominator of at least 3×10^{-18}).

In the case of identically zero collisionality, poles located on the positive real axis cause numeric instabilities. In these cases, PENT adjusts the integration path in order to avoid poles. This also has the benefit of eliminating the need to explicitly include the residues (4.36). The new path consists of a small step off the real axis ix_i along the imaginary axis, continuation to $\infty + ix_i$, and a final perpendicular step back to the real axis. The modified integration contour is shown schematically in Fig. D.1. This feature can be used to improve numerical stability in cases with extremely low collisionality, and is controlled by an input parameter specifying the step x_i.

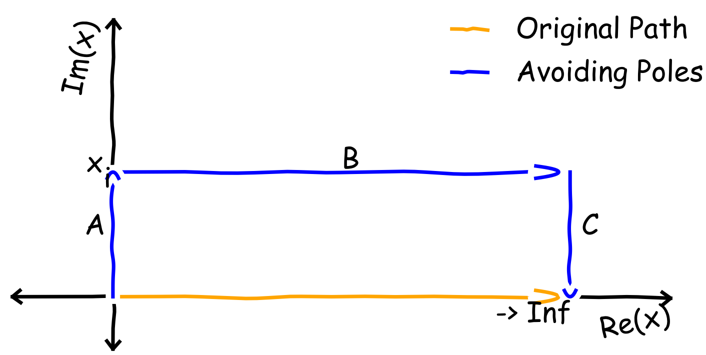

Figure D.1: Schematic of the energy integration path used in PENT for collisional (orange) and collisionless (blue) plasmas with kinetic resonances on the positive real axis. The final leg of the collisionless path, at positive ∞, does not contribute to the integral.

D.2.2 Treatment of Pitch Angle Resonances

Although possible poles in the space (x, Λ) are resolved using the integration techniques described above, δ-function resonances may exist in the Λ integrand as expressed in (4.20). This can be seen schematically,

$$\Re\left(\mathcal{R}_{T\ell}\right) \propto \frac{\nu}{\left(n\omega_D\right)^2 - \nu^2} \xrightarrow{\nu\to 0} \delta\left(\omega_D\right), \tag{D.16}$$

considering precession resonances only ($\ell = 0$) without electric precession.

These, as with the energy resonances, are of great importance at low collisionality and are responsible for distinct NTV regime transitions (see Sec. 4.4.2). As in the case of the energy resonances, PENT integrates over these resonances either by using cubic spline representations of the integrand on a cubic grid concentrated on pre-determined resonance locations or by using dynamic integration techniques. The resonance condition in Λ, approximately $\omega_D/x = 0$ in the superbanana plateau, is solved numerically in the former case since it is dependent of the geometry of the equilibrium flux surfaces.

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